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SUBJECT	BLUETOOTH NOTES THEORY CONCEPT
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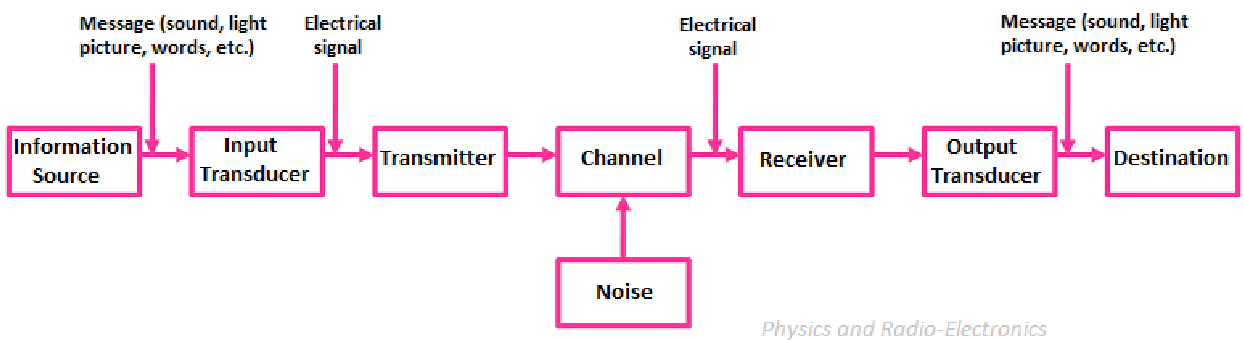
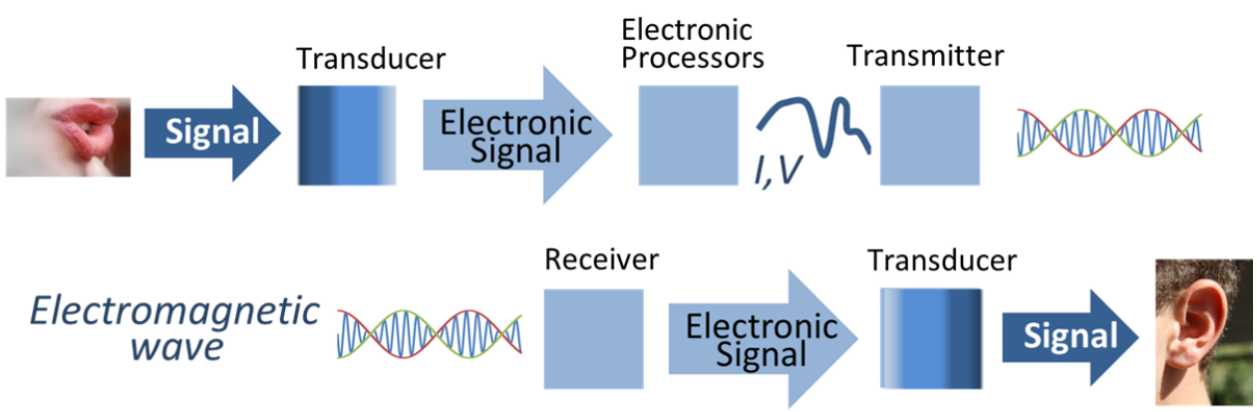
BLUETOOTH TECHNLOGY

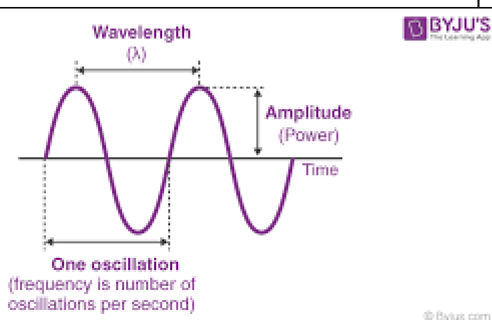
COMMUNICATION





Analog Communication	Digital Communication
Analog communication uses analog signals for the transmission of information.	Digital communication uses digital signals for the transmission of information.
Analog communication uses signals that can be	Digital communication uses signals that can be

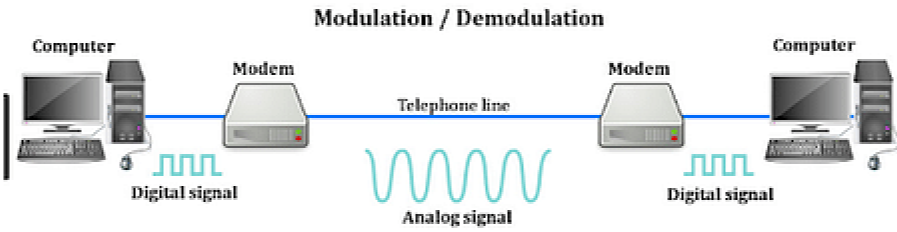
represented by sine waves.	represented by square waves.
Analog communication signals consist of continuous values.	Digital communication signals consist of discrete values.
Analog communications are affected by noise.	Digital communications are not affected by noise.
The probability of error in analog signals is high due to parallax.	Digital signals are virtually immune to parallax errors.
Analog communication requires complicated hardware.	Digital communication requires less complicated hardware.
Analog communication is relatively cheaper as compared to digital communication.	Digital communication is costly in nature.
Low bandwidth is required for analog communication.	High bandwidth is required for digital communication.
High power is required for analog communication.	Digital communication requires comparatively less power.
Analog communication systems are less portable as they contain heavy and complicated hardware.	Digital communication systems are more portable as compared to analog communication systems.
Examples: - Sound shows analog signals	Examples: - Computers use digital signals.

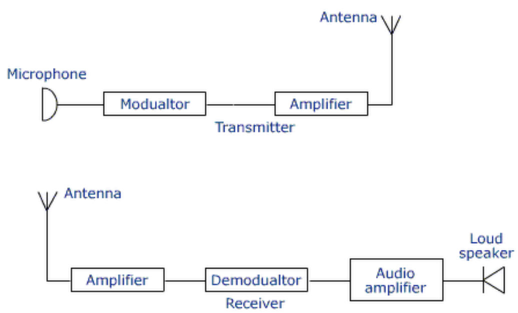


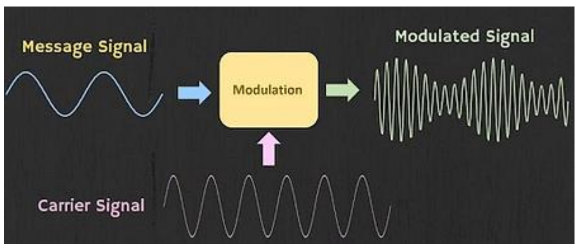
FREQUENCY = number of cycle per second , unit hz

Wavelength = The distance b/w the trough or crest of a wave motion

Modulation and demodulation







- **Modulation** is the process of encoding information in a transmitted signal,
- while **demodulation** is the process of extracting information from the transmitted signal.
- In order to transmit the message signal, we generally use a high-frequency signal called a **carrier signal.** The process is referred to as modulation**.**

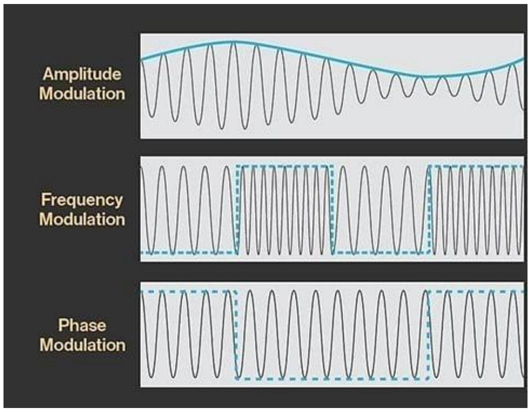
What is the purpose of carrier frequency?

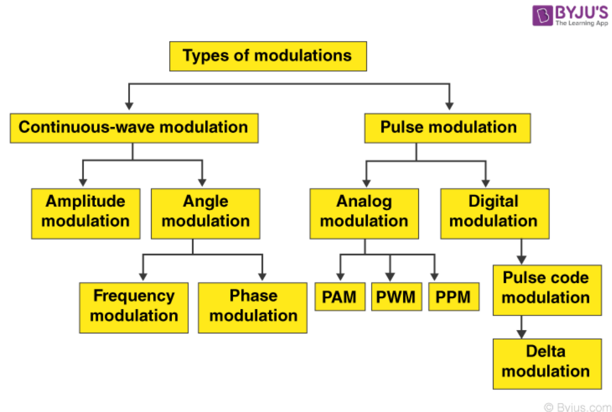
- Carrier frequency is defined as the frequency of a carrier wave, measured in cycles per second, or Hertz, that is modulated **to transmit signals**.

Types of Modulations

Modulation can be broken down into three different types:

1. **Frequency Modulation:** Frequency modulation occurs when a signal's amplitude and frequency have a constant state, but the carrier wave frequency changes or varies.
2. **Phase Modulation:** Phase modulation refers to a situation where the phase of a high-frequency carrier wave varies or changes due to a phase shift in the modulated signal, while the amplitude and frequency remain unchanged.
3. **Amplitude Modulation:** Amplitude modulation refers to a change in the amplitude of a carrier wave caused by a change in the modulation of the signal. The signal phase and frequency remain constant.





Advantages of Modulation

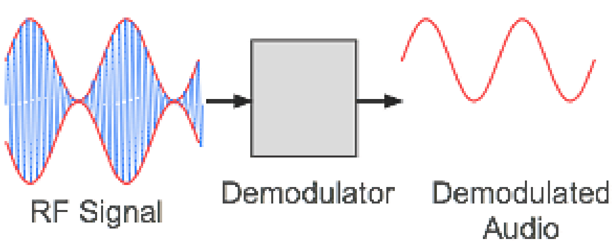
1. Modulation helps to reduce the size of the antenna.
2. In the case of Modulation, signals do not get mixed up.
3. It increases the communication ranges.
4. Necessary adjustments in the bandwidth signal can be done in modulation.
5. It also improves the reception quality.

BENEFITS of MODULATION

Allow multiple signals, to reduce antenna size, reduce interface, long distance communication.

What is Demodulation?

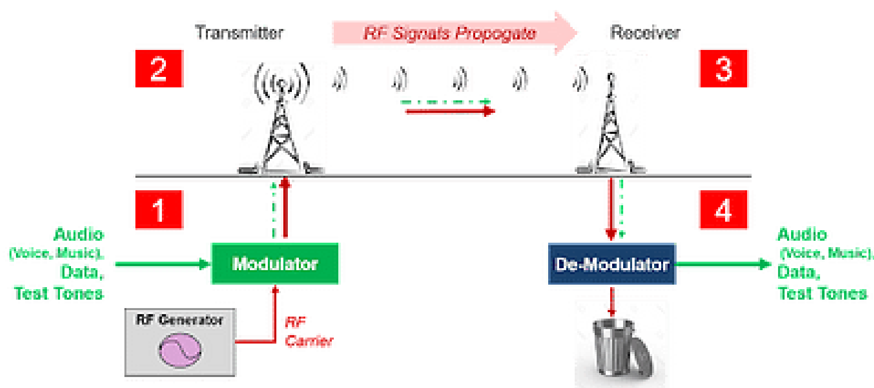
- When data modulated carrier waves are demodulated, you get their original form.
- It is a demodulator signal circuit that actually recovers data from a modulator carrier wave.
- Demodulators come in different types as well. Demodulators output signals that are generally described as audio, binary data, or pictures.



<u>**Need For Demodulation**</u>

Demodulation helps to extract information from modulated carrier waves. Waves of radio frequencies containing modulated audio frequencies are carried within wireless signals.

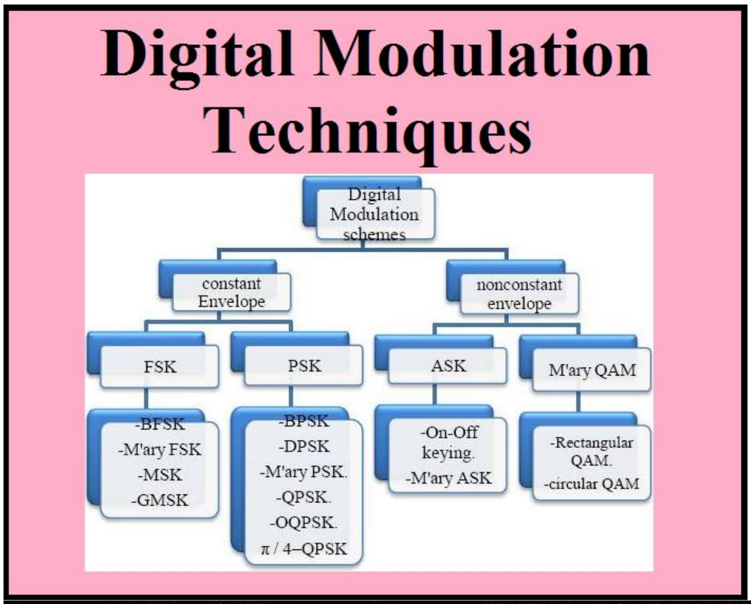
- It is impossible for the diaphragm of a telephone receiver to vibrate when a high frequency is involved.
- In addition, the human ear is not able to hear this frequency. **In order to distinguish the audio frequencies from the radio waves, a demodulator must be applied.**

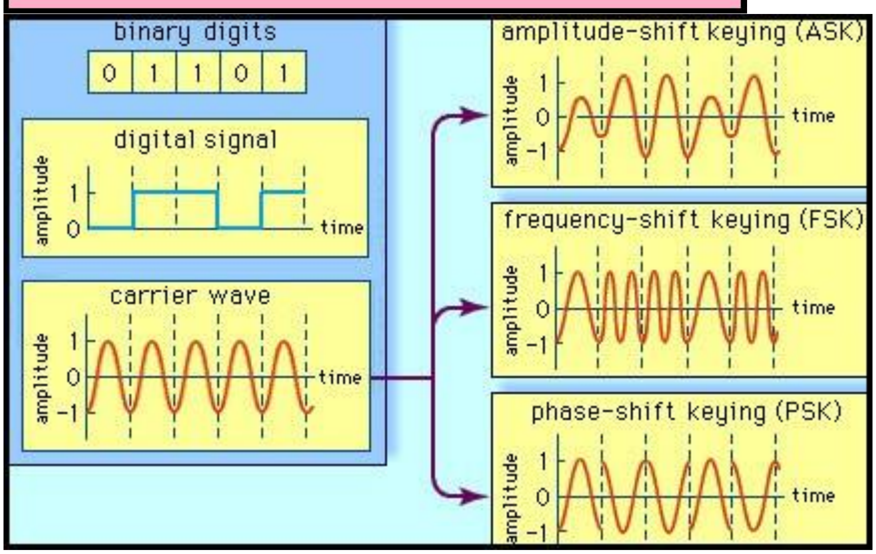


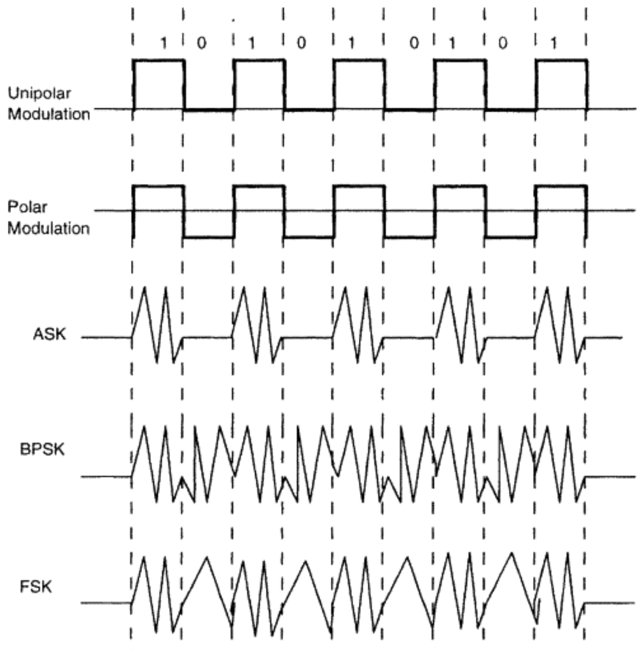
MODULATION	DEMODULATION
Data is collected and modified onto the carrier.	Data is recovered.
Modulation is carried out on the sender's side.	Demodulation takes place on the receiver's side.
Modulation is a process in which the original message signal is mixed with a carrier wave whose parameters need to be altered.	Demodulation takes place in order to create an original information signal by separating the carrier signal from the message signal.
Modulation is done to transmit data over longer distances	Demodulation is the process that prevents the signal from being modified.

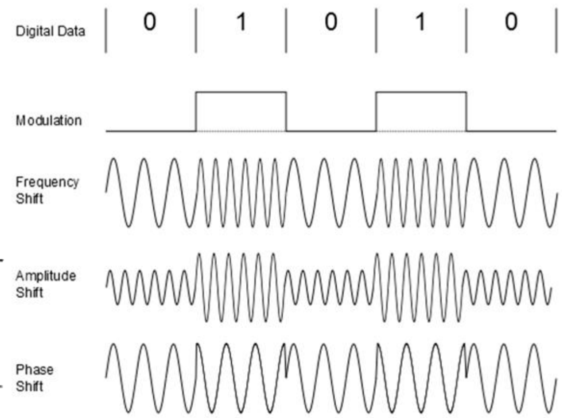
Parameters of Comparison	**Encoding**	**Decoding**
Definition	Encoding is referred to as creating or coding a message in an understandable form.	Decoding is referred to as interpreting the coded message.
Input	The applied signal or message is the input.	Coded binary data is the input.
Output	The generated data in the coded form is the output.	The message in understandable form is the output.
Usage	It is utilized in Emails, videos etc.	It is utilized in microprocessors,

		memory cards etc.
Installation	The encoder (sender or any software) used is installed at the transmitting end.	The decoder or receiver is there at the receiving end.
Complexity	It is a simpler process.	It is complex as it involves the interpretation of the codes.





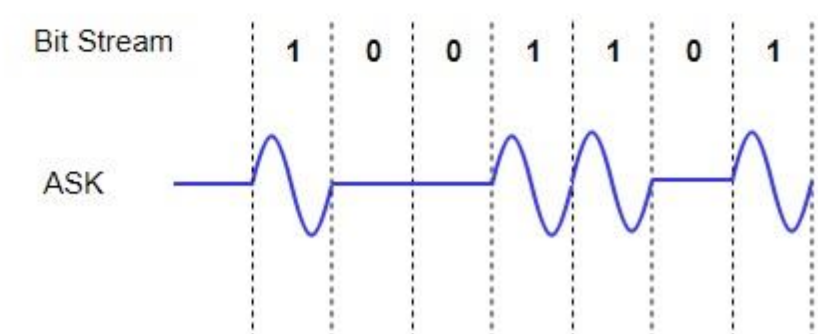




Basic Digital Modulation Techniques

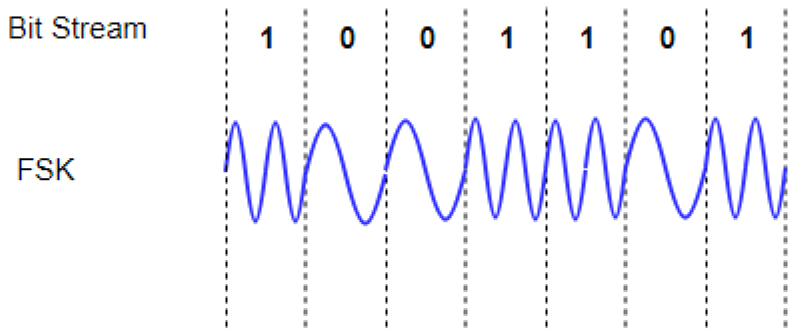
Amplitude Shift Keying (ASK)

In this method signal level is represented by variations in the amplitude of the signal. In ASK only the amplitude is varied keeping phase and frequency constant.



Frequency Shift Keying (FSK)

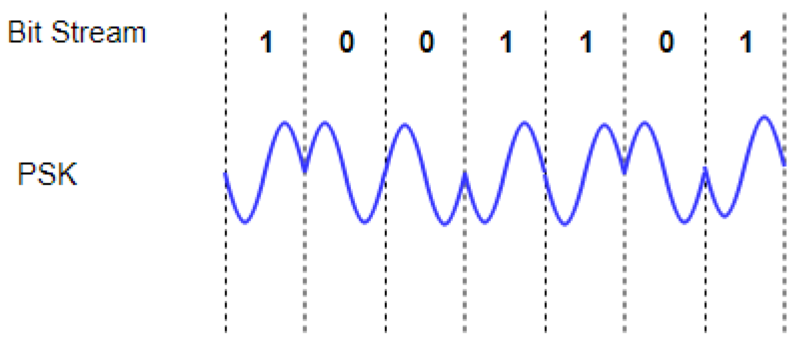
In this method also signal level is represented by variations in the frequency of the signal. In FSK only the frequency is varied keeping amplitude and phase constant. We then use different frequency levels to show the signal levels 0 and 1.



Phase Shift Keying (PSK)

The variation in phase of the signal in this method represents the signal level. In PSK only the frequency is varied keeping amplitude and frequency constant. Also we use different phases to show the signal levels 0 and 1.

The basic and simple form of PSK is Binary Phase Shift Keying (BPSK). In this modulation two different phases represent the two signals. Another method in PSK is Quadrature PSK (QPSK). In this method four different phases then transmit two bits of information.



Digital Modulation provides more information capacity, high data security, quicker system availability with great quality communication. Hence, digital modulation techniques have a greater demand, for their capacity to convey larger amounts of data than analog modulation techniques.

There are many types of digital modulation techniques and also their combinations, depending upon the need. Of them all, we will discuss the prominent ones.

ASK – Amplitude Shift Keying

The amplitude of the resultant output depends upon the input data whether it should be a zero level or a variation of positive and negative, depending upon the carrier frequency.

FSK – Frequency Shift Keying

The frequency of the output signal will be either high or low, depending upon the input data applied.

PSK – Phase Shift Keying

The phase of the output signal gets shifted depending upon the input. These are mainly of two types, namely Binary Phase Shift Keying (BPSK) and Quadrature Phase Shift Keying (QPSK), according to the number of phase shifts. The other one is Differential Phase Shift Keying (DPSK) which changes the phase according to the previous value.

M-ary Encoding

M-ary Encoding techniques are the methods where more than two bits are made to transmit simultaneously on a single signal. This helps in the reduction of bandwidth.

The types of M-ary techniques are −

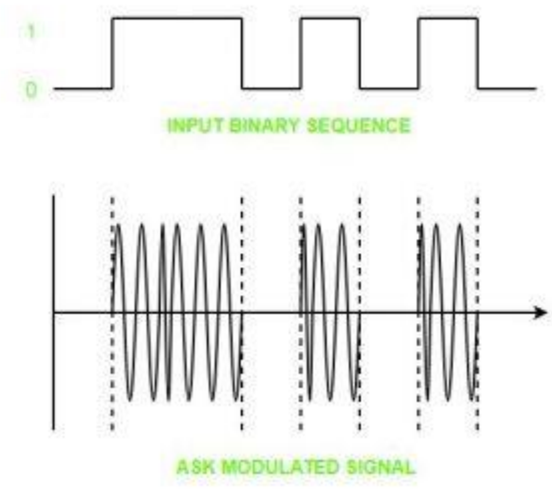
- M-ary ASK
- M-ary FSK
- M-ary PSK

Digital Signal – A digital signal is a signal that represents data as a sequence of discrete values; at any given time it can only take on one of a finite number of values.

Analog Signal – An analog signal is any continuous signal for which the time varying feature of the signal is a representation of some other time varying quantity i.e., analogous to another time varying signal.
The following techniques can be used for Digital to Analog Conversion:

1. Amplitude Shift keying – Amplitude Shift Keying is a technique in which carrier signal is analog and data to be modulated is digital. The amplitude of analog carrier signal is modified to reflect binary data.

The binary signal when modulated gives a zero value when the binary data represents 0 while gives the carrier output when data is 1. The frequency and phase of the carrier signal remain constant.



Advantages of amplitude shift Keying –

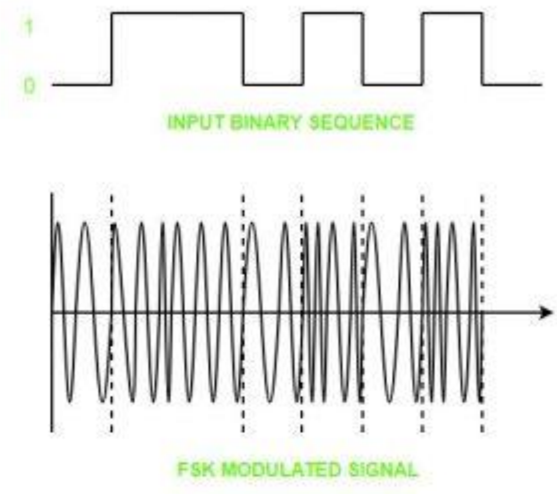
- It can be used to transmit digital data over optical fiber.
- The receiver and transmitter have a simple design which also makes it comparatively inexpensive.
- It uses lesser bandwidth as compared to FSK thus it offers high bandwidth efficiency.

Disadvantages of amplitude shift Keying –

- It is susceptible to noise interference and entire transmissions could be lost due to this.
- It has lower power efficiency.

2. Frequency Shift keying – In this modulation the frequency of analog carrier signal is modified to reflect binary data.

The output of a frequency shift keying modulated wave is high in frequency for a binary high input and is low in frequency for a binary low input. The amplitude and phase of the carrier signal remain constant.



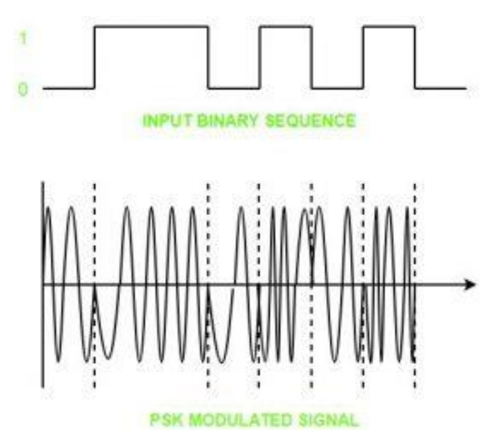
Advantages of frequency shift Keying –

- Frequency shift keying modulated signal can help avoid the noise problems beset by ASK.
- It has lower chances of an error.
- It provides high signal to noise ratio.
- The transmitter and receiver implementations are simple for low data rate application.

Disadvantages of frequency shift Keying –

- It uses larger bandwidth as compared to ASK thus it offers less bandwidth efficiency.
- It has lower power efficiency.

3. Phase Shift keying – In this modulation the phase of the analog carrier signal is modified to reflect binary data.The amplitude and frequency of the carrier signal remains constant.



It is further categorized as follows:

1. **Binary Phase Shift Keying (BPSK):**
   BPSK also known as phase reversal keying or 2PSK is the simplest form of phase shift keying. The Phase of the carrier wave is changed according to the two binary inputs. In Binary Phase shift keying, difference of 180 phase shift is used between binary 1 and binary 0.
   This is regarded as the most robust digital modulation technique and is used for long distance wireless communication.

2. **Quadrature phase shift keying:**
   This technique is used to increase the bit rate i.e we can code two bits onto one single element. It uses four phases to encode two bits per symbol. QPSK uses phase shifts of multiples of 90 degrees.
   It has double data rate carrying capacity compare to BPSK as two bits are mapped on each constellation points.

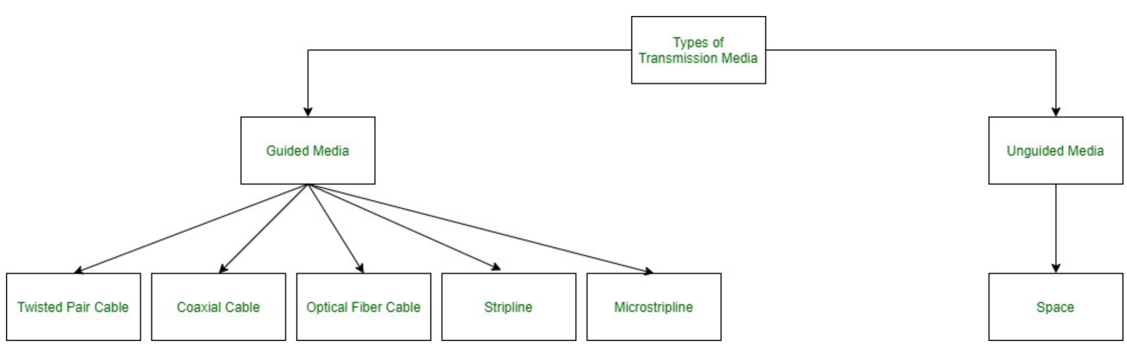
Advantages of phase shift Keying –

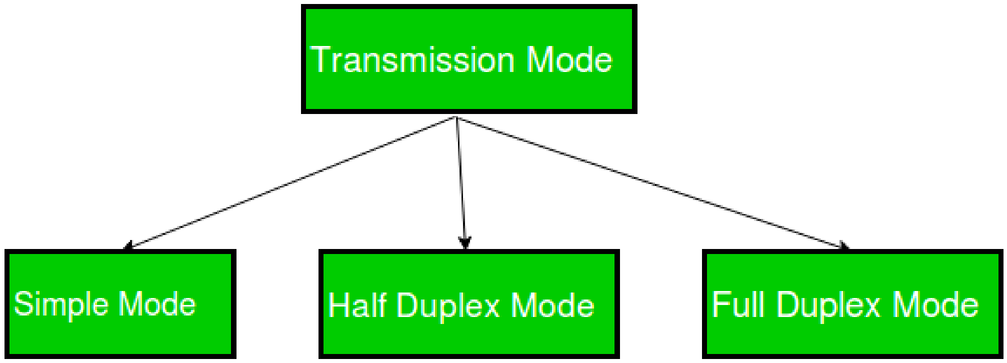
- It is a more power efficient modulation technique as compared to ASK and FSK.
- It has lower chances of an error.
- It allows data to be carried along a communication signal much more efficiently as compared to FSK.

Disadvantages of phase shift Keying –

- It offers low bandwidth efficiency.
- The detection and recovery algorithms of binary data is very complex.
- It is a non coherent reference signal.

<u>Types of Transmission Media</u>





Analog to Digital Conversion

Digital Signal: A digital signal is a signal that represents data as a sequence of discrete values; at any given time it can only take on one of a finite number of values.

Analog Signal: An analog signal is any continuous signal for which the time varying feature of the signal is a representation of some other time varying quantity i.e., analogous to another time varying signal.
The following techniques can be used for Analog to Digital Conversion:

a. PULSE CODE MODULATION:

The most common technique to change an analog signal to digital data is called pulse code modulation (PCM). A PCM encoder has the following three processes:

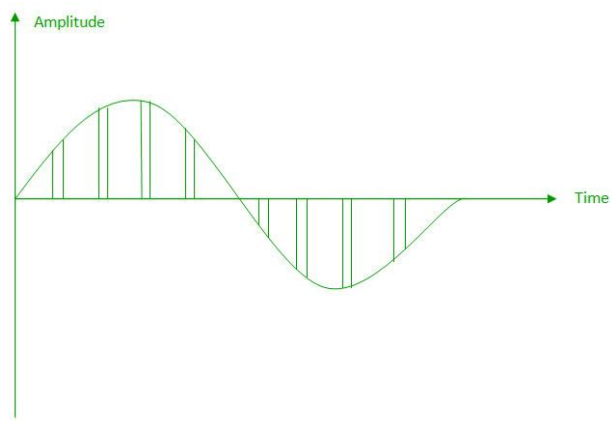
1. Sampling
2. Quantization
3. Encoding

Lowpass filter:
The low pass filter eliminates the high frequency components present in the input analog signal to ensure that the input signal to sampler is free from the unwanted frequency components. This is done to avoid aliasing of the message signal.

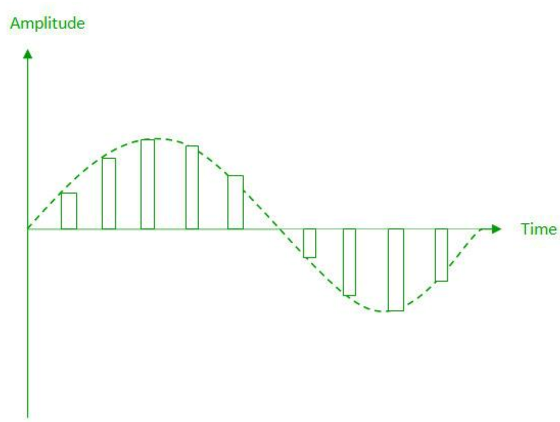
1. **Sampling –** The first step in PCM is sampling. Sampling is a process of measuring the amplitude of a continuous-time signal at discrete instants, converting the continuous signal into a discrete signal. There are three sampling methods:

   (i) Ideal Sampling: In ideal Sampling also known as Instantaneous sampling pulses from the analog signal are sampled. This is an ideal sampling method and cannot be easily implemented.

(ii) Natural Sampling: Natural Sampling is a practical method of sampling in which pulse have finite width equal to T.The result is a sequence of samples that retain the shape of the analog signal.



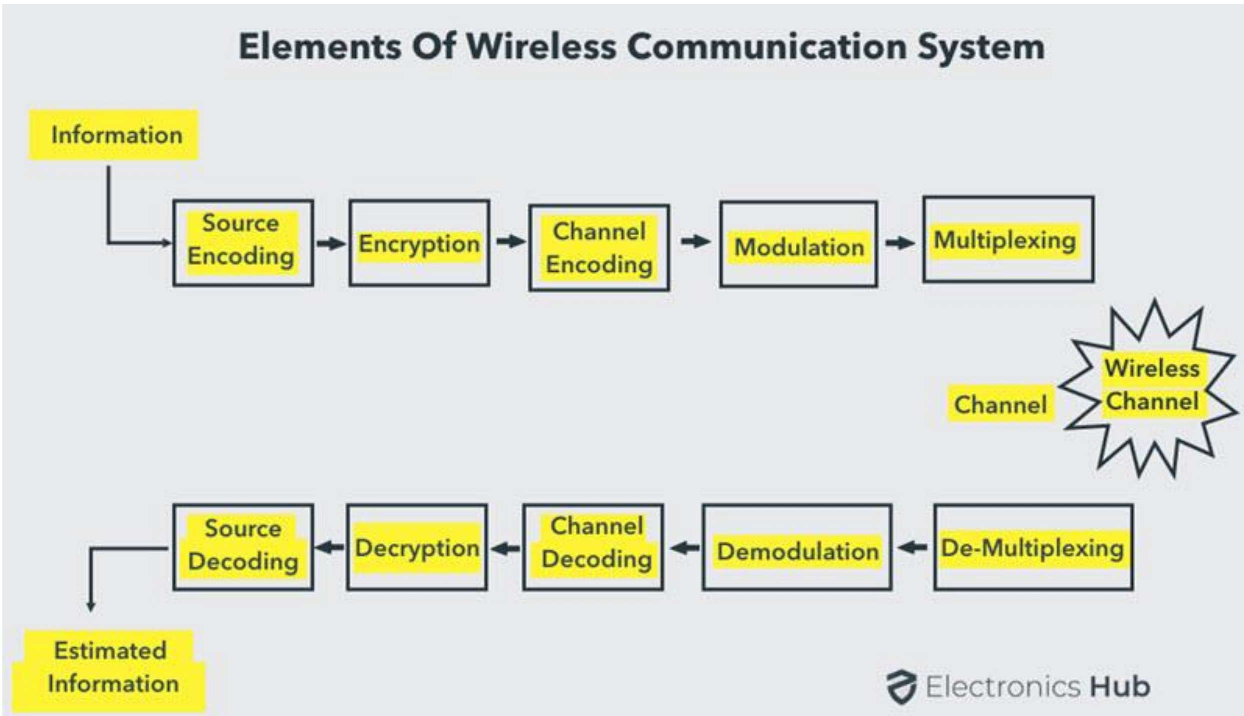
(iii) Flat top sampling: In comparison to natural sampling flat top sampling can be easily obtained. In this sampling technique, the top of the samples remains constant by using a circuit. This is the most common sampling method used.



2. **Quantization** –
   The result of sampling is a series of pulses with amplitude values between the maximum and minimum amplitudes of the signal. The set of amplitudes can be infinite with non-integral values between two limits.
   The following are the steps in Quantization:

3. **Encoding** –
   The digitization of the analog signal is done by the encoder. After each sample is quantized and the number of bits per sample is decided, each sample can be changed to an n bit code. Encoding also minimizes the bandwidth used.

Types of communication and wireless communication



Information = voice, audio, video is in analog signal

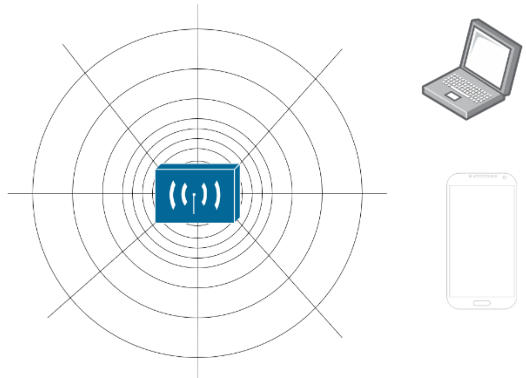
Source encoding = analog to digital signal and using a suitable form for applying signal producing techniques

Encryption = signals information 0 and 1 s

Wireless Communication is the fastest growing and most vibrant technological areas in the communication field. Wireless Communication is a method of transmitting information from one point to other, without using any connection like wires, cables or any physical medium.

Some of the commonly used Wireless Communication Systems in our day – to – day life are: Mobile Phones, GPS Receivers, Remote Controls, Bluetooth Audio and Wi-Fi etc.

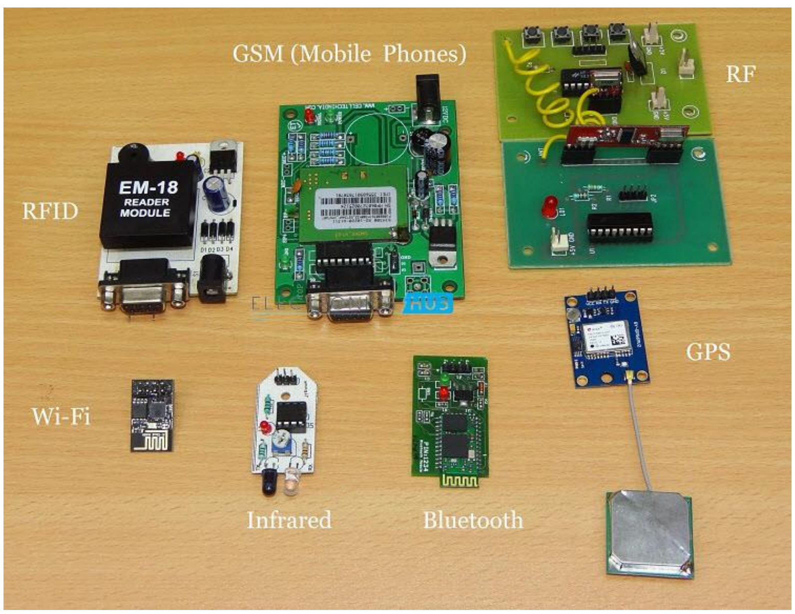
Electromagnetic waves do not travel in a straight line. they travel by expanding in all direction away from antenna. Like you have seen waves travelling in water when you drop or throw a stone in a water body.



Wave propagation with an antenna

S. No	Basis of Comparison	Baseband Transmission	Broadband Transmission
1.	Type of Signal	In baseband transmission, the type of signaling used is digital.	In broadband transmission, the type of signaling used is analog.
2.	Direction Type	Baseband Transmission is bidirectional in nature.	Broadband Transmission is unidirectional in nature.
3.	Signal Transmission	The Signal can be sent in both directions.	Sending of Signal in one direction only.
4.	Distance covered by the signal	Signals can only travel over short distances. For long distances, attenuation is required.	Signals can be traveled over long distances without being attenuated.
5.	Topology	It works well with bus topology.	It is used with a bus as well as tree topology.
6.	Device used to increase signal strength	Repeaters are used to enhance signal strength.	Amplifiers are used to enhance signal strength.
7.	Type of Multiplexing used	It utilizes Time Division Multiplexing.	It utilizes Frequency Division Multiplexing.
8.	Encoding Techniques	In baseband transmission, Manchester and Differential Manchester encoding are used.	Only PSK encoding is used.
9.	Transfer medium	Twisted-pair cables, coaxial cables, and wires are used as a transfer	Broadband signals were sent through optical fiber cables,

		medium for digital signals in baseband transmission.	coaxial cables, and radio waves.
10.	Impedance	Baseband transmission has a 50-ohm impedance.	Broadband transmission has a 70-ohm impedance.
11.	Data Streams	It can only transfer one data stream at a time in bi-directional mode.	It can send multiple signal waves at once but in one direction only.
12.	Installation and Maintenance	Baseband transmission is easy to install and maintain.	Broadband transmission is difficult to install and maintain.
13.	Cost	This transmission is cheaper to design.	This transmission is expensive to design.
14.	Application	Typically seen in Ethernet LAN networks.	Typically found in cable and telephone networks.



Wireless Communication Systems also provide different services like video conferencing, cellular telephone, paging, TV, Radio etc.

- Television and Radio Broadcasting
- Satellite Communication
- Radar
- Mobile Telephone System (Cellular Communication)
- Global Positioning System (GPS)
- Infrared Communication
- WLAN (Wi-Fi)

- Bluetooth
- ZigBee
- Paging
- Cordless Phones
- Radio Frequency Identification (RFID)

Television and Radio Broadcasting

Radio is considered to be the first wireless service to be broadcast. It is an example of a Simplex Communication System where the information is transmitted only in one direction and all the users receiving the same data.

Satellite Communication

Satellite Communication System is an important type of Wireless Communication. Satellite Communication Networks provide worldwide coverage independent to population density.

Satellite Communication Systems offer telecommunication (Satellite Phones), positioning and navigation (GPS), broadcasting, internet, etc. Other wireless services like mobile, television broadcasting and other radio systems are dependent of Satellite Communication Systems.

Mobile Telephone Communication System

Perhaps, the most commonly used wireless communication system is the Mobile Phone Technology. The development of mobile cellular device changed the World like no other technology. Today's mobile phones are not limited to just making calls but are integrated with numerous other features like Bluetooth, Wi-Fi, GPS, and FM Radio.

The latest generation of Mobile Communication Technology is 5G (which is indeed successor to the widely adapted 4G). Apart from increased data transfer rates (technologists claim data rates in the order of Gbps), 5G Networks are also aimed at Internet of Things (IoT) related applications and future automobiles.

Global Positioning System (GPS)

GPS is solely a subcategory of satellite communication. GPS provides different wireless services like navigation, positioning, location, speed etc. with the help of dedicated GPS receivers and satellites.

Bluetooth

Bluetooth is another important low range wireless communication system. It provides data, voice and audio transmission with a transmission range of 10 meters. Almost all mobile phones, tablets and laptops are equipped with Bluetooth devices. They can be connected to wireless Bluetooth receivers, audio equipment, cameras etc.

Paging

Although it is considered an obsolete technology, paging was a major success before the wide spread use of mobile phones. Paging provides information in the form of messages and it is a simplex system i.e. the user can only receive the messages.

Wireless Local Area Network (WLAN)

Wireless Local Area Network or WLAN (Wi-Fi) is an internet related wireless service. Using WLAN, different devices like laptops and mobile phones can connect to an access point (like a Wi-Fi Router) and access internet.

Wi-Fi is one of the widely used wireless network, usually for internet access (but sometimes for data transfer within the Local Area Network). It is very difficult to imagine the modern World without Wi-Fi.

Infrared Communication

Infrared Communication is another commonly used wireless communication in our daily lives. It uses the infrared waves of the Electromagnetic (EM) spectrum. Infrared (IR) Communication is used in remote controls of Televisions, cars, audio equipment etc.

What is Wireless Communication?

Communication Systems can be Wired or Wireless and the medium used for communication can be Guided or Unguided.

In Wired Communication, the medium is a physical path like Co-axial Cables, Twisted Pair Cables and Optical Fiber Links etc. which guides the signal to propagate from one point to other. Such type of medium is called Guided Medium.

On the other hand, Wireless Communication doesn't require any physical medium but propagates the signal through space. Since, space only allows for signal transmission without any guidance, the medium used in Wireless Communication is called Unguided Medium.

If there is no physical medium, then how does wireless communication transmit signals? Even though there are no cables used in wireless communication, the transmission and reception of signals is accomplished with Antennas.

Antennas are electrical devices that transform the electrical signals to radio signals in the form of Electromagnetic (EM) Waves and vice versa. These Electromagnetic Waves propagates through space. Hence, both transmitter and receiver consists of an antenna.

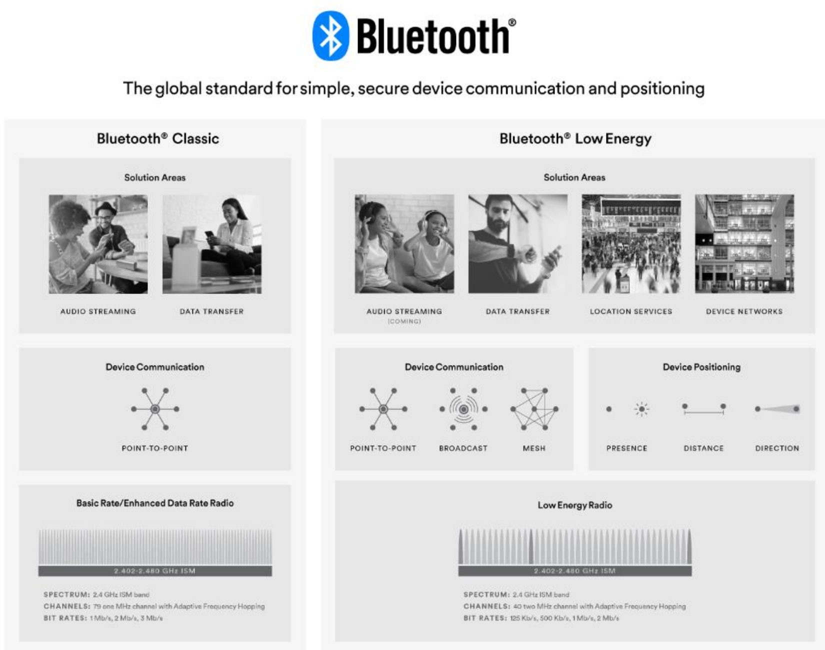
What is Electromagnetic Wave?
Electromagnetic Waves carry the electromagnetic energy of electromagnetic field through space. Electromagnetic Waves include Gamma Rays (γ – Rays), X – Rays, Ultraviolet Rays, Visible Light, Infrared Rays, Microwave Rays and Radio Waves. Electromagnetic Waves (usually Radio Waves) are used in wireless communication to carry the signals.

An Electromagnetic Wave consists of both electric and magnetic fields in the form of time varying sinusoidal waves.

Advantages of Wireless Communication
There are numerous advantage of Wireless Communication Technology, Wireless Networking and Wireless Systems over Wired Communication like Cost, Mobility, Ease of Installation, and Reliability

Disadvantages of Wireless Communication
Even though wireless communication has a number of advantages over wired communication, there are a few disadvantages as well. The most concerning disadvantages are Interference, S

BLUETOOTH INTRODUCTION (BR/EDR, BLE)



	Bluetooth Low Energy (LE)	Bluetooth Classic
Frequency Band	2.4GHz ISM Band (2.402 – 2.480 GHz Utilized)	2.4GHz ISM Band (2.402 – 2.480 GHz Utilized)
Channels	40 channels with 2 MHz spacing (3 advertising channels/37 data channels)	79 channels with 1 MHz spacing
Channel Usage	Frequency-Hopping Spread Spectrum (FHSS)	Frequency-Hopping Spread Spectrum (FHSS)
Modulation	GFSK	GFSK, π/4 DQPSK, 8DPSK
Data Rate	LE 2M PHY: 2 Mb/s LE 1M PHY: 1 Mb/s LE Coded PHY (S=2): 500 Kb/s LE Coded PHY (S=8): 125 Kb/s	EDR PHY (8DPSK): 3 Mb/s EDR PHY (π/4 DQPSK): 2 Mb/s BR PHY (GFSK): 1 Mb/s
Tx Power*	≤ 100 mW (+20 dBm)	≤ 100 mW (+20 dBm)
Rx Sensitivity	LE 2M PHY: ≤-70 dBm LE 1M PHY: ≤-70 dBm LE Coded PHY (S=2): ≤-75 dBm LE Coded PHY (S=8): ≤-82 dBm	≤-70 dBm
Data Transports	Asynchronous Connection-oriented Isochronous Connection-oriented Asynchronous Connectionless Synchronous Connectionless Isochronous Connectionless	Asynchronous Connection-oriented Synchronous Connection-oriented
Communication Topologies	Point-to-Point (including piconet) Broadcast Mesh	Point-to-Point (including piconet)

Bluetooth® technology [1], operating on the 2.4 GHz unlicensed industrial, scientific, and medical (ISM) frequency band, uses low-power radio frequency to enable short-range communication at a low cost. The two variants of the Bluetooth technology are –

- Bluetooth basic rate/enhanced data rate (BR/EDR) or classic Bluetooth
- Bluetooth low energy (LE) or Bluetooth Smart

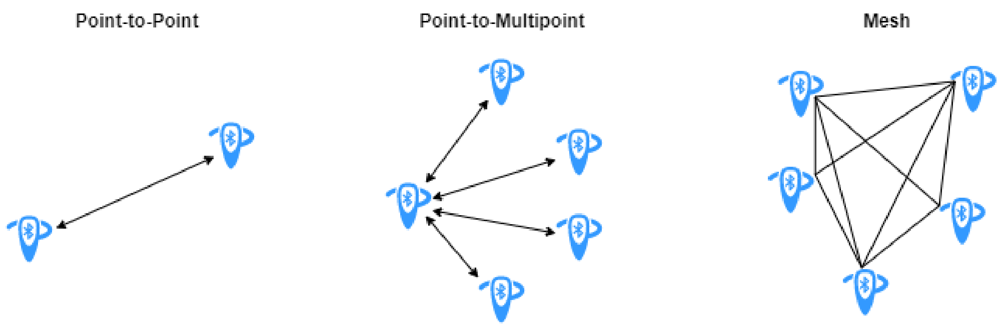
- Bluetooth wireless technology is a short-range communications system intended to replace the cable(s) connecting portable and/or fixed electronic devices.
- The key features of Bluetooth wireless technology are robustness, low power consumption, and low cost. Many features of the specification are optional, allowing product differentiation.

- The Basic Rate system offers synchronous and asynchronous connections with data rates of 721.2 kb/ s for Basic Rate and 2.1 Mb/s for Enhanced Data Rate.
- The Bluetooth core system consists of a Host and one or more Controllers
- An implementation of the Bluetooth Core has only one Controller which may be one of the following configurations:

- a BR/EDR Controller including the Radio, Baseband, Link Manager and optionally HCI.

- an LE Controller including the LE PHY, Link Layer and optionally HCI.
- a combined BR/EDR Controller portion and LE Controller portion (as identified in the previous two bullets) into a single Controller.

Bluetooth Connection Topologies

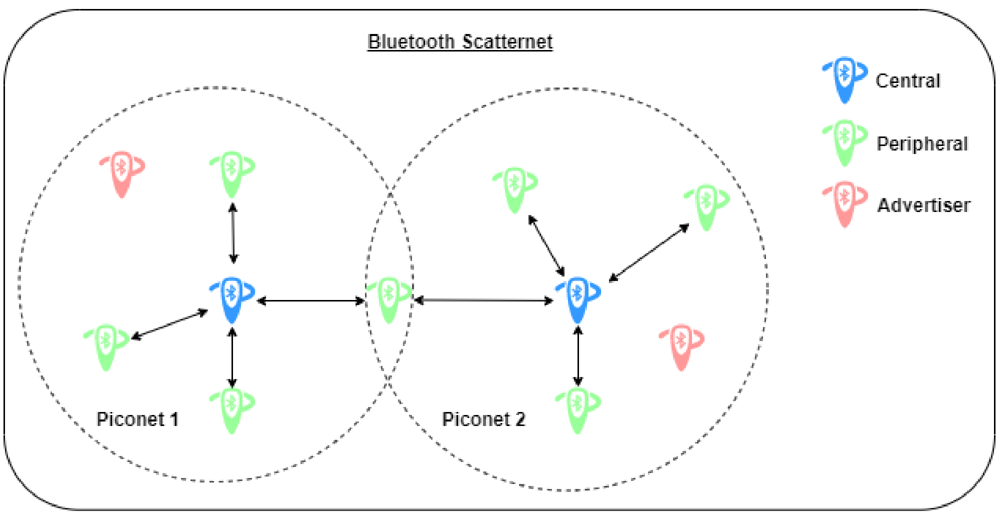


P to p = telephone, mobile

P to m = broadcast

Mesh = automatic home

- Devices using one-to-one communication operate in a Bluetooth *piconet*. As shown in this figure, each piconet consists of a device in the role of *Central*, with other devices in the *Peripheral* or *Advertiser* roles.
- Before joining the piconet, each *Peripheral* node is in an advertiser role.
- Multiple piconets connect to each other, forming a Bluetooth *scatternet*.



Piconet Network

Piconet is a wireless network that consists of one primary node, the master node, and seven energetic subsidiary nodes, referred to as slave nodes. As a result, we can say that there are a total of eight active nodes arranged at a distance of ten metres. The message transmission between these two nodes can be one-to-one or one-to-many. Only communication between master and slave is feasible, but communication between slave and slave is not. There are also 255 secondary nodes, which are parked nodes. These won't be able to communicate until the state is changed to active.

Scatternet Network

The Scatternet Network can be created using a variety of piconets. There is a slave on one piconet that functions as a master, but it can also be termed main in other piconets.
As a result, this sort of node receives a message from the master in one piconet and sends it to its slave in another piconet, where it operates as a slave. As a result, this type of node is known as a bridge-node. A station cannot be master in two piconets.

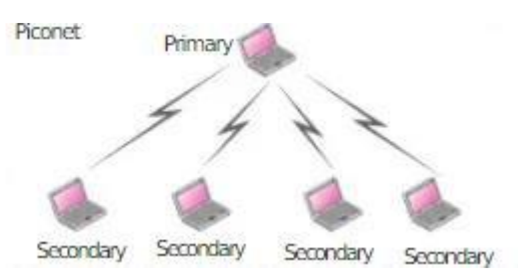
Bluetooth network technology establishes a personal area network by wirelessly connecting mobile devices over a short distance (PAN). Instead of following the usual OSI or TCP/IP models, the Bluetooth architecture has its own separate paradigm with a stack of protocols.

Piconet

A piconet is a network created by connecting multiple wireless devices using Bluetooth technology. In a piconet network a master device exists, this master device cm gets connected to 7 more slave devices.

It includes the master the number of devices that can be connected is limited to 8. Due to less number of devices active at a time the usage of channel band width is not more.

Number of devices that can be connected is limited to 8. It is applicable for devices belonging to small areas.



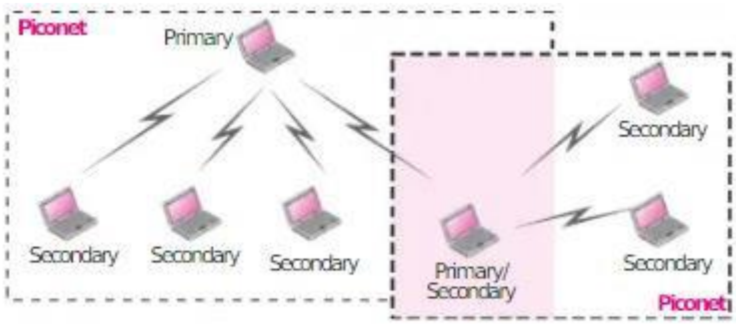
Scatternet

It is a network which connects multiple piconets using Bluetooth and it acts as a master and another type of piconet acts as a slave. It has more than 6 devices that can be connected.

Multiple devices are active, so there is an effective use of channel bandwidth.

It is a connection of multiple piconets therefore it is applicable for devices belonging to large areas.

Given below is the diagram of scatternet –



PICONET

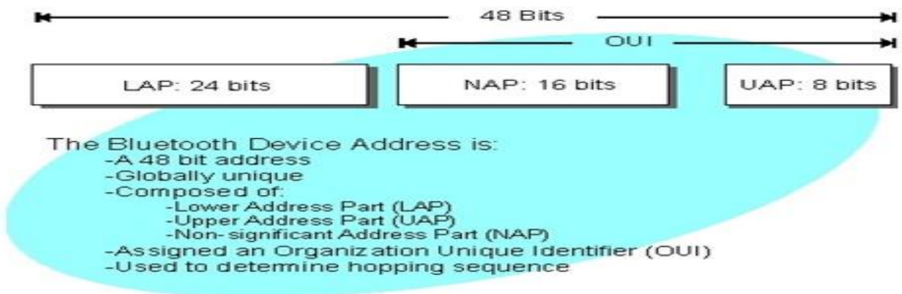
- One is called master and other is called are slave
- All slave stations synchronize their clocks with the master
- Possible communications one to one or one to many
- There may be one in parked state
- Each piconet has a unique hopping patter /id
- Each master can connect to 7 simultaneous

BLUETOOTH LOW ENERGY

- GFSKM= GAUSSIAN frequency shift key Modulation
- Frequency shift = changing of frequency of carrier
- Keying = from telegraphy
- FSK = modulation of frequency carrier and encode the data as a series of frequency changing

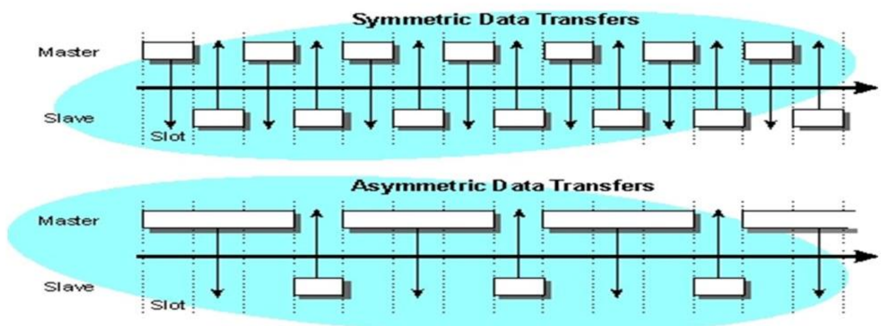
Bluetooth Address

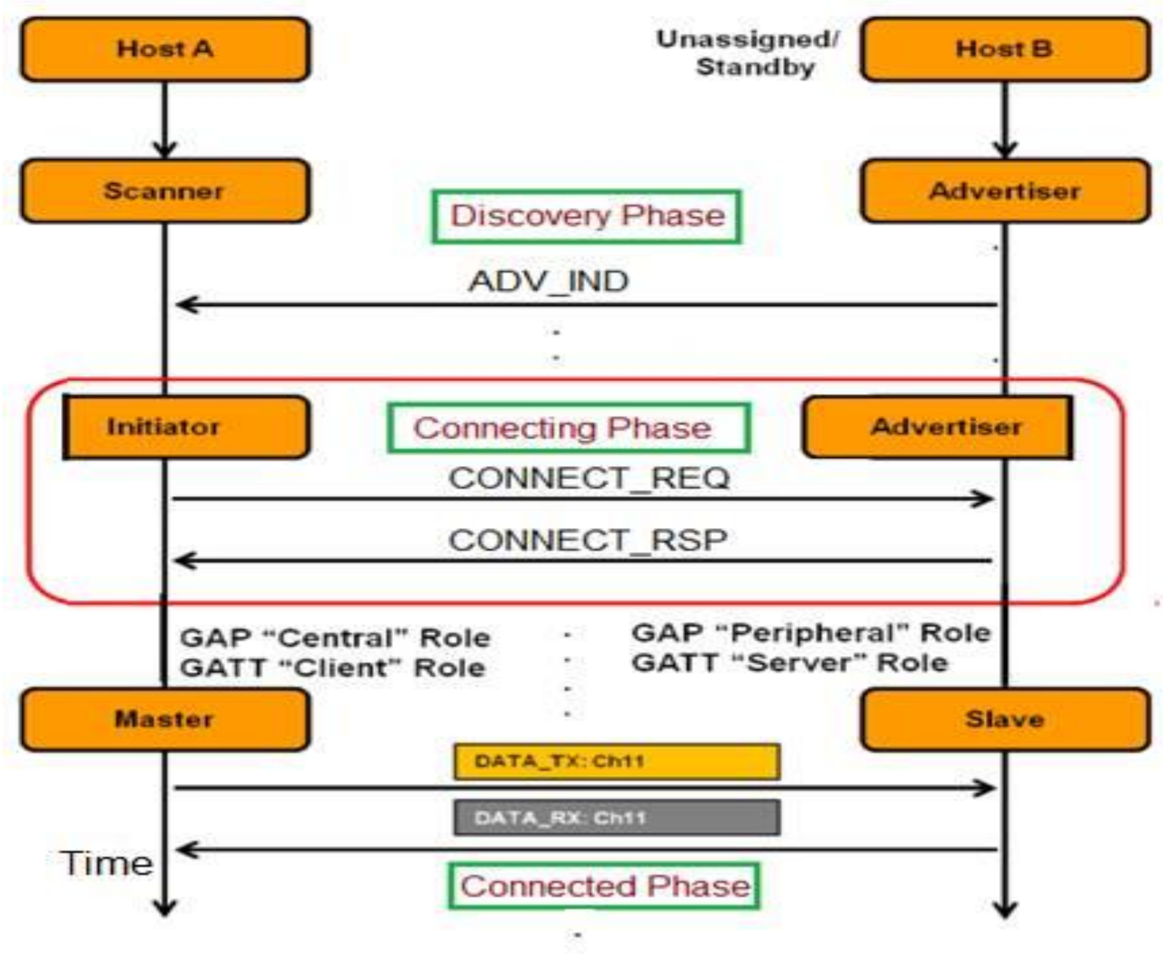
- Uniquely Identifies Bluetooth Device
- 48 Bits is Divided into Organization Part and Unique Part
- Used to Determine Frequency Hopping Sequence



Bluetooth Data Transmission

- 1 Mbps
- 2-3 Mbps EDR
- Symmetric
- Asymmetric





BLE Connection Establishment Procedure

Other Bluetooth Profiles

- Generic Access Profile (GAP)
- Advanced Audio Distribution Profile (A2DP)
- Audio Video Remote Control Profile (AVRCP)
- Basic Imaging Profile (BIP)
- Hard Copy Cable Replacement Profile (HCRP)
- Common ISDN Access Profile (CIP)
- Cordless Telephony Profile (CTP)
- Device ID Profile (DI)
- Dial-up Networking Profile (DUN)
- Extended Service Discovery Profile (ESDP)
- Fax Profile
- File Transfer Profile
- General Audio Visual Distribution Profile (GAVDP)
- General Object Exchange Profile (GOEP)
- Headset Profile
- Human Interface Device (HID)
- Intercom Profile (ICP)
- Object Push Profile (OPP)
- Personal Area Networking Profile (PAN)
- Phone Book Access Profile (PBAP)
- SIM Access Profile (SAP)
- Service Discovery Application Profile (SDAP)
- Serial Port Profile (SPP)
- Video Distribution Profile (VDP)

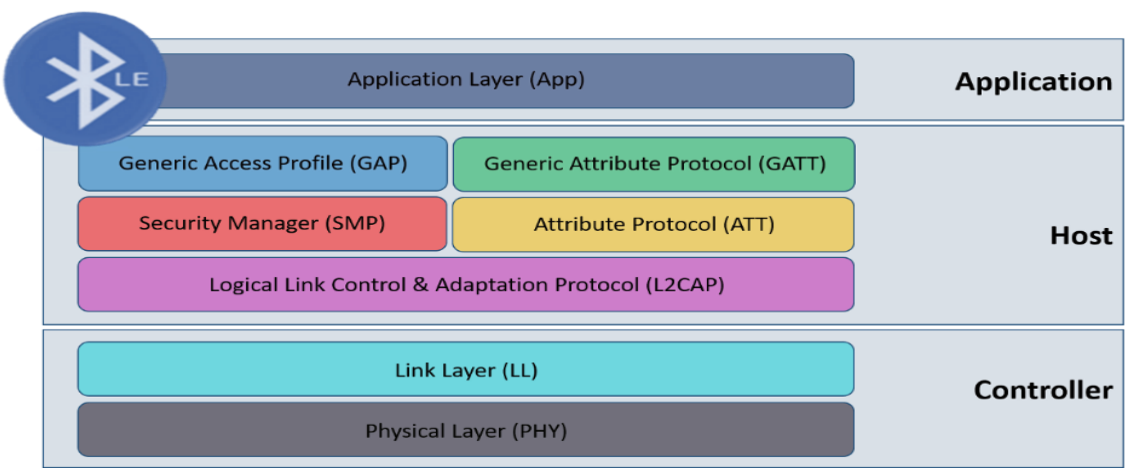
ALTHOS Simplifying Knowledge

Hundreds of Profiles!

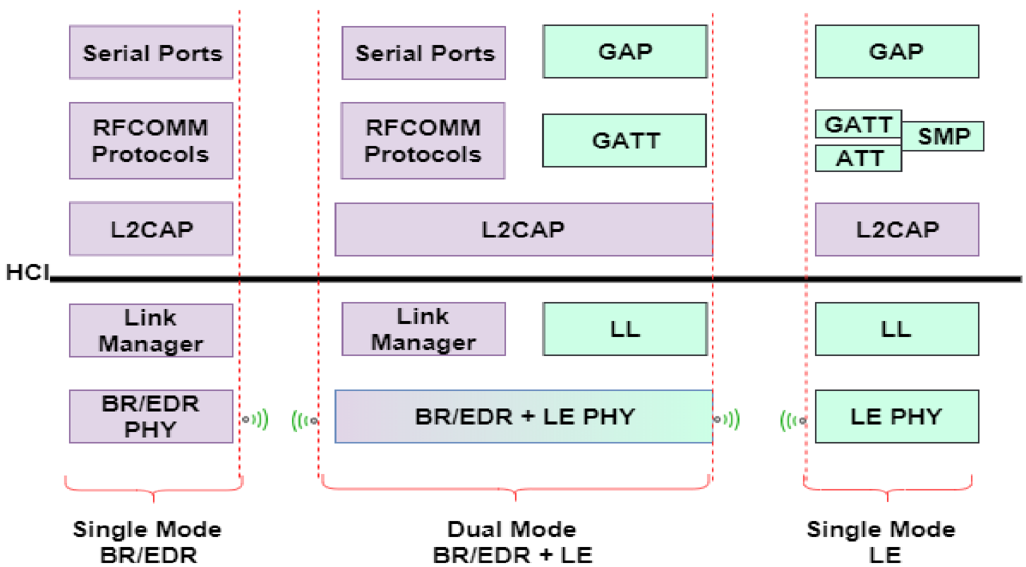
BLE PROFILE

    - Mesh profile
    - Health care profile
        - Blood pressure profile
        - Health Thermometer profile
        - Glucose profile
        - Continuous Glucose Monitor profile

- Sports and fitness profiles
    - Body composition service
    - Cycling speed and cadence profile
    - Cycling power profile
    - Heart rate profile
    - Location and navigation profile
    - Running speed and cadence profile
    - Weight scale profile
- Internet connectivity
    - Internet protocol support profile
- ALERTS AND TIME PROFILES
- BATTERY
- GENERIC SENSORS
- HDI CONNECTIVITY

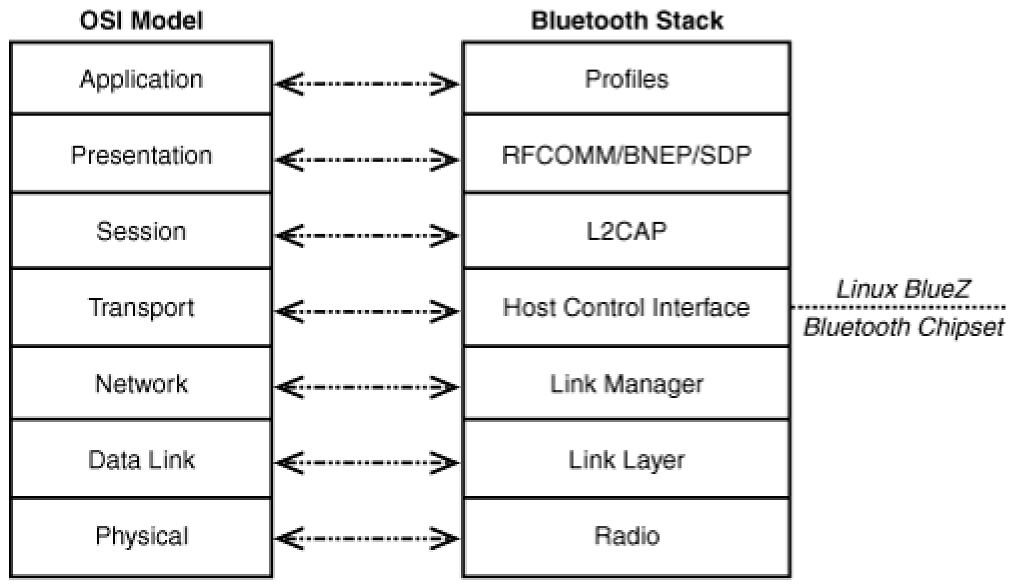


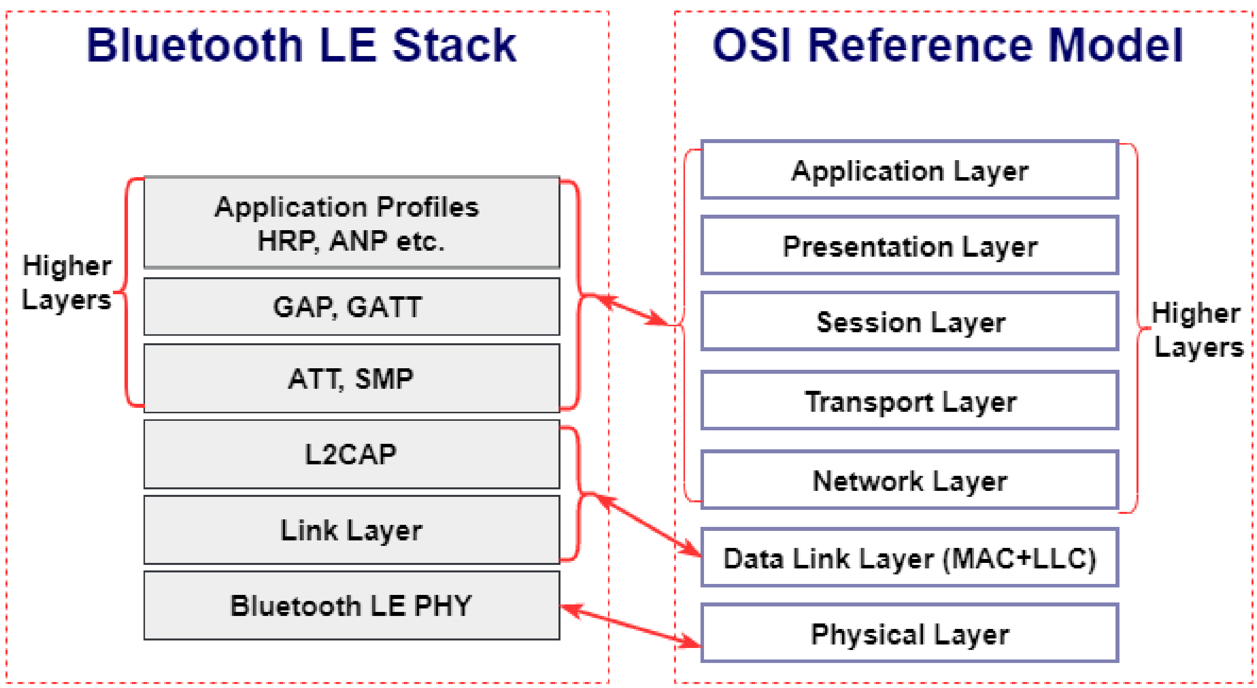
Bluetooth LE Protocol Stack

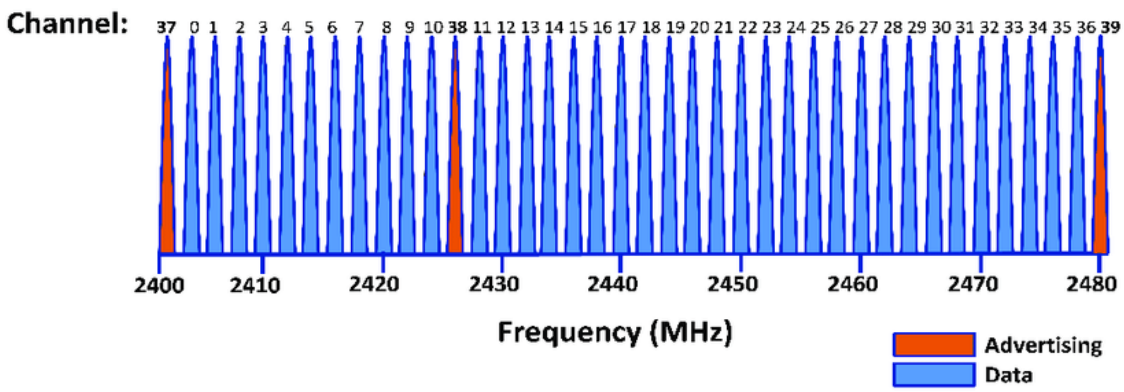


Bluetooth devices can be one of these two types:

- Single mode – Supports a BR/EDR or LE profile
- Dual mode – Supports BR/EDR and LE profiles


OSI Model
Application
Presentation
Session
Transport
Network
Data Link
Physical
Bluetooth Stack
Profiles
RFCOMM/BNEP/SDP
L2CAP
Host Control Interface
Link Manager
Link Layer
Radio
Linux BlueZ
Bluetooth Chipset


Bluetooth LE Stack
Higher Layers
Application Profiles
HRP, ANP etc.
GAP, GATT
ATT, SMP
L2CAP
Link Layer
Bluetooth LE PHY
OSI Reference Model
Application Layer
Presentation Layer
Session Layer
Transport Layer
Network Layer
Data Link Layer (MAC+LLC)
Physical Layer
Higher Layers

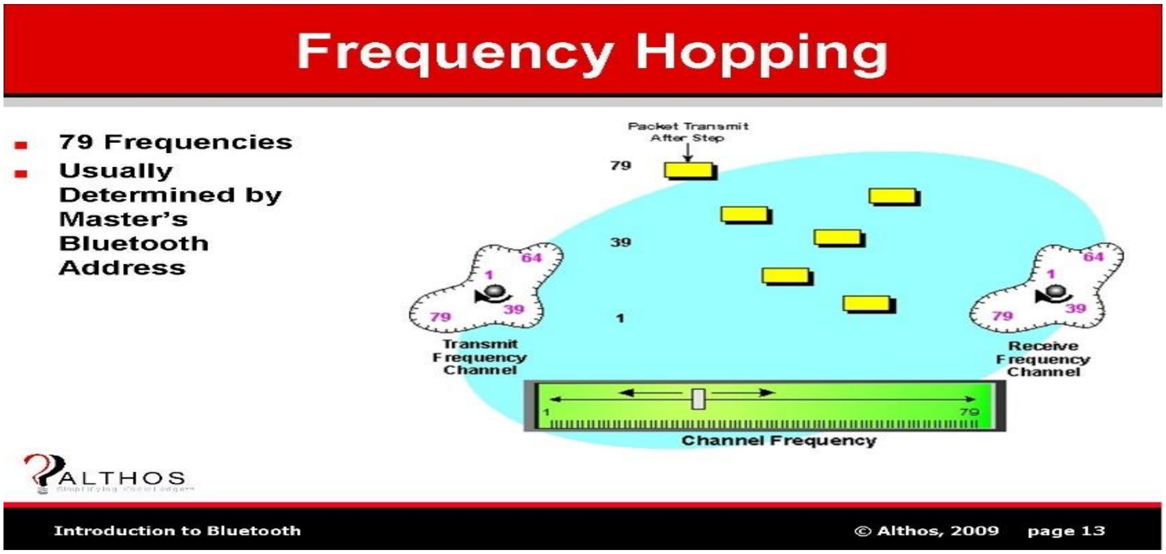


This table summarizes and compares different features of Bluetooth BR/EDR and Bluetooth LE.

Application	**Bluetooth BR/EDR**	**Bluetooth LE**
Audio streaming applications such as: • Bluetooth headphones or earbuds • Bluetooth speakers • Bluetooth watches	Supported	Supported
Data transmission applications such as: • Medical and health equipment's • Sports and fitness equipment's • Peripherals and accessories	Not supported	Supported
Device network applications such as: • Monitoring systems and services • Automation systems • Control systems	Not supported	Supported

Different Bluetooth Versions: What You Need to Know

What is frequency hopping?

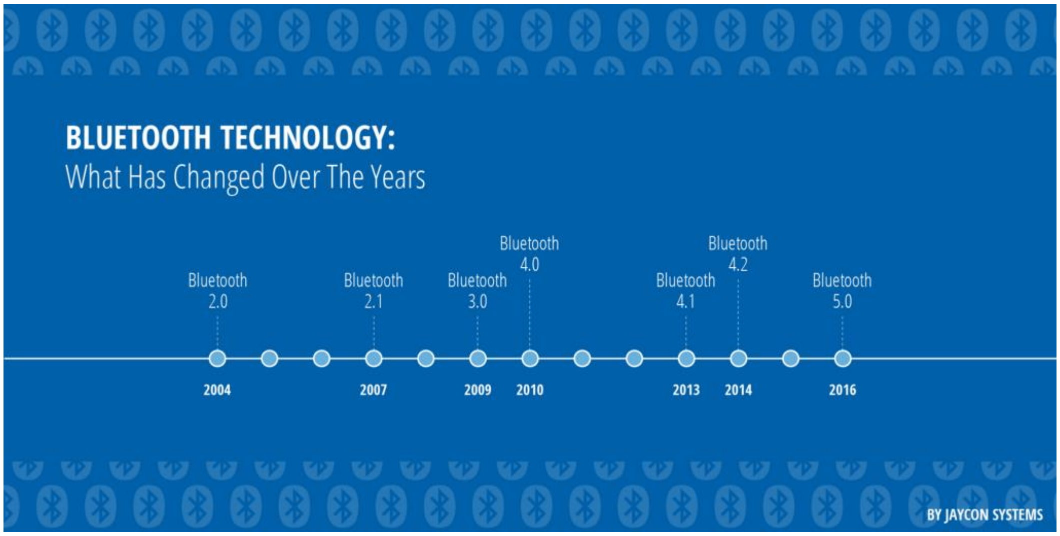


Frequency-hopping spread spectrum (FHSS) transmission is the **repeated switching of the carrier frequency during radio transmission to reduce interference and avoid interception**.
Simple radio with low interface

Frequency Hopping
• In order to mitigate interference, Bluetooth implements frequency hopping
• 1600 hops per second through 79, 1MHz channels
• Spreads Bluetooth traffic over the entire band
• All slaves in piconet follow the master for frequency hop sequence

- Frequency-hopping spread spectrum (FHSS) is **a method of transmitting radio signals by rapidly changing the carrier frequency among many distinct frequencies occupying a large spectral band**.

- What is frequency hopping and how does it work?
- Frequency hopping is **a technique mainly used to keep two or more RFID readers from interfering with each other**
- **while reading RFID tags in the same area**.
- FHSS is used to avoid interference, to prevent eavesdropping, and to enable code-division multiple access (CDMA) communications.



Year Introduced	Bluetooth Version	Feature
2004	2.0	Enhanced Data Rate
2007	2.1	Secure Simple Pairing
2009	3.0	High Speed with 802.11 Wi-Fi Radio
2010	4.0	Low-energy protocol
2013	4.1	Indirect IoT device connection
2014	4.2	IPv6 protocol for direct internet connection
2016	5.0	4x range, 2x speed, 8x message capacity + IoT

- Bluetooth is a short-range wireless technology for transferring data between devices.
- Bluetooth communication operates at frequencies ranging from 2.4 to 2.4835 gigahertz (GHz).
- It can now be found in practically all of our electronic devices, including smartphones, headphones, speakers, smartwatches, computers, and various other devices. It has become a necessary component of some people's daily lives.
- A closer look at the different Bluetooth versions, how they've changed, and how they affect your wireless listening experience.
- Bluetooth 5, the most recent version, has been a hot issue among developers and customers due to its promising internet of things (IoT) experience and advancements in speed, range, and data capacity.
- Bluetooth 5.0 alone has three new versions, given the developments between 2019 to 2021!

- Operating in the frequencies between 2.4 and 2.4835 gigahertz (GHz), Bluetooth connectivity is modeled on the packet-based protocol, which consists of dividing data into packets and transmitting each packet on one of the 79 designated Bluetooth channels.

- The Bluetooth Special Interest Group (SIG) formed in 1998 to oversee the development, licensing and trademarking of this technology. The SIG, originally comprised of five companies — Ericsson, IBM, Nokia, Toshiba and Intel — grew to 400 members by the end of its first year and now has 30,000 member companies.

- the **Vikings** did not discover Bluetooth. That's merely where we got the inspiration for Bluetooth's name and symbol.

- The direct answer to this question is the **Swedish telecommunications company Ericsson. Dr. Jaap Haartsen,** recognized as the inventor of this wireless technology, led the Swedish team, which worked for **Ericsson**.

FACTORS	**BLUETOOTH 1**	**BLUETOOTH 2**	**BLUETOOTH 3**	**BLUETOOTH 4**	**BLUETOOTH 5**
SPEED	732.2 kb/s	2.1 Mb/s Enhanced Data Rate (EDR	24 Mb/s (via Wi-Fi)	24 Mb/s (EDR)	50 Mb/s (EDR)
RANGE	10 meters	30 meters	30 meters	60 meters (200 feet)	240 meters (800 feet)
COMPATIBILITY	N/A	OK for any phone but expect some possible sound sync issues	OK for any phones	Good for any phone but best for models with the same Bluetooth version	Good for any phone but best for newer phone models
POWER REQUIREMENT	High	High	High	Mid-high	Low
RELIABLITY	Low	Low	Low	Mid-high	High

- This technology uses **short-wavelength UHF (ultrahigh frequency)** radio waves that range from **2.4 - 2.485 GHz**. Initially, it was invented as an alternative wireless technology that could replace **RS-232** data cables.
- **The Bluetooth Special Interest Group manages and upgrades the Bluetooth technology**. The group has more than 30,000-member companies working with it in its different telecommunication, processing, networking, and customer electronics industries.

BLUETOOTH CLASS::

Bluetooth class

Bluetooth classes indicate a Bluetooth device's power output and wireless range.

A higher power output means a more extended range. Class 1 and 2 Bluetooth devices typically include laptops and computers.

Type	Power	Max Power Level	Designed Operating Range	Sample Devices
Class 1	High	100 mW (20 dBm)	Up to 100 m (328 feet)	USB adapters, access points
Class 1.5 (low energy)[7]	Med-High	10 mW (10 dBm)	Up to 30 m (100 feet), but typically 5 m (16 feet)	Beacons, wearable sensors
Class 2	Medium	2.5 mW (4 dBm)	Up to 10 m (33 feet)	Mobile devices, Bluetooth adapters, smart card readers
Class 3	Low	1 mW (0 dBm)	Up to 1 m (3 feet)	Bluetooth adapters

Power control is mandatory to have in this Bluetooth power class. In class 1

Power control is not mandatory to have in this Bluetooth power class. In class 2

Lowest output power is possible with this power class, nominal is 1mW. CLASS 3

Difference Between Bluetooth and Wi-Fi

Parameters	**Wi-Fi**	**Bluetooth**
Full Form	Full form of Wi-Fi is Wireless Fidelity.	Bluetooth has no full form.
Component	Wi-Fi demands a wireless router to set up the connectivity and adapter on the devices.	Bluetooth demands a Bluetooth setting or adapter on all devices to set up connectivity.
Power Consumption	It consumes high power.	It consumes low power.
Security	Wi-Fi is more secure than Bluetooth.	Bluetooth is less secure than Wi-Fi.
Multi User	WiFi can hold a large number of users to be connected.	Bluetooth holds a smaller number of users to be connected.
Bandwidth	WiFi requires high bandwidth.	Bluetooth demands low bandwidth.
Coverage	WiFi can cover an area up to 32 meters.	Bluetooth can cover an area up to 10 meters.

Bluetooth 1.0-1.2 (1999)

- A serial port complying with the RS-232 recommended standard was once a standard feature of many types of computers. Personal computers used them for connections not only to modems, but also to printers, computer mice, data storage, uninterruptible power supplies, and other peripheral devices.
- Bluetooth was designed to replace the RS-232 computer serial port. Its main usage was in connecting PC peripherals like modems and printers. In the next few years, Bluetooth 1.2 was integrated into devices like hands-free headsets, mobile phones, and laptops.
- SEIRAL PORT COMMUICATION

Main features

Bluetooth 1.0a and 1.0b featured peak data transfer speeds of around 732.2 kb/s, with a connection range of 10m or 33ft. Version 1.2 improved this by increasing the data transfer speed to 1 Mbps. Other enhancements include:

Drawbacks

Despite the improvements, Bluetooth 1.2 didn’t have enough bandwidth to transmit high-quality audio. It also had security issues and was prone to high power consumption.

Bluetooth 2.0-2.1 (2005)

At this point, Bluetooth technology was becoming a standard in more consumer products and a common feature in people’s daily lives. This was evident in an increased global demand for Bluetooth-enabled devices

- Erroneous Data Reporting
- Encryption Pause and Resume
- Secure Simple Pairing

Introduced in the Bluetooth 2.1 specification, Secure Simple Pairing (SSP) **fixes issues of the original pairing method**. It was relatively easy to determine the PIN-code and determine the Link Key by passively 'sniffing' the pairing process. SSP also makes pairing Bluetooth devices simpler.

- What is link key in Bluetooth?
- The link key is **a 128-bit random number and is a “shared secret” between devices**; it is never transmitted over the air. It is created during the pairing process, is stored and kept secret, and is used at every connection for authentication and derivation of the encryption key.

Main features

The most notable improvement in version 2.0 was Enhanced Data Rate (EDR). This increased the data transfer rate up to 2.1 Mb/s. Other enhancements include:

- Increased connection range of 30m or 100ft.
- Lower power consumption than Bluetooth 1 and longer battery life for wireless devices.
- Improved security with a new pairing system called SSP (Secure Simple Pairing).

Drawbacks

Bluetooth codec

A Bluetooth codec compresses and decompresses audio data into a specific format. It then transmits it at a specified bitrate.

SBC - **Sub-band Coding** - The mandatory and default codec for all stereo Bluetooth headphones with the Advanced Audio Distribution Profile (A2DP). It is capable of bit rates up to 328 kbps with a sampling rate of 44.1kHz , SBC yields poor sound quality and is prone to audio delays.

Bluetooth 3.0 (2009)

Bluetooth 3 + HS (High Speed) used the power of Wi-Fi, allowing faster transfer speeds. As such, it could transmit larger amounts of audio and video data. Thus making it the most common Bluetooth version for headphones, speakers, and portable media players

- AMP ALTERNATE MAC PHY Manager protocol (A2MP)
- L2CAPf
- 802.11 Protocol Adaptation Layer
- Enhanced Power Control
- Unicast Connectionless Data

Main features

- One of the most significant changes in Bluetooth 3 was its ability to transfer data over Wi-Fi. And it did this at speeds of 24 Mb/s. Other enhancements include: L2CAP Enhanced modes and alternative MAC and PHYs for transferring large digital files.
- Enhanced Power Control lets wireless devices adjust power levels as needed. This helps maintain a good Bluetooth connection.
- Unicast Connectionless Data facilitated the quick transfer of smaller amounts of data

Drawbacks

The downside of Bluetooth 3 was its high-power consumption. It quickly drained the batteries of the Bluetooth-enabled devices

Bluetooth 4.0-4.2 (2010)

Bluetooth 4 brought forth Bluetooth Low Energy (BLE) or Bluetooth Smart.

It helped power smaller devices like fitness trackers, hearing aids, and headphones, allowing them to stay paired longer using less power.

Bluetooth Smart Ready turned primary devices like laptops, tablets, and smartphones into connection hubs. This allowed them to send and receive data from Smart devices.

For 4.0 version this

- Low Energy Physical Layer
- Low Energy Link Layer
- Enhancements to HCI for Low Energy
- Low Energy Direct Test Mode
- AES Encryption
- Enhancements to L2CAP for Low Energy
- Enhancements to GAP for Low Energy
- Attribute protocol (ATT)
- Generic Attribute profile (GATT)
- Security Manager (SM)

4.1 VERSION FOR THIS

- BR/EDR secure connections
- Train nudging
- Generalized interlaced scan
- Low duty cycle directed advertising
- 32-bit UUID support in LE
- LE dual mode topology
- Piconet clock adjustment
- LE L2CAP connection-oriented channel support
- LE privacy v1.1
- LE Link Layer topology
- LE ping

4.2 VESRION FOR THIS

- LE Data Packet Length Extension
- LE Secure Connections
- Link Layer privacy
- Link Layer Extended Scanner Filter policies

Main features

Bluetooth 4 improved more than its power consumption. The introduction of the aptX codec also enhanced audio data transmission. This is due to its higher bitrate and efficient lossy compression algorithm. Other enhancements include:

- Increased connection range to 60m or 200ft
- Less interference between Bluetooth and 4G/LTE signals
- Improved pairing and re-pairing of devices
- Increased packet capacity and data range for IoT devices
- Improved data transmission with Adaptive Frequency Hopping (AFH)

Drawbacks

Smart Ready devices can easily pair with devices using older Bluetooth models. But they are best paired with other Smart Ready devices. This is for them to benefit from the new energy-saving feature of BLE.

Bluetooth 5.0-5.3 (2016- present)

The latest version, Bluetooth 5, has been a hot topic for developers and customers for its promising internet of things (IoT) experience, along with its improvements in speed, range and data capacity.
The latest Bluetooth version, Bluetooth 5, was released in July 2016. Many of its enhancements focus on providing a better operation framework for IoT devices.

5.0 VERSION FEATURES

- LE Isochronous Channels
- Enhanced Attribute Protocol
- LE Power Control
- AdvDataInfo in Periodic Advertising
- Host to Controller Encryption Key Control Enhancements
- LE Enhanced Connection Update
- LE Channel Classification

Main features

Bluetooth 5 provides an increased transfer rate of 50 Mb/s. It also extended the connection range to up to 240m. What's more, the 5.2 version also comes with Low Complexity Communication Codec (LC3). This new codec transfers audio data at lower bitrates without sacrificing audio quality.

Other notable improvements include:

- Less power consumption
- Increased message capacity
- Backward compatibility with Bluetooth 4 versions
- Dual Audio feature that allows you to connect to two different devices at the same time
- Addition of Slot Availability Mask (SAM), which further lessens interference with LTE

Drawbacks

Bluetooth 5 offers five security-enhancing features. Still, no wireless technology is entirely impervious to risks. Despite the improvements, Bluetooth 5 still faces vulnerabilities like bluesnarfing.

The data transmission speeds and operating range have doubled and quadrupled, respectively. This makes it possible to hack devices from farther distances and download more data in a shorter amount of time. This is risky because Bluetooth 5 only uses device authentication, not user authentication.

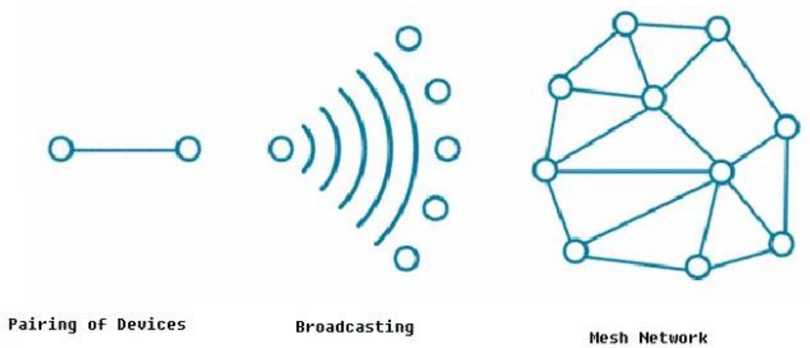


Fig. 4 – Working Principle of Bluetooth 5

REMOVE THE FEARURES

- Park State
- Unit keys
- Alternative MAC/PHY
- AMP Manager protocol (A2MP)
- L2CAP Enhancements for AMP
- 802.11 PAL
- 802.11n Enhancements to the 802.11 PAL

10. Glossary of Terms

- 2M PHY: a mode introduced in Bluetooth 5 in which the radio transmits data at a rate of 2 Mega symbols/second.

- Advertising: the link layer state in which the device is sending out advertising packets for other BLE to discover it.
- AES CCM: an authenticated encryption algorithm designed to provide both authentication and confidentiality.
- ATT (Attribute Protocol): a simple protocol for devices to list attributes that allow different operations such as write and read.
- Attribute: a generic term for any type of data exposed by a BLE server. It also defines the structure of this data.
- Authentication: the process or action of verifying the identity of a user, device, or process.
- Auxiliary Packets: this is the same as the secondary advertisement packets

- Beacon: a BLE device that broadcasts advertisement packets for the sole purpose of being discovered by other BLE devices. Usually, a beacon does not enter the connected state.
- Bluetooth Classic: a short-range wireless technology focused on streaming applications such as audio streaming.
- Bluetooth Low Energy: a short-range wireless technology focused on low-power and low bandwidth applications.
- Bluetooth mesh: a new Bluetooth specification that builds on top of BLE and allows BLE devices to form a many-to-many network topology.
- Bluetooth stack: software that refers to an implementation of the Bluetooth protocol.
- Broadcaster: a BLE device that sends out advertising packets and does not allow other BLE devices to connect to it.
- Central: a BLE device (usually a smartphone/tablet/PC) that listens for peripheral devices that are advertising. It is also capable of connecting to peripherals, and is responsible for managing the connection via its various parameters.
- Channel Hopping: the act of rapidly switching between different frequencies, using a pseudorandom frequency-selection algorithm agreed on by the transmitter and receiver.
- DTM (Direct Test Mode): a mode for performing RF tests, used during manufacturing and for certification tests of BLE devices.
- Encryption: the process of converting information or data into a code, especially to prevent unauthorized access.
- FHSS (Frequency Hopping Spread Spectrum): a method of transmitting radio signals by rapidly switching between different frequencies, using a pseudorandom frequency-selection algorithm agreed on by the transmitter and receiver.
- Gateway: a device that acts as both a BLE peripheral and a central connecting to multiple other peripheral devices
- Integrity: internal consistency or lack of corruption in digital data.
- ISM band: radio bands (portions of the radio spectrum) reserved internationally for the use of radio frequency (RF) energy for industrial, scientific and medical purposes other than telecommunications.
- Master: a role within the link layer that initiates a connection and controls the timings of data transmissions.
- MTU (Maximum Transmission Unit): the size of the largest protocol data unit (PDU) that can be communicated in a single network unit (packet).
- Observer: a BLE device that scans for advertising BLE devices, but is not capable of connecting to these devices.
- OOB (Out-of-Band): refers to a communication medium other than BLE, used for exchanging security keys between two BLE devices to secure the communication channel.

- Packet: a formatted unit of data carried by a network. A packet consists of control information and user data, which is also known as the payload.
- Pairing: the process by which two BLE devices exchange device information so that a secure link can be established.
- Payload: the user data portion of a packet.
- PDU (Protocol Data Unit): information that is transmitted as a single unit among peer entities in a network.
- Peripheral: a BLE device that sends out advertising packets and allows other BLE devices (specifically Centrals) to connect to it.
- Rand (Random Number): a random number used along with the EDIV (Encrypted Diversifier) in the process of creating and identifying the LTK (Long Term Key).
- Random Address: a device address that is programmed on the device or generated at runtime.
- Security Key: the alphanumeric key that's used to decrypt or encrypt data which could be exchanged between two BLE devices.
- Slave: a role within the link layer that accepts a connection from a master and abides to its timing requirements.
- Standby: the link layer state in which the device is not sending any BLE packets (not advertising, scanning, initiating, or connected to another BLE device).
- UUID (Universally Unique Identifier): a 128-bit number used to uniquely identify a device in a network, or entity within a device.

BLUETOOTH LOW ENERGRY

1.2 Technical Facts About BLE

- The frequency spectrum occupied is 2.400 - 2.4835 GHz
- The frequency spectrum is segmented into 40 “2 MHz”-wide channels.
- The maximum data rate supported by the radio (introduced in Bluetooth version 5) is 2 Mbps.
- A typical range is 10-30 meters (30-100 feet).
- The peak current consumption of a BLE chipset during radio transmission is typically under 15 mA.
- For all encryption operations, BLE uses AES CCM with a 128-bit key.
- BLE is designed for low-bandwidth data transfer applications.

Some of the notable differences are summarized in the following table:

Bluetooth Classic	BLE
Used for streaming applications such as audio streaming, file transfers, and headsets	Used for sensor data, control of devices, and low-bandwidth applications
Not optimized for low power, but has a higher data rate (3Mbps maximum compared to 2Mbps for BLE)	Meant for low power, low duty data cycles
Operates over 79 RF (radio frequency) channels	Operates over 40 RF channels.
Discovery occurs on 32 channels	Discovery occurs on 3 channels, leading to quicker discovery and connections than Bluetooth Classic

Table 1: Bluetooth Classic vs. BLE

ZigBee	Z-Wave
Data Rate: 250kb/s	Data Rate: 40kb/s
Power Consumption: ~40mA	Power Consumption: ~2.5mA
Range: 10-20 meters	Range: 30-65 meters
Operates at 2.4GHz	Operates at 908MHz
Chips and modules available from multiple manufacturers	Chips only sold by Silicon Labs
Variable certification process	Strict certification process
Supports over 65,000 end nodes	Supports over 232 end nodes
More difficult to configure and set up	More user-friendly and easier to set up
More cost-effective	More expensive than Zigbee

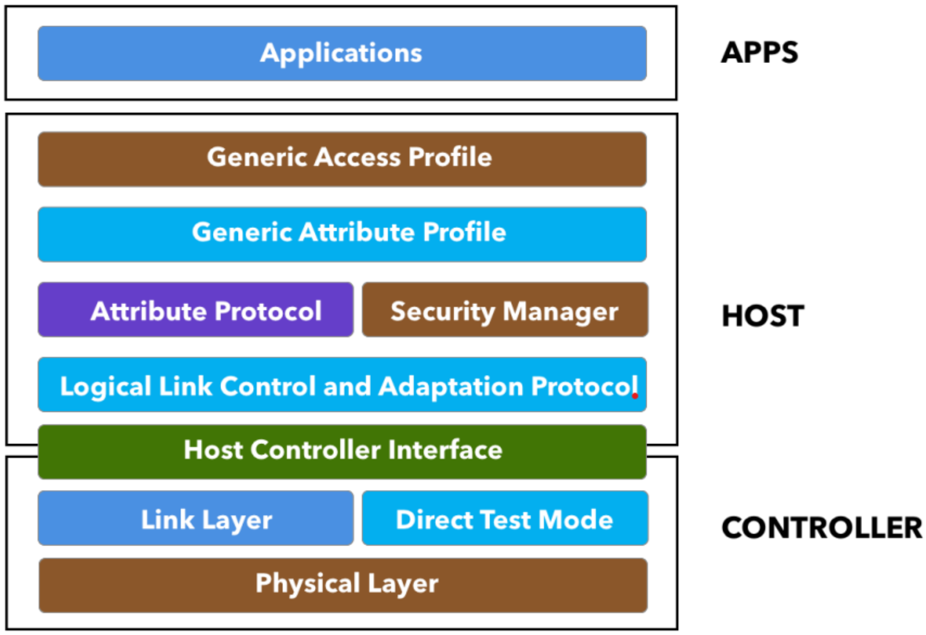


***Figure 3**: Architecture of BLE*

1.4.2. Advantages of BLE

- Lower power consumption
- No cost to access the official specification documents
- Lower cost of modules and chipsets

1.4.3. Applications Most Suitable for BLE

- Low-bandwidth data
- Device Configuration
- Using a smartphone as an interface
- Personal and wearable devices
- Broadcast-only devices
- Video streaming.
- High-quality audio streaming.
- Large data transfers for prolonged periods of time (if battery consumption is a concern).

1.5.2. Host

The **host** contains the following layers:

- Generic Access Profile (GAP)
- Generic Attribute Profile (GATT)
- Attribute Protocol (ATT)
- Security Manager (SM)
- Logical Link Control and Adaptation Protocol (L2CAP)
- Host Controller Interface (HCI) — Host side

1.5.3. Controller

The **controller** contains the following layers:

- Physical Layer (PHY)
- Link Layer
- Direct Test Mode
- Host Controller Interface (HCI) — Controller side

The physical layer (PHY)

refers to the radio hardware used for communication and for modulating/de-modulating the data.

BLE operates in the ISM band (2.4 GHz spectrum), which is segmented into 40 RF channels, each separated by 2 MHz (center-to-center), as shown in the following figure:

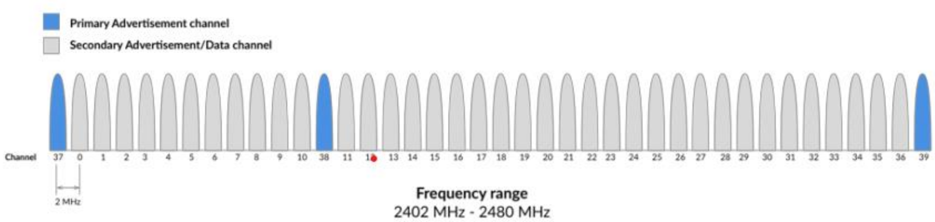


***Figure 4**: Frequency spectrum and RF channels in BLE*

- In older versions of Bluetooth (4.0, 4.1, and 4.2), the data rate was fixed at 1 Mbps.
- The physical layer radio (PHY) in this case is referred to as the 1M PHY and is mandatory in all versions including Bluetooth 5.
- With Bluetooth 5, however, two new optional PHYs were introduced: 2Mbps PHY, used to achieve twice the speed of earlier versions of Bluetooth. Coded PHY, used for longer range communication.

- Note: Well be covering these two new PHYs as well as the concept of coding in more detail in the chapter on Bluetooth 5.

1.5.4.2. Link Layer

a. The link layer is the layer that interfaces with the physical layer (radio) and provides the higher-level layers an abstraction and a way to interact with the radio (through an intermediary level called the HCI layer which we'll discuss shortly).
b. It is responsible for managing the state of the radio as well as the timing requirements necessary for satisfying the BLE specification. It is also responsible for managing hardware accelerated operations such as: CRC, random number generation, and encryption.

The three main states in which a BLE device operates in are:

- Advertising state
- Scanning state
- Connected state

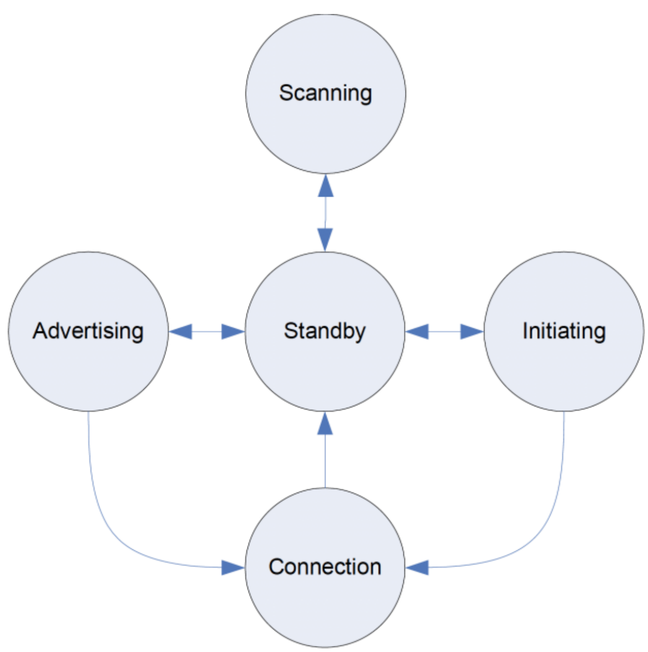


Figure 5: *Link layer states*

Standby: the default state in which the radio does not transmit or receive any data.

Advertising: the state in which the device sends out advertising packets for other devices to discover and read.

Scanning: the state in which the device scans for devices that are Advertising

Initiating: the state in which a scanning device decides to establish a connection with a device that is advertising.

Connected: the state in which a device has an established link with another device and regularly exchanges data with this other device.

This applies to both a device that was in the advertising state or one that was scanning for advertisements and then decided to initiate a connection with the advertising device.

In this connected state, the device that initiates the connection is called the master, and the device that was advertising is now called the slave.

Bluetooth Address Bluetooth devices

They are identified by a 48-bit address, similar to a MAC address.

- There are two main types of addresses:

Public Addresses and Random Addresses.

Public Address

This is a fixed address that does not change and is factory-programmed. It must be registered with the IEEE (similar to a WiFi or Ethernet device MAC address).

Random Address

Since manufacturers have a choice on what type of address to use (Random vs. Public), Random addresses are more popular since they do not require registration with the IEEE. A random address is programmed on the device or generated at runtime. It can be one of two sub-types:

- Static Address
  - Used as a replacement for Public addresses.
  - Can be generated at boot up OR stay the same during lifetime.
  - Cannot change until a power cycle.
- Private address
  This one is also split up into two additional sub-types:
  - Non-resolvable Private Address:
    - Random, temporary for a certain time.
    - Not commonly used.
  - Resolvable Private Address:
    - Used for privacy.
    - Generated using **Identity Resolving Key (IRK)** and a random number.
    - Changes periodically (even during the lifetime of the connection).
    - Used to avoid being tracked by unknown scanners
    - Trusted devices (or **Bonded**, which is described later in the chapter on **Security**) can resolve it using the previously stored IRK.

2.3. Observers and Broadcasters vs. Centrals and Peripherals

Let's go over some advantages and disadvantages of the four different types of device: Observers, Broadcasters, Centrals, and Peripherals.

Broadcaster	Peripheral	Observer	Central
No need for a radio receiver	Needs both a receiver and transmitter	No need for a transmitter	Needs both a receiver and transmitter
No bi-directional data transfer	Supports bi-directional data transfer	No bi-directional data transfer	Supports bi-directional data transfer
Reduced hardware, reduced BLE software stack	Requires the full BLE software stack	Reduced hardware, reduced BLE software stack	Requires the full BLE software stack

***Table 2**: Comparison between Observers, Broadcasters, Peripherals, and Centrals*

2.5. Multi-Role BLE Devices

- In some use cases, a BLE device would benefit from acting in multiple roles simultaneously.

- For example, a device may want to monitor multiple sensors (peripheral devices), and at the same time be able to advertise its presence to a smartphone to allow access to sensor data from a mobile app interface.

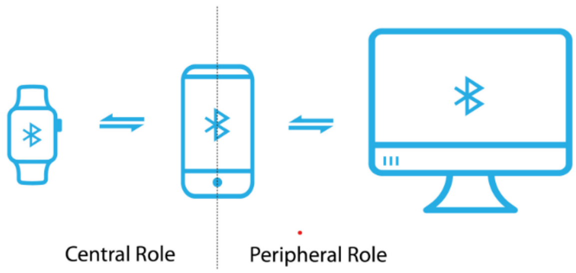


Figure 7: Multiple roles in BLE

3.1. Generic Access Profile (GAP)

The Generic Access Profile (GAP) provides the framework that defines how BLE devices interact with each other.

This includes the following aspects:

- Modes & Roles of BLE devices.
- Advertisements (advertising, scanning, advertising parameters, advertising data, scanning parameters).
- Connection establishment (initiating, accepting, connection parameters)
- Security

5. Services and Characteristics

5.1. Attribute Protocol (ATT)

It is the device that accepts incoming commands from a peer device, and sends responses, notifications and indications.

5.2. Generic Attribute Profile (GATT)

Now that we've covered the concept of attributes, we'll go over three important concepts in BLE that you will come across very often:

- Services
- Characteristics
- Profile

These concepts are used specifically to allow hierarchy in the structuring of the data exposed by the Server.

5.3.1. Services

- A service is a grouping of one or more attributes, some of which are characteristics.
- It's meant to group together related attributes that satisfy a specific functionality on the server.
- For example, the SIG-adopted battery service contains one characteristic called the battery level.
- A service also contains other attributes (non-characteristics) that help structure the data within a service (such as service declarations, characteristic declarations, and others).

Properties:

by a number of bits and which defines how a characteristic value can be used. Some examples include: read, write, write without response, notify, indicate.

Descriptors: used to contain related information about the characteristic Value. Some examples include: extended properties, user description, fields used for subscribing to notifications and indications, and a field that defines the presentation of the value such as the format and the unit of the value.

5.4. Profiles

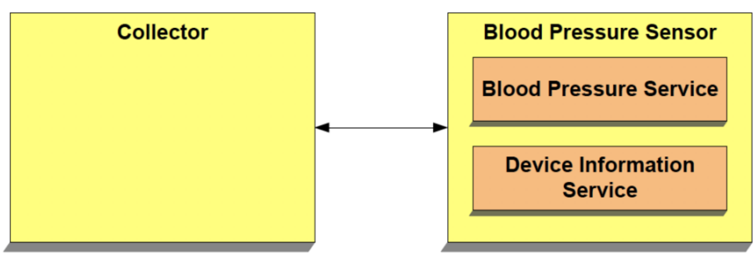


Figure 17: *Blood pressure profile*
(Source: Blood Pressure Profile specification document)

- Definition of roles and the relationship between the GATT server and client.
- Required Services.
- Service requirements. How the required services and characteristics are used.
- Details of connection establishment requirements including advertising and connection parameters.
- Security considerations.

5.6. Attribute Operations

Commands, request, response, notification, indication, confirmation

6. GATT Design Exercise

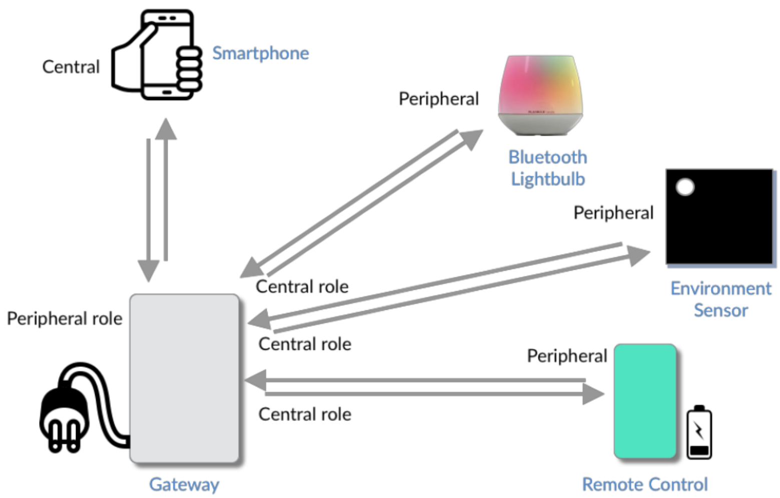


Figure 19: BLE home automation project example

8. Security

8.1. Security Concerns

- Authentication: Authentication is proof that the other side is who they claim they are. So if you're connecting to a BLE device, you want to be sure that you are actually connecting to the device of interest — and not some other malicious device that's pretending to be that device.
- Integrity: Integrity ensures us that the data received is free from corruption and tampering by unauthorized devices.
- Confidentiality: Confidentiality is concerned with making sure the data is not readable by unauthorized users or devices.
- Privacy: Privacy is concerned with how private the communication is, and whether a third party is able to track our device — especially by its Bluetooth address.

8.1.1. Types of Attacks

- Passive Eavesdropping: This describes when a malicious device listens in on the communication between two devices and is able to understand the data — usually by gaining access to the encryption key in the case the data is encrypted.
- Active Eavesdropping: This is also known as a Man-In-The-Middle (MITM) attack. In this attack, the malicious device impersonates both devices (the peripheral and the central). It could then intercept the communication between them, route it so they do not realize that the attack is happening, and possibly even injecting data into the packets.
- Privacy and Identity Tracking: In this attack, devices and users are tracked by the Bluetooth address — possibly revealing their location and correlating it with their behavior.

8.2. Security in BLE

The security manager defines the protocols and algorithms for generating and exchanging keys between two devices. It involves five security features:

- **Pairing**: the process of creating shared **secret keys** between two devices.
- **Bonding**: the process of creating and storing shared **secret keys** on each side (central and peripheral) for use in subsequent connections between the devices.
- **Authentication**: the process of verifying that the two devices share the same secret keys.
- **Encryption**: the process of encrypting the data exchanged between the devices. Encryption in BLE uses the 128-bit AES Encryption standard, which is a **symmetric-key** algorithm (meaning that the same key is used to encrypt and decrypt the data on both sides).
- **Message Integrity**: the process of **signing** the data, and **verifying** the signature at the other end. This goes beyond the simple integrity check of a calculated CRC.

The Security Manager addresses the different security concerns as follows: Confidentiality via encryption. Authentication via pairing & bonding. Privacy via resolvable private addresses. Integrity via digital signatures.

8.4. An Overview of the Different Security Keys

There are a number of keys and variables used during the different security procedures. Let's go over them one by one.

- **Temporary Key (TK)**:
  Generation of the temporary key (TK) depends on the pairing method chosen. The TK gets generated each time the pairing process occurs. The TK is used in legacy connections only.
- **Short Term Key (STK)**:
  This key is generated from the TK exchanged between the devices. The STK gets generated each time the pairing process occurs and is used to encrypt the data throughout the current connection. The STK is used in legacy connections only.
- **Long Term Key (LTK)**:
  This key gets generated and stored during phase three of the security process in legacy connections and during phase two in LE secure connections. It gets stored on each of the two devices that are bonded, and used in subsequent connections between the two devices.
- **Encrypted Diversifier (EDIV) and Random Number (Rand)**:
  These two values are used to create and identify the LTK. They also get stored during the bonding process.
- **Connection Signature Resolving Key (CSRK)**:
  Used to **sign** data and verify the **signature** attached to the data at the other end. This key is stored on each of the two bonded devices.
- **Identity Resolving Key (IRK)**:
  Used to resolve random private addresses. This key is unique per device, so the master's IRK will get stored on the slave side, and the slave's IRK will be stored on the master side.

8.5. Security Modes and Levels

Security Mode 1

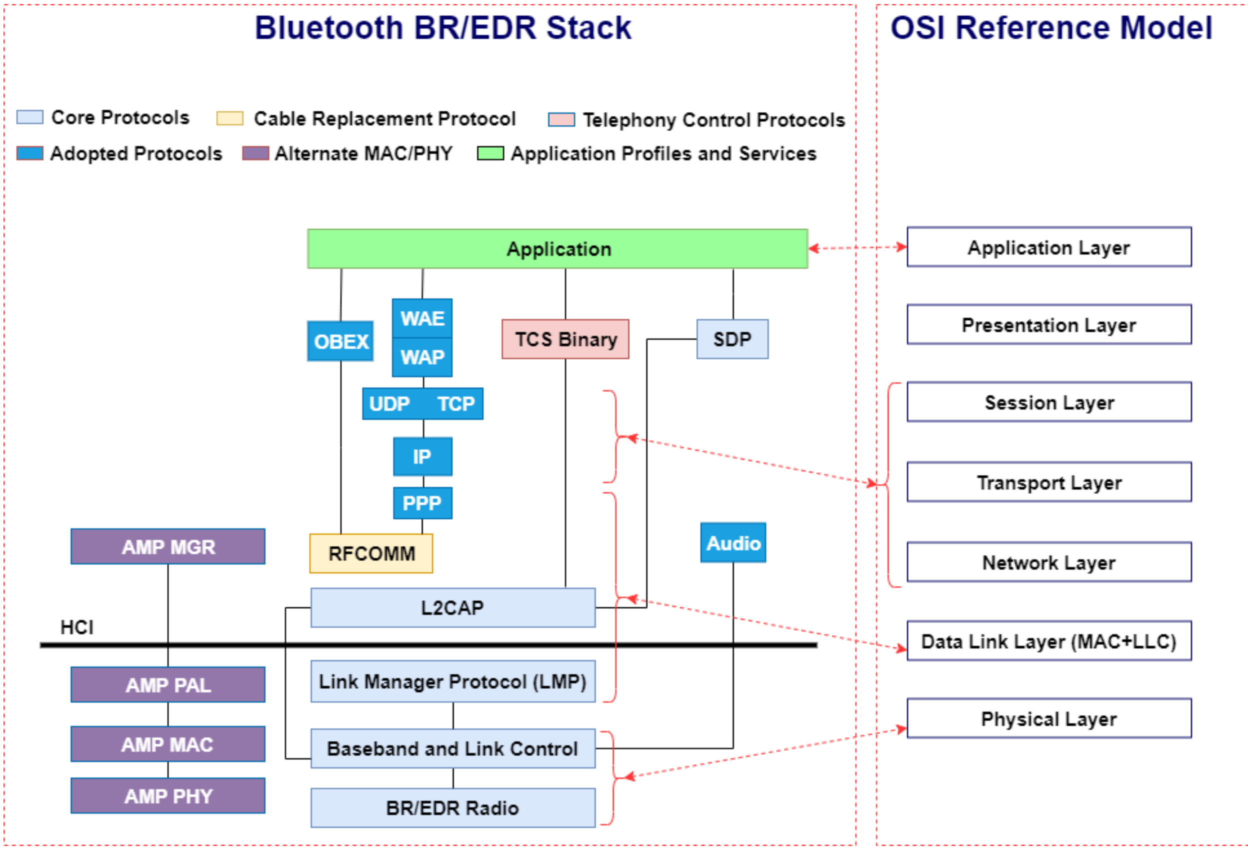
- **Level 1**: No security (no authentication and no encryption)
- **Level 2**: Unauthenticated pairing with encryption
- **Level 3**: Authenticated pairing with encryption
- **Level 4**: Authenticated LE secure connections pairing with encryption

Security Mode 2

- **Level 1**: Unauthenticated pairing with data signing
- **Level 2**: Authenticated pairing with data signing

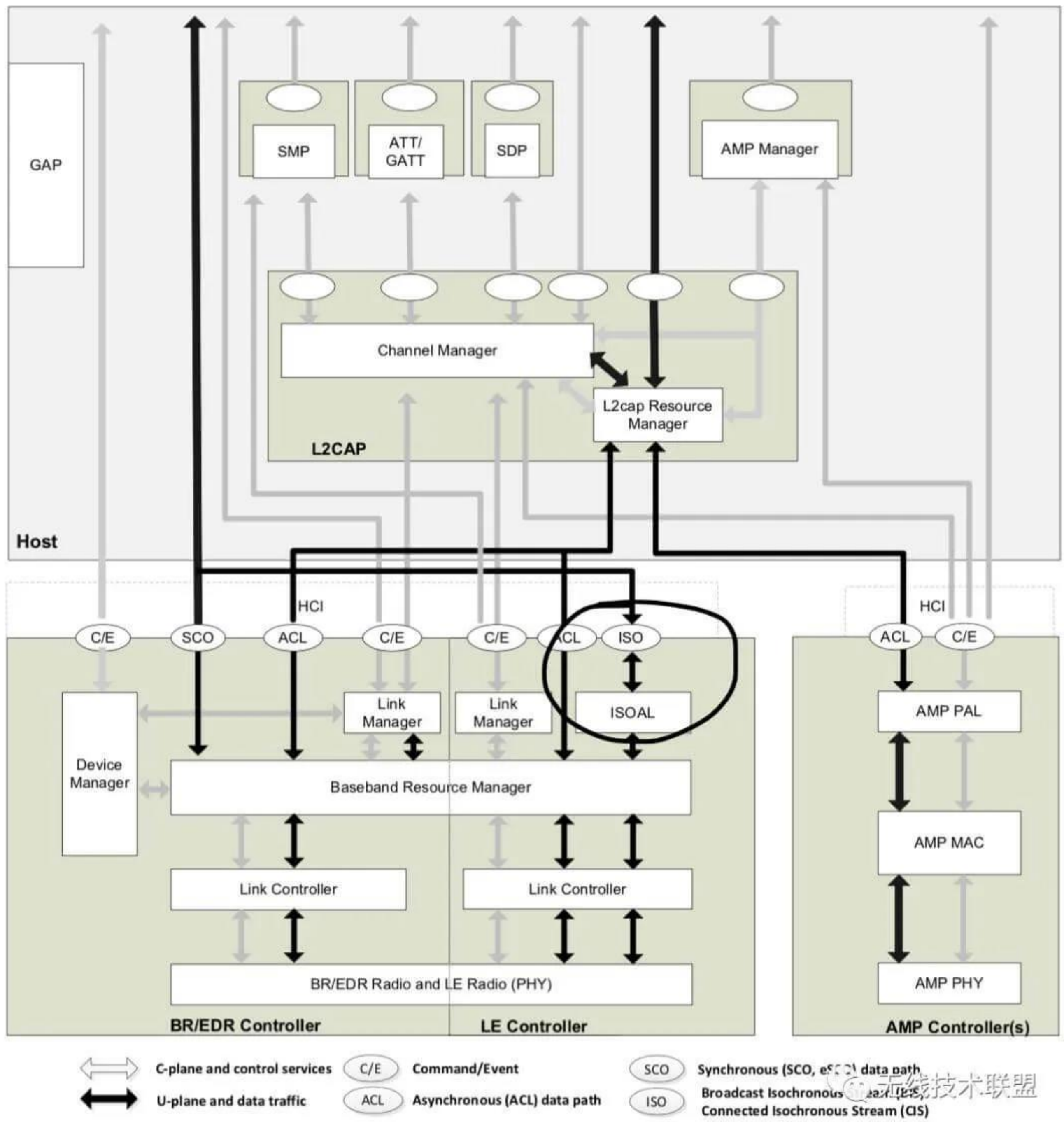
Communication Protocols

Applications	Provides what application are able to run.
RFCOMM / SDP	RFCOMM: Provides emulation of serial ports over the L2CAP. SDP: Provides a means for applications to discover which services are available and to determine the characteristics of those available services
L2CAP	Provides connection-oriented and connectionless data services to upper layer protocols with protocol multiplexing capability, segmentation, and reassembly operation and group abstractions.
Host Controller Interface (HCI)	Provides a command interface to the baseband controller and link manager, and access to hardware status and control registers.
Link Manager (LM)	Carries out link setup, authentication, link configuration, etc.
Baseband	Manages physical channels and links (asynchronous and synchronous), handles packets, paging and inquiry to access Bluetooth devices in the area.
Radio	Defines the requirements for the Bluetooth transceiver device.
Bluetooth	



Bluetooth BR/EDR Stack
OSI Reference Model
Core Protocols
Cable Replacement Protocol
Telephony Control Protocols
Adopted Protocols
Alternate MAC/PHY
Application Profiles and Services
Application
OBEX
WAE
WAP
TCS Binary
SDP
UDP
TCP
IP
PPP
RFCOMM
Audio
AMP MGR
HCI
L2CAP
AMP PAL
Link Manager Protocol (LMP)
AMP MAC
Baseband and Link Control
AMP PHY
BR/EDR Radio
Application Layer
Presentation Layer
Session Layer
Transport Layer
Network Layer
Data Link Layer (MAC+LLC)
Physical Layer

CORE SYSTEM ARCHITECTURE



What is generic access profile in Bluetooth?

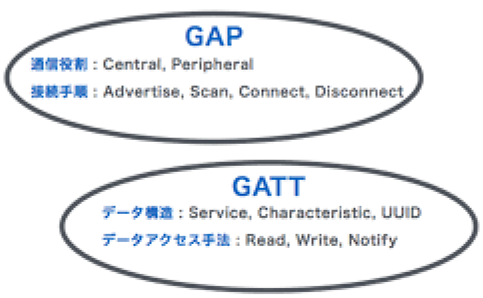
The GAP layer of the Bluetooth low energy protocol stack is responsible for connection functionality. This layer handles the access modes and procedures of the device including device discovery, link establishment, link termination, initiation of security features, and device configuration

What is Gap and GATT?

GAP provides a standard framework for controlling a BLE device, while GATT provides a standard framework for managing data in a BLE device. As a BLE firmware developer, these two layers are the most layers interacted with in the BLE protocol stack, that is why we will invest some time to master them.

- **GAP** defines the **general topology** of the BLE network stack.
- **GATT** describes in detail **how attributes (data) are transferred** once devices have a dedicated connection.

RFCOMM is a reliable stream-based protocol. L2CAP is a packet-based protocol that can be configured with varying levels of reliability. L2CAP actually serves as the transport protocol for RFCOMM, so every RFCOMM connection is actually encapsulated within an L2CAP connection



STUDY MORE ON =====LMP, HCI, L2CAP, SDP, GAP, HFP, A2DP,

What Bluetooth profile means?

A Bluetooth profile is **a wireless interface specification for Bluetooth-based communication between devices**, such as the Hands-Free profile. For a mobile device to connect to a wireless headset, both devices must support the Hands-Free profile.

How do Bluetooth profiles work?

Bluetooth Profiles are **a set of rules that allow the technology to complete a particular task**. For example, to connect a pair of headphones to another device, a specific Bluetooth profile (or rules) is used. A different Bluetooth profile is needed to transfer files from one device to another

The Bluetooth Core system consists of a Host, a Primary Controller and zero or more Secondary Controllers.

Architecture Figure 2.1: Bluetooth core system architecture

CONTROLLER

- LINK MANGER, LINK CONTROLLER, BR/EDR RADIO BLOCK == BR/EDR CONTROLLER
- LINK MANGER, LINK CONTROLLER, LE RADIO BLOCK == LE CONTROLLER

HOST

- L2CAP, SDP, GAP BLOCKS == BR/EDR HOST
- L2CAP, SMP, ATTRIBUTE PROTOCOL, GAP, GATT == LE HOST

This table summarizes and compares different features of Bluetooth BR/EDR and Bluetooth LE.

Feature	**Bluetooth BR/EDR**	**Bluetooth LE**
Frequency band	Operates on a 2.4 GHz Industrial, Scientific, and Medical (ISM) band, with the values in the range from 2.4000 GHz to 2.4835 GHz	Operates on 2.4 GHz ISM band, with the values in the range from 2.4000 GHz to 2.4835 GHz

Channels	79 channels	40 channels (37 data channels and 3 advertising channels)
Channel bandwidth	1 MHz	2 MHz
Spread spectrum technique	1600 hops/sec frequency-hopping spread spectrum (FHSS)	FHSS
Modulation scheme	• Gaussian frequency shift keying (GFSK) • π/4 differential quadrature phase shift keying (DQPSK) • 8 differential phase shift keying (DPSK)	GFSK
Power usage	1 W (reference value)	~0.01x W to 0.5x W of reference (depending on the use case scenario)
Maximum transmission power	• Class 1: 100 mW (20 dBm) • Class 2: 2.5 mW (4 dBm) • Class 3: 1 mW (0 dBm)	• Class 1: 100 mW (20 dBm) • Class 1.5: 10 mW (10 dBm) • Class 2: 2.5 mW (4 dBm) • Class 3: 1 mW (0 dBm)
Device discovery	Inquiry or paging	Advertising
Device address privacy	None	Private device address supported
Encryption algorithm	E0/SAFER+	AES-CCM
Audio capable	Yes	Yes (Bluetooth LE audio is introduced in Bluetooth Core Specification 5.2)
Network topology	Point-to-point (including piconet)	• Point-to-point (including piconet) • Broadcast • Mesh

BR/EDR/LE Controller architectural blocks

BR/EDR Radio.

- The BR/EDR radio is the lowest defined layer of the Bluetooth specification.
- The BR mode is mandatory, whereas the EDR mode is optional.
- This layer defines the requirements of the Bluetooth transceiver device operating in the 2.4 GHz ISM frequency band.
- It implements a 1600 hops/sec FHSS technique. (Frequency-hopping spread spectrum (FHSS) transmission is the repeated switching of the carrier frequency during radio transmission to reduce interference and avoid interception.

FHSS is useful to counter eavesdropping, as well as to obstruct the frequency jamming of telecommunications and to enable code-division multiple access communications. It can also minimize the effects of unintentional interference.)

- The radio hops in a pseudo-random way on **79 designated** Bluetooth channels.
- Each Bluetooth channel has a bandwidth of 1 MHz's Each frequency is located at (2402 + *k*) MHz, where *k* = 0,1,...78.
- The modulation technique for BR and EDR mode is GFSK and differential phase shift keying (DPSK), respectively.

Differential phase shift keying (DPSK) is **a common form of phase modulation that conveys data by changing the phase of the carrier wave.**

What is GFSK Bluetooth?

- The Bluetooth radio interface also uses a modulation technique called **Gaussian Frequency Shift Keying**, GFSK. This form of modulation is spectrally efficient and also enables the use of efficient radio power amplifiers, thereby saving on battery life.

Bluetooth LE PHY.

The Bluetooth LE PHY air interface operates in the same unlicensed 2.4 GHz Industrial, Scientific, and Medical (ISM) frequency band as Wi-Fi®. The Bluetooth LE PHY air interface also includes these characteristics:

- Operating radio frequency (RF) is in the range 2.4000 GHz to 2.4835 GHz, inclusive.
- The channel bandwidth is 2 MHz. The operating band is divided into 40 channels, k = 0, …, 39. The center frequency of the kth channel is 2402 + k × 2 MHz.
    - User data packets are transmitted using channels in the range [0, 36].
    - Advertising data packets are transmitted in channels 37, 38, and 39.
- Gaussian frequency shift-keying (GFSK) modulation scheme is implemented.
- The Bluetooth LE PHY uses frequency-hopping spread spectrum (FHSS) to reduce interference and to counter the impact of fading channels. The time between

frequency hops can vary from 7.5 ms to 4 s and is set at the connection time for each Peripheral.

- Support for throughput at 1 Mbps is mandatory for specification version 4.x compliant devices. At a data rate of 1 Mbps, the transmission is uncoded.
- Optionally, devices compliant with the Bluetooth Core Specification version 5.1 support these additional data rates:
  - Coded transmission at bit rates of 500 kbps or 125 kbps
  - Uncoded transmission at a bit rate of 2 Mbps

- Radio layer :
  – defines the requirements for a Bluetooth transceiver
- Baseband :
  – manages physical channels, links, error correction, hop selection, etc.
- LMP : Link Manager Protocol
  – used by Link Managers for link set up and control, authentication, encryption
- HCI : Host Controller Interface
  – provides a command interface to Baseband Link Controller and Link Manager
- L2CAP : Logical Link Control and Adaptation Protocol
  – provides connection-oriented and connectionless data services, protocol multiplexing, packet segmentation and reassembly, QoS info. etc.
- RFCOMM :
  – provides emulation of serial ports over the L2CAP protocol
- SDP : Service Discovery Protocol
  – allows applications to discover available services and their characteristics

Device manager

- The device manager is the functional block in the baseband that controls the general behavior of the Bluetooth device.
- It is responsible for all operations of the Bluetooth system that are not directly related to data transport, such as inquiring for the presence of nearby Bluetooth devices, connecting to Bluetooth devices, or making the local Bluetooth device discoverable or connectable by other devices.

- The device manager also controls local device behavior implied by a number of the HCI commands, such as managing the device local name, any stored link keys, and other functionality.

Link Manager protocol

- The link manager is responsible for the **creation, modification and release of logical links**, as well as the update of parameters related to physical links between devices.
- The LM or LL protocol allows the creation of new logical links and logical transports between devices when required.
- as well as the general control of link and transport attributes such as *the adjustment of QoS settings in BR/EDR for a logical link.*

- Piconet Management = Piconet creation, Attach and detach slaves, Master-slave switch, Establishing SCO and ACL links, Handling of low power modes (Sniff, Hold, Park)
- Link Configuration = packet type negotiation (bring about by discussion), power control
- Security functions = Authentication, Encryption

 Procedures defined for LMP are grouped into 24 functional areas, which include

  - Authentication
  - Pairing
  - Encryption
  - Clock offset request
  - Switch master/slave
  - Name request
  - Hold or park or sniff mode

Baseband resource manager

- The baseband resource manager is responsible for **all access to the radio medium**.
- It has two main functions. At its heart is a scheduler that grants time on the physical channels to all of the entities that have negotiated an access contract.
- The other main function is to negotiate (bring about by discussion) access contracts with these entities.
- The access contract and scheduling function must take account of any behavior that requires use of the Primary Controller.

Link Controller

- The Link Controller is responsible for the **encoding and decoding** of Bluetooth packets from the data payload and parameters related to the physical channel, logical transport and logical link.

- The Link Controller carries out the link control protocol signaling in BR/EDR and Link Layer protocol in LE, which is used to communicate **flow control and acknowledgment and retransmission request signals.**
- Interpretation and control of the link control signaling is normally associated with the resource manager's scheduler.

PHY

- The PHY block is responsible for transmitting and receiving packets of information on the physical channel.
- A control path between the baseband and the PHY block allows the baseband block to control the timing and frequency carrier of the PHY block.
- The PHY block transforms a stream of data to and from the physical channel and the baseband into required formats.

Isochronous Adaptation Layer

- The Isochronous Adaptation Layer (ISOAL) enables the upper layer to send or receive isochronous data to or from the Link Layer in a flexible way such that the size and interval of data packets in the upper layer can be different from the size and interval of data packets in the Link Layer.
- The ISOAL uses fragmentation/recombination or segmentation/reassembly operations to convert upper layer data units into lower layer data units (or the other way around).

HOST ARCHITECTURAL BLOCKS

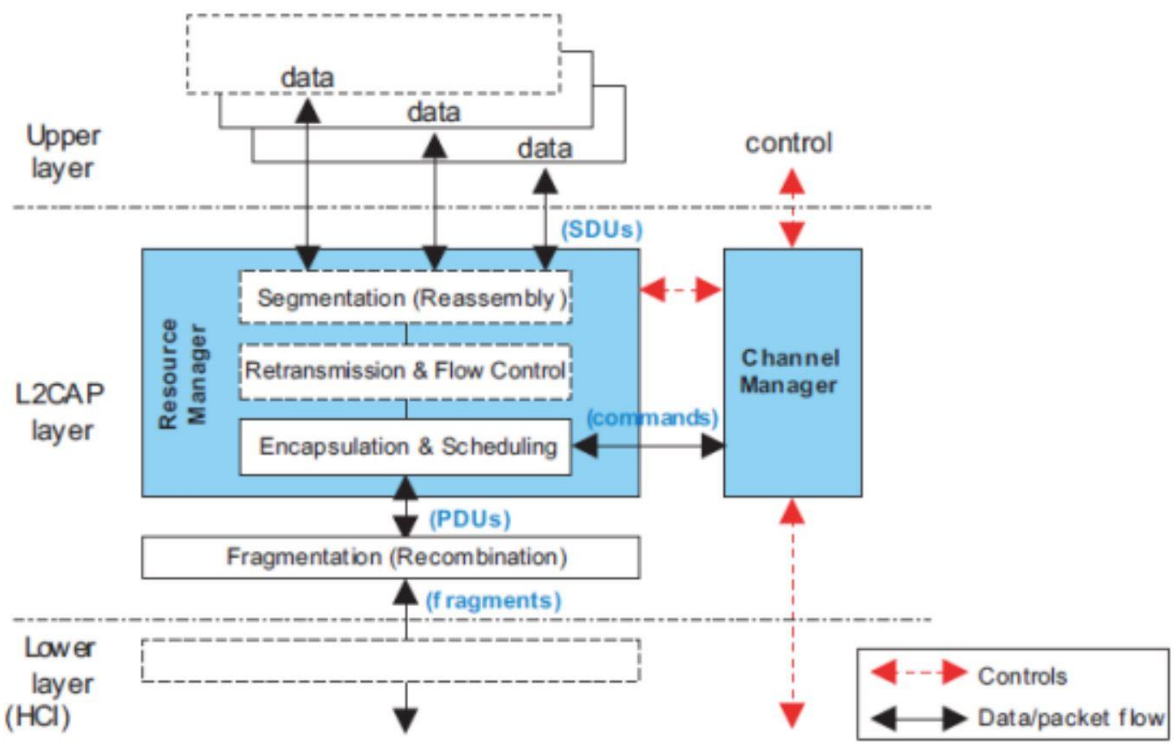
CHANNEL MANGER

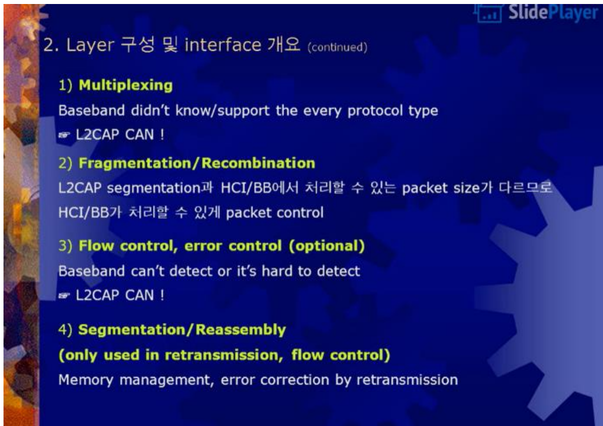
- The channel manager is responsible for ***creating, managing and closing L2CAP channels*** for the transport of **service protocols and application data streams**.
- The channel manager uses the L2CAP protocol to interact with a channel manager on a peer device to create these L2CAP channels and connect their endpoints to the appropriate entities.
- The channel manager interacts with its local link manager to create new logical links (if necessary) and to configure these links to provide the required quality of service for the type of data being transported.

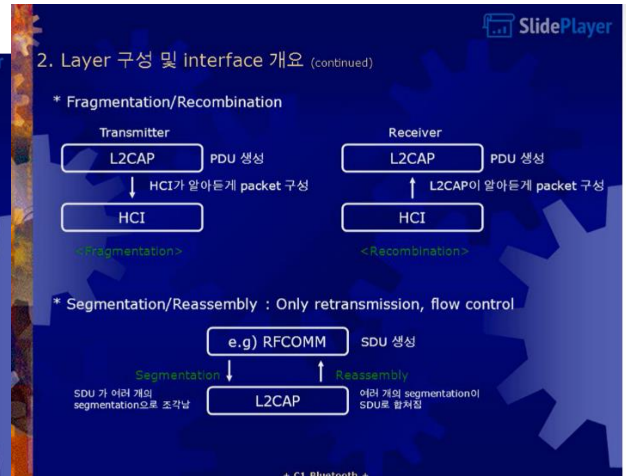
L2CAP resource manager

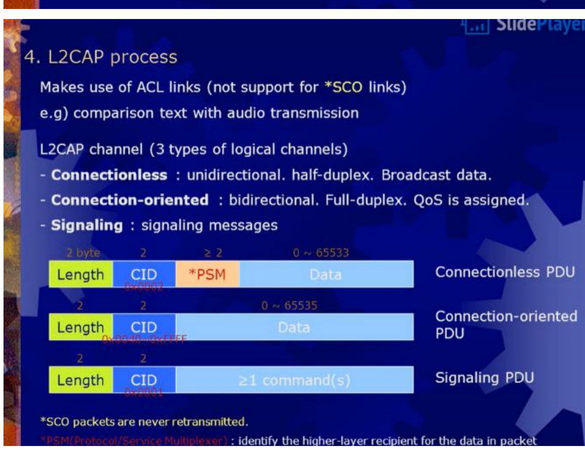
L2CAP is used within the Bluetooth protocol stack. It passes packets to either the Host Controller Interface (HCI) or, on a hostless system, directly to the Link Manager/ACL link.

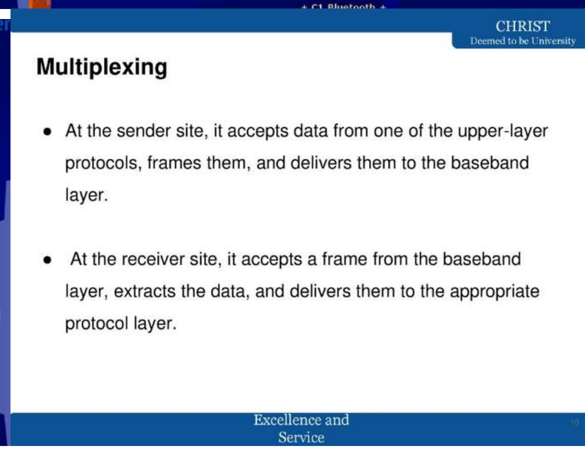
- The Logical Link Control and Adaptation Layer Protocol (L2CAP) is layered over the Baseband Protocol and resides in the data link layer.
- L2CAP provides connection-oriented and connectionless data services to upper layer protocols with protocol multiplexing capability, segmentation and reassembly operation, and group abstractions.
- L2CAP permits higher level protocols and applications to transmit and receive L2CAP data packets up to 64 kilobytes in length.
- Two link types are supported for the Baseband layer: Synchronous Connection-Oriented (SCO) links and Asynchronous Connection-Less (ACL) links.
- SCO links support real-time voice traffic using reserved bandwidth.
- ACL links support best effort traffic. The L2CAP Specification is defined for only ACL links and no support for SCO links is planned.











7.3.1 Segmentation of L2CAP

- SDUs In Flow Control, Streaming, or Retransmission modes, incoming SDUs may be broken down into segments, which shall then be individually encapsulated with L2CAP protocol elements (header and checksum fields) to form I-frames. Iframes are subject to flow control and may be subject to retransmission procedures.

7.3.2 Reassembly of L2CAP SDUs

- The receiving side uses the SAR field of incoming 'I-frames' for the reassembly process.
- The L2CAP SDU length field, present in the "start of SDU" I-frame, is an extra integrity check, and together with the sequence numbers may be used to indicate lost L2CAP SDUs to the application.

7.3 ENCAPSULATION OF SDUs

- All SDUs are encapsulated into one or more L2CAP PDUs.
- In Basic L2CAP mode, an SDU shall be encapsulated with a minimum of L2CAP protocol elements, resulting in a type of L2CAP PDU called a Basic Information Frame (B-frame).
- Segmentation and Reassembly operations are only used in Enhanced Retransmission mode, Streaming mode, Retransmission mode, and Flow Control mode.
- SDUs may be segmented into a number of smaller packets called SDU segments.

- Each segment shall be encapsulated with L2CAP protocol elements resulting in an L2CAP PDU called an Information Frame (Iframe).
- The maximum size of an SDU segment shall be given by the Maximum PDU Payload Size (MPS).
- The MPS parameter may be exported using an implementation specific interface to the upper layer.

7.2.1 Fragmentation of L2CAP PDUs

- An L2CAP implementation may fragment any L2CAP PDU for delivery to the lower layers.
- If L2CAP runs directly over the Controller without HCI, then an implementation may fragment the PDU into multiple Controller packets for transmission over the air.
- If L2CAP runs above the HCI, then an implementation may send HCI transport sized fragments to the Controller.
- All L2CAP fragments associated with an L2CAP PDU shall be processed for transmission by the Controller before any other L2CAP PDU for the same logical transport shall be processed.

7.2.2 Recombination of L2CAP PDUs

- Controllers will attempt to deliver packets in sequence and without errors.
- Recombination of fragments may occur in the Controller but ultimately it is the responsibility of L2CAP to reassemble PDUs and SDUs and to check the length field of the SDUs.
- As the Controller receives packet fragments, it either signals the L2CAP layer on the arrival of each fragment or accumulates a number of fragments (before the receive buffer fills up or a timer expires) before passing packets to the L2CAP layer.

TERMINOLOGY

Upper layer

- The system layer above the L2CAP layer, which exchanges data with L2CAP in the form of SDUs**. (Service data unit)** or segments of SDUs

- The interface of the L2CAP layer with the upper layer is not specified.

Lower layer

- The system layer below the L2CAP layer, which exchanges data with the L2CAP layer in the form of PDUs, or fragments of PDUs**. (Protocol data unit)**
- The lower layer is mainly represented within the Controller,
- Except for the HCI functional specification (in case HCI is involved) the interface between L2CAP and the lower layer is not specified.

L2CAP channel

The logical connection between two endpoints in peer devices, characterized by their Channel Identifiers (CID), which is multiplexed over one Controller based logical link.

SDU, or L2CAP SDU

Service Data Unit: a packet of data that L2CAP exchanges with the upper layer and transports transparently over an L2CAP channel using the procedures specified here.

Segment, or SDU segment

A part of an SDU, as resulting from the Segmentation procedure. An SDU may be split into one or more segments.

Segmentation

A procedure used in the L2CAP Retransmission and Flow Control Modes, resulting in an SDU being split into one or more smaller units, called Segments, as appropriate for the transport over an L2CAP channel.

Reassembly

- The reverse procedure corresponding to Segmentation, resulting in an SDU being re-established from the segments received over an L2CAP channel, for use by the upper layer. ISOAL
- The interface between the L2CAP and the upper layer is not specified; therefore, reassembly may actually occur within an upper layer entity although it is concept part of the L2CAP layer.

PDU, or L2CAP PDU

- **Protocol Data Unit:** a packet of data containing L2CAP protocol information fields, control information, and upper layer information data.
- A PDU is always started by a Basic L2CAP header.
- Types of PDUs are B-frames, I-frames, S-frames, C-frames, G-frames, and LE - Frames.

- **Basic information frame (B-frame)** A B-frame is a PDU used in the Basic L2CAP mode for L2CAP data packets. It contains a complete SDU as its payload, encapsulated by a Basic L2CAP header.
- **Information frame (I-frame)** An I-frame is a PDU used in Enhanced Retransmission Mode, Streaming mode, Retransmission mode, and Flow Control Mode. It contains an SDU segment and additional protocol information, encapsulated by a Basic L2CAP header
- **Supervisory frame (S-frame)** An S-frame is a PDU used in Enhanced Retransmission Mode, Retransmission Mode, and Flow Control Mode. It contains protocol information only, encapsulated by a Basic L2CAP header, and no SDU data.
- **Control frame (C-frame)** A C-frame is a PDU that contains L2CAP signaling messages exchanged between the peer L2CAP entities. C-frames are exclusively used on the L2CAP signaling channels.
- **Group frame (G-frame)** A G-frame is a PDU exclusively used on the Connectionless L2CAP channel. It is encapsulated by a Basic L2CAP header and contains the PSM followed by the completed SDU. G-frames may be used to broadcast data to active slaves via Active Broadcast or to send unicast data to a single remote device.

*Fragment (*a small part broken off or separated from something.)

- A part of a PDU ***(PROTOCAL DATA UNIT),*** as resulting from a fragmentation operation.
- Fragments are used only in the delivery of data to and from the lower layer.
- They are not used for peer-to-peer transportation.
- A fragment may be a Start or Continuation Fragment with respect to the L2CAP PDU.
- A fragment does not contain any protocol information beyond the PDU; the distinction of start and continuation fragments is transported by lower layer protocol provisions.

Fragmentation

- A procedure used to split L2CAP PDUs to smaller parts, named fragments, appropriate for delivery to the lower layer transport.
- Although described within the L2CAP layer, fragmentation may actually occur in an HCI Host driver, and/or in a Controller, to accommodate the L2CAP PDU transport to HCI Data packet or Controller packet sizes.

- Fragmentation of PDUs may be applied in all L2CAP modes.

Recombination

- The reverse procedure corresponding to fragmentation, resulting in an L2CAP PDU re-established from fragments.
- In the receive path, full or partial recombination operations may occur in the Controller and/or the Host, and the location of recombination does not necessarily correspond to where fragmentations occur on the transmit side.

Maximum Transmission Unit (MTU)

The maximum size of payload data, in octets, that the upper layer entity is capable of accepting, i.e. the MTU corresponds to the maximum SDU size.

Maximum PDU Payload Size (MPS)

The maximum size of payload data in octets that the L2CAP layer entity can accept (that is, the MPS corresponds to the maximum PDU payload size).

- The Logical Link Control and Adaptation Layer Protocol (L2CAP) is layered over the Baseband Protocol and resides in the data link layer.
- L2CAP provides connection-oriented and connectionless data services to upper layer protocols with **protocol multiplexing capability, segmentation and reassembly operation, and group abstractions.**
- L2CAP permits higher level protocols and applications to transmit and receive L2CAP data packets up to 64 kilobytes in length.
- **Two link types are supported for the Baseband layer**:
- **Synchronous Connection-Oriented (SCO) links :**
  - ✓ point to point full Duplex between master and slave
  - ✓ established once by master and kept alive till released by master
  - ✓ Typically used for voice connection (to guarantee continuity
  - ✓ Master reserves slots used for SCO link on the channel to preserve time sensitive information

- **Asynchronous Connection-Less (ACL) links. Controlling data link**
  - ✓ It is a momentary link between master and slave
  - ✓ No slots are reversed
  - ✓ It is a point to multipoint connection
  - ✓ Symmetric and asymmetric link possible
  - ✓ Designed for data traffic
  - ✓ Packet switched connection where data is exchanged at irregular intervals. as when data is availability from higher up the stack

- ✓ Data integrity is checked though error checking and retransmission
- ✓ Once ALC link between a master and a slave
- ✓ **Differences between SCO and ACL**

	SCO	ACL
1	SCO provides a circuit switched connection, where a dedicated, point-to-point link is established between the master device and the slave device before communication starts.	ACL is a packet oriented link, i.e. the link establishes a packet – switched network.
2	SCO is a symmetric link, i.e. fixed slots are allocated for each direction.	Both symmetric and asymmetric traffic are supported. The master device controls the bandwidth of the ACL link.
3	SCO radio links are used for time critical data transfer, mainly voice data.	ACL is used for transmission of data traffic which are delivered at irregular intervals.
4	A master device can support three SCO links with the same or different slaves. A slave device can have a maximum of three SCO links with its master device.	One master device which connects with maximum seven slave devices via ACL links to form a Piconet.
5	The focus is minimization of time latency.	The main objective is to maintain data integrity rather than time latency.
6	The maximum data rate of SCO link is 64,000 bps (bits per second).	The maximum data rates of ACL links can reach a 57.6 Kbps in downlink and 721 bps in uplink.
7	Packet retransmissions are not allowed, for assuring real-time transfer of voice traffic.	Packet retransmissions are allowed to ensure data integrity.
8	Forward Error Correction (FEC) is applied for data reliability.	Both FEC as well as Backward Error Correction with retransmissions are adopted for data reliability.

- SCO links support real-time voice traffic using reserved bandwidth.
- ACL links support best effort traffic. The L2CAP Specification is defined for only ACL links and no support for SCO links is planned.

- The logical connection between two endpoints in peer devices, characterized by their Channel Identifiers (CIDs)

There are three types of channels in L2CAP:

- Connection-oriented
- Connectionless data
- L2CAP signaling

L2CAP Functional Requirements

Protocol Multiplexing:

- L2CAP supports multiplexing over individual Controllers and across multiple Controllers.
- An L2CAP channel shall operate over one Controller at a time.
- During channel setup, protocol multiplexing capability is used to route the connection to the correct upper layer protocol.
- L2CAP must be able to distinguish between upper layer protocols such as the Service Discovery Protocol, RFCOMM, and Telephony Control.
- L2CAP must support protocol multiplexing because the Baseband Protocol does not support any ’type? field identifying the higher layer protocol being multiplexed above it.

Segmentation & Reassembly:

- Segmentation will allow the interleaving of application data units in order to satisfy latency requirements.
- Memory and buffer management is easier when L2CAP controls the packet size.
- Error correction by retransmission can be made more efficient.
- The amount of data that is destroyed when an L2CAP PDU is corrupted or lost can be made smaller than the application's data unit.
- The application is decoupled from the segmentation required to map the application packets into the lower layer packets.

Quality of Service: The L2CAP connection establishment process allows the exchange of information regarding the quality of service (QoS) expected between two Bluetooth units. WITH the help of ACCES contract

Flow control per L2CAP channel

- Controllers provide error and flow control for data going over the air and HCI flow control exists for data going over an HCI transport.
- When several data streams run over the same Controller using separate L2CAP channels, each channel requires individual flow control. A window-based flow control scheme is provided.

Error control and retransmissions:

- When L2CAP channels are moved from one Controller to another data can be lost.
- Also, some applications require a residual error rate much smaller than some Controllers can deliver.
- L2CAP provides error checks and retransmissions of L2CAP PDUs.
- L2CAP error checking and retransmission also protect against loss of packets due to flushing by the Controller.
- The error control works in conjunction with flow control in the sense that the flow control mechanism will throttle retransmissions as well as first transmissions.

Support for Streaming

- Streaming applications such as audio set up an L2CAP channel with an agreed-upon data rate and do not want flow control mechanisms, including those in the Controller, to alter the flow of data.
- A flush timeout is used to keep data flowing on the transmit side.
- Streaming mode is used to stop HCI and Controller based flow control from being applied on the receiving side.

Fragmentation and Recombination

Some Controllers may have limited transmission capabilities and may require fragment sizes different from those created by L2CAP segmentation.

Therefore, layers below L2CAP may further **fragment(incomplete)** and recombine L2CAP PDUs to create fragments which fit each layer's capabilities.

During transmission of an L2CAP PDU, many different levels of fragmentation and recombination may occur in both peer devices.

Groups: L2CAP group abstraction permits implementations to efficiently map protocol groups on to piconets.

SCOPE

• L2CAP does not transport synchronous data designated for SCO or eSCO ((extended synchronous connection orientated) links) logical transports.

eSCO is **a circuit-switch connection which allows for retransmissions of corrupted packets and higher data rates compared to SCO links**.

• L2CAP does not support a reliable broadcast channel.

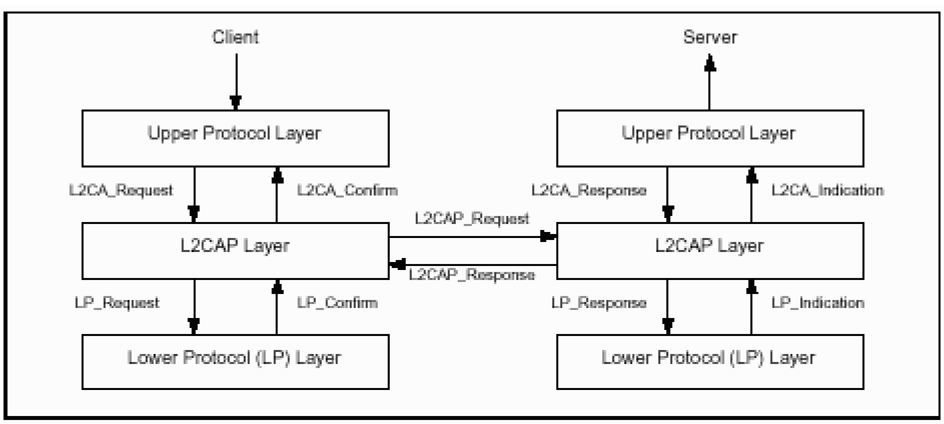
L2CAP General Operation

Channel Identifiers: Channel identifiers (CIDs) are local names representing a logical channel endpoint on the device.

Operation between Devices

- The connection-oriented data channels represent a connection between two devices, where a CID identifies each endpoint of the channel.
- The connectionless channels restrict data flow to a single direction.
- This channel is used to create and establish connection-oriented data channels and to negotiate changes in the characteristics of these channels.
- Another CID is reserved for all incoming connectionless data traffic.

Operation between Layers



- L2CAP implementations follow the general architecture described here:
- L2CAP implementations must transfer data between higher layer protocols and the lower layer protocol.
- Each implementation must also support a set of signaling commands for use between L2CAP implementations.
- L2CAP implementations should also be prepared to accept certain types of events from lower layers and generate events to upper layers. How these events are passed between layers is an implementation-dependent process.

MODES OF OPERATION

- Basic L2CAP Mode
- Flow Control Mode
- Retransmission Mode
- Enhanced Retransmission Mode
- Streaming Mode
- LE Credit Based Flow Control Mode
- Enhanced Credit Based Flow Control Mode

Other L2CAP Features

Data Packet Format:

- A channel represents a data flow between L2CAP entities in remote devices.
- Channels may be connection-oriented or connectionless. All packet fields use Little Endian byte order. (**when the least significant bytes are stored before the more significant bytes)**

Signaling

- Various signaling commands can be passed between two L2CAP entities on remote devices.
- All signaling commands are sent to CID 0x0001 (the signaling channel). The L2CAP implementation must be able to determine the Bluetooth address (BD_ADDR) of the device that sent the commands.
- Multiple commands may be sent in a single (L2CAP) packet and packets are sent to CID 0x0001.
- MTU Commands take the form of Requests and Responses. For a complete list see the L2CAP specs.

Configuration Parameter Options

- Options are a mechanism to extend the ability to negotiate(bring about by discussion.) different connection requirements.
- Options are transmitted in the form of information elements comprised an option type, an option length, and one or more option data fields.

Service Primitives

- Connection: setup, configure, disconnect
- Data: read, write
- Group: create, close, add member, remove member, get membership
- Information: ping, get info, request a call-back at the occurrence of an event

- Connection-less Traffic: enable, disable

- The L2CAP resource manager block is responsible for managing the ordering of submission of PDU (In networking, a protocol data unit (PDU) is the basic unit of exchange...) cause to break into fragments to the baseband.
- L2CAP Resource Managers may also carry out traffic conformance policing to ensure that applications are submitting L2CAP SDUs within the bounds of their negotiated QoS settings.
- Packets received by a layer are called Service Data Unit (SDU)

7 GENERAL PROCEDURES

7.1 CONFIGURATION PROCESS

7.1.1 Request Path

The Request Path can configure the following:

- requester’s incoming MTU
- requester’s outgoing flush timeout
- requester’s outgoing QoS
- requester’s incoming and outgoing flow and error control information
- requester's incoming and outgoing Frame Check Sequence option
- requester's outgoing QoS using Extended Flow Specification option
- requester's incoming Extended Window size option plus incoming and outgoing frame format.

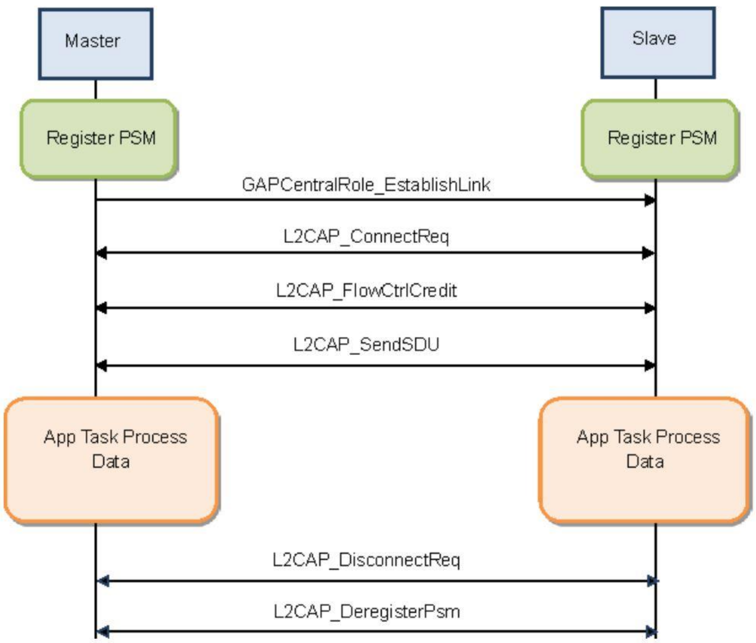
7.1.2 Response Path

The Response Path can configure the following:

- responder’s outgoing MTU, that is the remote side’s incoming MTU
- remote side’s flush timeout
- responder’s incoming QoS Flow Specification (remote side’s outgoing QoS Flow Specification)
- responder’s incoming and outgoing Flow and Error Control information

• responder's incoming QoS Extended Flow Specification (remote side's outgoing QoS Flow Specification)

• responder's outgoing Extended Window size.



Security Manager Protocol

- The Security Manager Protocol (SMP) is the peer-to-peer protocol used to generate encryption keys and identity keys.
- The protocol operates over a dedicated fixed L2CAP channel.
- The SMP block also manages storage of the encryption keys and identity keys and is responsible for generating random addresses and resolving random addresses to known device identities.
- This block is only used in LE systems.
- SMP functionality is in the Host on LE systems to reduce the implementation cost of the LE only Controllers.

- **Services and Characteristics**

Attribute Protocol (ATT).

3.3 ATTRIBUTE PDU

Attribute PDUs have one of six types, which are indicated by the suffix to the PDU name as shown in Table 3.1:

Type	Purpose	Suffix
Commands	PDUs sent to a server by a client that do not invoke a response.	CMD
Requests	PDUs sent to a server by a client that invoke a response.	REQ
Responses	PDUs sent to a client by a server in response to a request.	RSP
Notifications	Unsolicited PDUs sent to a client by a server that do not invoke a confirmation.	NTF
Indications	Unsolicited PDUs sent to a client by a server that invoke a confirmation.	IND
Confirmations	PDUs sent to a server by a client to confirm receipt of an indication.	CFM

Table 3.1: Attribute PDUs

- The Attribute Protocol (ATT) block implements the peer-to-peer protocol between an attribute server and an attribute client.
- The ATT client sends commands, requests, and confirmations to the ATT server. The ATT server sends responses, notifications, and indications to the client
- These ATT client commands and requests provide a means to read and write values of attributes on a peer device with an ATT.

- **Server** – The server acts as a database and stores data that can be read or written by the client. E.g., a fitness tracker will store the latest heart rate value.
- **Client –** which is the device that requests the data from the server. E.g., a smartphone connected to a fitness tracker.

Attributes

The ATT protocol defines a data structure called an attribute as a standard method for effectively organizing the data stored, accessed, and updated on the server.

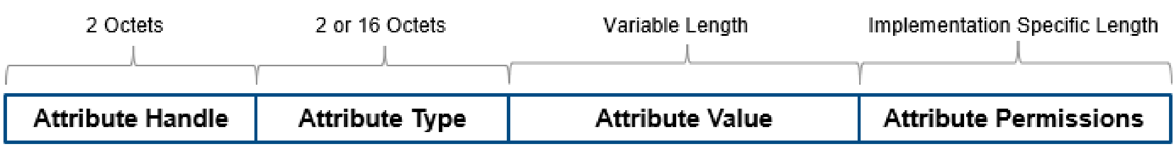
The attribute data structure has four fields:

Handel = 16 bit unique identifier , 0x0001 to 0 xffff

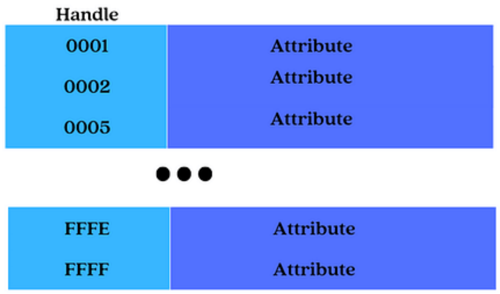
UUID = universal unique identifier, 16 bit = sig adopted, 128 bit = custom , No central registry.

Value = hold data, length, format based on type.

Permission = access, security



- **An Attribute Handle** – This is a unique unsigned 16-bit identifier that is used by the client to reference an attribute on the server. It makes the attribute "addressable," and it does not change during a single connection.
- **An Attribute Type (UUID)** – This is a globally unique 2-byte or 16-bit UUID identifier There are two types of UUIDs: Service UUID and Characteristic UUID.
- **An Attribute Value** – This is the actual data being stored. For example, the heart rate sensor reading value or temperature reading from a temperature sensor.
- **An Attribute Permissions Field** – It specifies the various methods that can be used to access the attribute value, as well as the security level required to access the attribute value. For example, an attribute may require no permissions to read but may require authentication to modify (write to) it.



A table of Attributes in a Server's database

5.2. Generic Attribute Profile (GATT)

A device can act in both roles at the same time.

An example of configurations illustrating the roles for this profile is depicted in Figure 2.2.

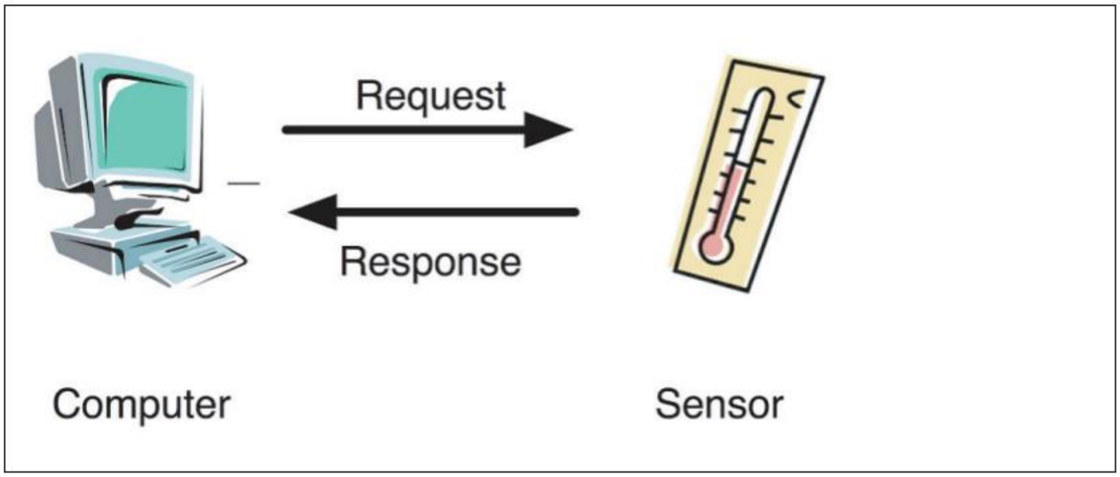


Figure 2.2: Examples of configuration

2.3 USER REQUIREMENTS AND SCENARIOS

The following scenarios are covered by this profile:

- Exchanging configuration
- Discovery of services and characteristics on a device
- Reading a characteristic value
- Writing a characteristic value
- Notification of a characteristic value
- Indication of a characteristic value

- The Generic Attribute Profile (GATT) block represents the functionality of the attribute server and, optionally, the attribute client.
- The profile describes the hierarchy of services, characteristics and attributes used in the attribute server.
- The block provides interfaces for discovering, reading, writing and indicating of service characteristics and attributes.
- GATT is used on LE devices for LE profile service discovery

Format of services and characteristics

Procedure to interface with these attributes: discovery, red, write, notification, indication.

- Same roles as ATTS
- Roles not set per device
- Determined by transaction

- Services = battery services
- Characteristics = Properties = read, write, write without response, notify
  Descriptors = user description, enabling notifications, presentation format, unit
- Profile =behavior services characteristics, connections, security

Attribute protocol packets

- Commands = no response required
- requests = response is required
- response = response to a request
- notifications = no response required
- indications = response is required
- confirmations = response to an indication

5.6. Attribute Operations

- Commands, request, response, notification, indication, confirmation

data operations

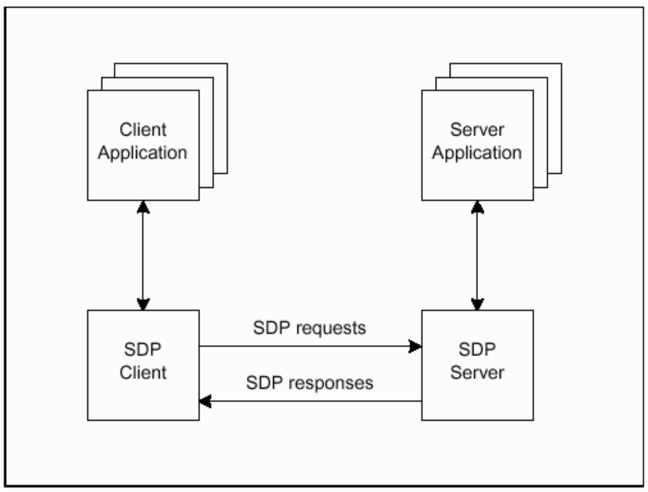
- reading attributes = all reads are requests
- writing attributes = writes can be request or commands

Generic Access Profile

- This profile defines the generic procedures related to discovery of Bluetooth devices (idle mode procedures)
- link management aspects of connecting to Bluetooth devices (connecting mode procedures).
- The Generic Access Profile (GAP) block represents the base functionality common to all Bluetooth devices such as modes and access procedures used by the transports, protocols and application profiles.
- GAP services **include device discovery, connection modes, security, authentication, association models and service discovery**.

- The purpose of the connection establishment procedure is to establish a connection between applications on two Bluetooth devices.

Service Discovery Protocol (SDP)



This protocol **allows profile drivers to search or browse for services that are offered by Bluetooth devices that are in range of the local radio**.

- The SDP or service discovery protocol is a key element of the Bluetooth ad-hoc networking capability.
- The Bluetooth SDP allows a Bluetooth device to discover and make many connections during the course of its life.
- It enables the devices to discover what services other Bluetooth devices support, and also lists everything that the Bluetooth device is capable of supporting.
- The Bluetooth SDP uses the Universal Unique Identifier, UUID.
- Services supported by the Bluetooth Sig are given a short form UUID of 16 bits rather than the complete 128 bits that would otherwise be required.
- A profile known as the Service Discovery Applications Profile, or SDAP, is often confused with the Bluetooth SDP.
- This defines how devices can interrogate each other's SDP after and L2CAP link has been established.
- All Bluetooth devices implement the features of an SDP client as well as having and SDP server database.

- The service discovery protocol (SDP) provides a means for applications to discover which services are available and to determine the characteristics of those available services.

- A specific Service Discovery protocol is needed in the Bluetooth environment, as the set of services that are available changes dynamically based on the RF proximity of devices in motion, qualitatively different from service discovery in traditional network-based environments.

2.2 SERVICE RECORD

- A service is any entity that can provide information, perform an action, or control a resource on behalf of another entity.
- A service may be implemented as software, hardware, or a combination of hardware and software.
- All of the information about a service that is maintained by an SDP Server is contained within a single service record.
- The service record shall only be a list of service attributes.
- A service record handle is a 32-bit number that shall uniquely identify each service record within an SDP Server.
- In general, each handle is unique only within each SDP Server.
- If SDP Server S1 and SDP Server S2 both contain identical service records (representing the same service), the service record handles used to reference these identical service records are completely independent.
- The handle used to reference the service on S1 will be meaningless if presented to S2.

2.3 SERVICE ATTRIBUTE

Each service attribute describes a single characteristic of a service. Some examples of service attributes are:

ServiceClassIDList	Identifies the type of service represented by a service record. In other words, the list of classes of which the service is an instance
ServiceID	Uniquely identifies a specific instance of a service
ProtocolDescriptorList	Specifies the protocol stack(s) that may be used to utilize a service
ProviderName	The textual name of the individual or organization that provides a service
IconURL	Specifies a URL that refers to an icon image that may be used to represent a service
ServiceName	A text string containing a human-readable name for the service
ServiceDescription	A text string describing the service

2.4 SERVICE CLASS

Each service is an instance of a service class. The service class definition provides the definitions of all attributes contained in service records that represent instances of that class. Each attribute definition specifies the numeric value of the attribute ID, the intended use of the attribute value, and the format of the attribute value. A service record contains attributes that are specific to a service class as well as universal attributes that are common to all services.

Each service class is assigned a unique identifier ,this service class identifier is contained in the attribute value for the ServiceClassIDList attribute, and is represented as a UUID. A UUID is a universally unique identifier that is guaranteed to be unique across all space and all time. UUIDs can be independently created in a distributed fashion. No central registry of assigned UUIDs is required. A UUID is a 128-bit value.

2.5 SEARCHING FOR SERVICES

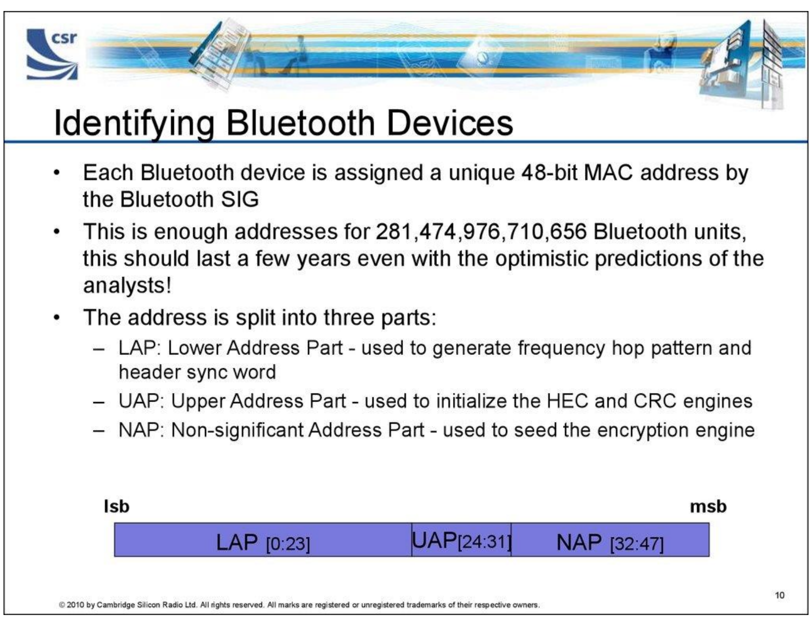
The Service Search transaction allows a client to retrieve the service record handles for particular service records based on the values of attributes contained within those service records.

The capability search for service records based on the values of arbitrary attributes is not provided. Rather, the capability is provided to search only for attributes whose values are Universally Unique Identifiers (UUIDs). Important attributes of services that can be used to search for a service are represented as UUIDs. Service search pattern are used to locate the desired service. A service search pattern is a list of UUIDs (service attributes) used to locate matching service records.

2.6 BROWSING FOR SERVICES

This process of looking for any offered services is termed browsing. In SDP, the mechanism for browsing for services is based on an attribute shared by all service classes. This attribute is called the BrowseGroupList attribute. The value of this attribute contains a list of UUIDs. Each UUID represents a browse group with which a service may be associated for the purpose of browsing.

When a client desires to browse an SDP server's services, it creates a service search pattern containing the UUID that represents the root browse group. All services that may be browsed at the top level are made members of the root browse group by having the root browse group's UUID as a value within the BrowseGroupList attribute.



Bluetooth Channels

- A master can create two types of logical channel with a slave device:
  - Asynchronous Connection Less (ACL): Packet Switched System provides a reliable data connection with a best effort bandwidth; depends on radio performance and number of devices in the piconet
  - Synchronous Connection Oriented (SCO): Circuit Switched System provides real time unreliable connection with a guaranteed bandwidth; usually used for voice based applications

- The Bluetooth connections are limited to 1Mbps across the air (without EDR)
- This gives a theoretical maximum of ~723kbps of useable data

Adaptive Frequency Hopping

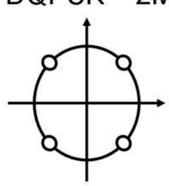
- Introduced in Bluetooth v1.2
- Bluetooth shares the 2.4GHz ISM band with:
  - 802.11b/g Wi-Fi Systems
  - 2.4GHz cordless phones
  - Microwave ovens
- More devices = More interference.
- 802.11b/g does not work well with BT interferers.
- AFH allows BT to avoid known ‘bad’ channels.
- Increased bandwidth, reduced lost data.

Adaptive Frequency Hopping

- Benefits:
  - Fewer lost packets = better audio quality
  - Less degradation to Bluetooth and 802.11b/g networks
  - Greater bandwidth efficiency
  - Not backward compatible with v1.1 systems

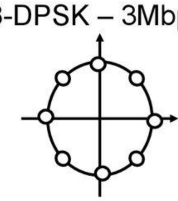
EDR Modulation Schemes

π/4-DQPSK – 2Mbps



1MSps => 2Mbps

8-DPSK – 3Mbps



1MSps => 3Mbps

Bluetooth 2.0 modulation schemes fully backwards compatible w/ Standard Rate

- Same packet timing & structure, major spectral characteristics, and packet negotiation
- Same radio used for all modulation schemes (FSK and PSK)
- Master devices support mixed Piconets by using appropriate packets for each slave

EDR devices must support 1x and 2x data rate, 3x data rate is optional

EDR Packets

- v1.2 Packets:

Header - GFSK	Payload - GFSK

- v2.0 EDR Packets:

Header - GFSK	Payload – DQPSK or 8-DPSK

Forward Error Correction

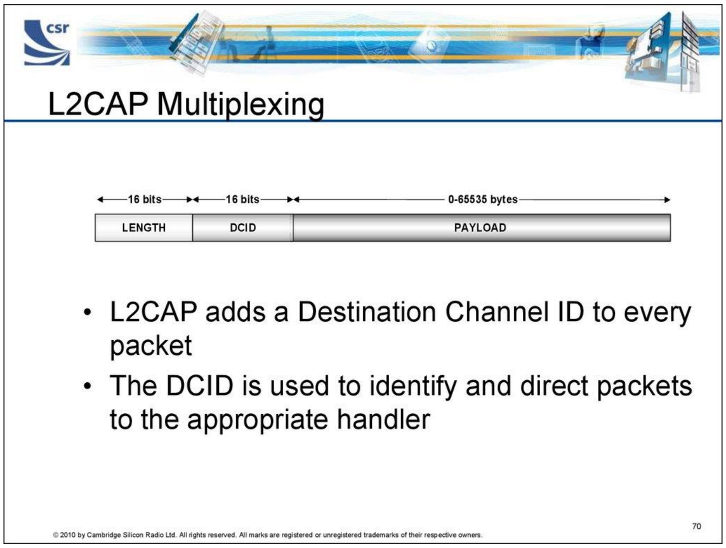
- Bluetooth defines three levels of forward error correction
- No Error Correction:
  - There is no error correction!
  - Data is just put in the payload and sent
- 1/3 FEC:
  - Each bit is repeated 3 times
  - Majority voting decides bit value
- 2/3 FEC:
  - The data is encoded using a (15,10) shortened hamming code
  - Every 10 bits of data are encoded into 15 bits of data
  - Can correct single bit errors and detect double bit errors

Secure Simple Pairing (SSP)

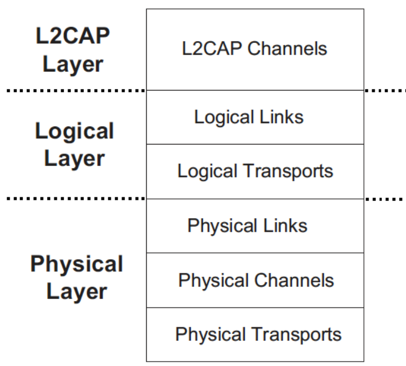
- Feature of Bluetooth 2.1
- Enables easier connectivity between devices and better use of security features

Logical Link Control and Adaptation Protocol (L2CAP)

- Logical Link Control
  – Multiplexing: many logical links onto one physical link
- Adaptation
  – Segmentation & reassembly: adapts large packets to baseband size
- Protocol
  – A well defined set of signaling rules understood by all devices



DATA TRANSPORT ARCHITECTURE



Inquiry , paging, discovery, = connection

Paging is the process of forming a connection between two Bluetooth devices

Connection Process

Creating a Bluetooth connection between two devices is a multi-step process involving three progressive states:

1. **Inquiry** -- If two Bluetooth devices know absolutely nothing about each other, one must run an inquiry to try to **discover** the other. One device sends out the inquiry request, and any device listening for such a request will respond with its address, and possibly its name and other information.
2. **Paging (Connecting)** -- Paging is the process of forming a connection between two Bluetooth devices. Before this connection can be initiated, each device needs to know the address of the other (found in the inquiry process).

3. **Connection** -- After a device has completed the paging process, it enters the connection state. While connected, a device can either be actively participating or it can be put into a low power sleep mode.
    - **Active Mode** -- This is the regular connected mode, where the device is actively transmitting or receiving data.
    - **Sniff Mode** -- This is a power-saving mode, where the device is less active. It'll sleep and only listen for transmissions at a set interval (e.g. every 100ms).
    - **Hold Mode** -- Hold mode is a temporary, power-saving mode where a device sleeps for a defined period and then returns back to active mode when that interval has passed. The master can command a slave device to hold.
    - **Park Mode** -- Park is the deepest of sleep modes. A master can command a slave to "park", and that slave will become inactive until the master tells it to wake back up.
    - Bonds are created through one-time a process called **pairing**. Pairing usually requires an **authentication process**

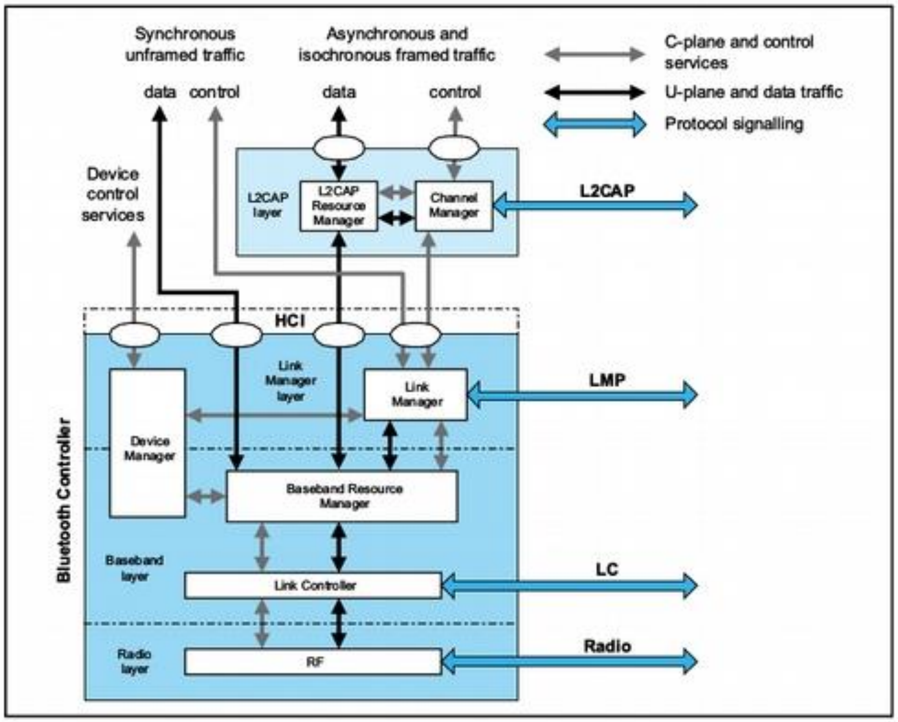


Figure 2.1: Bluetooth core system architecture

THE PHYSIOLOGY OF BLUETOOTH:

BLUETOOTH COMMUNICATION ARCHITECTURE

Bluetooth communication architecture can be divided into:

Physical channel, Logical layer, L2CAP layer

1) Physical Transports

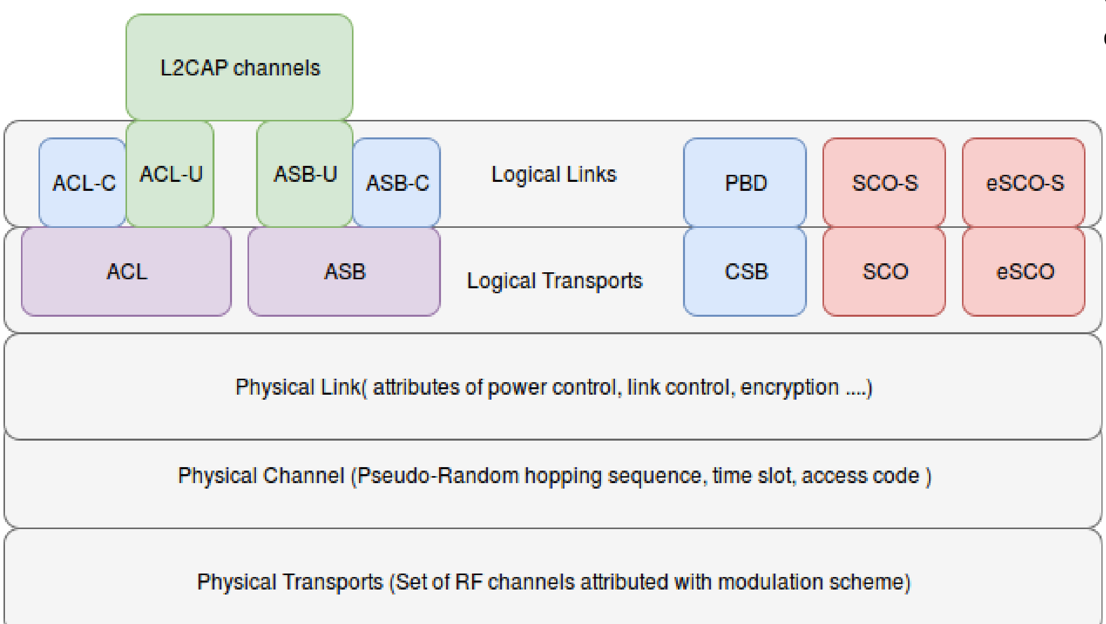
2) Physical Channels

3) Physical Link

4) Logical Transports

5) Logical Links

6) L2CAP channels



3.1 Physical Transport:

The set of Radio frequency bands that are used for communication between the Master and Slave devices within a piconet is called ***Physical Transport****.*

Since ISM band with central frequency of 2.4 GHz (BW = 79 MHz) is divided into 79 Channels of 1 MHz each, the set of 79 RF carriers can be considered as Physical Transports.

3.2 Physical Channel:

- ***Physical Channel*** *is characterized by pseudo-random RF channel hopping sequence, time slot for packet transmission and access code.*

- Two devices wishing to communicate must be present in the same physical channel and their transceivers must be tuned to the same radio frequency matching the phase.

    - All packets transmitted between the master and slaves include the **channel access code**.

- This is used to identify communications on a particular physical channel, and to exclude or ignore packets on a different physical channel the same RF carrier in physical proximity.
- Channel access code is always present in the packets transmitted in a piconet and is a property of the physical channel.
    - A device is said to be part of a piconet when it is connected to the physical channel of the piconet.
- When the device is synchronized to the timing, frequency and the access code, it is said to be **connected** to the physical channel which is unique to the piconet.

There are 5 types of Physical channels defined,

3.2.1 Basic Piconet Physical Channel:

- Basic Piconet Physical Channel is used during communication between Bluetooth devices in a piconet.
- Master and its slave devices will be hopping through 79 RF channels, phase of hopping is according to the master's clock and the access code is derived from the BD_ADDR of the master.
- *All the devices participating in the piconet are time and hop synchronized to the physical channel.*

3.2.2 Adapted Piconet Physical Channel:

- Adapted Piconet Physical Channel can also be used during communication but differs from basic piconet channel in 2 ways
- a.) Slave transmits in the same frequency in which the master addressed it, in the preceding even slot.
- b.) Adapted piconet may be based on less than 79 RF channels, where at least 20 RF channels are to be present according to the specification.

3.2.3 Page Scan Physical Channel:

- Page Scan Physical Channel is used during connection establishment using subset 32 Wakeup frequencies or RF channels. Here aspiring master and aspiring slave do not hop at the same rate.
- Paging device (Aspiring master) hops to new pseudo-random frequency (of 32 wakeup frequencies with period 32) for every 312.5 microseconds but Page Scan device (Aspiring slave) will stay in the same frequency for 1.28 seconds (2048 timeslots) for increasing the chances of connection establishment.
- Access code which is the property of Page Scan Physical Channel is called **Device Access Code** and is derived from the BD_ADDR of page scanning device (aspiring slave).
- This is so because, only particular Bluetooth device has to respond to connection establishment packets, and this makes other devices to reduce overhead of analyzing packet and payload headers not addressed to them.

3.2.4 Inquiry Scan Physical Channel:

- Inquiry Scan Physical Channel is used during device discovery using subset 32 Wakeup frequencies or RF channels (Out of 79 RF frequencies).
- Here aspiring master and aspiring slave do not hop at the same rate.
- Inquiring device (Aspiring master) hops to new pseudo-random frequency (of 32 wakeup frequencies with period 32) for every 312.5 microseconds but Inquiry Scanning device (Aspiring slave) will stay in the same frequency for 1.28 seconds (2048 timeslots) for increasing the chances of device discovery.
- Access code which is the property of Inquiry Scan Physical Channel is called **Inquiry Access Code** and uses Generic Inquiry Access Code (GIAC) (0x9E8B33) for general inquiries.
- This is so because enquiry procedure is a broadcast message which is received and responded by all the devices which are in enquiry scan state (the state in discoverable mode).

3.2.5 Synchronization Scan Physical Channel:

Synchronization Scan Physical Channel is used by the devices to obtain timing and frequency information about Connection-less Slave Broadcast physical link and to recover piconet clock.

Physical Channel	Access Code Used	Derived From
Basic Piconet Physical Channel	Channel Access Code	Master's BD_ADDR
Adapted Piconet Physical Channel	Channel Access Code	Master's BD_ADDR
Page Scan Physical Channel	Device Access Code	Aspiring Slave's BD_ADDR
Inquiry Scan Physical Channel	Inquiry Access Code (Generic)	GIAC (0x9E8B33)

3.3 Physical Link:

Physical Link *describes the common attributes of transmission of all the logical transports at the link level on a physical channel.*

Physical Link *can also be looked as, time slots describing baseband connection between master and a particular slave in a piconet confirming to the* ***attributes of link control, power control, encryption and many others, where time slots are dynamically allocated for ACL logical transport and statically allocated for SCO and eSCO.***

A Physical link is always associated with the single physical channel and supports the common features of all the logical transports that are multiplexed over it.

Some Physical links have modifiable properties such as transmit power

for the link, while others have no modifiable properties. Link manager updates the properties of physical links.

3.4 Logical Transports:

Logical Transports can exist between a master and a particular slave in a piconet (connection oriented) or between master and all the slaves in the piconet (connection-less).

3.4.1 If Logical Transport is **Synchronous Connection Oriented (SCO)**, pre-determined time slots are to be allocated by baseband resource manager to SCO logical link, where there is no retransmission of error-prone or unreached packets.

This logical transport exists between the master and a particular slave in the piconet. SCO logical transport shares LT_ADDR of the default ACL logical transport.

3.4.2 If Logical Transport is **extended Synchronous Connection Oriented (e-SCO)**, pre-determined time slots are to be allocated by baseband resource manager to eSCO logical link, with an added benefit of retransmission of error-prone or unreached packets.

This logical transport exists between the master and a particular slave in the piconet. eSCO logical transport gets a new LT_ADDR.

3.4.3 If Logical Transport is **Asynchronous Connection Oriented (ACL)**, time slots are dynamically allocated to ACL-C and ACL-U logical links by the baseband resource manager. Re-transmission of error-prone or unreached packets is present.

This logical transport exists between the master and a particular slave. LT_ADDR of default ACL logical transport is called as **primary LT_ADDR** and must be shared with SCO logical transport (if exists).

3.4.4 If Logical Transport is **Active Slave Broadcast (ASB),** time slots are dynamically allocated for LMP control signaling and connectionless L2CAP user data to all the slaves in the piconet. There is no response or re-transmission of packets, to increase reliability the same data packets are transmitted multiple times.

This logical transport exists between the master and all the slaves in the piconet. Since ASB logical transport is for broadcasting, LT_ADDR for ASB is 0 (and is called **reserved LT_ADDR**).

3.4.5 If Logical Transport is **Connectionless 'Slave Broadcast' (CSB)**, time slots are fixed and periodic for profile broadcast data and the messages are received by all configured connectionless broadcast slaves.

  - There is no response or re-transmission of packets, to increase reliability the same data packets are transmitted multiple times.

- This logical transport exists between master and 'connection-less broadcast configured' slaves.
- There is unique LT_ADDR given by master to identify the CSB transport.
- This unique LT_ADDR is shared between master and all CSB configured slaves. CSB logical transport can only exist on Adaptive Piconet physical channel.

3.5 Logical Links:

***Logical Links** define the type of data that is being transceiver.*

- SCO logical transport supports (User Synchronous Data) **SCO-S logical link**, which carries isochronous unframed user data. Data is delivered in constant sized units at a constant rate.

- e-SCO logical transport supports (User Extended Synchronous Data) **e-SCO logical link,** which carries isochronous unframed user data. Data is delivered in constant sized units at a constant rate.

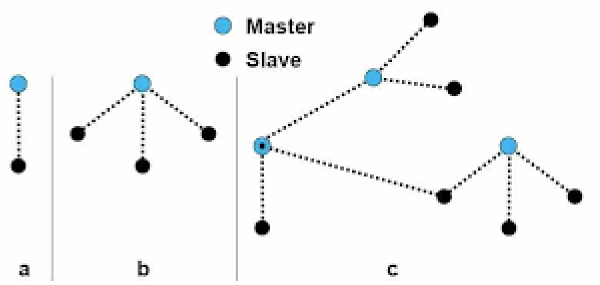
- CSB logical transport supports (Profile Broadcast Data) **PBD logical link** which is used to broadcast isochronous unframed data to multiple receivers. Profile directly talks to HCI to transfer unframed profile data skipping L2CAP layer.
- ACL logical transport supports **ACL-C and ACL-U logical links**, where ACL-C logical link carries LMP signalling messages and ACL-U logical link carries L2CAP user data and L2CAP signalling messages. Variable rate isochronous data also has to be transported using ACL logical transport.

- ASB logical transport supports **ASB-C** and **ASB-U** logical links, where ASB-C logical link carries LMP signalling and ASB-U logical link carries L2CAP user data and L2CAP signalling messages. It's important to note that a subset of LMP signalling and only L2CAP connectionless user traffic can be sent over ASB logical transport.
- **Note**: Since both SCO and ACL share the same LT_ADDR, default ACL logical transport can be identified with the combination of **LT_ADDR** field and Packet **TYPE** field in the packet header.

- **ACL-C** and **ACL-U** logical links in ACL logical transport are identified using **LLID** (Logical Link ID) field in the payload header. Similarly, **ASB-C** and **ASB-U** logical links in ASB logical transport are identified using LLID field in the payload header. **PBD** logical link of CSB logical transport, which is only used for broadcast of profile data, will contain LLID of 10 in payload header, where CSB logical transport having own LT_ADDR. There is no LLID field in the SCO and eSCO packets because these logical transports support single logical links.

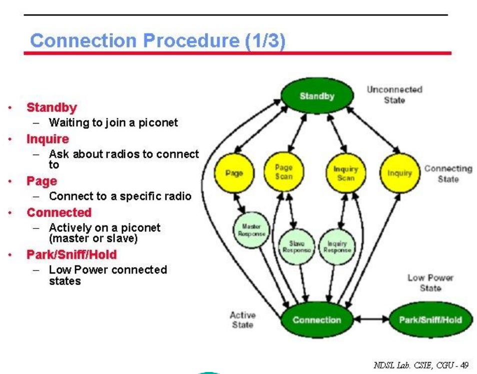
BASEBANDSPECIFICATION (BLUETOOTH CLOCK, BLUETOOTH DEVICE ADDRESSING , ACCESS CODES, PHYSICAL CHANNELS AND PHYSICAL LINKS)(PAGENO:407).

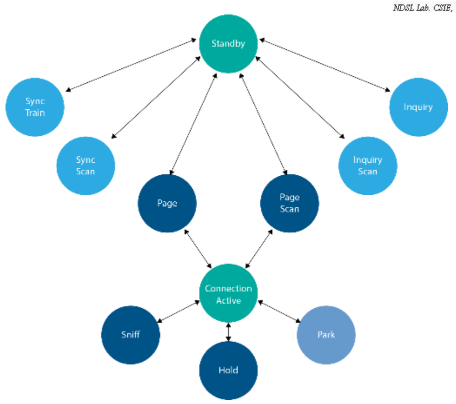
- The Bluetooth system provides a point-to-point connection or a point-to multipoint connection
- In a point-to-point connection the physical channel is shared between two Bluetooth devices.
- In a point-to-multipoint connection, the physical channel is shared among several Bluetooth devices.
- Two or more devices sharing the same physical channel form a piconet. One Bluetooth device acts as the master of the piconet, whereas the other device(s) act as slave(s).
- Up to seven slaves can be active in the piconet. The channel access is controlled by the master
- An unlimited number of slaves can receive data on the physical channel carrying the Connectionless Slave Broadcast physical link.
- Each piconet only has a single master, however, slaves can participate in different piconets on a time-division multiplex basis.
- Piconets shall not be frequency synchronized and each piconet has its own hopping sequence.



- Data is transmitted over the air in packets.
- Two modes are defined: a mandatory mode called Basic Rate and an optional mode called Enhanced Data Rate.
- The symbol rate for all modulation modes is 1 Msym/s.
- The gross air data rate is 1 Mb/s for Basic Rate. Enhanced Data Rate has a primary modulation mode that provides a gross air data rate of 2 Mb/s, and a secondary modulation mode that provides a gross air data rate of 3 Mb/s.

Bluetooth States and Link Controller



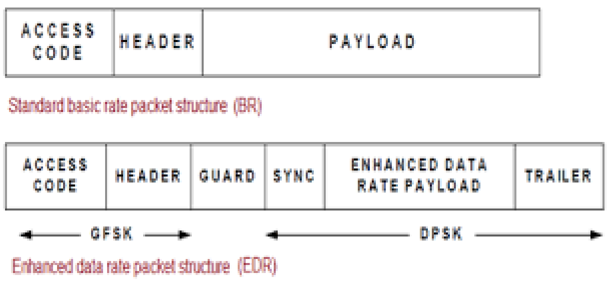


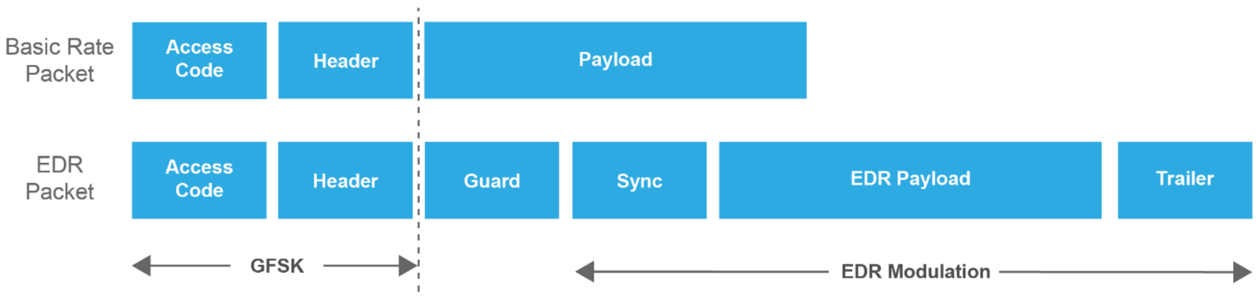
<u>The baseband is the digital engine of a Bluetooth system. It is responsible for</u>

- constructing and decoding packets,
- encoding and managing error correction,
- encrypting and decrypting for secure communications,
- calculating radio transmission frequency patterns,

- maintaining synchronization,
- controlling the radio,
- All of the other low-level details necessary to realize Bluetooth communications.

The general Basic Rate packet format is shown in Figure 1.2. Each packet consists of 3 entities: the access code, the header, and the payload.





The diagram above shows the packet format for both Basic Rate and Enhanced Data Rate. As we mentioned before, the EDR packet sends the Access Code and Header using the basic rate, then the guard time gives it time to change modulation. The Sync, payload and Trailer are sent using the modulation format.

<u>What is meant by DPSK?</u>

Differential Phase Shift Keying (DPSK) means differential phase modulation of the radio frequency carrier with relative phase states of 0 degree or 180 degrees.

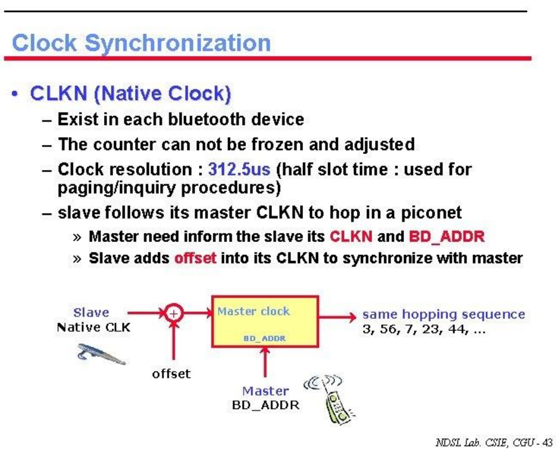
<u>Gaussian frequency-shift keying:</u>

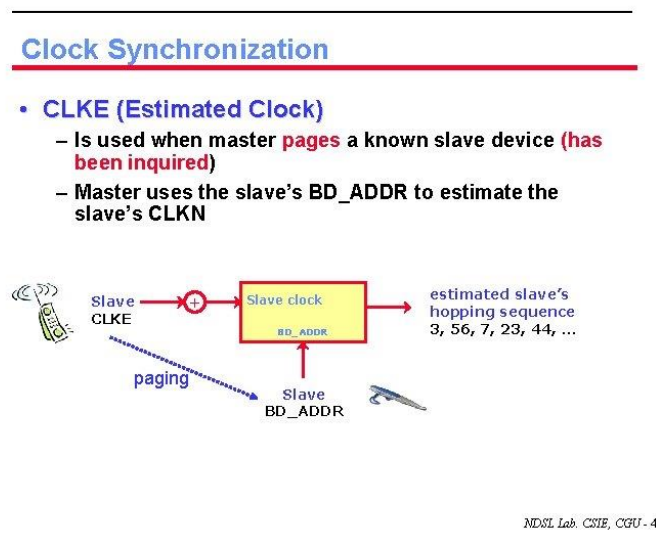
A type of FSK modulation which uses a Gaussian filter to shape the pulses before they are modulated. This reduces the spectral bandwidth and out-of-band spectrum, to meet adjacent-channel power rejection requirements. Bluetooth uses GFSK.

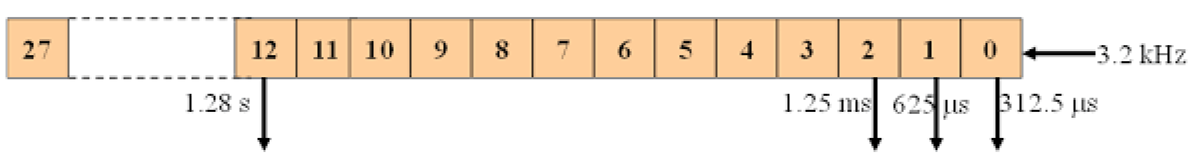
The general Enhanced Data Rate packet format is shown in Figure 1.3. Each packet consists of 6 entities:

- the access code, the header, the guard period, the synchronization sequence, the Enhanced Data Rate payload and the trailer.
- The *access code* and *header* use the **same modulation** mode as for Basic Rate packets while the *synchronization sequence, the Enhanced Data Rate payload and the trailer* use the **Enhanced Data Rate modulation mode.**
- The guard time allows for the transition between the modulation modes.

BLUETOOTH CLOCK







- One of the most critical aspects of Bluetooth is the Clock.
- The master device in a Bluetooth connection has a clock that is used to split the time on each physical channel.
- All slaves in a connection have clocks that are synchronized to the Master clock.
- Synchronizing the clocks on Bluetooth is critical because the radios need to sync up on when they transmit.

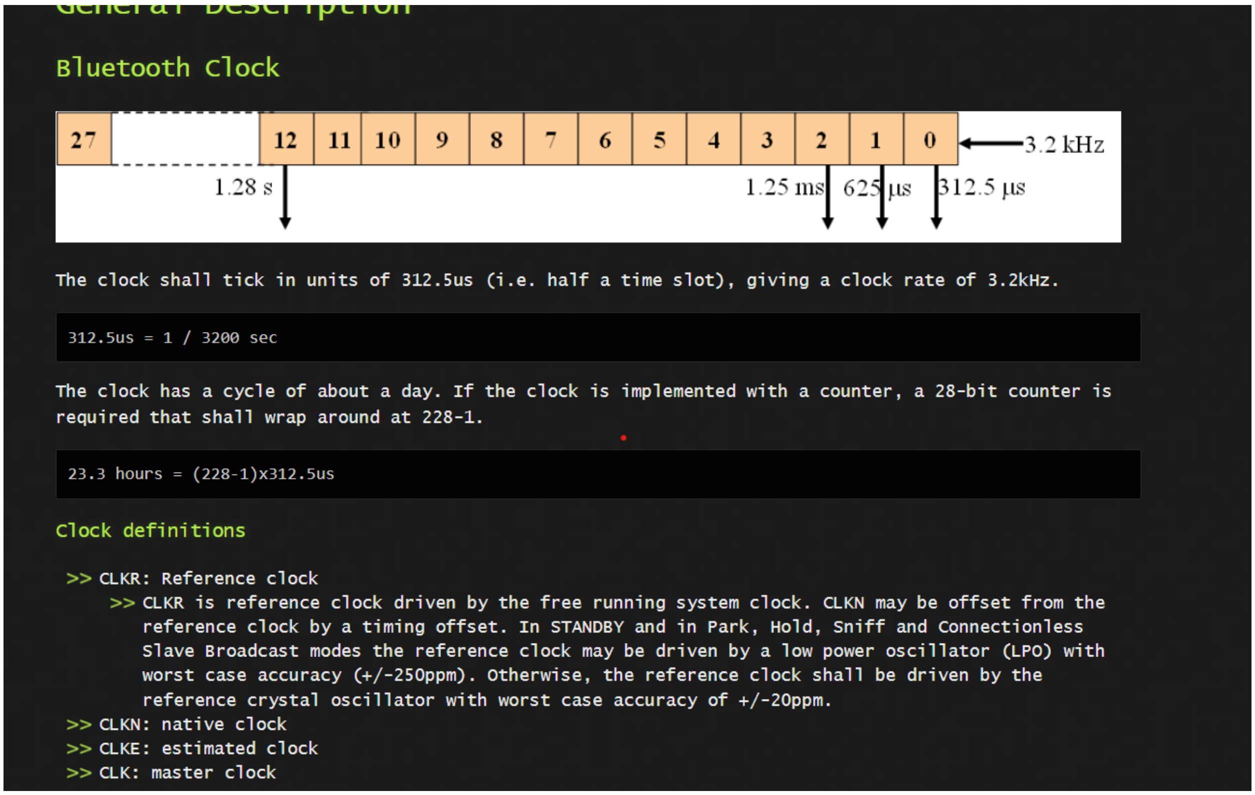
- Because Bluetooth uses precise timeslots for transmissions with the devices alternating, if the clocks are not synchronized there could be issues with devices transmitting at the wrong time.

  - Every Bluetooth device shall have a native clock that shall be derived from a free running reference clock.
  - Offsets may be added to the reference clock to synchronize the native clock with other non-Bluetooth systems.
  - For synchronization with other Bluetooth devices, offsets are used that, when added to the native clock, provide temporary Bluetooth clocks that are mutually synchronized.
  - It should be noted that the Bluetooth clock has no relation to the time of day; it may therefore be initialized to any value. The clock has a cycle of about a day.
  - If the clock is implemented with a counter, a 28-bit counter is required that shall wrap around at 228-1.
  - The least significant bit (LSB) shall tick in units of 312.5 µs (i.e. half a time slot), giving a clock rate of 3.2 kHz.
  - The clock determines critical periods and triggers the events in the device.
  - Four periods are important in the Bluetooth system: 312.5 µs, 625 µs, 1.25 ms, and 1.28 s; these periods correspond to the timer bits CLK0, CLK1, CLK2, and CLK12,

  - **In the different modes and states a device can reside in, the clock has different appearances**:
  - CLKR reference clock
  - CLKN native clock
  - CLKE estimated clock
  - CLK master clock
- Each device has an internal clock (CLKN), known as the native clock
- Use to determine the transceiver timing
- Piconet synchronization determines by the master’s clock of the piconet (CLK)
- CLK is identical to master’s native clock (CLKN)

  - **CLKR is the reference clock** driven by the free running system clock.
  - **CLKN** may be offset from the reference clock by a timing offset.
  - In Standby state and in Hold, Sniff, and Connectionless Slave Broadcast modes the reference clock shall have a worst-case accuracy of ±250 ppm.

- In all other circumstances, it shall have a worst-case accuracy of ±20 ppm; this accuracy shall also be used by the piconet master device while performing Piconet Clock Adjustment



General Description

Bluetooth Clock

The clock shall tick in units of 312.5us (i.e. half a time slot), giving a clock rate of 3.2kHz.

```
312.5us = 1 / 3200 sec
```

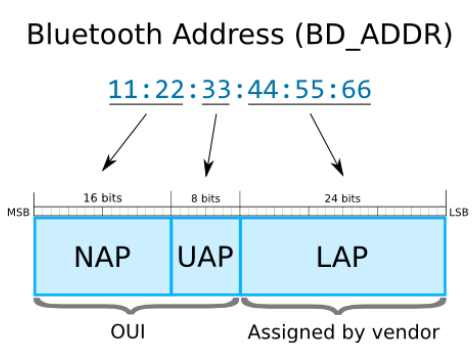
The clock has a cycle of about a day. If the clock is implemented with a counter, a 28-bit counter is required that shall wrap around at 228-1.

```
23.3 hours = (228-1)x312.5us
```

Clock definitions

>> CLKR: Reference clock
    >> CLKR is reference clock driven by the free running system clock. CLKN may be offset from the reference clock by a timing offset. In STANDBY and in Park, Hold, Sniff and Connectionless Slave Broadcast modes the reference clock may be driven by a low power oscillator (LPO) with worst case accuracy (+/-250ppm). Otherwise, the reference clock shall be driven by the reference crystal oscillator with worst case accuracy of +/-20ppm.
>> CLKN: native clock
>> CLKE: estimated clock
>> CLK: master clock

BLUETOOTH DEVICE ADDRESSING



- Bluetooth Device Address (or BD_ADDR) is a unique 48-bit identifier assigned to each Bluetooth device by the manufacturer.

- Bluetooth Address is usually displayed as 6 bytes written in hexadecimal and separated by colons (example - 00:11:22:33: FF:EE).
- The upper half of a Bluetooth Address (most-significant 24 bits) is so called Organizationally Unique Identifier (OUI). It can be used to determine the manufacturer of a device (Bluetooth MAC Address Lookup form). OUI prefixes are assigned by the Institute of Electrical and Electronics Engineers (IEEE).
- Additionally, to identification, Bluetooth Device Address is used to determine the frequency hopping pattern in radio communication between Bluetooth devices.

Bluetooth Address Structure

NAP

Non-significant Address Part (2 bytes). Contains first 16 bits of the OUI. The NAP value is used in Frequency Hopping Synchronization frames.

UAP

Upper Address Part (1 byte). Contains remaining 8 bits of the OUI. The UAP value is used for seeding in various Bluetooth specification algorithms.

LAP

Lower Address Part (3 bytes). This portion of Bluetooth Address is allocated by the vendor of device. The LAP value uniquely identifies a Bluetooth device as part of the Access Code in every transmitted frame. It is used to find access code in every packet..

The LAP and the UAP make the significant address part (SAP) of the Bluetooth Address.

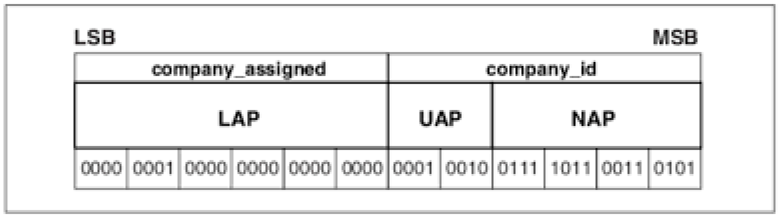


Figure 1.5: Format of BD_ADDR.

- Each Bluetooth device shall be allocated a unique 48-bit Bluetooth device address (BD_ADDR). The address shall be a 48-bit extended unique identifier (EUI-48) created in accordance with section 8.2 ("Universal addresses") of the IEEE 802-2014 standard

Creation of a valid EUI-48 requires one of the following MAC Address Block types to be obtained from the IEEE Registration Authority:

- MAC Address Block Large (MA-L)
- MAC Address Block Medium (MA-M)
- MAC Address Block Small (MA-S)

The BD_ADDR may take any values except those that would have any of the 64 reserved LAP values for general and dedicated inquiries

Reserved addresses

- A block of 64 contiguous LAPs is reserved for inquiry operations; one LAP common to all devices is reserved for general inquiry, the remaining 63 LAPs are reserved for dedicated inquiry of specific classes of devices.

Logical transport	**Links supported**	**Supported by**	**Bearer**	**Overview**
Asynchronous Connection-Oriented (ACL)	Control (LMP) ACL-C or (PAL) AMP-C User (L2CAP) ACL-U or AMP-U	BR/EDR active physical link, BR/EDR basic or adapted piconet physical channel, AMP physical link, AMP physical channel	BR/EDR, AMP	Reliable or timebounded, bi-directional, point-to-point

Synchronous Connection-Oriented (SCO)	Stream (unframed) SCO-S	BR/EDR active physical link, BR/EDR basic or adapted piconet physical channel	BR/EDR	Bi-directional, symmetric, point-topoint, AV channels. Used for 64 kb/s constant rate data
Extended Synchronous Connection-Oriented (eSCO)	Stream (unframed) eSCO-S	BR/EDR active physical link, BR/EDR basic or adapted piconet physical channel	BR/EDR	Bi-directional, symmetric or asymmetric, point-to-point, general regular data, limited retransmission. Used for constant rate data synchronized to the master Bluetooth clock
Active Slave Broadcast (ASB)	Control (LMP) ASB-C User (L2CAP) ASB-U	BR/EDR active physical link, basic or adapted physical channel	BR/EDR	Unreliable, uni-directional broadcast to any devices synchronized with the physical channel. Used for broadcast L2CAP groups and certain LMP messages.
Connectionless Slave Broadcast (CSB)	Profile Broadcast Data (PBD)	Connectionless Slave Broadcast physical link, BR/EDR adapted piconet physical channel	BR/EDR	Unreliable, unidirectional, point-to-multipoint, periodic transmissions to zero or more devices
LE asynchronous connection (LE ACL	Control (LL) LE-C, User (L2CAP) LE-U	LE active physical link, LE piconet physical channel	LE	Reliable, bi-directional, point-to-point.
LE Advertising Broadcast (ADVB)	Control (LL) ADVB-C, User (LL) ADVB-U	LE advertising physical link, LE advertising physical channel	LE	Unreliable, uni-directional broadcast to all devices in a given area or directed to one recipient. Used to carry data and Link Layer signaling between unconnected devices.
LE Periodic Advertising Broadcast (PADVB)	Control (LL) ADVB-C, User (LL) ADVB-U	LE periodic physical link, LE periodic physical channel	LE	Unreliable, periodic, unidirectional broadcast to all devices in a given area.
Connected Isochronous Stream	Low Energy Stream (LE-S) and Low Energy Framed data (LE-F)	LE isochronous physical link	LE	Unidirectional or bidirectional transport in a point-topoint connection for transferring isochronous data.
Broadcast Isochronous	Low Energy Stream (LE-S), Low Energy	LE isochronous physical link	LE	Unidirectional transport for broadcasting data in a

Stream	Framed data (LE-F) and Low Energy Broadcast Control (LEBC)			point to multipoint configuration and unidirectional transport for controlling the broadcast data.

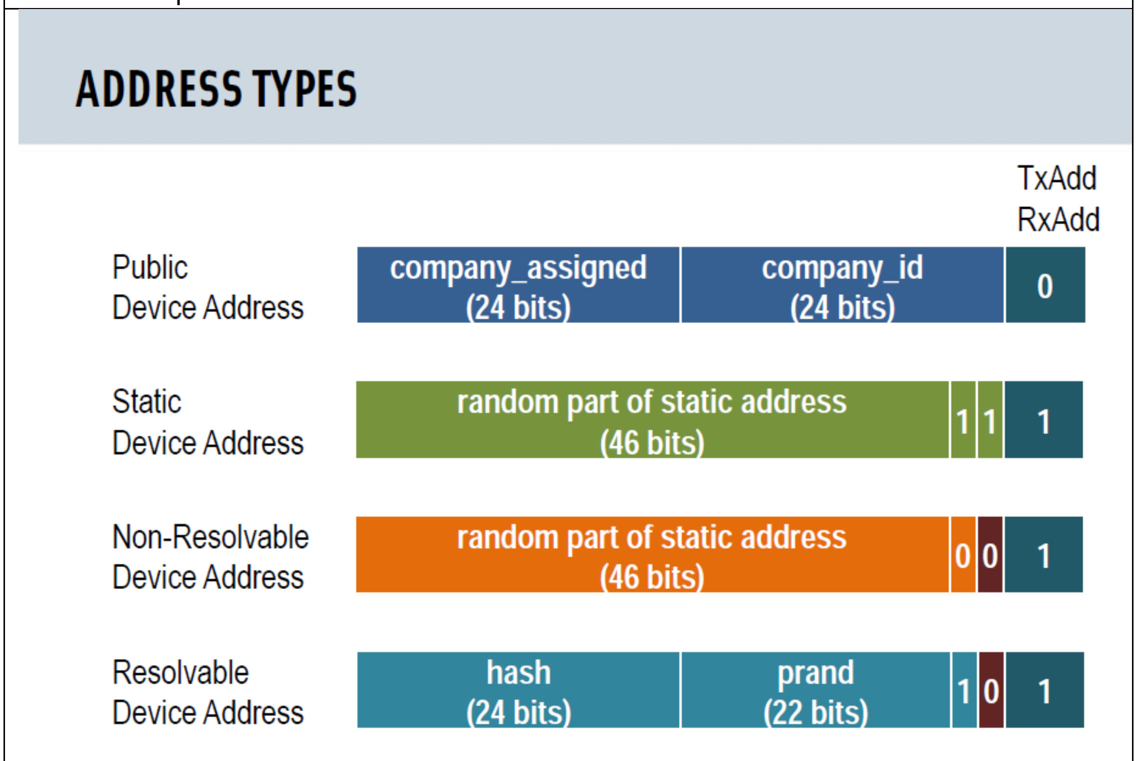
LOGICAL LINKS

- Link Control (LC)
- ACL Control (ACL-C and ASB-C) c = control
- User Asynchronous/Isochronous (ACL-U and ASB-U) u = user data
- User Synchronous (SCO-S) s = stream data
- User Extended Synchronous (eSCO-S)
- Profile Broadcast Data (PBD)

- The control logical link LC is used at the **link control level.**
- The ACL-C and ASB-C logical links are used at **the link manager level.**
- The ACL-U and ASB-U logical links are used to carry either **asynchronous or isochronous** user information.
- The SCO-S, and eSCO-S logical links are used to carry **synchronous user information.**
- The PBD logical link is used to carry **profile broadcast data.**
- The ACL-C, ACL-U, ASB-C, and ASB-U logical links are indicated in the ***logical link ID, LLID, field in the payload header.***

- The SCO-S and eSCO-S logical links are carried by the synchronous logical transports only.
- the ACL-U link is normally carried by the ACL logical transport.
- The ACL-C link may be carried either by the SCO or the ACL logical transport.
- The ASB-C and ASB-U links are carried by the ASB logical transport.
- The PBD logical link is carried by the CSB (connectionless slave broadcast) logical transport.

0b00	Non-resolvable private address
0b01	Resolvable private address
0b10	Reserved for future use
0b11	Static device address



1. LINK CONTROL LOGICAL LINK (LC)

- The LC control logical link shall be mapped onto the **packet header.**
- This logical link carries low level link control information like **ARQ** (ARQ is an error-control strategy used in a two-way communication system. It is a group of error-control protocols to achieve reliable data), **flow control, and payload characterization.**
- The LC logical link is carried in every packet **except in the ID packet** which does not have a packet header.

2. ACL CONTROL LOGICAL LINKS (ACL-C AND ASB-C)

- The ACL-C and ASB-C logical links shall carry control information exchanged between the **link managers of the master and the slave(s).**
- The ACL-C logical link shall use **DV (data voice)** packets. DV packets shall only be used on the ACL-C link if the ACL-C message is less than or equal to 9 bytes and an **HV1 (high quality voice)** synchronous logical transport is in use.
- The ASB-C logical link shall use DM1 (medium rate) packets. The ACL-C and ASB-C logical links are indicated by the LLID code 0b11 in the payload header.
    - **Logical Link Identifier** is a 2-byte tag in the preamble of an Ethernet frame. This 2-byte tag uses *1 bit as a mode indicator (point-to-point or broadcast mode) and the remaining 15 bits as the ONU ID. (optical network unit identifier)*
    - ONU (**Optical Network Unit**) is also called optical modem, which is a device similar to the baseband modem. But the difference is ONU accesses optical fiber. It converts the digital signal into analog signal through modulation at the transmitting end and transmits the analog signal into digital signal.

3. USER ASYNCHRONOUS/ISOCHRONOUS LOGICAL LINKS (ACL-U AND ASB-U)

- The ACL-U and ASB-U logical links shall carry L2CAP asynchronous and isochronous user data.
- These messages may be transmitted in one or more Baseband packets.
- For fragmented messages, the start packet shall use an LLID code of 0b10 in the payload header.
- Remaining continuation packets shall use LLID code 0b01. If there is no fragmentation, all packets shall use the LLID start code 0b10.

4. USER SYNCHRONOUS DATA LOGICAL LINK (SCO-S)

The SCO-S logical link carries transparent synchronous user data. This logical link is carried over the synchronous logical transport SCO.

5. USER EXTENDED SYNCHRONOUS DATA LOGICAL LINK (eSCO-S)

The eSCO-S logical link also carries transparent synchronous user data. This logical link is carried over the extended synchronous logical transport eSCO.

6. LOGICAL LINK PRIORITIES

- The ACL-C logical link shall have a **higher priority** than the ACL-U logical link when **scheduling traffic** on the shared **ACL logical transport**, except in the case when retransmissions of unacknowledged ACL packets shall be given priority over traffic on the ACL-C logical link.
- The ASB-C logical link shall have a ***higher priority*** than the ASB-U logical link when scheduling traffic on the shared ASB logical transport.

- The ACL-C logical link should also have priority over traffic on the SCO-S and eSCO-S logical links .
- The ACL-C, SCO-S, and eSCO-S logical links should have priority over traffic on the PBD logical link.

7. PROFILE BROADCAST DATA LOGICAL LINK

The PBD logical link carries profile broadcast data. Messages shall not be fragmented and shall always use LLID start code 0b10.

LOGICAL TRANSPORTS

Between master and slave(s), different types of logical transports may be established. Five logical transports have been defined:

• Synchronous Connection-Oriented (SCO) logical transport

• Extended Synchronous Connection-Oriented (eSCO) logical transport

• Asynchronous Connection-Oriented (ACL) logical transport

• Active Slave Broadcast (ASB) logical transport

• Connectionless Slave Broadcast (CSB) logical transport.

- **The synchronous logical transports** are point-to-point logical transports between a master and a single slave in the piconet. The synchronous logical transports typically like voice.
- The master maintains the synchronous logical transports by using reserved slots at regular intervals.
- In addition to the reserved slots the eSCO logical transport may have a retransmission window after the reserved slots.
- The ACL logical transport is also a point-to-point logical transport between the master and a slave.
- In the slots not reserved for synchronous logical transport(s), the master can establish an ACL logical transport on a per-slot basis to any slave, including the slave(s) already engaged in a synchronous logical transport.

The ASB logical transport is used by a master to communicate with active slaves.

<u>The CSB logical transport</u> is used by a master to send profile broadcast data to zero or more slaves.

SYNCHRONOUS LOGICAL TRANSPORTS

- The first type of synchronous logical transport, the SCO logical transport, is a symmetric, point-to-point transport between the master and a specific slave.
- The SCO logical transport reserves slots and can therefore be considered as a circuit-switched connection between the master and the slave.
- The master may support up to three SCO links to the same slave or to different slaves.
- SCO packets are never retransmitted.
- The second type of synchronous logical transport, **the eSCO logical transport**, is a point-to-point logical transport between the master and a specific slave.
- eSCO logical transports may be symmetric or asymmetric. Similar to SCO, eSCO reserves slots and can therefore be considered a circuit-switched connection between the master and the slave.
- In addition to the reserved slots, eSCO supports a retransmission window immediately following the reserved slots.
- Together, the reserved slots and the retransmission window form the complete eSCO window.

ASYNCHRONOUS LOGICAL TRANSPORT

- In the slots not reserved for synchronous logical transports, the master may exchange packets with any slave on a per-slot basis.
- The ACL logical transport provides a packet-switched connection between the master and all active slaves participating in the piconet.
- Both asynchronous and isochronous services are supported.
- Only a single ACL logical transport shall exist between any two devices.
- For most ACL packets, packet retransmission is applied to assure data integrity.
- ACL packets not addressed to a specific slave (LT_ADDR=0) are considered as broadcast packets and should be received by every slave except slaves with only a CSB logical transport.
- If there is no data to be sent on the ACL logical transport and no polling (Bluetooth [1] is a wireless access mechanism where polling is **used to share bandwidth among the participants)** is required, no transmission is required.

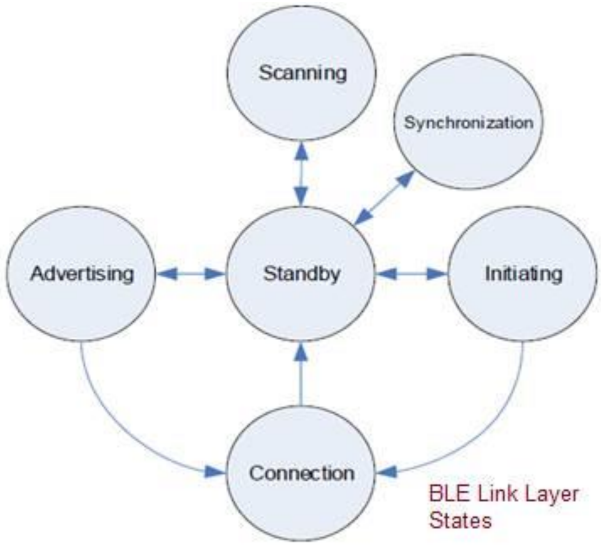
ACTIVE SLAVE BROADCAST TRANSPORT

- The active slave broadcast logical transport is used to transport L2CAP user traffic and certain LMP traffic to all devices in the piconet that are currently connected to the piconet physical channel that is used by the ASB.
- There is no acknowledgment protocol and the traffic is unidirectional from the piconet master to the slaves.
- The ASB logical transport may only be used for L2CAP group traffic and shall never be used for L2CAP connection-oriented channels or L2CAP control signaling.
- The ASB logical transport is unreliable.
- To improve reliability somewhat each packet is transmitted a number of times.
- An identical sequence number is used to assist with filtering retransmissions at the slave device. The ASB logical transport is identified by the reserved, all-zero, LT_ADDR. Packets on the ASB logical transport may be sent by the master at any time.

CONNECTIONLESS SLAVE BROADCAST LOGICAL TRANSPORT

- The CSB logical transport is used to transport profile broadcast data from a master to multiple slaves. There is no acknowledgment scheme and the traffic is unidirectional.

Note: The CSB logical transport is not used for L2CAP connection-oriented channels, L2CAP control signaling, or LMP control signaling. The CSB logical transport is unreliable. To improve reliability each packet payload may be transmitted a number of times.



- **Standby state:** There is no transmission or reception of packet in this state. This state can be entered from any other BLE states.

  • **Advertising state:** A BLE device in this state is called "Advertiser". This state can be reached from "standby state". Link layer will transmit advertising packets and listens to and responds to responses triggered due to advertising packets.

  • **Scanning state:** In this state, link layer listens for advertising packets from other advertising devices. The BLE device in this state is known as "scanner". This state can be reached from "standby state".

  • **Initiating state:** The BLE device in this state is known as initiator. This BLE state can be reached from "standby state". Link layer listens for advertising physical packets from specific BLE devices. It also responds to these packets to initiate a connection from another device.

  • **Connection State:** This state can be reached either from "initiating state" or "advertising state". In this connection state, there are two roles of BLE device either master or slave. When it has entered from "initiating state" it will be in master role. When it has entered from "advertising state" it will be in slave role. In master role, link layer will communicate with device in the slave role and defines timings of transmissions.

  • **Synchronization state:** This state can be entered from "standby state". The link layer in this state listens for periodic channel packets from specified device transmitting periodic advertising.

PACKETS

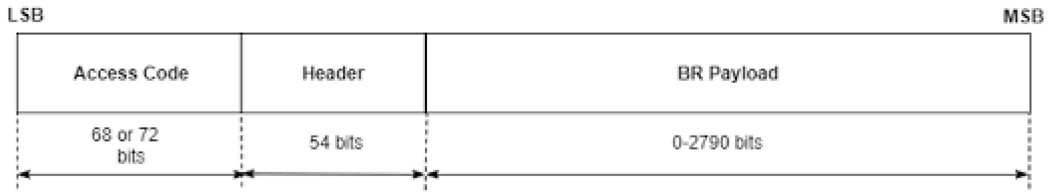
- ***GENERAL FORMAT***

Bluetooth BR/EDR Packet Structure

- ***Basic Rate***

- **Basic rate (BR)** - Mandatory mode, uses Gaussian frequency shift keying (GFSK) modulation with a data rate of 1 Mbps.

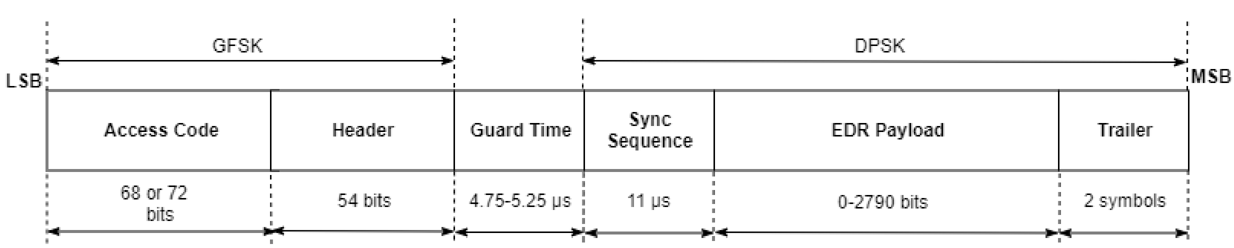
- The general packet format of Basic Rate packets is shown in Figure 6.1. Each packet consists of 3 entities: the access code, the header, and the payload.



General Basic Rate packet format

- The access code is 72 or 68 bits and the header is 54 bits. The payload ranges from 0 to a maximum of 2790 bits. Different packet types have been defined. A packet may consist of:
    - the shortened access code only (see Section 6.5.1.1)
    - the access code and the packet header
    - the access code, the packet header and the payload.
- the access code is 72 bits long, otherwise the access code is 68 bits long and is known as a shortened access code. The shortened access code does not contain a ***trailer***. There is no header. The shortened access code is used in paging and inquiry.
- This access code is used for synchronization, DC offset compensation and identification. The access code itself is used as a signaling message and neither a header nor a payload is present. The access code consists of a preamble, a sync word, and possibly a trailer.

Enhanced Data Rate



Enhanced data rate (EDR) - Optional mode, uses phase shift keying (PSK) modulation with these two variants:

Phase-shift keying (**PSK**) is a digital modulation process which conveys data by changing (modulating) the phase of a constant frequency reference signal

EDR2M: Uses pi/4-DQPSK with a data rate of 2 Mbps. differential quadrature phase-shift keying (**DQPSK**)

EDR3M: Uses 8-DPSK with a data rate of 3 Mbps. Differential phase-shift keying (**DPSK**)

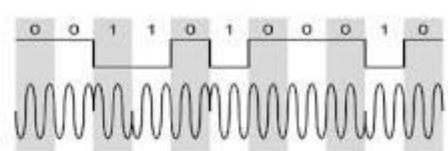
- Following the header field, the EDR packets have a guard time in the range [4.75, 5.25] µs, a sync sequence (11 µs), payload in the range [0, 2790] bits, and trailer (two symbols) fields.

<u>Gaussian Frequency Shift Keying</u>

- The Bluetooth radio interface also uses a modulation technique called **Gaussian Frequency Shift Keying**, GFSK. This form of modulation is spectrally efficient and also enables the use of efficient radio power amplifiers, thereby saving on battery life.
- *A binary one is represented by a positive frequency deviation and a binary zero is represented by a negative frequency deviation.*

<u>What is meant by DPSK?</u>

Differential Phase Shift Keying (DPSK) means **differential phase modulation of the radio frequency carrier with relative phase states of 0 degree or 180 degrees**.
What is the principle of DPSK?



DPSK **encodes two distinct signals, i.e., the carrier and the modulating signal with 180° phase shift each**.

<u>BIT ORDERING</u>

ARM processors are a family of central processing units (CPUs) based on a reduced instruction set computer (RISC) architecture. ARM stands for **Advanced RISC Machine**.

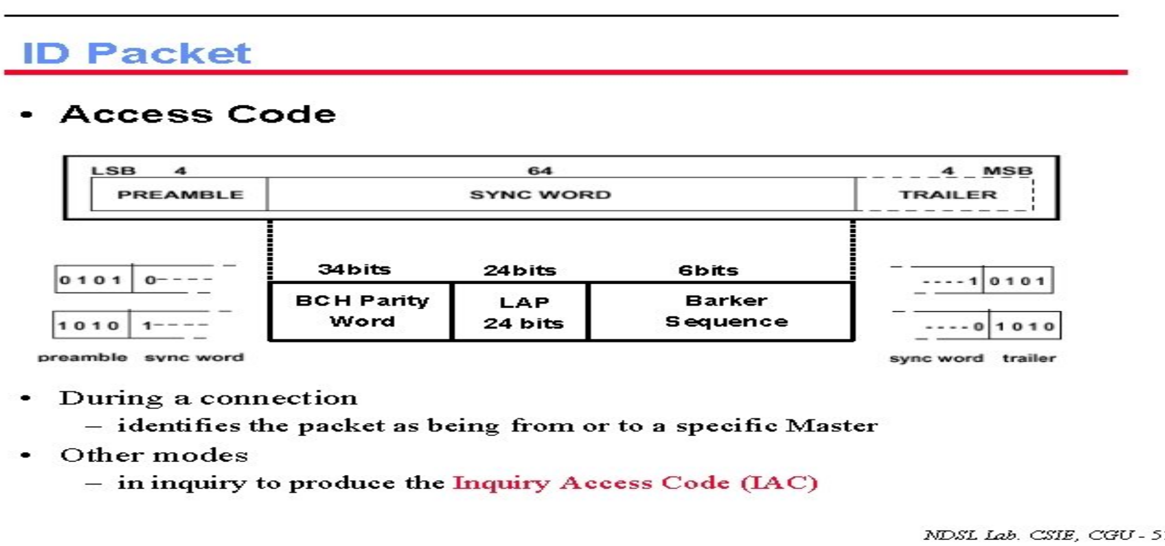
- **Little-endian** is the **default memory format for ARM processors**. In little-endian format, the byte with the lowest address in a word is the least-significant byte of the word. The byte with the highest address in a word is the most significant.

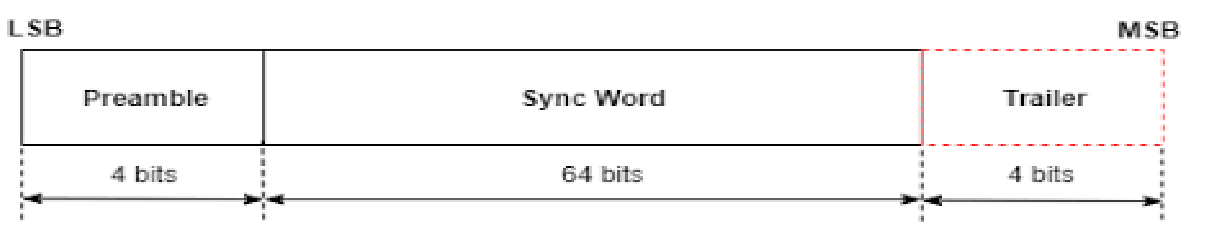
- *The bit ordering when defining packets and messages in the Baseband Specification, follows the little-endian format.*

When defining packets and messages in the baseband specification, the bit ordering follows the little-endian format. In this format, these rules apply:

- The least significant bit (LSB) corresponds to b_0.
- LSB is the first bit sent over the air.
- When illustrating the packet structure, the LSB is shown on the left side.
  - Moreover, data fields generated internally at the baseband must be transmitted with the LSB first.
  - For example, *a 3-bit parameter is sent as: $b_0b_1b_2$ = 110 over the air, where 1 is sent first and 0 is sent last.*

ACCESS CODE

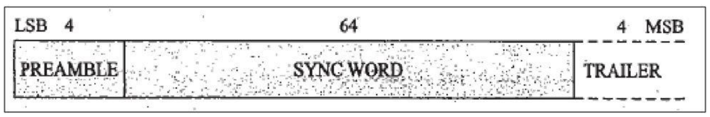


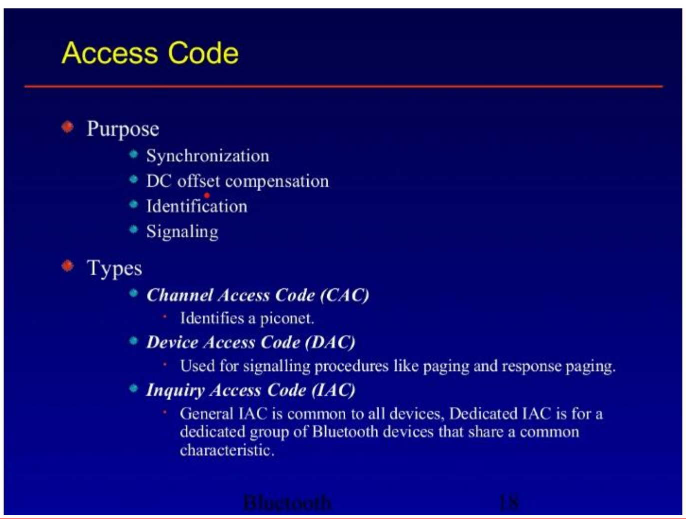


➡ Packet Formats

→ Access Code Types

- CAC (Channel Access Code)
  - identify & sync piconet, present in every packet
- DAC (Device Access Code)
  - signaling paging & paging response
- IAC (Inquiry Access Code)
- GIAC(General IAC)–discover bluetooth devices range
- DIAC(Dedicated IAC) – discover bluetooth devices in range & with common characteristics





DIAC (Dedicated IAC), which is used for specific devices, or GIAC (Generic IAC) for all devices.

- **Preamble:** The preamble is a fixed zero-one pattern of four symbols.

- Used for DC compensation
- 0101 is if LSB of syn word is 0
- 1010 is if LSB of syn word is 1
- ‘1010’ or ‘0101’ (in transmission order)

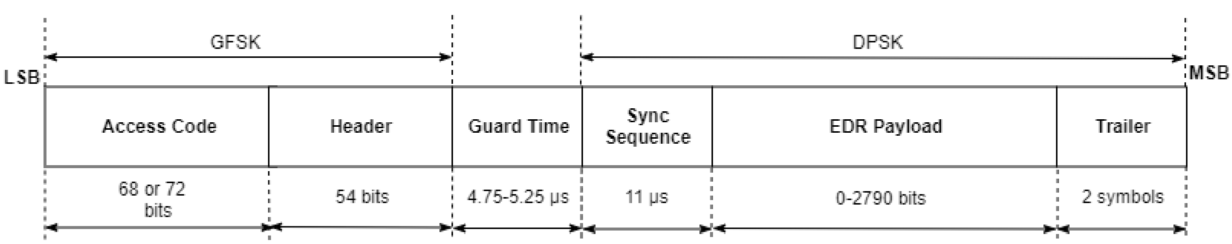
- **Sync** = It is a **64-bit code** word derived from a **24-bit LAP** address. The construction guarantees a large Hamming distance between sync words based on different LAPs. The *autocorrelation properties of the sync word improve timing acquisition.*

- **Trailer** = For EDR packets, the trailer bits must be all zero pattern of two symbols, {00,00} for pi/4-DQPSK and {000,000} for 8DPSK. Differential phase-shift keying (**DPSK**) differential quadrature phase-shift keying (**DQPSK**)

    - The trailer together with the three MSBs of the sync word form a 7-bit pattern of alternating ones and zeroes which can be used for extended DC compensation
    - 0101 is if MSB of syn word is 0
    - 1010 is if MSB of syn word is 1
    - This is typically the case with the CAC, *but the trailer is also used in the DAC and IAC when these codes are used in FHS packets exchanged during page response and inquiry response.*

    - FHS: **Frequency Hop Synchronizations** (FHS) is used to synchronize hop frequencies during piconet establishment. DM1: Data Medium Rate 1-slot is used to carry data packets and control packets. Used by the Link Manager and Link Controller for control packet exchange.

- **ACCESS CODE** = Used for timing *synchronization, DC off set compensation, paging, inquiry, synchronization and identification.* Each packet starts with an access code. If a packet header follows, the access code is 72 bits long, otherwise the access code is 68 bits long. The access code consists of these fields:
    - **Header** = used to identify packet type and carry protocol control information.
    - **Payload** = contains user voice or data payload header is present. Actual data
    - **Guard**: For EDR packets, guard time allows the Bluetooth radio to prepare for the change in modulation from GFSK to DPSK. The guard time must be **between 4.75 to 5.25 microseconds**. 2400 – 24835 = 83.5 security is used



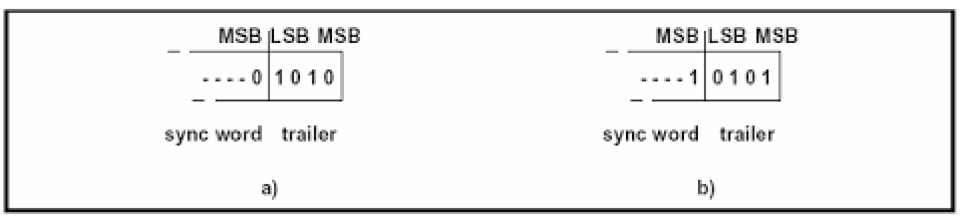
Lap = lower address part   substate = subdivision

Access Code Type	LAP	Access Code Length (Bits)	Description

Channel access code (CAC)	Central or Master	72	This access code is used in the connection state, synchronization train substate, and synchronization scan substate. It is derived from the LAP of the Central's BD_ADDR.
Device access code (DAC)	Paged device	68 or 72	This access code is used during page, page scan, and page response substates. It is derived from the paged devices' BD_ADDR.
Dedicated inquiry access code (DIAC)	Dedicated	68 or 72	This access code is used in the inquiry substate for dedicated inquiry operations.
General inquiry access code (GIAC)	Reserved	68 or 72	This access code is used in the inquiry substate for general inquiry operations.

- For DAC, DIAC, and GIAC access code types, the access code length of 72 bits is used only in combination with *frequency hopping sequence (FHS) packets*.
- When used as self-contained messages without a header, the DAC, DIAC and GIAC do not include trailer bits and are of length 68 bits.
- The CAC consists of a preamble, sync word, and trailer and its total length is 72 bits. *When used as self-contained messages without a header, the DAC and IAC do not include the trailer bits and are of length 68 bits.*

- *length 72 is only used in combination with FHS packets*



- Every packet starts with an access code.
- If a packet header follows, the access code is 72 bits long, otherwise the access code is 68 bits long and is known as a shortened access code. The shortened access code does not contain a trailer.
- This access code is used for synchronization, DC offset compensation and identification.
- The access code identifies all packets exchanged on a physical channel: all packets sent in the same physical channel are preceded by the same access code.
- In the receiver of the device, a sliding correlator correlates against the access code and triggers when a threshold is exceeded.
- This trigger signal is used to determine the receive timing.
- The shortened access code is used in paging and inquiry.
- The access code itself is used as a signaling message and neither a header nor a payload is present. The access code consists of a preamble, a sync word, and possibly a trailer.

PACKET HEADER

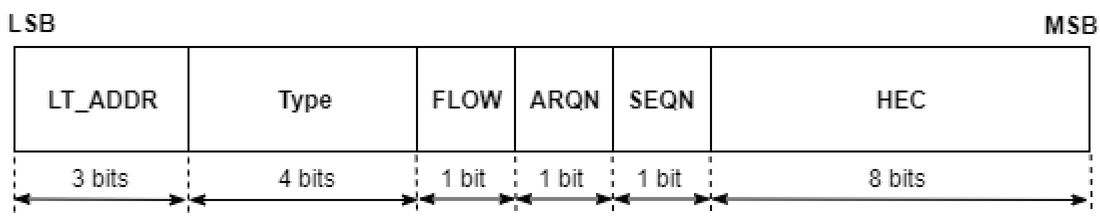
Packet Type	TYPE Code	Slot Occupancy	Payload Header (Bytes)	User Payload (Bytes)	FEC	CRC	Logical Transport Types Supported
ID	N/A	1	N/A	N/A	N/A	N/A	N/A
NULL	0000	1	N/A	N/A	N/A	N/A	SCO, eSCO, ACL, CSB
POLL	0001	1	N/A	N/A	N/A	N/A	SCO, eSCO, ACL
FHS	0010	1	N/A	18	2/3	Yes	SCO, ACL
DM1	0011	1	1	0-17	2/3	Yes	SCO, ACL, CSB
DH1	0100	1	1	0-27	No	Yes	ACL, CSB
DM3	1010	3	2	0-121	2/3	Yes	ACL, CSB
DH3	1011	3	2	0-183	No	Yes	ACL, CSB
DM5	1110	5	2	0-224	2/3	Yes	ACL, CSB
DH5	1111	5	2	0-339	No	Yes	ACL, CSB
2-DH1	0100	1	2	0-54	No	Yes	ACL, CSB
2-DH3	1010	3	2	0-367	No	Yes	ACL, CSB
2-DH5	1110	5	2	0-679	No	Yes	ACL, CSB
3-DH1	1000	1	2	0-83	No	Yes	ACL, CSB
3-DH3	1011	3	2	0-552	No	Yes	ACL, CSB
3-DH5	1111	5	2	0-1021	No	Yes	ACL, CSB
HV1	0101	1	N/A	10	1/3	No	SCO
HV2	0110	1	N/A	20	2/3	No	SCO
HV3	0111	1	N/A	30	No	No	SCO
DV	1000	1	1 D	10+(0-9) D	2/3 D	Yes D	SCO
EV3	0111	1	N/A	1-30	No	Yes	eSCO
EV4	1100	3	N/A	1-120	2/3	Yes	eSCO
EV5	1101	3	N/A	1-180	No	Yes	eSCO
2-EV3	0110	1	N/A	1-60	No	Yes	eSCO
2-EV5	1100	3	N/A	1-360	No	Yes	eSCO

3-EV3	0111	1	N/A	1-90	No	Yes	eSCO
3-EV5	1101	3	N/A	1-540	No	Yes	eSCO

Header (54 bits). Actually, consists of 18 bits, but each bit is transmitted three times.

Payload.

Only for Enhanced Data Rate packets: trailer (2 symbols), containing all zero data.



The header contains link control (LC) information and consists of 6 fields:

LT_ADDR	3-bit Logical Transport Address
TYPE	4-bit packet type code
	• LT_ADDR and TYPE fields in combination identify Default ACL Logical Transport, as the same LT_ADDR is shared for ACL and SCO logical transports**.** • which specifies the packet type used for transmission? It can be one of {ID, NULL, POLL, FHS, HV1, HV2, HV3, DV, EV3, EV4, EV5, 2-EV3, 2-EV5, 3-EV3, 3-EV5, DM1, DH1, DM3, DH3, DM5, DH5, AUX1, 2-DH1, 2-DH3, 2-DH5, 3-DH1, 3-DH3, 3-DH5}.
FLOW	1-bit flow control (Master -> Slave or Slave -> Master)
ARQN	1-bit acknowledgement indicator
SEQN	1-bit sequence number
	SEQN for ordering of received packets and ARQN for re-transmission for unreached packets
HEC	Header error check code.

ID = identifier

HV = HIGH-RATE VOICE

EV =

DV = DATA VOICE

FHS = Frequency Hopping Synchronization

medium-rate (DM) and high-rate (DH) packets

- The total header, including the HEC, consists of **18 bits**.

Packet Header Field	**Size of the Field (Bits)**	**Description**
Logical transport address (LT_ADDR)	3	This field indicates the destination Peripheral(s) for a packet in a Central-to-Peripheral transmission slot and indicates the source Peripheral for a Peripheral-to-Central transmission slot.
Type	4	▪ This field specifies the type of packet used. ▪ The Bluetooth Core Specification defines 16 different types of BR/EDR packets. ▪ The value in this field depends on the value of LT_ADDR field in the packet. ▪ This field determines the number of slots occupied by the current packet.
Flow control (FLOW)	1	▪ This field implements the flow control of BR/EDR packets over the asynchronous connection-oriented logical (ACL) transport. ▪ When the receive buffer for the ACL logical transport is full, a 'STOP' indication (FLOW = 0) is returned to stop the other device from transmitting data temporarily. ▪ When the receive buffer can accept data, a 'GO' indication (FLOW = 1) is returned.
Automatic repeat request number (ARQN)	1	▪ This field informs the source of a successful transfer of payload data with the CRC. ▪ This field is reserved for future use on the connectionless Peripheral broadcast (CSB) logical transport. This field provides a sequential numbering scheme to order the data packet stream. ▪ This field is reserved for future use on the ▪ ▪ transport. Acknowledgement indication; 0=NAK, 1=ACK.
Sequence number	1	Sequence number. This bit is toggled for consecutive packets.

(SEQN)		
Header error check (HEC)	8	▪ This field checks the packet header integrity. ▪ Before generating the HEC, the HEC generator is initialized with an 8-bit value. ▪ These 8 bits correspond to the upper address part (UAP). ▪ After the initialization, the HEC generator calculates the HEC value7 for the 10 header bits. ▪ Before checking the HEC, the receiver initializes the HEC check circuitry with the appropriate 8-bit UAP. ▪ If the HEC does not check the packet header integrity, the entire packet is discarded.

6.1 Default ACL Logical Transport Packet Structure:

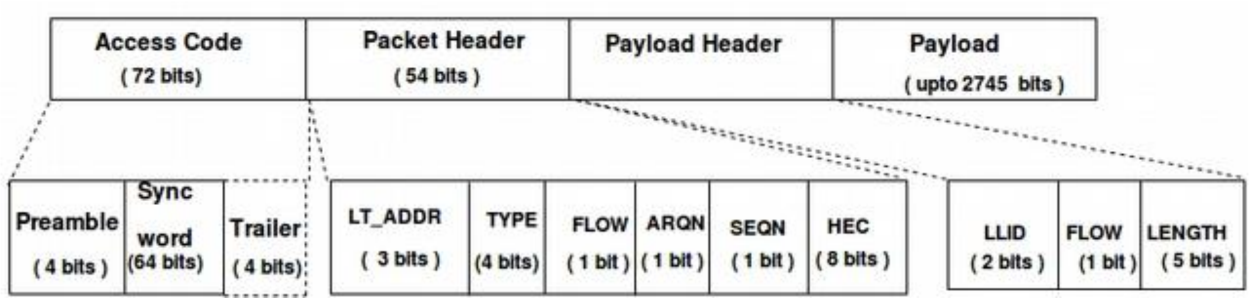


Fig: ACL logical transport packet structure (general)

The above figure represents packet structure of default ACL logical transport, which is used to carry both LMP signalling and, L2CAP signalling and user data.

ACL logical transport contain payload header in addition to packet header which is having 3 fields as described below:

LLID: 2-bit field

LLID	Logical Link	Information
00	N/A	Reserved for future use
01	ACL-U and ASB-U	Continuation fragment of an L2CAP message
10	ACL-U and ASB-U	Start of an L2CAP message / No fragmentation
	PBD	Profile Broadcast Data, No fragmentation (Will not pass through L2CAP, profile directly interacts with HCI)
11	ACL-C and ASB-U	LMP message

FLOW: L2CAP layer flow control bit to control flow of L2CAP messages.

FLOW = 0	STOP	Flow off
FLOW = 1	GO	Flow on

LENGTH: This field specifies length of payload.

PACKET TYPES

The packets used on the piconet are related to the logical transports on which they are used.

Four logical transports with distinct packet types are defined (see Section 4):

- SCO logical transport
- eSCO logical transport
- ACL logical transport
- CSB logical transport.

Seg-ment	TYPE code $b_3b_2b_1b_0$	Slot occu-pancy	SCO logical trans-port (1 Mbps)	eSCO logical trans-port (1 Mbps)	eSCO logical trans-port (2-3 Mbps)	ACL logical trans-port (1 Mbps) ptt=0	ACL logical trans-port (2-3 Mbps) ptt=1
1	0000	1	NULL	NULL	NULL	NULL	NULL
	0001	1	POLL	POLL	POLL	POLL	POLL
	0010	1	FHS	reserved	reserved	FHS	FHS
	0011	1	DM1	reserved	reserved	DM1	DM1
2	0100	1	undefined	undefined	undefined	DH1	2-DH1
	0101	1	HV1	undefined	undefined	undefined	undefined
	0110	1	HV2	undefined	2-EV3	undefined	undefined
	0111	1	HV3	EV3	3-EV3	undefined	undefined
	1000	1	DV	undefined	undefined	undefined	3-DH1
	1001	1	undefined	undefined	undefined	AUX1	AUX1
3	1010	3	undefined	undefined	undefined	DM3	2-DH3
	1011	3	undefined	undefined	undefined	DH3	3-DH3
	1100	3	undefined	EV4	2-EV5	undefined	undefined
	1101	3	undefined	EV5	3-EV5	undefined	undefined
4	1110	5	undefined	undefined	undefined	DM5	2-DH5
	1111	5	undefined	undefined	undefined	DH5	3-DH5

Table 6.2: Packets defined for synchronous and asynchronous logical transport types.

6.2 Logical Transport Packet types:

6.2.2 SCO Packets:

SCO (synchronous connection orientated) links

Key facts:

- Following a request from either the master or slave device, the master may establish a SCO link to that device.
- The master can support a maximum of three SCO links, which can be to the same or different slaves.
- SCO links are symmetric: the master sends a SCO packet, then the slave sends a SCO packet in the following slot, so the data rate in each direction is the same.
- SCO packets are always transmitted in defined time slots, at regular intervals (every 2, 4, or 6 slots).
- SCO packets are always 1 slot in length.
- SCO packets are **not** acknowledged.

ID = identifier

HV = HIGH RATE VOICE

EV = Extended Voice

DV = DATA VOICE

FHS = Frequency Hopping Synchronization

medium-rate (DM) and high-rate (DH) packets

The packet type is determined by the TYPE OF code in the header:

0000	NULL	1 slot	No payload. Used for acknowledgements or flow control.			
0001	POLL	1 slot	No payload. Used by the master to poll slaves. Requires acknowledgement.			
0010	FHS	1 slot	Frequency Hopping Sequence. Contains address, clock, etc. information			
0011	DM1	1 slot	Data, Medium rate. An ACL packet, which can interrupt the regular flow of SCO data.			
0101	HV1	1 slot	High quality Voice	(1/3 rate FEC)	10 data bytes	transmitted every 2 slots (data rate = 64,000 bit/sec)
0110	HV2	1 slot	High quality Voice	(2/3 rate FEC)	20 data bytes	transmitted every 4 slots (data rate = 64,000 bit/sec)
0111	HV3	1 slot	High quality Voice	(No error correction)	30 data bytes	transmitted every 6 slots (data rate = 64,000 bit/sec)
1000	DV	1 slot	Combined Data/Voice. Can be transmitted in place of an HV1 packet.			

The DV packet payload contains:

No error correction: 10 voice data bytes.

2/3 rate FEC: 1 payload header byte (as for an ACL packet);
0...9 data bytes;
2 CRC bytes.

HV and DV packets are used on the SCO logical transport. HV packets do not include CRC and these packets will not be re-transmitted.

- **HV1 Packet**

  The HV1 packet has 10 information bytes. The bytes are protected with a rate 1/3 FEC. No MIC is present. No CRC is present. The payload length is fixed at 240 bits. There is no payload header present.

Audio (10 Bytes)	FEC (1/3)

- **HV2 Packet**

Audio (20 Bytes)	FEC (2/3)

- **HV3 Packet**

Audio (30Bytes)

- **DV Packet**

Audio (10 Bytes)	Header (1 Byte)	Payload (0 – 9 Bytes)	2/3 FEC	CRC (16 bits)

6.2.3 eSCO Packet Types:

eSCO (extended synchronous connection orientated) links

Key facts:

- eSCO links were added in version 1.2 of the Bluetooth specification.
- Following a request from either the master or slave device, the master may establish an eSCO link to that device.
- eSCO packets are always transmitted in predetermined time slots: the regular interval between eSCO packets is specified when the link is established.
- eSCO packets can be 1 or 3 slots in length.

- eSCO packets to/from a specific slave are acknowledged, and may be retransmitted if not acknowledged.

eSCO packets

The packet type is determined by the TYPE OF code in the header:

0000	NULL	No payload. Used for acknowledgements or flow control.
0001	POLL	No payload. Used by the master to poll slaves. Requires acknowledgement.
0111	EV3	Extended Voice (no error correction), 1 slot: maximum 30 data bytes plus a 16-bit CRC.
1100	EV4	Extended Voice (2/3 rate FEC), 3 slots: maximum 120 data bytes plus a 16-bit CRC.
1101	EV5	Extended Voice (no error correction), 3 slots: maximum 180 data bytes plus a 16-bit CRC.

EV packets are used on the synchronous eSCO logical transport. The packets includes CRC and retransmission is applied if acknowledgment is not received.

- **EV3 packet**

Audio (30 Bytes)	**CRC (16 bits)**

- **EV4 packet**

Audio (120 Bytes)	**FEC (2/3)**	**CRC (16 bits)**

- **EV5 packet**

Audio (180 Bytes)	**CRC (16 bits)**

- **2-EV3 packet**

Audio (60 Bytes)	**CRC (16 bits)**

- **2-EV5 packet**

Audio (360 Bytes)	CRC (16 bits)

- **3-EV3 packet**

Audio (90 Bytes)	CRC (16 bits)

- **3-EV5 packet**

Audio (540 Bytes)	CRC (16 bits)

6.2.1 ACL packets

ACL (asynchronous connectionless) links

Key facts:

- The master device establishes one ACL link to each slave device.
- ACL packets can only be transmitted in time slots not being used for SCO packets.
- ACL packets can be 1, 3, or 5 slots in length.
- All ACL packets to/from a specific slave are always acknowledged. Broadcast ACL packets from the master are **not** acknowledged but may be transmitted several times for reliability.

ACL packets

The packet type is determined by the TYPE code in the header and the currently selected packet type table (as selected by the LMP_packet_type_table_req message).

Packet type table 0 (default):

0000	NULL	1 slot	No payload. Used for acknowledgements or flow control.		
0001	POLL	1 slot	No payload. Used by the master to poll slaves. Requires acknowledgement.		
0010	FHS	1 slot	Frequency Hopping Sequence. Contains address, clock, etc. information.		
0011	DM1	1 slot	Data, Medium rate	(2/3 rate FEC)	maximum 17 data bytes.
0100	DH1	1 slot	Data, High rate	(no error correction)	maximum 27 data bytes.
1001	AUX1	1 slot	Like DH1, but without a CRC; maximum 29 data bytes. Should not be used.		

1010	DM3	3 slots	Data, Medium rate	(2/3 rate FEC)	maximum 121 data bytes.
1011	DH3	3 slots	Data, High rate	(no error correction)	maximum 183 data bytes.
1110	DM5	5 slots	Data, Medium rate	(2/3 rate FEC)	maximum 224 data bytes.
1111	DH5	5 slots	Data, High rate	(no error correction)	maximum 339 data bytes.

Packet type table 1 (added to support Enhanced Data Rate modulation, at 2 or 3 MBit/sec):

0000	NULL	1 slot	No payload. Used for acknowledgements or flow control.			
0001	POLL	1 slot	No payload. Used by the master to poll slaves. Requires acknowledgement.			
0010	FHS	1 slot	Frequency Hopping Sequence. Contains address, clock, etc. information.			
0011	DM1	1 slot	Data, Medium rate	(2/3 rate FEC)	maximum 17 data bytes.	
0100	2-DH1	1 slot	Data, High rate	(no error correction)	maximum 54 data bytes,	using 2 MBit/sec modulation.
1000	3-DH1	1 slot	Data, High rate	(no error correction)	maximum 83 data bytes,	using 3 MBit/sec modulation.
1001	AUX1	1 slot	Like DH1, but without a CRC; maximum 29 data bytes. Should not be used.			
1010	2-DH3	3 slots	Data, High rate	(no error correction)	maximum 367 data bytes,	using 2 MBit/sec modulation.
1011	3-DH3	3 slots	Data, High rate	(no error correction)	maximum 552 data bytes,	using 3 MBit/sec modulation.
1110	2-DH5	5 slots	Data, High rate	(no error correction)	maximum 679 data bytes,	using 2 MBit/sec modulation.
1111	3-DH5	5 slots	Data, High rate	(no error correction)	maximum 1021 data bytes,	using 3 MBit/sec modulation.

For single slot packets using basic rate (1 MBit/sec) modulation, the payload contains:

Header (1 byte), least significant bit first:

L_CH (2 bits): Logical Channel.

01 = continuation of an L2CAP PDU
10 = start of an L2CAP PDU
11 = LMP PDU

FLOW (1 bit): flow control on the ACL link; 0=stop, 1=go.
LENGTH (5 bits): number of bytes of data.

Data (*LENGTH* bytes).

CRC (16 bits): Cyclic Redundancy Check.

For multi-slot packets, and single slot packets using Enhanced Data Rate (2 or 3 MBit/sec) modulation, the payload contains:

Header (2 bytes), least significant bit first:

L_CH (2 bits): Logical Channel.

01 = continuation of an L2CAP PDU
10 = start of an L2CAP PDU
11 = LMP PDU

FLOW (1 bit): flow control on the ACL link; 0=stop, 1=go.
LENGTH (10 bits): number of bytes of data.
Padding (3 bits): undefined value.

Data (*LENGTH* bytes).

CRC (16 bits): Cyclic Redundancy Check.

ACL packets are used on the asynchronous logical transport. The information carried may be user data or control data.

6.2.1.1 ACL packets for Basic Rate Operation:

- **DM1 packet -** Data Medium Rate (1 slot)

Header (1byte)	Payload (18 bytes)	2/3 FEC	CRC (16 bit)

- **DH1 packet - Data** High Rate (1 slot)

Header (1byte)	Payload (28 bytes)	CRC (16 bit)

- **DM3 packet –** Data Medium Rate (3 slots)

Header (2byte)	Payload (123 bytes)	2/3 FEC	CRC (16 bit)

- **DH3 packet –** Data High Rate (3 slots)

Header (2byte)	Payload (185 bytes)	CRC (16 bit)

- **DM5 packet –** Data Medium slot (5 slots)

Header (2byte)	Payload (226 bytes)	2/3 FEC	CRC (16 bit)

- **DH5 packet –** Data High Rate (5 slots)

Header (2byte)	Payload (341 bytes)	CRC (16 bit)

- **AUX1 packet**

Header (1byte)	Payload (30 bytes)

6.2.1.2 ACL packets for Enhanced Data Rate operation:

- **2-DH1 packet**

Header (2byte)	Payload (54 bytes)	CRC (16 bit)

- **2-DH3 packet**

Header (2byte)	Payload (367 bytes)	CRC (16 bit)

- **2-DH5 packet**

Header (2byte)	Payload (679 bytes)	CRC (16 bit)

- **3-DH1 packet**

Header (2byte)	Payload (83 bytes)	CRC (16 bit)

- **3-DH3 packet**

Header (2byte)	Payload (552 bytes)	CRC (16 bit)

- **3-DH5 packet**

Header (2byte)	Payload (1021bytes)	CRC (16 bit)

6.2.4 Common Packet Types:

6.2.4.1 NULL Packet

The NULL packet has no payload and consists of the channel access code and Packet header only. Its total (fixed) length is 126 bits.

Channel Access Code	Packet Header

6.2.4.2 POLL Packet

POLL packet is similar to that of NULL packet. It does not have payload. POLL packets shall not be sent on Connectionless slave broadcast.

Channel Access Code	Packet Header

6.2.4.3 ID Packet

The identity or ID packet consists of the device access code (DAC) or inquiry access code (IAC). It has a fixed length of 68 bits.

For Inquiry:

Preamble (4b)	Sync word (64b)

x

- **Preamble**: Depends on the first (i.e. least significant) bit of the sync word: either 0101(0...) or 1010(1...).
- **Sync Word**: Depends upon the type of access code.
- **IAC** (Inquiry Access Code): It is used for inquiring. For general inquiries, the GIAC (General IAC) is used, generated from the reserved LAP of 9E8B33.

For Paging:

Preamble (4b)	Sync word (64b)

- **Preamble:** Depends on the first (i.e. least significant) bit of the sync word: either 0101(0...) or 1010(1...).

- **Sync Word:** Depends upon the type of access code.

- **DAC** (Device Access Code): used for paging, generated from the LAP of the paged device.

FHS Packet

The FHS packet is a special control packet containing, among other things, the Bluetooth device address and the clock of the sender.

The payload contains 144 information bits plus a 16-bit CRC code. The payload is coded with a rate 2/3 FEC with a gross payload length of 240 bits.

The FHS packet is not encrypted. No payload header or MIC is present.

34	24	1	1	2	2	8	16	24	3	26	3
Parity bits	**LAP**	**EIR**	**Reserved**	**SR**	**SP**	**UAP**	**NAP**	**Class of device**	**LT_ADDR**	**CLK 2 7 -2**	**Page Scan Mode**

FHS packet for Inquiry:

- **Parity bits:** This 34-bit field contains the parity bits that form the first part of the sync word of the access code of the device that sends the FHS packet.

- **LAP:** (Lower Address Part) This 24-bit field shall contain the lower address part of the device that sends the FHS packet (here, aspiring slave).

- **EIR:** (Extended Inquiry Response) this bit shall indicate that an extended inquiry response packet may follow.

- **Undefined:** This 1-bit field is reserved for future use and shall be set to zero.

- **SR:**

SR bit format b1b0	SR (page scan repetition) mode
00	R0
01	R1
10	R2

11	Reserved for future use

- **Reserved:** This 2-bit field shall be set to 10.

- **UAP:** (Upper Address Part) This 8-bit field shall contain the upper address part of the device that sends the FHS packet (here, aspiring slave).

- **NAP:** (Non-Significant Address Part) This 16-bit field shall contain the non-significant address part of the device that sends the FHS packet (here, aspiring slave).
- **Class of device:** This 24-bit field shall contain the class of device of the device that sends the FHS packet (here, aspiring slave).

- **LT_ADDR:** Zero (Since it's an inquiry packet)

- **CLK 27-2:** This 26-bit field shall contain the value of the native clock of the device that sends the FHS packet (here, aspiring slave). The value contained is the sampled value at the beginning of the transmission of the access code of this FHS packet.

- **Page scan mode:**

Bit format b2b1b0	Page scan mode
000	Mandatory scan mode
All other values	Reserved for future use

FHS packet for Paging:

- **Parity bits :** This 34-bit field contains the parity bits that form the first part of the sync word of the access code of the device that sends the FHS packet(here, aspiring master).

- **LAP:** This 24-bit field shall contain the lower address part of the device that sends the FHS packet (here, aspiring master).

- **EIR:** This bit shall indicate that an extended inquiry response packet may follow.

- **Undefined:** This 1-bit field is reserved for future use and shall be set to zero.

- **SR:** This 2-bit field is the scan repetition field and indicates the interval between two consecutive page scan windows.

- **Reserved:** This 2-bit field shall be set to 10.

- **UAP:** This 8-bit field shall contain the upper address part of the device that sends the FHS packet (here, aspiring master).

- **NAP:** This 16-bit field shall contain the non-significant address part of the device that sends the FHS packet.

- **Class of device:** This 24-bit field shall contain the class of device of the device that sends the FHS packet (here, aspiring master).

- **LT_ADDR:** This 3-bit field shall contain the logical transport address the recipient and shall use if the FHS packet is used at connection setup or role switch. A slave responding to a master or a device responding to an inquiry request message shall include an all-zero LT_ADDR field if it sends the FHS packet.

- **CLK 27-2:** This 26-bit field shall contain the value of the native clock of the device that sends the FHS packet (here, aspiring master). The value contained is the sampled value at the beginning of the transmission of the access code of this FHS packet.

- **Page scan mode:** This 3-bit field shall indicate which scan mode is used by default by the sender of the FHS packet.

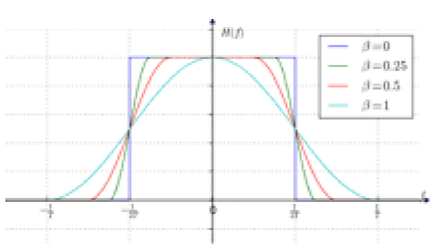
Aspect	Encryption	Decryption
Definition	Encryption is the process in which a sender converts the original information to another form and transmits the resulting unintelligible message over the network.	Decryption inverts the encryption process in order to convert the message back to its original form.
Procedure	The data is encrypted/coded automatically with the help of a secret key when data is being transferred between two different machines.	The data receiver lets you to automatically decipher the code in its actual form using the data that was sent.
Location	The sender while sending the data to the destination will convert it.	The receiver while receiving the data would convert it.
Algorithm	The algorithm used is the same for both encryption and decryption process. The keys used depend on the cryptosystem used, symmetric vs asymmetric key.	
Functionality	To transform easily understandable and human decipherable messages into a non – decipherable and obscure form that is almost incomprehensible to interpret	It is the transformation of an obscure message into a decipherable form which is understood by a human.

BIT STREAM PROCESSING

What is FEC in Bluetooth?

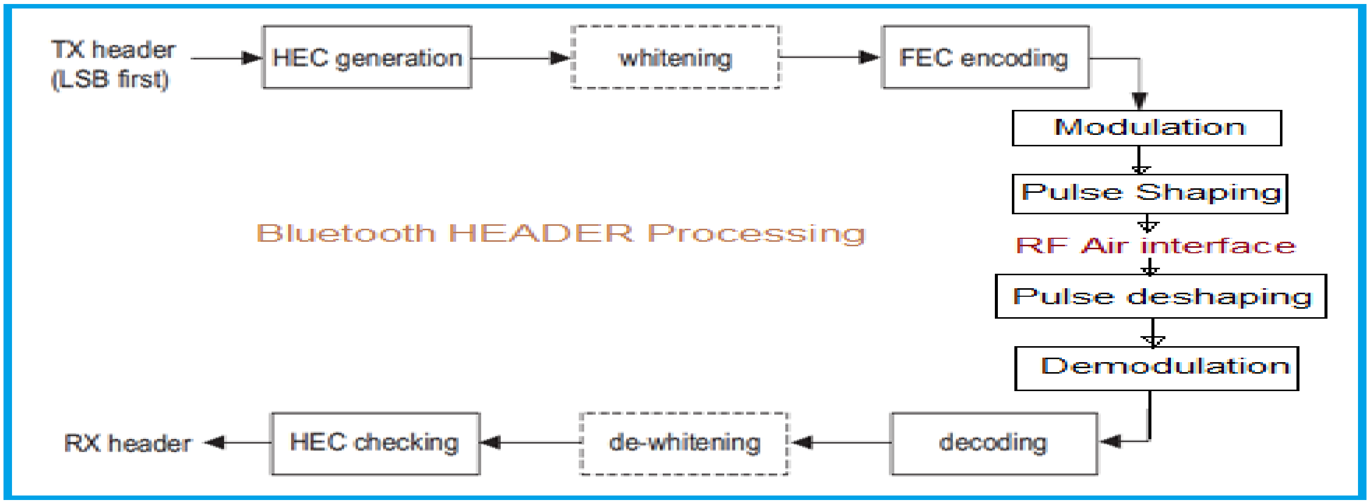
The LE Coded PHY uses **Forward Error Correction** (FEC), a particular approach to error correction.

What is meant by pulse shaping?



In electronics and telecommunications, pulse shaping is **the process of changing the waveform of transmitted pulses**. Its purpose is to make the transmitted signal better suited to its purpose or the communication channel, typically by limiting the effective bandwidth of the transmission.

Header bit processes



Following section describes header processing through Bluetooth Physical Layer.

➤(Transmitter Part :)

• FEC encoded data bits are modulated.

• The pulse shaping is applied before RF up conversion.

• The pulse shaped data is upconverted at 2.4 GHz frequency and transmitted over the air.

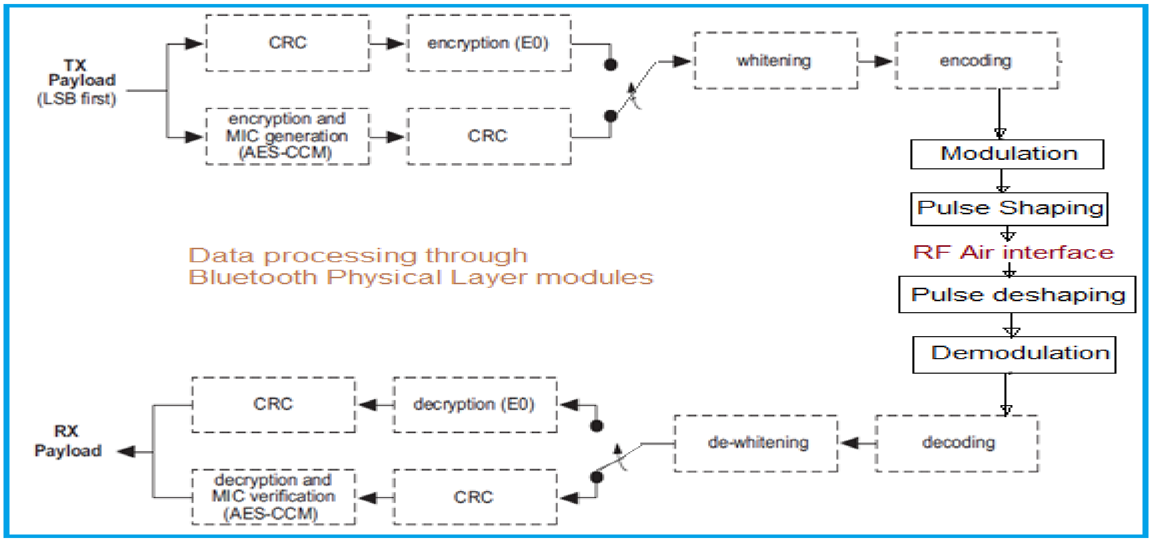
➤(Receiver Part :)

• The reverse operations are performed at the Bluetooth receiver part viz. RF down conversion, pulse de-shaping, demodulation, FEC decoding, de-whitening, HEC checking before the header is decoded.

- Before the payload is sent over the air interface, several bit manipulations are performed in the transmitter to increase reliability and security.
- An HEC is added to the packet header, the header bits are scrambled with a whitening word, and FEC coding is applied.
- In the receiver, the inverse processes are carried out.
- All header bit processes are mandatory except that whitening and de-whitening shall not be performed on synchronization train packets.

DATA PAYLOAD PROCESSING THROUGH BLUETOOTH PHYSICAL LAYER

Payload bit processes



- The payload processing uses all the modules as applied for the header processing.
- In addition, it uses encryption module either E0 or AES-CCM.
- When E0 encryption is applied, the entire payload should be encrypted.
- When AES-CCM encryption is applied, only payload body and MIC shall be encrypted, whereas **payload header and CRC shall not be encrypted.**
- Advanced Encryption Standard-Counter

Error Checking:

- **HEC generation and checking, CRC generation** is defined in the standard with their respective generator polynomials.
- The packet can be checked for errors or wrong delivery using the channel access code, the HEC in the header, and the CRC in the payload.
- At packet the access code is checked first.
- Since the 64-bit sync word in the channel access code is derived from the 24-bit master LAP, this check if the LAP is correct, and prevents the receiver from accepting a packet of another piconet.
- Before calculating the HEC or CRC, the shift registers in the HEC/CRC generators shall be initialized with the 8-bit UAP valve.
- Then the header and payload information shall be shifted into the HEC and CRC generators, respectively (with the LSB first).
- A packet has a valid header if the access code is correct for the piconet and the HEC has the correct value, irrespective of the values of the other header fields.

7.1.1 HEC generation

- Initially this circuit shall be pre-loaded with the 8-bit UAP such that the LSB of the UAP (denoted UAP0) goes to the left-most shift register element, and, UAP7 goes to the right-most element.
- The initial state of the HEC LFSR is depicted in Figure 7.4.
- Then the data shall be shifted in with the
- switch S set in position 1. When the last data bit has been clocked into the LFSR,
- the switch S shall be set in position 2, and, the HEC can be read out from the register.
- The LFSR bits shall be read out from right to left (i.e., the bit in position 7 is the first to be transmitted, followed by the bit in position 6, etc.).

Position:	0	1	2	3	4	5	6	7
LFSR:	UAP	UAP	UAP	UAP	UAP	UAP	UAP	UAP

Linear Feedback Shift Registers (**LFSRs**)

What is linear feedback shift register used for?

LFSRs are used in **circuit testing for test-pattern generation (for exhaustive testing, pseudo-random testing or pseudo-exhaustive testing) and for signature analysis**.

7.1.2 CRC generation

<u>What is meant by Cyclic Redundancy Check?</u>

An error detection technique using a polynomial to generate a series of two 8-bit block check characters that represent the entire block of data. These block check characters are incorporated into the transmission frame and then checked at the receiving end.

Why is CRC used?

A cyclic redundancy check (CRC) is an error-detecting code commonly used in digital networks and storage devices **to detect accidental changes to digital data**. Blocks of data entering these systems get a short check value attached, based on the remainder of a polynomial division of their contents.

Can CRC detect all errors?

The CRC error detection scheme **cannot detect all errors at all times**. In the case of an error pattern being the multiples of the polynomial constant, the error cannot be detected.

What is a CRC failure?

What is a CRC error? A CRC error **informs you that the cyclic redundancy check has detected damaged or incomplete files**. In that case, the matched value of the data and the attached CRC value do not match, resulting in an aborted action (save, copy, read access, etc.)

How reliable is CRC?

The good thing about CRC is that it is **very accurate**. If a single bit is incorrect, the CRC value will not match up.
Both checksum and CRC are good for preventing random errors in transmission but provide little protection from an intentional attack on your data.

- For this case, the 8 left-most bits shall be initially loaded with the 8-bit UAP (UAP0 to the left and UAP7 to the right) while the 8 right-most bits shall be reset to zero.

- The initial state of the 16-bit LFSR is specified in Figure 7.7.
- The switch S shall be set in position 1 while the data is shifted in. After the last bit has entered the LFSR,
- the switch shall be set in position 2, and, the register's contents shall be transmitted, from right to left

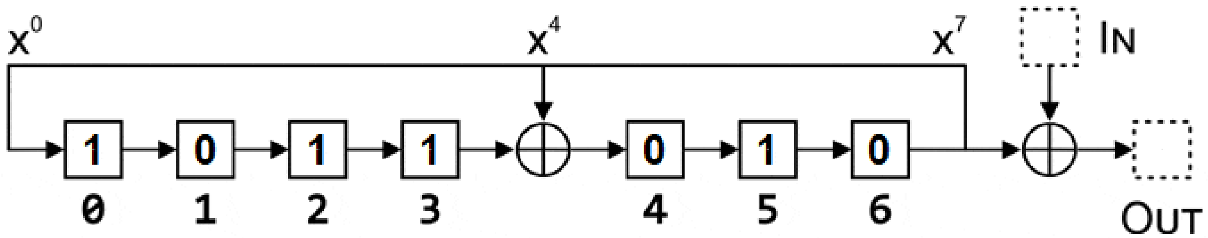
- The 24 bit (3 Octet) Cyclic Redundancy Check (CRC) is calculated with the bits from the PDU 2564. It is transmitted at the end of the packet.

Position:	0	1	2	3	4	5	6	7
LFSR:	UAP0	UAP1	UAP2	UAP3	UAP4	UAP5	UAP6	UAP7
	8	9	10	11	12	13	14	15
	0	00	0	0	0	00	0	0

Data Whitening:

- The data whitening shall not be applied to synchronization train packets.
- Both header and payload is scrambled with data whitening word in order to randomize the data from highly redundant patterns.
- This *helps in minimizing DC bias in the Bluetooth packet before being applied to FEC encoder module.*
- Whitening word is generated with polynomial g(D) = D7 + D4 + 1 which 221 in octal representation.

- Data whitening prevents long sequences of repetitive bits (00000000 or 11111111).
- It is applied to the PDU and CRC fields of all Link-Layer packets after the CRC of the transmitter.
- De-whitening is performed before the CRC in the receiver.
- Both the whitener and de-whitener use a 7-bit linear feedback shift register (LFSR) with taps at bit 4 and bit 7.
- The shift register is initialized with a sequence that is derived from the channel index.
- Position 0 is set to 1, and positions 1 to 6 are set to the channel index of the channel used when transmitting or receiving, with the MSB in position 1 and the LSB in position 62601.
- Bits are shifted along the shift register from 0→1, 1→2, 2→3, 4→5, 5→6, 6→0. Bit 3 and bit 6 are processed with the XOR ⊕ operator to determine bit 4 (0⊕0=0,0⊕1=1,1⊕0=1,1⊕1=0)



Error Correction:

- The three error correction schemes used in bluetooth are 1/3 rate FEC, 2/3 rate FEC and ARQ scheme.
- Using ARQ scheme different packet types are transmitted until ack of successful reception is returned by the destination (or timeout is exceeded)

7.6 ARQ SCHEME

- With an automatic repeat request scheme, DM packets, DH packets, the data field of DV packets, and EV packets shall be transmitted until acknowledgment of a successful reception is returned by the destination.
- The acknowledgment information shall be included in the header of the return packet.
- The ARQ scheme is only used on the payload in the packet and only on packets that have a CRC.
- The packet header and the synchronous data payload of HV and DV packets are not protected by the ARQ scheme.

Why we use FEC

Longer span with less amplifier, lower launch power, compensation of system loss, Increased system gain, High channel density, Better system performance monitoring, effective error free channel,

TABLE II. CODING METHODS IN WIRELESS COMMUNICATION APPLICATIONS

Wireless Communication Technology	**FEC Methods**
Third Generation Wireless Technology (3G)	Convolutional coding: Turbo coding
Fourth Generation Wireless Technology (4G)	Concatenated coding
Worldwide Interoperability for Microwave Access (WiMAX)	Convolutional coding: Turbo coding and Low-Density Parity Check (LDPC); Reed-Solomon coding
Bluetooth	1/3 FEC: repeating data sequence 3 times 2/3 FEC: (15,10) shortened Hamming code
Orthogonal Frequency-Division Multiplexing (OFDM)	Inner code: Convolutional coding Outer code: Concatenated with Reed-Solomon coding
Ultra-Wide Band (UWB)	Convolutional coding: LDPC with Reed-Solomon coding

ERROR CORRETION IN BLUETOOTH

- 1/3 Rate FEC

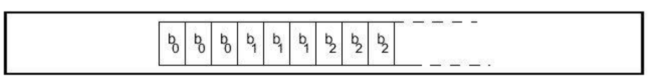
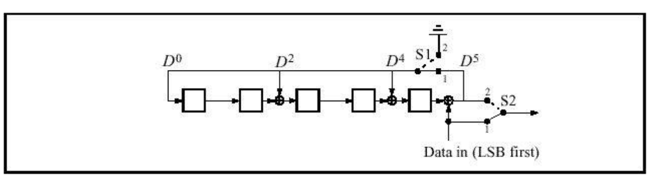


Figure 5.1: Bit-repetition encoding scheme.

- 2/3 Rate FEC



- CRC-ARQ Scheme

- **1/3 rate FEC (Forward Error Correcting)**

When using this coding scheme, each bit is simply sent three times, over three consecutive bits.

- **2/3 rate FEC (Forward Error Correcting)**

In computer science and telecommunication, Hamming codes are a family of linear error-correcting codes. Hamming codes **can detect one-bit and two-bit errors or correct one-bit errors without detection of uncorrected errors**.

What is the importance of hamming code?

Hamming code is an error correction system that **can detect and correct errors when data is stored or transmitted**. It requires adding additional parity bits with the data. It is commonly used in error correction code (ECC) RAM.

What is the limitation of Hamming code?

The biggest drawback of the hamming code method is that **it can solve only single bits issues**. We can perform the process of encrypting and decoding the message with the help of hamming code.

The packet payload (header, data, and CRC (Cyclic redundancy check)) must be padded out with trailing zeroes to make it a multiple of 10 bits in length.

Modulation:

- There are two main modulation modes viz. basic rate and enhanced data rate.
- Binary FM modulation is applied to basic rate where as PSK modulation is applied to enhanced data rate.
- There are two variants of PSK modulation viz. π/4-DQPSK, 8-DPSK.
- Binary FM modulation is also referred as GFSK (Gaussian Frequency Shift Keying).
- The symbol rate for all the three modulation types are same i.e. 1 MSym/Sec.
- The different gross data rates viz. 1 Mbps (for basic rate), 2 Mbps (for EDR using π/4-DQPSK) and 3 Mbps (for EDR using 8-DPSK) can be achieved.
- For transmission of access code and packet header, GFSK is used.
- For transmission of sync. sequence, payload and trailer sequence, PSK modulation with data rate of 2 Mbps or 3 Mbps is used.

Pulse Shaping:

- The square-root raised cosine pulse shaping is applied to generate the equivalent low pass information-bearing signal v(t).

RF frequency conversion:

- The pulse shaped data signal is up converted as per ISM band carrier frequency (2.4 GHz) out of the 40 assigned bluetooth frequency channels >> as per requirement (i.e. advertising channel or data channel).

ERRONEOUS SYNCHRONOUS DATA REPORTING

Erroneous data reporting may be enabled for synchronous links. When enabled, synchronous data shall be processed as follows:

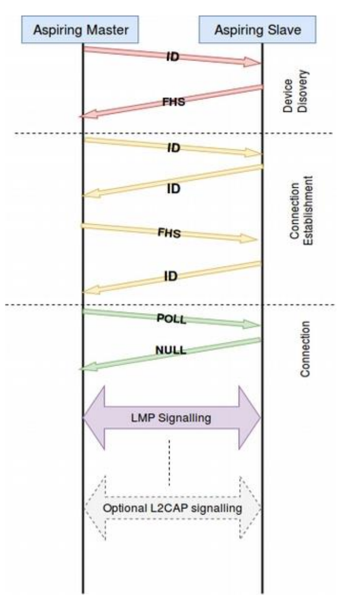
• If, during an (e)SCO interval, an eSCO packet was received with valid HEC, an allowed TYPE for the connection, and valid CRC, or a SCO packet was received with valid HEC, the received payload data shall be sent to the upper-edge of the Controller with a "good data" indication.

• If, during an eSCO interval, eSCO packets with valid HEC were received, but none of them had an allowed TYPE for the connection and a valid CRC, the best known data available (e.g., data derived from the received data or the actual payload data that was received with a CRC error) shall be sent to the upper-edge of the Controller with a "data with possible errors" indication.

• If, during an (e)SCO interval, no SCO or eSCO packet with valid HEC has been received, a "lost data" indication shall be sent to the upper edge of the Controller.

MESSAGE INTEGRITY CHECK

When encryption with AES-CCM is enabled, a Message Integrity Check (MIC) is added to the payload before the CRC for packets that are defined to include the MIC. The MIC is sent Most Significant Octet (MSO) first.



Procedure of Inquiry:

Baseband: Inquiry procedure

- To discover other units in range.
- ID packets containing GIAC are transmitted by inquiring device.
- ID packets sent on inquiry hopping sequence derived from GIAC.
- Inquirer sends 2 ID packets at different frequencies in even slots and waits for response(s) in the odd slots.
- 32 inquiry hop frequencies are split in two 16 hop parts (trains) A and B. Each train lasts 10msec (16 slots).
- A scanning device listens at one of 32 inquiry frequencies for 11.25 msec at least once every 2.56 sec.
- A/B trains of ID packets are repeated 256 times each.

Baseband Resource Manager enters into inquiry state when device manager asks to

- InquiryTO is turned ON.

  - Aspiring master will send the ID packets in two different wake-up frequencies {f (k), f (k+1)} for 312.5 µsecs each in TX slots (of 625 µsecs).
  - ID packet contains the **Generic Inquiry Access Code** (GIAC) 0x9E8B33.
  - Master device tunes back to the same frequency set {f (k), f (k+1)} and waits for the response in RX slot of 625 µsecs (312.5 µsecs in f1 and 312.5 µsec in f2).
  - Hence, aspiring master tunes to new frequency, every 312.5 µsecs but aspiring slave remains in one frequency for 1.28 sec (2048 slots). Inquiry is happened in 1.28 secs …

  - If an aspiring slave is tuned to the same frequency as that of the aspiring master, the aspiring slave will receive the ID packet.
  - In response to the ID packet, aspiring slave has to send FHS packet after 625 µsecs, but since ID packet was **not addressed to a particular slave** and is a broadcast, generates a random number 'b' from the closed interval [0, 1023] .

- This process is introduced to avoid collision of responses in the case when more than one device is listening on the same frequency.
- Once the timeout generated has expired, the device re-enters the inquiry scan mode and responds to the ID packet.
- The response packet is a so-called FHS packet containing the Bluetooth device address and clock values of the responding Bluetooth device.

- ID packets are transmitted twice in each TX slot in two hopping frequencies (one at the beginning and one at the middle of TX slot).

- Aspiring master takes 8 TX slots to transmit whole Train of 16 frequencies.

- Because TX and RX slots are interleaved a total length of time to cover each train is 16 slots (8 TX slots, 8 RX slots).

Total time taken by the Train A to cover 16 slots = 16 * 625 µsec = 10 msec.

The inquiry happens 256 times for each train. For error free response collection, at least four Trains must be used. Hence total time taken for Inquiry = 4 Trains * 256 times * 10 msec = 10.24 sec.

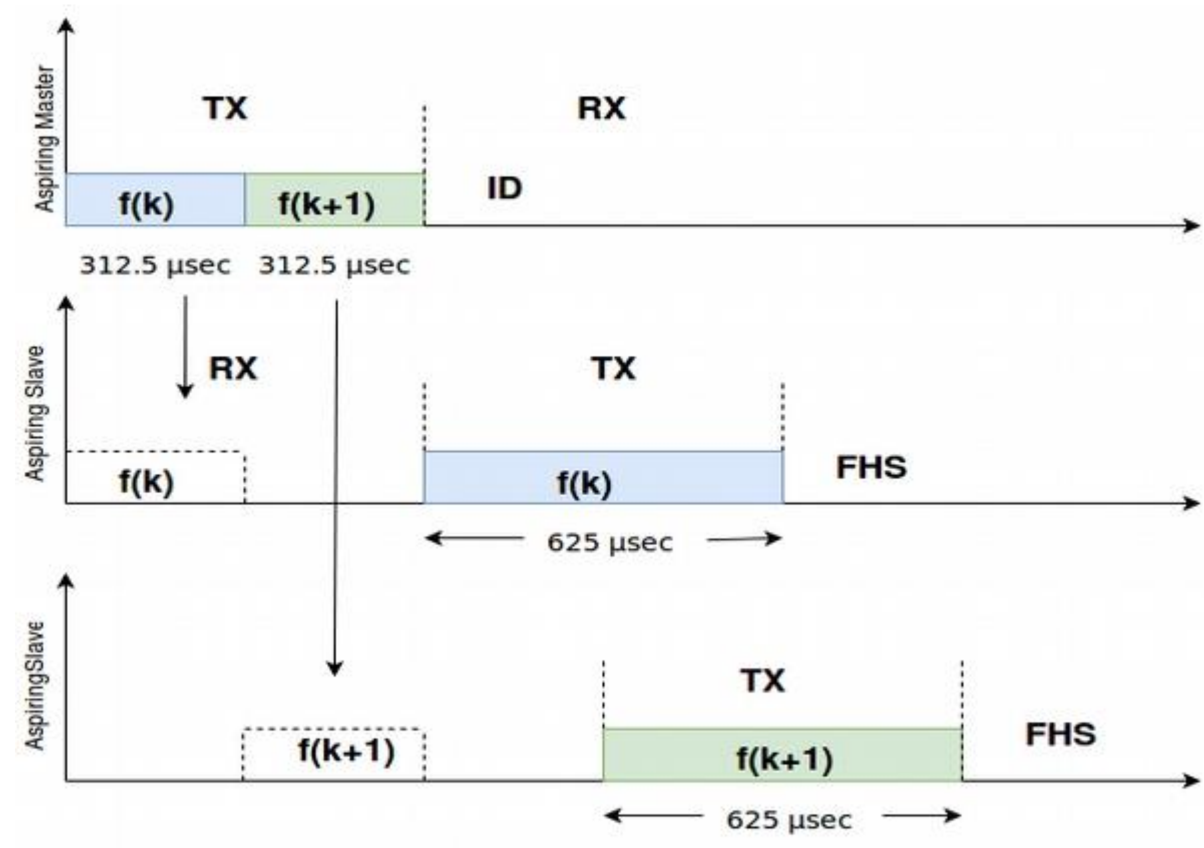
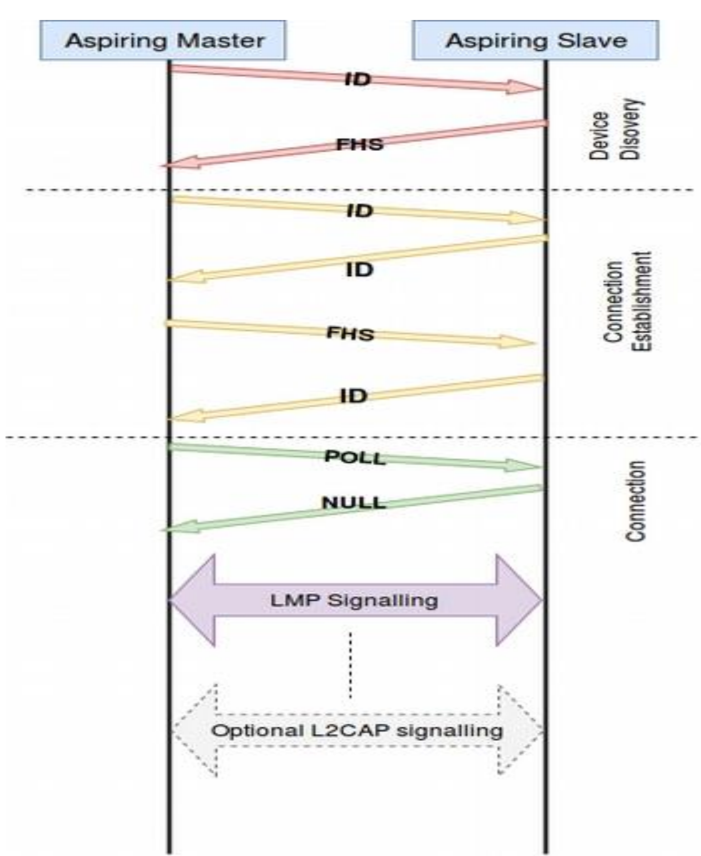
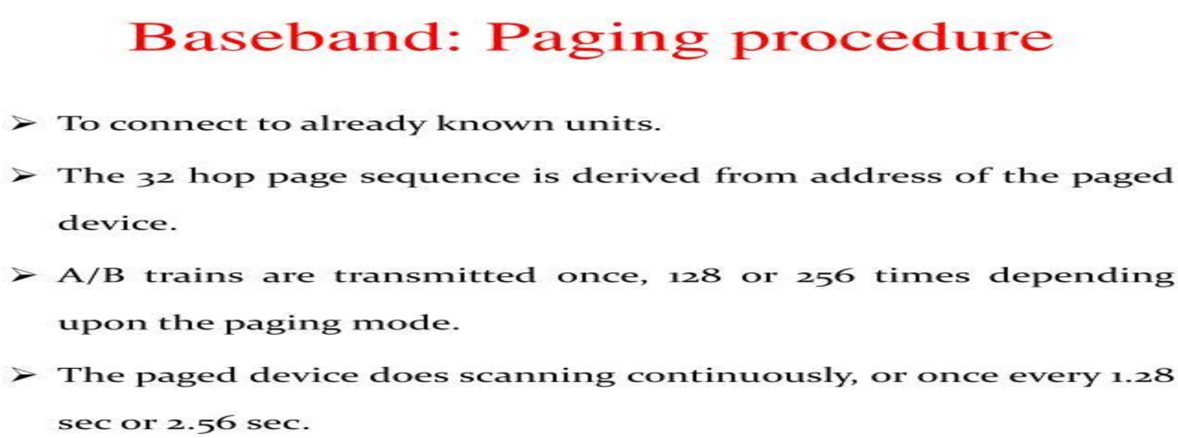


Fig: Packet transaction for inquiry procedure



5.3 Connection Establishment Procedure:



Connection Establishment Procedure:

5.4.1 Paging Device:

#c.1: Baseband Resource Manager enters into paging state when device manager asks toPageTO is turned ON.

- For faster inquiry, a subset of 79 frequency bands i.e. only 32 equi-spaced frequency bands (with a period of 32) are used and these are called as **wake-up frequencies**.

- These 32 wake-up frequencies are again divided into 2 trains of 16 frequencies each, **Train A** (16 frequency band subset) and **Train B** (16 frequency band subset)**.**

- Aspiring master sends ID packets in two different wake-up frequencies {f (k), f (k+1)} for 312.5 µsecs each in TX slots (of 625 µsecs).
- ID packet contains Device Access Code (**DAC**) of a particular slave.
- Master device tunes back to the same frequency set {f (k), f (k+1)} and waits for the response in RX slot of 625 µsecs (312.5 µsecs in f1 and 312.5 µsec in f2).

- Hence, aspiring master tunes to new frequency, every 312.5 µsecs but aspiring slave remains in one frequency for 1.28 sec (2048 slots). Paging is happened in 1.28 secs …

- If an aspiring slave is tuned to the same frequency as that of the aspiring master, the aspiring slave will receive the ID packet.

- The first packet transmitted by the master to the slave is the POLL packet, on default ACL logical transport for which it expects any packet type in response.

- Slave responds with NULL packet which confirms the establishment of connection between the master and the slave.

ID packet	Paging Device to Page Scanning Device (Aspiring Master to Aspiring Slave)	Contains Device Access Code (DAC) of the particular Slave
ID packet	Page Scanning device to Paging Device (Aspiring Slave to Aspiring Master)	Contains Device Access Code (DAC) of itself (as an ACK for ID packet).
FHS packet	Paging Device to Page Scanning Device (Aspiring Master to Aspiring Slave)	Contain BD_ADDR, Class of Device, and CLK27–2 of paging device (Master). Also contain LT_ADDR, which is taken as

		primary LT_ADDR by the slave for default ACL logical transport.
ID packet	Page Scanning device to Paging Device (Aspiring Slave to Aspiring Master)	Contains Device Access Code (DAC) of itself (as an ACK for FHS packet).

Paging

Page scan (slave)

- An unconnected Bluetooth device must periodically enter the **page scan** state; in this state, the device activates its receiver and listens for a master device that might be trying to page it.
- A device operates in one of three-page SR (Scan Repetition) modes:

R0: the device listens continuously for a master paging it.
R1: the device listens at least every 1.28 seconds (2048 slots).
R2: the device listens at least every 2.56 seconds (4096 slots).

Page (master)

The master repeats the 16-slot sequence for at least long enough for the paged device to enter the page scan state:

R0: at least once.
R1: at least 128 times (i.e. for at least 1.28 seconds).
R2: at least 256 times (i.e. for at least 2.56 seconds).

If the master doesn't receive a response, it will then try the other 16 channels.

Page sequence

- Slot n+0: The master transmits one ID packet in the first half slot, then a second ID packet (on a different frequency) in the second half slot.
- Slot n+1: The slave responds with an ID packet, in either the first half or second half of the slot.
- Slot n+2: The master transmits an FHS packet.
- Slot n+3: The slave responds with an ID packet, in the first half of the slot.
- Slot n+4: The master switches to the normal hopping scheme. The first packet it transmits is a POLL packet.

6.2.4.1 NULL Packet

The NULL packet has no payload and consists of the channel access code and

Packet header only. Its total (fixed) length is 126 bits. The NULL packet does not require acknowledgment.

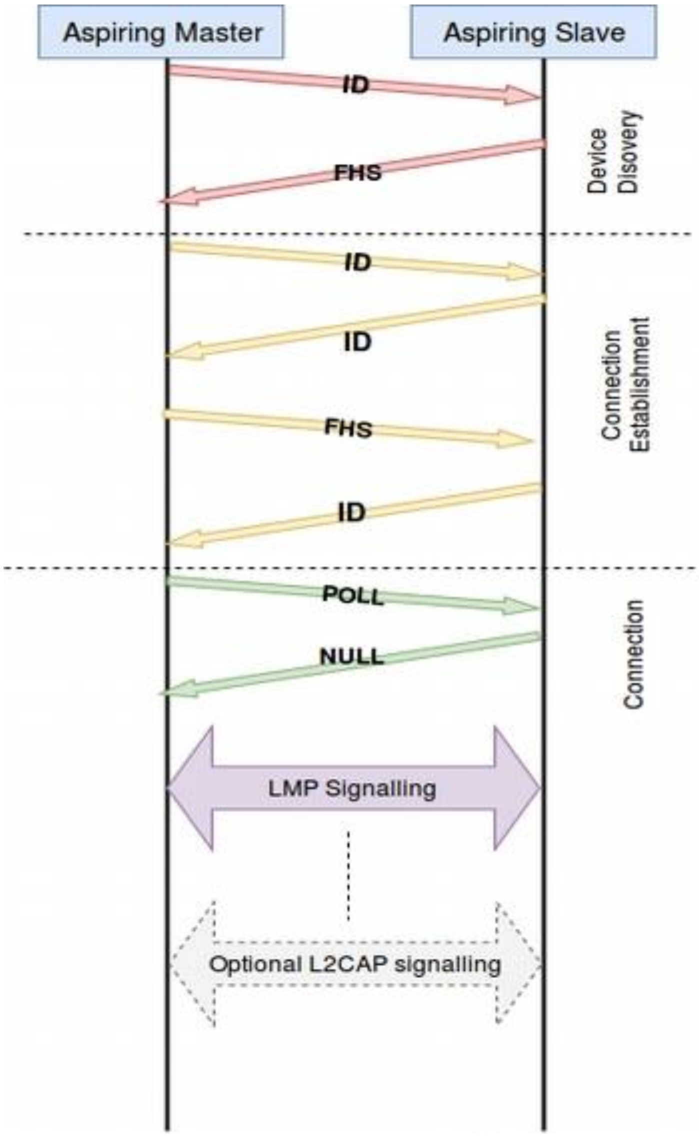
Channel Access Code	Packet Header

6.2.4.2 POLL Packet

POLL packet is similar to that of NULL packet. It does not have payload. POLL packets shall not be sent on Connectionless slave broadcast.

Channel Access Code	Packet Header

Packets used for Paging are:



BLUETOOTH LOW ENERGY PACKET

Bluetooth LE Uncoded Physical Layer (PHY)

The Bluetooth Core Specification [2] defines two physical layer (PHY) transmission modes (LE 1M and LE 2M) for uncoded PHY. This figure shows the packet structure for the Bluetooth LE uncoded PHY operating on LE 1M and LE 2M.

LSB MSB

Preamble	Access Address	Protocol Data Unit (PDU)	Cyclic Redundancy Check (CRC)	Constant Tone Extension
1 or 2 octets	4 octets	2-258 octets	3 octets	16 to 160 µs

- Each packet contains four mandatory fields (preamble, access-address, protocol data unit (PDU), and cyclic redundancy check (CRC)) and one optional field (constant tone extension (CTE)).
- The preamble is transmitted first, followed by the access address, PDU, CRC, and CTE (if present) in that order.
- The entire packet is transmitted at the same symbol rate of 1 Msym/s or 2 Msym/s.

Preamble.

- All link layer (LL) packets contain a preamble, which is used in the receiver to perform frequency synchronization, automatic gain control (AGC) training, and symbol timing estimation.
- The preamble is a fixed sequence of alternating 0 and 1 bits.
- For the Bluetooth LE packets transmitted on the LE 1M PHY and LE 2M PHY, the preamble size is 1 octet and 2 octets, respectively.

Access address.

- The access address is a 4-octet value.
- Each LL connection between any two devices and each periodic advertising train has a distinct access address.
- Each time the Bluetooth LE device needs a new access address, the LL generates a new random value adhering to these requirements:

❖ The value must not be the access address for any existing LL connection on this device.
❖ The value must not be the access address for any enabled periodic advertising train.
❖ The value must have no more than six successive 1s or 0s.
❖ The value must not be the access address for any advertising channel packets.
❖ The value must not be a sequence that differs from the access address of advertising physical channel packets by only 1 bit.
❖ All four octets for the value must not be equal.
❖ The value must have a minimum of two transitions in the most significant 6 bits.

- If the random value does not satisfy the above requirements, a new random value is generated until it meets all the requirements.

PDU.

- When a Bluetooth LE packet is transmitted on either the primary or secondary advertising physical channel or the periodic physical channel, the PDU is defined as the Advertising Physical Channel PDU.
- When a packet is transmitted on the data physical channel, the PDU is defined as the Data Physical Channel PDU.

CRC.

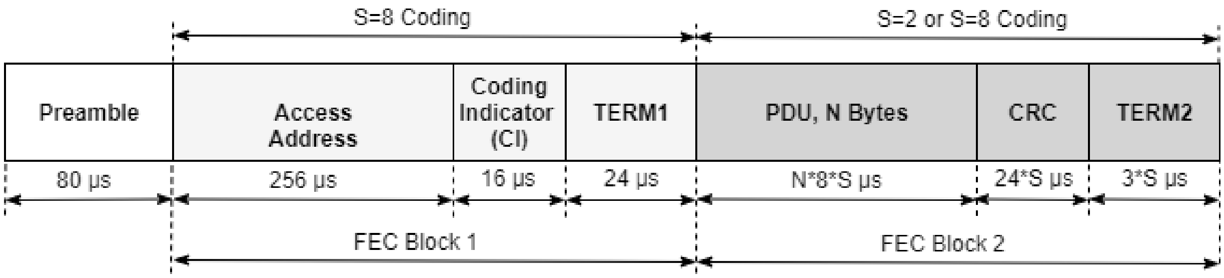
- The size of the CRC is 3 octets and is calculated on the PDU of all LL packets.
- If the PDU is encrypted, then the CRC is calculated after encryption of the PDU is complete. The CRC polynomial has the form $x^{24}+x^{10}+x^9+x^6+x^4+x^3+x+1$.

CTE.

- The CTE consists of a constantly modulated series of unwhitened 1s. This field has a variable length that ranges from 16 µs to 160 µs.

Bluetooth LE Coded PHY

This figure shows the packet structure for the Bluetooth LE coded PHY and is implemented for Bluetooth LE packets on all the physical channels.



Each Bluetooth LE packet consists of a preamble and these two forward error correcting (FEC) blocks:

- FEC block 1— This block contains three fields: access address, coding indicator (CI), and TERM1. This block implements an *S*=8 coding scheme, where each bit represents eight symbols. This gives a data rate of 125 Kbps.
- FEC block 2— This block contains these three fields: PDU, CRC, and TERM2. This block implements an *S*=8 or *S*=2 coding scheme. In the *S*=2 coding scheme, each bit represents two symbols. Therefore, the data rate is 500 Kbps.

The Bluetooth LE coded PHY does not contain the CTE.

Preamble.

- The Bluetooth LE coded PHY preamble is 80 symbols in length and contains 10 repetitions of the symbol pattern '00111100' (in the transmission order).

Access address.

- The length of Bluetooth LE coded PHY access address is 256 symbols.
- The value must have at least three 1s in the last significant bits.
- The value must have no more than 11 transitions in the least significant 16 bits.

The CI field consists of two bits as shown in this table:

Bits in CI	**Description**
00b	FEC block 2 coded using *S*=8
01b	FEC block 2 coded using *S*=2
All other values	Reserved for future use

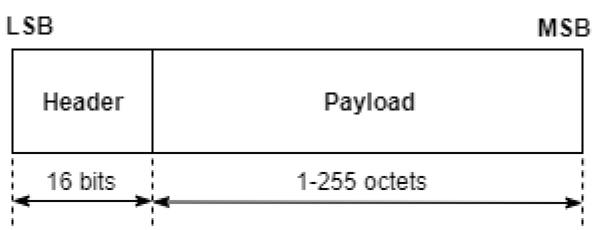
PDU. The PDU in the Bluetooth LE coded PHY packet structure has the same formatting as the PDU in the Bluetooth LE uncoded PHY packet.

CRC. The CRC in the Bluetooth LE coded PHY packet structure has the same formatting as the CRC in the Bluetooth LE uncoded PHY packet.

TERM1 and TERM2. Each FEC block contains a terminator at the end of the block. That terminator is referred to as TERM1 and TERM2. Each terminator is 3 bits long and forms the termination sequence during the FEC encoding process.

Advertising Physical Channel PDU

The packet structure format of the advertising physical channel PDU is shown in this figure.



- The advertising physical channel PDU has a 16-bit header and a variable-size payload.

The 16-bit header field of the advertising physical channel PDU is shown in this figure.

LSB MSB

PDU Type	RFU	ChSel	RFU	ChSel	Length
4 bits	1 bit	1 bit	1 bit	1 bit	8 bits

- The PDU type field in the advertising channel PDU header defines different types of PDUs that can be transmitted on the Bluetooth LE coded PHY.
- This table maps different types of PDUs with the physical channels and the PHYs on which the Bluetooth LE packet might appear.
- The table also indicates the PHY transmission modes supported for each type of advertising physical channel PDU.

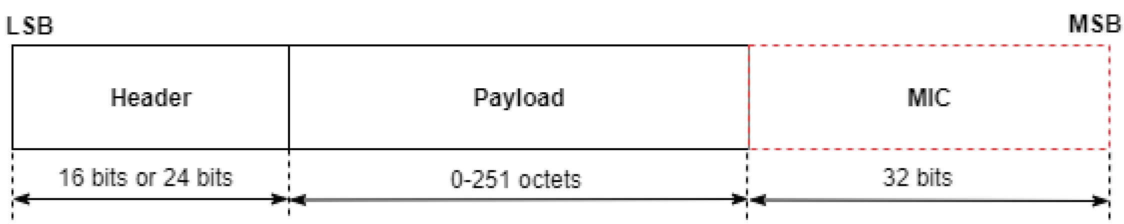
PDU Type	**PDU Name**	**Physical Channel**	**LE 1M Support**	**LE 2M Support**
0000b	ADV_IND	Primary Advertising	Yes	
0001b	ADV_DIRECT_IND	Primary Advertising	Yes	
0010b	ADV_NONCONN_IND	Primary Advertising	Yes	
0011b	SCAN_REQ	Primary Advertising	Yes	
	AUX_SCAN_REQ	Secondary Advertising	Yes	Yes
0100b	SCAN_RSP	Primary Advertising	Yes	
0101b	CONNECT_IND	Primary Advertising	Yes	
	AUX_CONNECT_REQ	Secondary Advertising	Yes	Yes
0110b	ADV_SCAN_IND	Primary Advertising	Yes	
0111b	ADV_EXT_IND	Primary Advertising	Yes	
	AUX_ADV_IND	Secondary Advertising	Yes	Yes
	AUX_SCAN_RSP	Secondary Advertising	Yes	Yes
	AUX_SYNC_IND	Periodic	Yes	Yes
	AUX_CHAIN_IND	Secondary Advertising and Periodic	Yes	Yes
1000b	AUX_CONNECT_RSP	Secondary Advertising	Yes	Yes
All other values	Reserved for future use			

- The RFU field is reserved for future use.

- The ChSel, TxAdd, and RxAdd fields of the advertising physical channel PDU header contain information specific to the type of PDU defined for each advertising physical channel PDU separately.
- If the ChSel, TxAdd, or RxAdd fields are not defined as used in a given PDU, then they are considered as reserved for future use.
- The Length field of the advertising physical channel PDU header denotes the length of the payload in octets.
- The valid range of the Length field is 1 to 255 octets.
- The Payload field in the advertising physical channel PDU packet structure is specific to the type of PDUs listed in the preceding table.

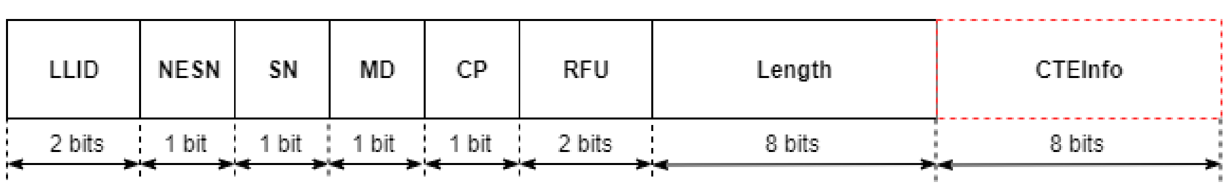
Data Physical Channel PDU

The packet structure format of the data physical channel PDU is shown in this figure.



- The data physical channel PDU has a 16-bit or 24-bit header, a variable length payload in the range [0, 251] octets, and can include a 32-bit message integrity check (MIC) field.
- The MIC is not included in an unencrypted LL connection or in an encrypted LL connection with a data channel PDU containing an empty payload.
- The MIC is included in an encrypted LL connection with a data channel PDU containing a nonzero length payload.
- In this case, the MIC is calculated as specified in Volume 6, Part E, Section 1 of the Bluetooth Core Specification [2].

The header field of the data physical channel PDU is shown in this figure.



The data physical channel PDU header includes these fields:

1. Link layer identifier (LLID) — This field indicates whether the packet is an LL data PDU or LL control PDU.
   - 00b — Reserved for future use
   - 01b — LL Data PDU, which can be a continuation fragment of an logical link control and adaptation (L2CAP) message or an empty PDU
   - 10b — LL Data PDU, which can be a start of an L2CAP message or a complete L2CAP message with no fragmentation
   - 11b — LL control PDU
2. Next expected sequence number (NESN):
   - The LL uses this field to either acknowledge the last data physical channel PDU sent by the peer or to request the peer to resend the last data physical channel PDU.
3. Sequence number (SN):
   - The LL uses this field to identify the Bluetooth LE packets sent by it.
4. More data (MD):
   - This field indicates that the Bluetooth LE device has more data to send.
   - If neither of Central and Peripheral Bluetooth LE device has set the MD bit in their packets, the packet from the Peripheral closes the connection event.
   - If the Central and Peripheral devices have set the MD bit, the Central can continue the connection event by sending another packet, and the Peripheral must listen after sending its packet.
5. CTEInfo present (CP):
   - This field indicates whether the data physical channel PDU header has a CTEInfo field and, subsequently whether the data physical channel packet has a CTE.
6. Length:
   - This field indicates the size, in octets, of the payload and MIC, if present. The size of this field is in the range [0, 255] octets.
7. CTEInfo: This field indicates the type and length of the CTE.

The two types of data physical channel PDUs are: LL Data PDU and LL Control PDU.

LL Data PDU.

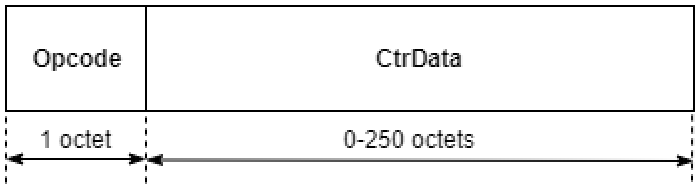
- The LL uses the LL data PDU to send L2CAP data. The LLID field in the LL data channel PDU header is set to either 01b or 10b. An LL data PDU is referred to as an empty PDU if

- The LLID field of the LL data channel PDU header is set to 01b.
- The Length field of the LL data channel PDU header is set to 00000000b.

An LL data PDU with the LLID field in the header set to 10b does not have the Length field set to 00000000b.

LL Control PDU.

- The LL uses the LL data PDU to control the LL connection. If the LLID field of data physical channel PDU header is set to 11b, the data physical channel PDU contains an LL control PDU. This figure shows the LL control PDU payload.



The Opcode field defines different types of LL control PDUs as shown in this table.

Opcode	**LL Control PDU**
0x00	LL_CONNECTION_UPDATE_IND
0x01	LL_CHANNEL_MAP_IND
0x02	LL_TERMINATE_IND
0x03	LL_ENC_REQ
0x04	LL_ENC_RSP
0x05	LL_START_ENC_REQ
0x06	LL_START_ENC_RSP
0x07	LL_UNKNOWN_RSP
0x08	LL_FEATURE_REQ
0x09	LL_FEATURE_RSP
0x0A	LL_PAUSE_ENC_REQ
0x0B	LL_PAUSE_ENC_RSP
0x0C	LL_VERSION_IND
0x0D	LL_REJECT_IND
0x0E	LL_SLAVE_FEATURE_REQ
0x0F	LL_CONNECTION_PARAM_REQ
0x10	LL_CONNECTION_PARAM_RSP
0x11	LL_REJECT_EXT_IND
0x12	LL_PING_REQ
0x13	LL_PING_RSP
0x14	LL_LENGTH_REQ
0x15	LL_LENGTH_RSP
0x16	LL_PHY_REQ
0x17	LL_PHY_RSP
0x18	LL_PHY_UPDATE_IND
0x19	LL_MIN_USED_CHANNELS_IND
0x1A	LL_CTE_REQ
0x1B	LL_CTE_RSP
0x1C	LL_PERIODIC_SYNC_IND
0x1D	LL_CLOCK_ACCURACY_REQ
0x1E	LL_CLOCK_ACCURACY_RSP

All other values	Reserved for future use

LINK MANAGER PROTOCOL

Basics of Link manager transactions and understanding commands for
1. Features request and response
2. Data rate and packets selection
3. Power management
4. Low power modes switching
5. Creation and deletion of logical links

- The Link Manager Protocol (LMP) is used to control and negotiate all aspects of the operation of the Bluetooth connection between two devices.

- This includes the set-up and control of logical transports and logical links, and for control of physical links.

- All LMP messages shall apply solely to the physical link and associated logical links and logical transports between the sending and receiving devices.

- The protocol is made up of a series of messages which shall be transferred over the ACL-C or ASB-C logical link between two devices.

- LMP messages shall be interpreted and acted-upon by the LM and shall not be directly propagated to higher protocol layers.

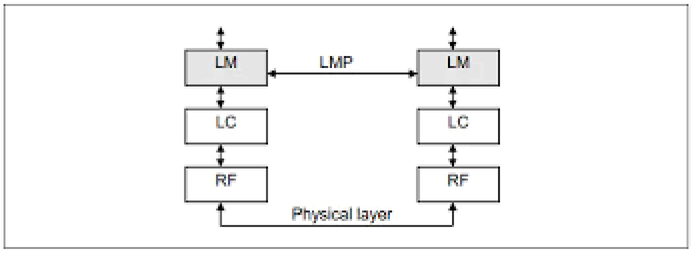


Figure 1.1: Link Manager Protocol signaling layer

- The Link Manager carries out link setup, authentication, link configuration and other protocols.

Link Manager Protocol (LMP)

- Used for link set-up, security and control.
- All LMP messages are single slot packets.
- Priority higher than user data (L2CAP).
- Payload body for LM PDUs:

LSB	MSB
OpCode and transaction Id	Content

BLUETOOTH – LMP functions

- **Authentication :** Challenge-Response scheme
  one device is called verifier.
  The other is called claimant.
- **Encryption** :
- **Clock offset request**: Synchronization of clock
- **Timing accuracy information request :**
- To ensure synchronization, the master can request the slaves for timing accuracy information.
- **LMP version**
- **Type of packets supported**

BLUETOOTH – LMP functions

- **Switching Master/Slave role**

  In a piconet, a device will act a s a Master and other devices will act as Slaves.

  The Master and the Slave in a piconet can switch roles using the LMP messages.
- **Name request**

  User-friendly name having a maximum of 248 bits in ASCII format.
- **Detach**
- **Hold mode**
- To place ACL link in hold for a specified time when there is no data to send. This feature is mainly to save power.

BLUETOOTH – LMP functions

- **Park mode:** To be in synchronization with the Master but not participate in data exchange
- **Power control:** To request for transmitting less power. This is useful particularly for class I devices which are capable of transmitting 100 m W power.
- **Quality of Service (QoS) parameters exchange:** In applications that require good quality transmission link, quality of service para meters can be specified. These parameters include number of repetitions for broadcast packets, delay and bandwidth allocation.
- **Request SCO link:** To request for an SCO link after the ACL link is established**.**
- **Multi-slot packet control:** To control the procedure when data is sent in consecutive packets.

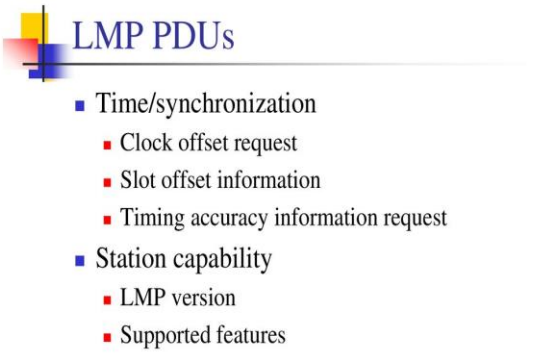
BLUETOOTH – LMP functions

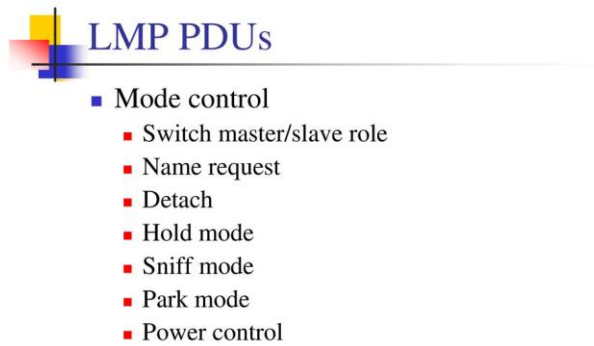
- **Link Supervision:** To monitor link when device goes out of range (through a time-out mechanism).
- **Connection establishment:**

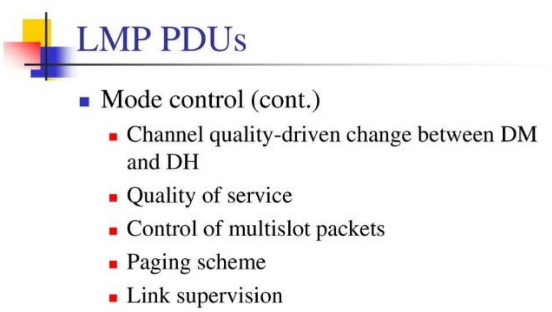
  After paging is successfully completed, to establish the connection.

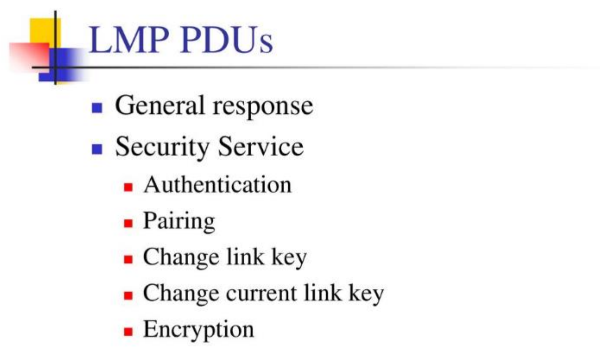
LMP: Other procedures

- LMP exchanges are also used for:
  - Authentication, pairing, encryption.
  - Exchanging clock/slot offset information.
  - Switching of master/slave roles.
  - Changing power modes.
  - QoS negotiation.









The following is a brief list of the types of PDU's available and their function/operation. These PDU are either Mandatory : M (must be supported), or Optional : O (optionally supported)

Table Of Contents

3.1 LMP PDUs

In telecommunications, a **protocol data unit** (**PDU**) is a single unit of information transmitted among peer entities of a computer network.

PDU. When a Bluetooth LE packet is transmitted on either the primary or secondary advertising physical channel, the PDU is defined as the Advertising Physical Channel PDU. When a packet is transmitted on the data physical channel, the PDU is defined as the Data Physical Channel PDU.

3.1.1 General Response

- (M) LMP_accepted
- LMP_not_accepted

These PDU's are used as response messages to other PDU's in a number of different procedures, containing the opcode of the message that is being responded to.

3.1.2 Authentication

How is the authentication performed in Bluetooth network?

In Bluetooth, authentication is achieved **through a challenge-response scheme** whose purpose is to verify that the device requesting access has knowledge of the secret link key. The requesting device first sends its unique device address to the verifying device.

- (M) LMP_au_rand
- LMP_sres

3.1.3 Pairing

- (M) LMP_in_rand
- LMP_au_rand
- LMP_sres
- LMP_comb_key
- LMP_unit_key

3.1.4 Change Link Key

What is link key in Bluetooth?

The link key is **a 128-bit random number and is a "shared secret" between devices**; it is never transmitted over the air. It is created during the pairing process, is stored and kept secret, and is used at every connection for authentication and derivation of the encryption key.

- (M) LMP_comb_key

3.1.5 Change the Current Link Key

- (M) LMP_temp_rand
- , LMP_temp_key ,

3.1.6 Encryption

Data encryption is used to prevent passive and active—man-in-the-middle (MITM) —eavesdropping attacks on a Bluetooth low energy link. Encryption is the means to make the data unintelligible to all but the Bluetooth master and slave devices forming a link.

- (O) LMP_encryption_mode_req ,
- LMP_encryption_key_size_req ,
- LMP_start_encryption_req ,
- LMP_stop_encryption_req

3.1.7 Clock Offset Request

What is clock offset in Bluetooth?

When interacting with the master it adds it to its clock, obtaining the clock of the master. For this reason, the difference is called clock offset. **This method produces a sort of virtual clock that is synchronized with that of the master**.

- (M) LMP_clkoffset_req,
- LMP_clkoffset_res

3.1.8 Slot Offset Request

- (O) LMP_slot_offset

3.1.9 Timing Accuracy Information Request

- (O) LMP_timing_accuracy_req
- , LMP_timing_accuracy_res

3.1.10 LMP Version

- (M) LMP_version_req ,
- LMP_version_res

3.1.11 Supported Features

- (M) LMP_features_req ,
- LMP_features_res

3.1.12 Switch of Master-Slave Role

- (O) LMP_switch_req ,
- LMP_slot_offset

Sin

3.1.13 Name Request

- (M) LMP_name_req ,
- LMP_name_res

3.1.14 Detach

The connection between two Bluetooth devices can be closed anytime by the master or the slave. A reason parameter is included in the message to inform the other party of why the connection is closed.

(M) LMP_detach

1. **Connection** -- After a device has completed the paging process, it enters the connection state. While connected, a device can either be actively participating or it can be put into a low power sleep mode.
   - **Active Mode** -- This is the regular connected mode, where the device is actively transmitting or receiving data.
   - **Sniff Mode** -- This is a power-saving mode, where the device is less active. It'll sleep and only listen for transmissions at a set interval (e.g. every 100ms).

- **Hold Mode** -- Hold mode is a temporary, power-saving mode where a device sleeps for a defined period and then returns back to active mode when that interval has passed. The master can command a slave device to hold.
- **Park Mode** -- Park is the deepest of sleep modes. A master can command a slave to "park", and that slave will become inactive until the master tells it to wake back up.

3.1.15 Hold Mode

- (O) LMP_hold ,
- LMP_hold_req

3.1.16 Sniff Mode

- (O) LMP_sniff_req ,
- LMP_unsniff_req

3.1.17 Park Mode

- (O) LMP_park_req ,
- LMP_unpark_PM_ADDR_req ,
- LMP_unpark_BD_clock
- ADDR_req ,
- LMP_set_broadcast_scan_window ,

3.1.18 Power Control

- (O) LMP_incr_power_req
- , LMP_decr_power_req ,
- LMP_max_power ,
- LMP_min_power

3.1.19 Channel Quality-Driven Change (between DM and DH)

- (O) LMP_auto_rate ,
- LMP_preferred_rate

3.1.20 Quality of Service

- (M) LMP_quality_of_service ,
- LMP_quality_of_service_req

3.1.21 SCO Links

- (O) LMP_SCO_link_req ,
- LMP_remove_SCO_link_req

3.1.22 Control of Multi-Slot Packets

- (M) LMP_max_slot ,
- LMP_max_slot_req

3.1.23 Paging Scheme

- (O) LMP_page_mode_req ,
- LMP_page_scan_mode_req

3.1.24 Link Supervision

- (M) LMP_supervision_timeout

3.1.25 Connection Establishment

- (M) LMP_host_connection_req ,
- LMP_setup_complete

.

3.1.26 Test Mode

- (M) LMP_test_activate ,
- LMP_test_control

3.1.27 Error Handling

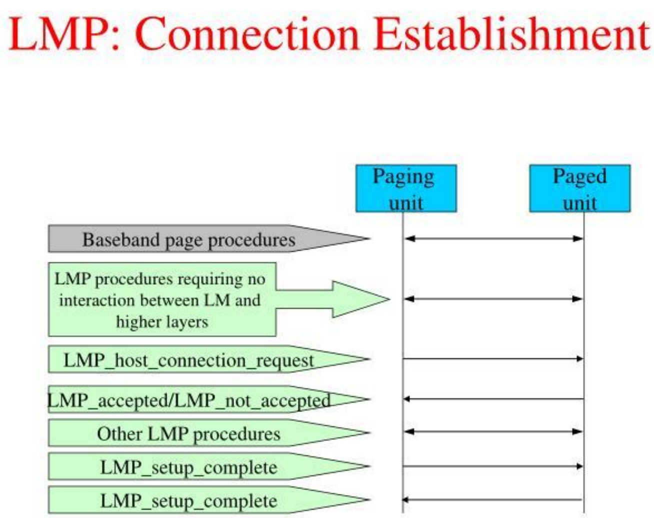
- (M) LMP_not_accepted

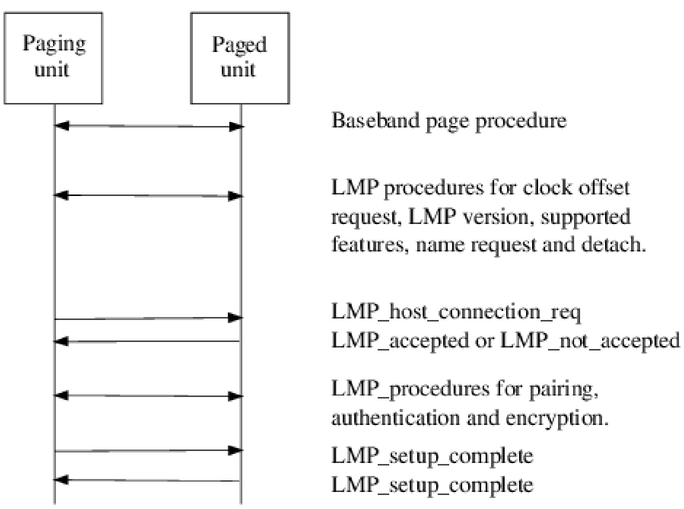
4 PROCEDURE RULES

1.1 CONNECTION CONTROL

4.1.1 Connection establishment

After the paging procedure, LMP procedures for clock offset request, LMP version, supported features, name request and detach may then be initiated





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2.3 PACKET FORMAT

operation code
What is meant by opcode?

The opcode is **the instruction that is executed by the CPU** and the operand is the data or memory location used to execute that instruction.

What is an escape OpCode?

An "escape" code in general is **one that modifies the meaning of the next byte / symbol, instead of meaning something on its own**.

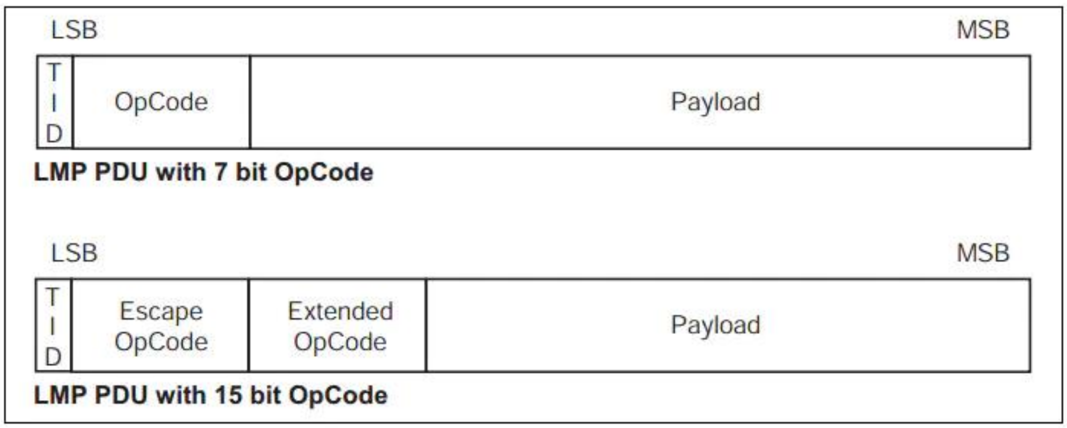


Figure 2.2: Payload body when LMP PDUs are sent

- Each PDU is assigned either a 7 or a 15-bit opcode used to uniquely identify different types of PDUs.
- The first 7 bits of the opcode and a transaction ID are located in the first byte of the payload body.

- The FLOW bit in the packet header is always 1 and shall be ignored on reception.

- LMP messages shall be transmitted using DM1 packets, however if an HV1 SCO link is in use and the length of the payload is no greater than 9 bytes then DV packets may be used.

2.4 TRANSACTIONS

- The LMP operates in terms of transactions.
- A transaction is a connected set of message exchanges which achieve a particular purpose.
- PDU sent by the master will have transaction ID = 0.

- PDU sent by the salves will have transaction ID = 1.

- For connection establishment,
- LMP_HOST_CONNECTION_REQ and the response with LMP_ACCEPTED or LMP_NOT_ACCEPTED shall form one transaction and have the transaction ID of 0.
- LMP_SETUP_COMPLETE is a stand-alone PDU, which forms a transaction by itself.

HCI (HOST CONTROLLER INTERFACE)

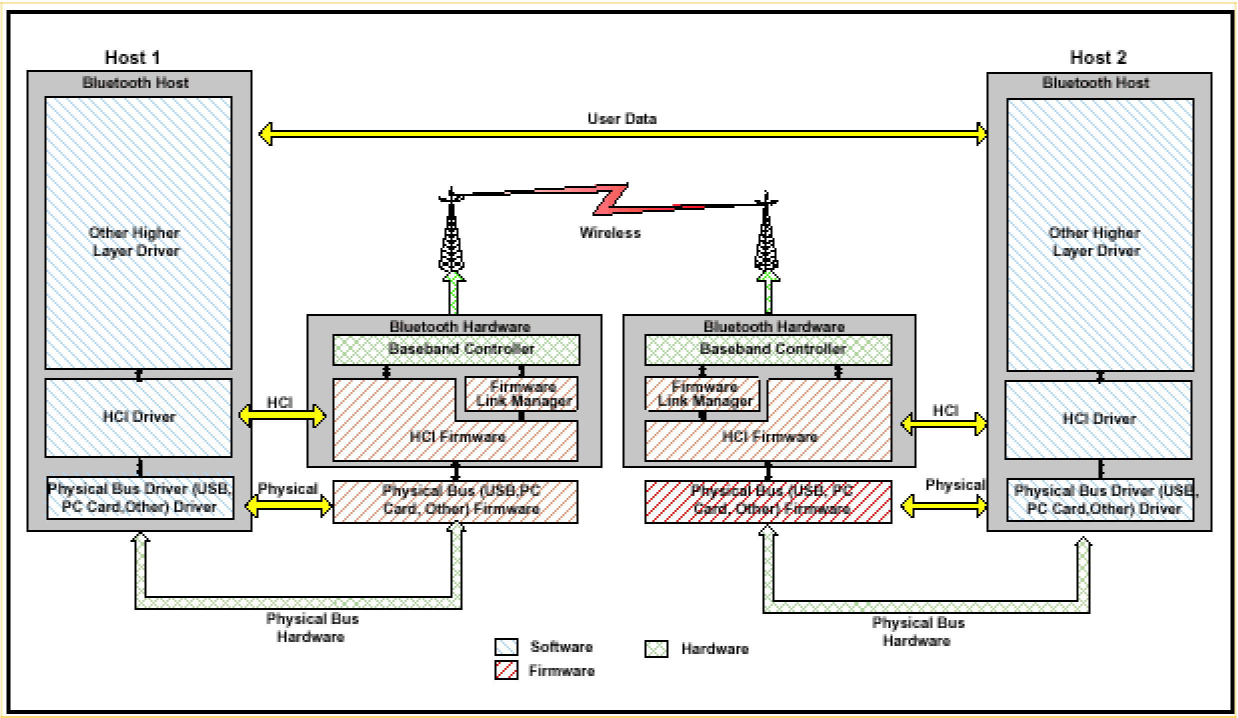
- The HCI provides a command interface to the baseband controller and link manager, and access to hardware status and control registers.
- Essentially this interface provides a uniform method of accessing the Bluetooth baseband capabilities.
- The HCI exists across 3 sections, the Host - Transport Layer - Host Controller.
- Each of the sections has a different role to play in the HCI system.

HCI Interface

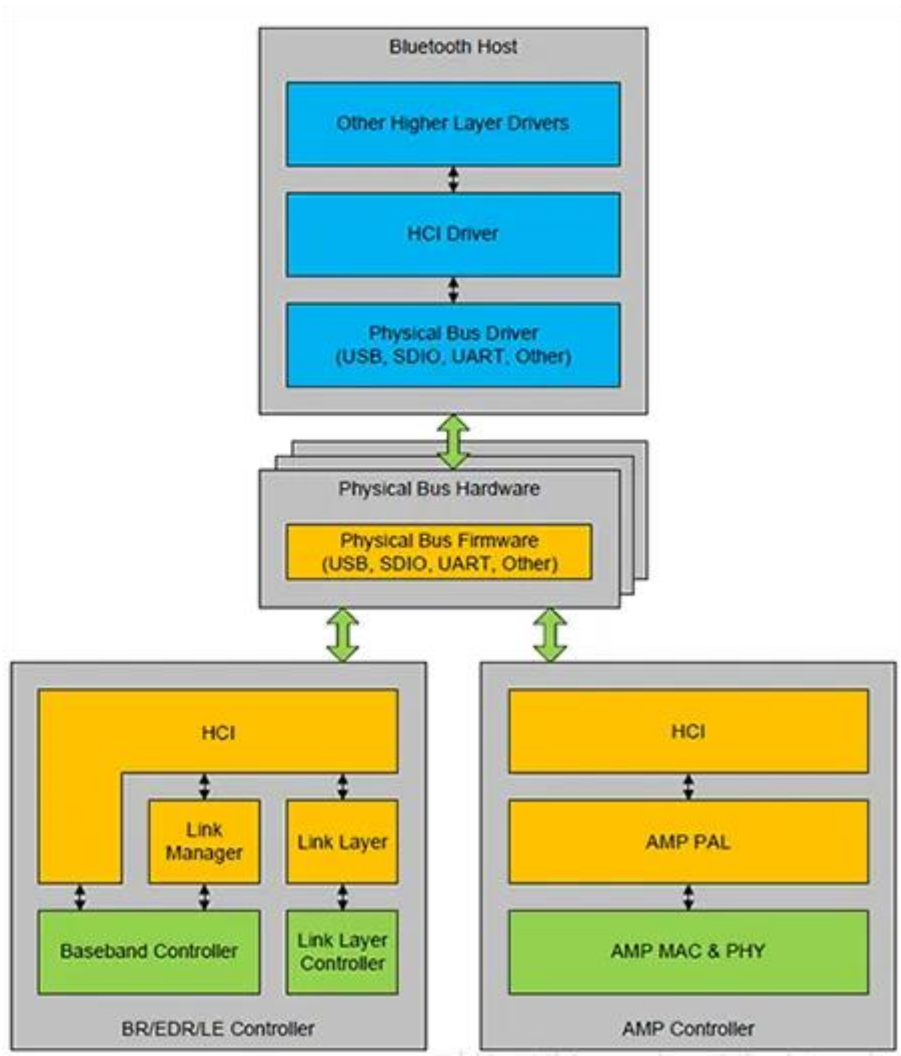
- The communication between a Host (a computer or an MCU) and a Host Controller (the actual Bluetooth chipset) follows the Host Controller Interface (HCI).

- HCI defines how commands, events, asynchronous and synchronous data packets are exchanged.

- Asynchronous packets (ACL) are used for data transfer, while synchronous packets (SCO) are used for Voice with the Headset and the Hands-Free Profiles.



Host 1
Bluetooth Host
Other Higher Layer Driver
HCI Driver
Physical Bus Driver (USB, PC Card,Other) Driver
User Data
Wireless
Bluetooth Hardware
Baseband Controller
Firmware Link Manager
HCI Firmware
Physical Bus (USB,PC Card, Other) Firmware
HCI
Physical
Physical Bus Hardware
Host 2
Bluetooth Host
Other Higher Layer Driver
HCI Driver
Physical Bus Driver (USB, PC Card,Other) Driver
Bluetooth Hardware
Baseband Controller
Firmware Link Manager
HCI Firmware
Physical Bus (USB, PC Card, Other) Firmware
HCI
Physical
Physical Bus Hardware
Software
Firmware
Hardware



HCI Commands

5.4.1 HCI Command packet

- The HCI Command packet is used to send commands to the Controller from the Host.
- Controllers shall be able to accept HCI Command packets with up to 255 bytes of data excluding the HCI Command packet header.
- The HCI Command packet header is the first 3 octets of the packet.
- Each command is assigned a 2 byte Opcode used to uniquely identify different types of commands.
- The Opcode parameter is divided into two fields, called the Opcode Group Field (OGF)

and Opcode Command Field (OCF).

- The OGF occupies the upper 6 bits of the Opcode, while the OCF occupies the remaining 10 bits. Any opcode not mentioned in this Part is reserved for future use.

OGF = opcode group field
OCF = opcode command field

All need to add the hci commands ; like = INQURIY = HCI_Inquiry

Group Name	OGF
Link Control Commands	0x01
Link Policy Commands	0x02
Controller & Baseband Commands	0x03
Informational Parameters	0x04
Status Parameters	0x05
Testing Commands	0x06
LE Controller Commands	0x08
Vendor Specific Commands	0x3F

4.2 HCI Commands

- The HCI provides a uniform command method of accessing the Bluetooth hardware capabilities.
- The HCI Link commands provide the Host with the ability to control the link layer connections to other Bluetooth devices.
- These commands typically involve the Link Manager (LM) to exchange LMP commands with remote Bluetooth devices.
- The HCI Policy commands are used to affect the behavior of the local and remote LM.
- These Policy commands provide the Host with methods of influencing how the LM manages the piconet.
- The Host Controller and Baseband commands, Informational commands, and Status commands provide the Host access to various registers in the Host Controller.

4.2.1 HCI-Specific Information Exchange

- The Host Controller Transport Layer provides transparent exchange of HCI-specific information.
- These transporting mechanisms provide the ability for the Host to send HCI

commands, ACL data, and SCO data to the Host Controller.

- These transport mechanisms also provide the ability for the Host to receive HCI events, ACL data, and SCO data from the Host Controller.
- Since the Host Controller Transport Layer provides transparent exchange of HCI-specific information, the HCI specification specifies the format of the commands, events, and data exchange between the Host and the Host Controller.

4.2.2 Link Control Commands

- The Link Control commands allow the Host Controller to control connections to other Bluetooth devices.
- When the Link Control commands are used, the Link Manager (LM) controls how the Bluetooth piconets and scatternets are established and maintained.
- These commands instruct the LM to create and modify link layer connections with Bluetooth remote devices, perform Inquiries of other Bluetooth devices in range, and other LMP commands.

4.2.3 Link Policy Commands

- The Link Policy Commands provide methods for the Host to affect how the Link Manager manages the piconet.
- When Link Policy Commands are used, the LM still controls how Bluetooth piconets and scatternets are established and maintained, depending on adjustable policy parameters.
- These policy commands modify the Link Manager behaviour that can result in changes to the link layer connections with Bluetooth remote devices.

4.2.4 Host Controller & Baseband Commands

- The Host Controller & Baseband Commands provide access and control to various capabilities of the Bluetooth hardware.
- These parameters provide control of Bluetooth devices and of the capabilities of the Host Controller, Link Manager, and Baseband.
- The host device can use these commands to modify the behaviour of the local device.

4.2.5 Informational Parameters

- The Informational Parameters are fixed by the manufacturer of the Bluetooth hardware.
- These parameters provide information about the Bluetooth device and the capabilities of the Host Controller, Link Manager, and Baseband.
- The host device cannot modify any of these parameters.

4.2.6 Status Parameters

- The Host Controller modifies all status parameters.
- These parameters provide information about the current state of the Host Controller, Link Manager, and Baseband.
- The host device cannot modify any of these parameters other than to reset certain specific parameters.

4.2.7 Testing Commands

- The Testing commands are used to provide the ability to test various functionality's of the Bluetooth hardware.
- These commands provide the ability to arrange various conditions for testing.

Link Control Commands (OGF = 0x01) 7.1 LINK CONTROL COMMANDS			
No	**Command Name**	**OpCode**	**Description**
1	HCI_Inquiry	0x0001	Command used to enter Inquiry mode where it discovers other Bluetooth devices.
2	HCI_Inquiry_Cancel	0x0002	Command to cancel the Inquiry mode in which the Bluetooth device is in.
3	HCI_Periodic_Inquiry_Mode	0x0003	Command to set the device to enter Inquiry modes periodically according to the time interval set.
4	HCI_Exit_Periodic_Inquiry_Mode	0x0004	Command to exit the periodic Inquiry mode
5	HCI_Create_Connection	0x0005	Command to create an ACL connection to the device specified by the BD_ADDR in the parameters.
6	HCI_Disconnect	0x0006	Command to terminate the existing connection to a device
7	HCI_Add_SCO_Connection	0x0007	Create an SCO connection defined by the connection handle parameters.
8	HCI_Accept_Connection_Request	0x0009	Command to accept a new connection request
9	HCI_Reject_Connection_Request	0x000A	Command to reject a new connection request
10	HCI_Link_Key_Request_Reply	0x000B	Reply command to a link key

			request event sent from controller to the host
11	HCI_Link_Key_Request_Negative_Reply	0x000C	Reply command to a link key request event from the controller to the host if there is no link key associated with the connection.
12	HCI_PIN_Code_Request_Reply	0x000D	Reply command to a PIN code request event sent from a controller to the host.
13	HCI_PIN_Code_Request_Negative_Reply	0x000E	Reply command to a PIN code request event sent from the controller to the host if there is no PIN associated with the connection.
14	HCI_Change_Connection_Packet_Type	0x000F	Command to change the type of packets to be sent for an existing connection.
15	HCI_Authentication_Requested	0x0011	Command to establish authentication between two devices specified by the connection handle.
16	HCI_Set_Connection_Encryption	0x0013	Command to enable or disable the link level encryption.
17	HCI_Change_Connection_Link_Key	0x0015	Command to force the change of a link key to a new one between two connected devices.
18	HCI_Master_Link_Key	0x0017	Command to force two devices to use the master's link key temporarily.
19	HCI_Remote_Name_Request	0x0019	Command to determine the user friendly name of the connected device.
20	HCI_Read_Remote_Supported_Features	0x001B	Command to determine the features supported by the connected device.
21	HCI_Read_Remote_Version_Information	0x001D	Command to determine the version information of the connected device.
22	HCI_Read_Clock_Offset	0x001F	Command to read the clock offset of the remote device.
HCI Policy Command (OGF=0x02)			
1	HCI_Hold_Mode	0x0001	Command to place the current or remote device into the Hold mode

			state.
2	HCI_Sniff_Mode	0x0003	Command to place the current or remote device into the Sniff mode state.
3	HCI_Exit_Sniff_Mode	0x0004	Command to exit the current or remote device from the Sniff mode state.
4	HCI_Park_Mode	0x0005	Command to place the current or remote device into the Park mode state.
5	HCI_Exit_Park_Mode	0x0006	Command to exit the current or remote device from the Park mode state.
6	HCI_QoS_Setup	0x0007	Command to setup the Quality of Service parameters of the device.
7	HCI_Role_Discovery	0x0009	Command to determine the role of the device for a particular connection.
8	HCI_Switch_Role	0x000B	Command to allow the device to switch roles for a particular connection.
9	HCI_Read_Link_Policy_Settings	0x000C	Command to determine the link policy that the LM can use to establish connections.
10	HCI_Write_Link_Policy_Settings	0x000D	Command to set the link policy that the LM can use for a particular connection.
Host Controller and Baseband Commands (OGF=0x03)			
1	HCI_Set_Event_Mask	0x0001	Command to set which events are generated by the HCI for the host.
2	HCI_Reset	0x0003	Command to reset the host controller, link manager and the radio module.
3	HCI_Set_Event_Filter	0x0005	Command used by host to set the different types of event filters that the host needs to receive.
4	HCI_Flush	0x0008	Command used to flush all pending data packets for transmission for a particular connection handle.
5	HCI_Read_PIN_Type	0x0009	Command used by host to determine if the link manager assumes that the host requires a

			variable PIN type or fixed PIN code. PIN is used during pairing.
6	HCI_Write_PIN_Type	0x000A	Command used by host to write to the host controller on the PIN type supported by the host.
7	HCI_Create_New_Unit_Key	0x000B	Command used to create a new unit key.
8	HCI_Read_Stored_Link_Key	0x000D	Command to read the link key stored in the host controller.
9	HCI_Write_Stored_Link_Key	0x0011	Command to write the link key to the host controller.
10	HCI_Delete_Stored_Link_Key	0x0012	Command to delete a stored link key in the host controller.
11	HCI_Change_Local_Name	0x0013	Command to modify the user friendly name of the device.
12	HCI_Read_Local_Name	0x0014	Command to read the user friendly name of the device.
13	HCI_Read_Connection_Accept_Timeout	0x0015	Command to determine the timeout session before the host denies and rejects a new connection request.
14	HCI_Write_Connection_Accept_Timeout	0x0016	Command to set the timeout session before a device can deny or reject a connection request.
15	HCI_Read_Page_Timeout	0x0017	Command to read the timeout value where a device will wait for a connection acceptance before sending a connection failure is returned.
16	HCI_Write_Page_Timeout	0x0018	Command to write the timeout value where a device will wait for a connection acceptance before sending a connection failure is returned.
17	HCI_Read_Scan_Enable	0x0019	Command to read the status of the Scan_Enable configuration.
18	HCI_Write_Scan_Enable	0x001A	Command to set the status of the Scan_Enable configuration.
19	HCI_Read_Page_Scan_Activity	0x001B	Command to read the value of the Page_Scan_Interval and Page_Scan_Window configurations.
20	HCI_Write_Page_Scan_Activity	0x001C	Command to write the value of

			the Page_Scan_Interval and Page_Scan_Window configurations.
21	HCI_Read_Inquiry_Scan_Activity	0x001D	Command to read the value of the Inquiry_Scan_Interval and Inquiry_Scan_Window configurations.
22	HCI_Write_Inquiry_Scan_Activity	0x001E	Command to set the value of the Inquiry_Scan_Interval and Inquiry_Scan_Window configurations.
23	HCI_Read_Authentication_Enable	0x001F	Command to read the Authentication_Enable parameter.
24	HCI_Write_Authentication_Enable	0x0020	Command to set the Authentication_Enable parameter.
25	HCI_Read_Encryption_Mode	0x0021	Command to read the Encryption_Mode parameter.
26	HCI_Write_Encryption_Mode	0x0022	Command to write the Encryption_Mode parameter.
27	HCI_Read_Class_Of_Device	0x0023	Command to read the Class_Of_Device parameter.
28	HCI_Write_Class_Of_Device	0x0024	Command to set the Class_Of_Device parameter.
29	HCI_Read_Voice_Setting	0x0025	Command to read the Voice_Setting parameter. Used for voice connections.
30	HCI_Write_Voice_Setting	0x0026	Command to set the Voice_Setting parameter. Used for voice connections.
31	HCI_Read_Automatic_Flush_Timeout	0x0027	Command to read the Flush_Timeout parameter. Used for ACL connections only.
32	HCI_Write_Automatic_Flush_Timeout	0x0028	Command to set the Flush_Timeout parameter. Used for ACL connections only.
33	HCI_Read_Num_Broadcast_Retransmissions	0x0029	Command to read the number of time a broadcast message is retransmitted.
34	HCI_Write_Num_Broadcast_Retransmissions	0x002A	Command to set the number of time a broadcast message is retransmitted.
35	HCI_Read_Hold_Mode_Activity	0x002B	Command to set the Hold_Mode activity to instruct the device to

			perform an activity during hold mode.
36	HCI_Write_Hold_Mode_Activity	0x002C	Command to set the Hold_Mode_Activity parameter.
37	HCI_Read_Transmit_Power_Level	0x002D	Command to read the power level required for transmission for a connection handle.
38	HCI_Read_SCO_Flow_Control_Enable	0x002E	Command to check the current status of the flow control for the SCO connection.
39	HCI_Write_SCO_Flow_Control_Enable	0x002F	Command to set the status of the flow control for a connection handle.
40	HCI_Set_Host_Controller_To_Host_Flow_Control	0x0031	Command to set the flow control from the host controller to host in on or off state.
41	HCI_Host_Buffer_Size	0x0033	Command set by host to inform the host controller of the buffer size of the host for ACL and SCO connections.
42	HCI_Host_Number_Of_Completed_Packets	0x0035	Command set from host to host controller when it is ready to receive more data packets.
43	HCI_Read_Link_Supervision_Timeout	0x0036	Command to read the timeout for monitoring link losses.
44	HCI_Write_Link_Supervision_Timeout	0x0037	Command to set the timeout for monitoring link losses.
45	HCI_Read_Number_Of_Supported_IAC	0x0038	Command to read the number of IACs that the device can listen on during Inquiry access.
46	HCI_Read_Current_IAC_LAP	0x0039	Command to read the LAP for the current IAC.
47	HCI_Write_Current_IAC_LAP	0x003A	Command to set the LAP for the current IAC.
48	HCI_Read_Page_Scan_Period_Mode	0x003B	Command to read the timeout session of a page scan.
49	HCI_Write_Page_Scan_Period_Mode	0x003C	Command to set the timeout session of a page scan.
50	HCI_Read_Page_Scan_Mode	0x003D	Command to read the default Page scan mode.
51	HCI_Write_Page_Scan_Mode	0x003E	Command to set the default page scan mode.

7.4 INFORMATIONAL PARAMETERS

Command Name	OpCode	Description
HCI_Read_Local_Version_-Information	0x0001	This command reads the values for the version information for the local Controller.
HCI_Read_Local_Supported_-Commands	0x0002	This command reads the list of HCI commands supported for the local Controller.
HCI_Read_Local_Supported_Features	0x0003	This command requests a list of the supported features for the local BR/EDR Controller. This command will return a list of the LMP features.
HCI_Read_Local_-Extended_Features	0x0004	The HCI_Read_Local_Extended_Features command returns the requested page of the extended LMP features.
HCI_Read_Buffer_Size	0x0005	The HCI_Read_Buffer_Size command is used to read the maximum size of the data portion of HCI ACL and Synchronous Data packets sent from the Host to the Controller.
HCI_Read_BD_ADDR	0x0009	On a BR/EDR Controller, this command reads the Bluetooth Controller address (BD_ADDR).
HCI_Read_Data_- Block_Size	0x000A	The HCI_Read_Data_Block_Size command is used to read values regarding the maximum permitted data transfers over the Controller and the data buffering available in the Controller.

7.5 STATUS PARAMETERS

HCI_Read_Failed_-Contact_Counter	0x0001	This command reads the value for the Failed_Contact_Counter parameter for a particular connection to another device.
HCI_Reset_Failed_-Contact_Counter	0x0002	This command resets the value for the

		Failed_Contact_Counter parameter for a particular connection to another device.
HCI_Read_Link_Quality	0x0003	This command returns the value for the Link_Quality for the specified Handle
HCI_Read_RSSI	0x0005	This command reads the Received Signal Strength Indication (RSSI) value from a Controller.
HCI_Read_AFH_-Channel_Map	0x0006	This command returns the values for the AFH_Mode and AFH_Channel_Map for the specified Connection_Handle. The Connection_Handle shall be a Connection_Handle for an ACL connection.
HCI_Read_Clock	0x0007	This command reads the estimate of the value of the Bluetooth Clock from the BR/EDR Controller.
HCI_Read_-Encryption_Key_Size	0x0008	This command reads the current encryption key size associated with the Connection_Handle. The Connection_Handle shall be a Connection_Handle for an active ACL connection. All BR/EDR Controllers shall implement this command.

7.6 TESTING COMMANDS

HCI_Read_Loopback_Mode	0x0001	
HCI_Write_Loopback_Mode	0x0002	
HCI_Enable_Device_Under_-Test_Mode	0x0003	
HCI_Write_Simple_Pairing_-	0x0004	

Debug_Mode		

HCI Events

4.3 HCI Events/ Error Codes/ Flow Control

4.3.1 Flow Control

- Flow control is used in the direction from the Host to the Host Controller to avoid filling up the Host Controller data buffers with ACL data destined for a remote device (connection handle) that is not responding.
- It is the Host that manages the data buffers of the Host Controller.

4.3.2 HCI Events

- A number of different events are defined for the HCI layer.
- The events provide a method to return parameters and data associated for each event.
- 32 HCI different events have been implemented so far, they range from Inquiry Complete Event to Page Scan Repetition Mode Change Event.

4.3.3 HCI Error Codes

- A large number of error codes have been defined for the HCI layer.
- When a command fails, Error codes are returned to indicate the reason for the error.
- 35 HCI error codes have so far been defined, from Unknown HCI Command to LMP PDU Not Allowed.

No	Command Name	Event Code	Description
1	HCI_Inquiry_Complete_Event	0x01	Indicates the Inquiry has finished.
2	HCI_Inquiry_Result_Event	0x02	Indicates that Bluetooth device(s) have responded for the inquiry.
3	HCI_Connection_Complete_Event	0x03	Indicates to both hosts that the new connection

			has been formed.
4	HCI_Connection_Request_Event	0x04	Indicates that a new connection is trying to be established.
5	HCI_Disconnection_Complete_Event	0x05	Occurs when a connection has been disconnected.
6	HCI_Authentication_Complete_Event	0x06	Occurs when an authentication has been completed.
7	HCI_Remote_Name_Request_Complete_Event	0x07	Indicates that the request for the remote name has been completed.
8	HCI_Encryption_Change_Event	0x08	Indicates that a change in the encryption has been completed.
9	HCI_Change_Connection_Link_Key_Complete_Event	0x09	Indicates that the change in the link key has been completed.
10	HCI_Master_Link_Key_Complete_Event	0x0A	Indicates that the change in the temporary link key or semi permanent link key on the master device is complete.
11	HCI_Read_Remote_Supported_Features_Complete_Event	0x0B	Indicates that the reading of the supported features on the remote device is complete.
12	HCI_Read_Remote_Version_Complete_Event	0x0C	Indicates that the version number on the remote device has been read and completed.
13	HCI_Q0S_Setup_Complete_Event	0x0D	Indicates that the Quality of Service setup has been complete.
14	HCI_Command_Complete_Event	0x0E	Used by controller to send status and event parameters to the host for the particular command.
15	HCI_Command_Status_Event	0x0F	Indicates that the command has been received and is being

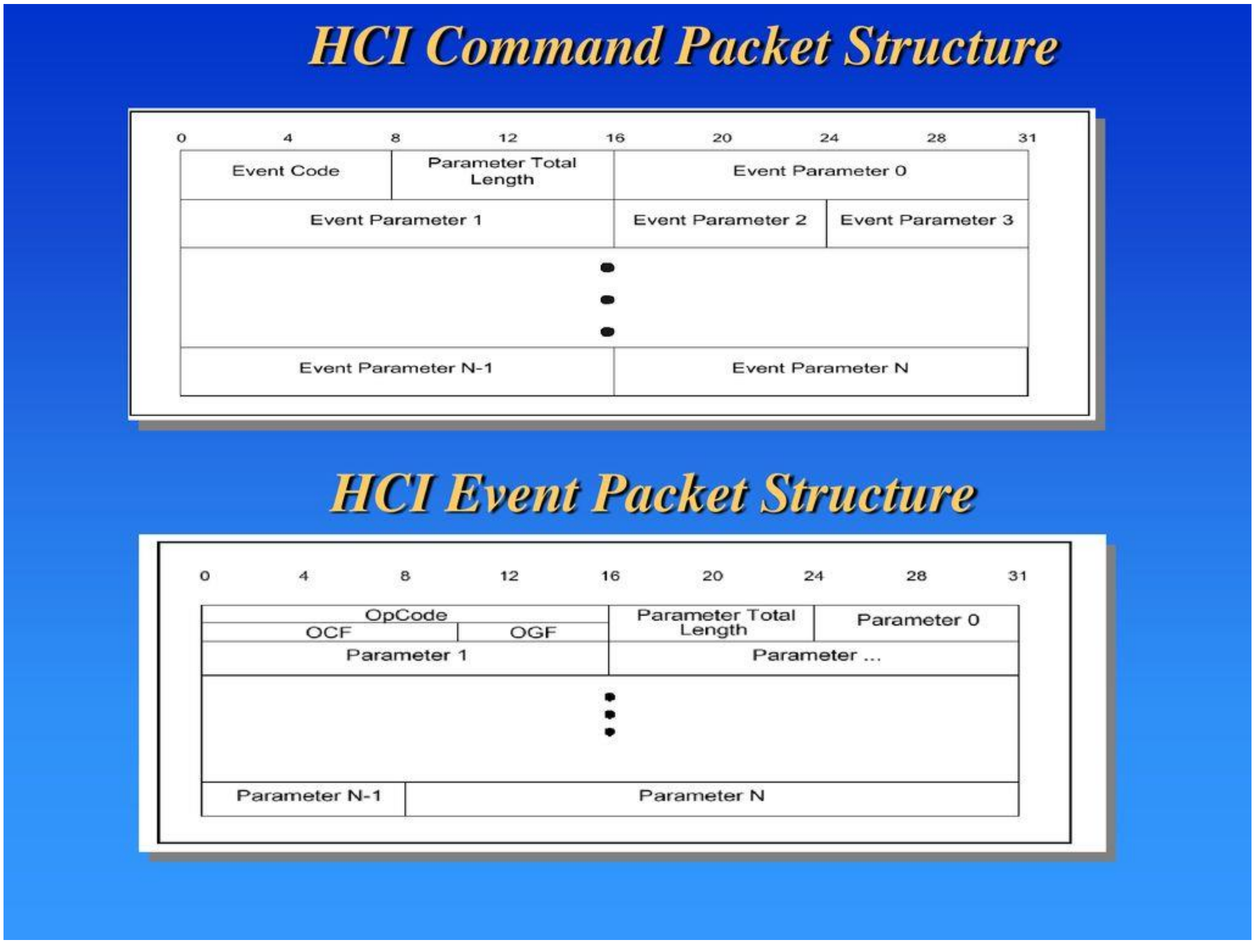
			processed in the host controller.
16	Hardware_Error_Event	0x10	Indicates a hardware failure of the Bluetooth device.
17	Flush_Occured_event	0x11	Indicates that the data has been flushed for a particular connection.
18	Role_Change_Event	0x12	Indicates that the current bluetooth role for a connection has been changed.
19	Number_Of_Completed_Packets_Event	0x13	Indicates to the host the number of data packets sent compared to the last time the same event was sent.
20	Mode_Change_Event	0x14	Indicates the change in mode from hold, sniff, park or active to another mode.
21	Return_Link_Keys_Event	0x15	Used to return stored link keys after a Read_Stored_Link_Key command was issued.
22	PIN_Code_Request_Event	0x16	Indicates the a PIN code is required for a new connection.
23	Link_Key_Request_Event	0x17	Indicates that a link key is required for the connection.
24	Link_Key_Notification_Event	0x18	Indicates to the host that a new link key has been created.
25	Loopback_Command_Event	0x19	Indicates that command sent from the host will be looped back.
26	Data_Buffer_Overflow_Event	0x1A	Indicates that the data buffers on the host has overflowed.
27	Max_Slots_Change_Event	0x1B	Informs the host when the LMP_Max_Slots parameter changes.

28	Read_Clock_Offset_Complete_Event	0x1C	Indicates the completion of reading the clock offset information.
29	Connection_Packet_Type_Changed_Event	0x1D	Indicate the completion of the packet type change for a connection.
30	QoS_Violation_Event	0x1E	Indicates that the link manager is unable to provide the required Quality of Service.
31	Page_Scan_Mode_Change_Event	0x1F	Indicates that the remote device has successfully changed the Page Scan mode.
32	Page_Scan_Repetition_Mode_Change_Event	0x20	Indicates that the remote device has successfully changed the Page Scan Repetition mode.

HCI Error Codes

No	**Command Name**	**Error Code**
	Success	0x00
1	Unknown HCI Command	0x01
2	No Connection	0x02
3	Hardware Failure	0x03
4	Page Timeout	0x04
5	Authentication Failure	0x05
6	Key Missing	0x06
7	Memory Full	0x07
8	Connection Timeout	0x08
9	Max Number Of Connections	0x09
10	Max Number Of SCO Connections To A Device	0x0A
11	ACL Connection Already Exists	0x0B
12	Command Disallowed	0x0C
13	Host Rejected Due To Limited Resources	0x0D
14	Host Rejected Due To Security Reasons	0x0E
15	Host Rejected Due To A Remote Device Only A Personal Device	0x0F
16	Host Timeout	0x1
17	Unsupported Feature Or Parameter Value	0x11
18	Invalid HCI Command Parameters	0x12

19	Other End Terminated Connection: User Ended Connection	0x13
20	Other End Terminated Connection: Low Resources	0x14
21	Other End Terminated Connection: About To Power Off	0x15
22	Connection Terminated By Local Host	0x16
23	Repeated Attempts	0x17
24	Pairing Not Allowed	0x18
25	Unknown LMP PDU	0x19
26	Unsupported Remote Feature	0x1A
27	SCO Offset Rejected	0x1B
28	SCO Interval Rejected	0x1C
29	SCO Air Mode Rejected	0x1D
30	Invalid LMP Parameters	0x1E
31	Unspecified Error	0x1F
32	Unsupported LMP Parameter	0x20
33	Role Change Not Allowed	0x21
34	LMP Response Timeout	0x22
35	LMP Error Transaction Collision	0x23
36	LMP PDU Not Allowed	0x24
37	Encryption Mode Not Acceptable	0x25
38	Unit Key Used	0x26
39	QoS Not Supported	0x27
40	Instant Passed	0x28
41	Pairing With Unit Key Not Supported	0x29
42	Reserved For Future Use	0x2A-0xFF



- The HCI Command packet is used to send commands to the Controller from the Host.
- Controllers shall be able to accept HCI Command packets with up to 255 bytes of data excluding the HCI Command packet header.
- The HCI Command packet header is the first 3 octets of the packet.
- Each command is assigned a 2 byte Opcode used to uniquely identify different types of commands.
- The Opcode parameter is divided into two fields, called the OpCode Group Field (OGF) and OpCode Command Field (OCF).
- The OGF occupies the upper 6 bits of the Opcode, while the OCF occupies the remaining 10 bits.
- Any opcode not mentioned in this Part is reserved for future use.
- The OGF of 0x3F is reserved for vendor-specific debug commands.
- The organization of the opcodes allows additional information to be inferred without fully decoding the entire Opcode.
- Note: The OGF composed of all 'ones' has been reserved for vendor-specific debug commands.

- These commands are vendor-specific and are used during manufacturing, for a possible method for updating firmware, and for debugging.
- On receipt of a Vendor Specific Debug command the Controller should respond with either:

- The HCI Event packet is used by the Controller to notify the Host when events occur.
- If the Controller sends an HCI Event Packet containing an Event Code or an LE subevent code that the Host has not masked out and does not support, the Host shall ignore that packet.
- Any event code or LE subevent code not mentioned in this Part is reserved for future use.
- The Host shall be able to accept HCI Event packets with up to 255 octets of data excluding the HCI Event packet header.
- The format of the HCI Event packet is shown in Figure 5.4, and the definition of each field is explained below.
- The HCI Event packet header is the first 2 octets of the packet.
- Note: An LE Controller uses a single Event Code (see Section 7.7.65) to transmit all LE specific events from the Controller to the Host.
- The first Event Parameter is always a subevent code identifying the specific event.

What is BlueZ?

BlueZ is the official Linux Bluetooth stack. It provides, in it's modular way, support for the core Bluetooth layers and protocols.

Currently BlueZ consists of many separate modules:

- Bluetooth kernel subsystem core
- L2CAP and SCO audio kernel layers
- RFCOMM, BNEP, CMTP and HIDP kernel implementations
- HCI UART, USB, PCMCIA and virtual device drivers

- General Bluetooth and SDP libraries and daemons
- Configuration and testing utilities
- Protocol decoding and analysis tools

What is BlueZ?

BlueZ is the Bluetooth stack for Linux. It handles both Bluetooth BR/EDR as well as BLE. To communicate with BlueZ, we'll be using something called D-Bus.

What is D-Bus?

- Before we get into BlueZ, we'll want to have a cursory understanding of d-bus and how we use it to talk to BlueZ.
- D-Bus is essentially a message bus that allows different processes running on a Linux system to talk to each other.

What is Bluetooth daemon?

- bluetoothd daemon, which manages all the Bluetooth devices.
- bluetoothd itself does not accept many command-line options, as most of its configuration is done in the /etc/bluetooth/main.conf file, which has its own man page.
- bluetoothd can also provide a number of services via the D-Bus message bus system.

What is PulseAudio?

PulseAudio **acts as a sound server, where a background process accepting sound input from one or more sources (processes, capture devices, etc.) is created**. The background process then redirects these sound sources to one or more sinks (sound cards, remote network PulseAudio servers, or other processes).

What is HCIDUMP?

hcidump **reads raw HCI data coming from and going to a Bluetooth device** (which can be specified with the option -i, default is the first available one) and prints to screen commands, events and data in a human-readable form.

What is Bluetoothctl?

The Bluetoothctl command will keep your Bluetooth devices talking to Linux. bluetoothctl [1] is **the main command for configuring Bluetooth devices on Linux**.

What is BTMON?

The btmon command provides access to the Bluetooth subsystem monitor infrastructure for reading HCI traces.

btmon is Bluetooth monitor

What is TWS in Bluetooth?

It's called **True Wireless Stereo** (TWS), and it's a unique Bluetooth feature that will let you enjoy true stereo sound quality without the use of cables or wires.

RFCOMM

- The RFCOMM protocol provides emulation of serial ports over the L2CAP protocol [2]. The protocol is based on the ETSI standard GSM 07.10. (European Telecommunications Standards Institute) (Global System for Mobile Communications)
- RFCOMM is a simple transport protocol, with additional provisions for emulating the nine circuits of RS-232 (ITU-T V.24) serial ports.

What do you mean by RS-232?

Recommended Standard 232

- The term RS232 stands for "**Recommended Standard 232**" and it is a type of serial communication used for transmission of data normally in medium distances.
- many applications like computer printers, factory automation devices etc.
- The RFCOMM protocol supports up to 60 simultaneous connections between two Bluetooth devices.

1.2 Byte Ordering

This document uses the same byte ordering as the GSM 07.10 specification, i.e. all binary numbers are in Least Significant Bit to Most Significant Bit order, reading from left to right.

1.3 Device Types

For the purposes of RFCOMM, a complete communication path involves two applications running on different devices (the communication endpoints) with a communication segment between them.

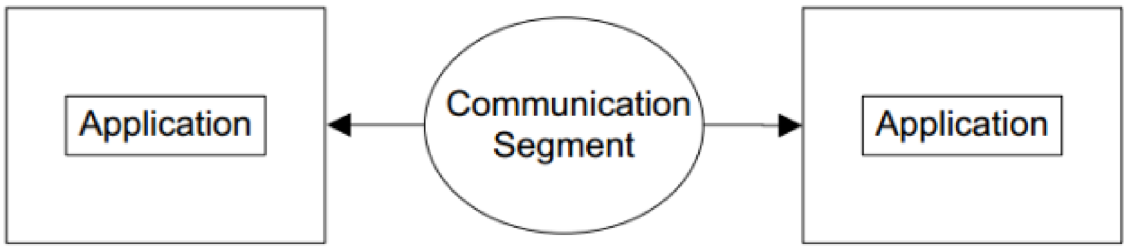


Figure 1.1: RFCOMM Communication Segment

This specification supports implementation of the following two device types:

• Type 1 devices are communication endpoints such as computers and printers.

• Type 2 devices are those that are part of the communication segment; e.g. modems.

Though RFCOMM does not make a distinction between these two device types in the protocol, accommodating both types of devices impacts the RFCOMM protocol.

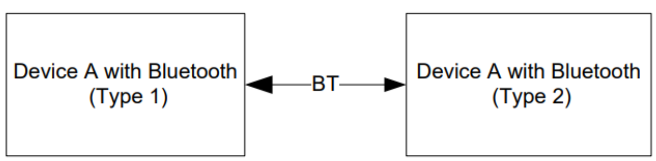


Figure 1.2: RFCOMM Direct Connect

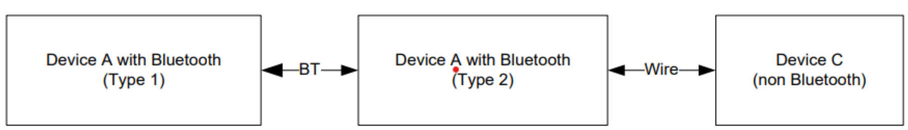


Figure 1.3: RFCOMM used with legacy COMM device

RFCOMM Service Overview

- RFCOMM emulates RS-232 (ITU-T V.24) serial ports.
- The emulation includes transfer of the state of the non-data circuits.
- RFCOMM has a built-in scheme for null modem emulation.

- In the event that a baud rate is set for a particular port through the RFCOMM service interface, that will not affect the actual data throughput in RFCOMM; i.e. RFCOMM does not incur artificial rate limitation or pacing.
- However, if either device is a type 2 device (relays data onto other media), or if data pacing is done above the RFCOMM service interface in either or both ends, actual throughput will, on average, reflect the baud rate setting.
- RFCOMM supports emulation of multiple serial ports between two devices and emulation of serial ports between multiple devices

2.1 RS-232 Control Signals

Pin	Circuit Name
102	Signal Common
103	Transmit Data (TD)
104	Received Data (RD)
105	Request to Send (RTS)
106	Clear to Send (CTS)
107	Data Set Ready (DSR)
108	Data Terminal Ready (DTR)
109	Data Carrier Detect (CD)
125	Ring Indicator (RI)

Table 2.1: Emulated RS-232 circuits in RFCOMM

6.1.3 Null Modem Emulation

- RFCOMM is based on TS 07.10. When it comes to transfer of the states of the non-data circuits, TS 07.10 does not distinguish between DTE and DCE devices.
- The RS-232 control signals are sent as a number of DTE/DCE independent signals.
- The way in which TS 07.10 transfers the RS-232 control signals creates an implicit null modem when two devices of the same kind are connected together.
- No single null-modem cable wiring scheme works in all cases; however the null modem scheme provided in RFCOMM should work in most cases.

2.3 Multiple Emulated Serial Ports

2.3.1 Multiple Emulated Serial Ports between two Devices

- Two BT devices using RFCOMM in their communication may open multiple emulated serial ports.
- RFCOMM supports up to 60 open emulated ports; however the number of ports that can be used in a device is implementation-specific.

- A **Data Link Connection Identifier (DLCI)** identifies an ongoing connection between a client and a server application.
- The DLCI is unique within one RFCOMM session between two devices.

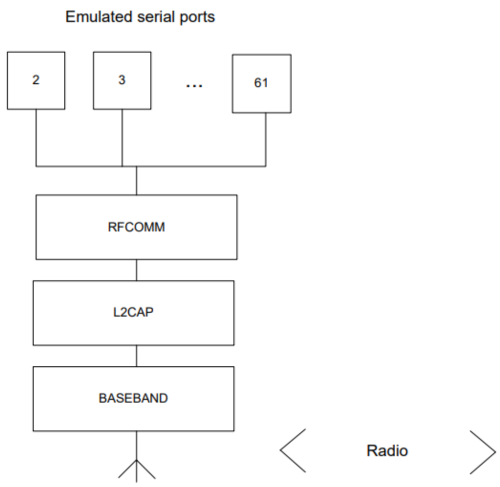


Figure 2.2: Multiple Emulated Serial Ports.

2.3.2 Multiple Emulated Serial Ports and Multiple Bluetooth Devices

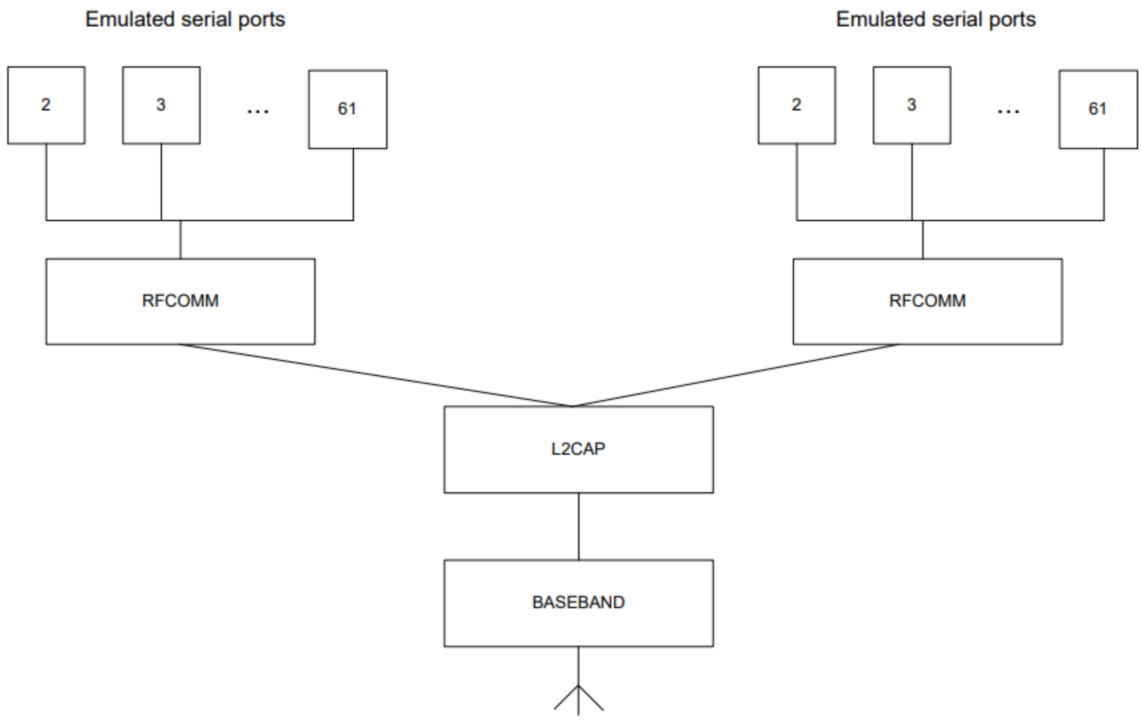


Figure 2.3: Emulating serial ports coming from two Bluetooth devices

6.2 TS 07.10 Adaptations for RFCOMM

6.2.1 Media Adaption

The opening flag and the closing flags in the 07.10 basic option frame are not used in RFCOMM, instead it is only the fields contained between the flags that are exchanged between the L2CAP layer and RFCOMM layer. There is always exactly one RFCOMM frame contained in each L2CAP frame.

6.2.2 TS 07.10 Multiplexer Startup & Closedown Procedure

- At any time, there must be at most one RFCOMM session between any pair of devices.
- When establishing a new DLC, the initiating entity must check if there already exists an RFCOMM session with the remote device, and if so, establish the new DLC on that.
- A session is identified by the Bluetooth BD_ADDR of the two endpoints.

6.2.3 DLCI Allocation with RFCOMM Server Channels

To account for the fact that both client and server applications may reside on both sides of an RFCOMM session, with clients on either side making connections independent of each other, the DLCI value space is divided between the two communicating devices using the concept of RFCOMM server channels and a direction bit. The RFCOMM server channel number is a subset of the bits in the DLCI part of the address field in the TS 07.10 frame.

Server applications registering with an RFCOMM service interface are assigned a Server Channel number in the range 1?0. For an RFCOMM session, the initiating device is given the direction bit D=1 (and conversely, D=0 in the other device). When establishing a new data link connection on an existing RFCOMM session, the direction bit is used in conjunction with the Server Channel to determine the DLCI to use to connect to a specific application. This DLCI is thereafter used for all packets in both directions between the endpoints.

An RFCOMM entity making a new DLC on an existing session forms the DLCI by combining the Server Channel for the application on the other device, and the inverse of its own direction bit for the session.

6.2.4 Multiplexer Control Commands

Note that in TS 07.10, some Multiplexer Control commands pertaining to specific DLCIs may be exchanged on the control channel (DLCI 0) before the corresponding DLC has been established.

Remote Port Negotiation (RPN) Command:	The RPN command can be used before a new DLC is opened and should be used whenever the port settings change.
Remote Line Status (RLS) Command:	This command is used for indication of remote port line status.
DLC Parameter Negotiation (PN) Command:	This is mandatory to use for RFCOMM implementations conforming to the Bluetooth specification version 1.1 and later. This command must be used at least before creation of the first DLC on an RFCOMM session, and the initiator has to try to turn on the use of credit based flow control,

6.3 Flow Control Methods Used

- Wired ports commonly use flow control such as RTS/CTS [RTS/CTS (**Request To Send / Clear To Send**)] to control communications.

What is CTS and RTS?

RTS / CTS Flow Control is another flow control mechanism that is part of the RS232 standard. It makes use of two further pins on the RS232 connector, RTS (Request to Send) and CTS (Clear to Send). These two lines **allow the receiver and the transmitter to alert each other to their state**.

- On the other hand, the flow control between RFCOMM and the lower layer L2CAP depends on the service interface supported by the implementation.
- In addition, RFCOMM has its own flow control mechanisms.

The following describes the different flow control mechanisms.

6.3.1 L2CAP Flow Control

L2CAP relies on the flow control mechanism provided by the Link Manager layer in the baseband. The flow control mechanism between the L2CAP and RFCOMM layers is implementation specific.

6.3.2 Wired Serial Port Flow Control

Wired Serial port flow control falls into two camps

Software flow control using characters such as XON/XOFF

Hardware flow control using RTS/CTS or DTR/DSR circuits.

Data terminal ready (DTR), another form of hardware flow control, is normally generated by the devices, such as printers to indicate that they are ready to communicate with the system. This signal is used in conjunction with data set ready (DSR) generated by the system to control data flow.

These methods may be used by both sides of a wired link, or may be used only in one direction.

6.3.3 RFCOMM Flow Control

The RFCOMM protocol provides two flow control mechanisms:

The RFCOMM protocol contains flow control commands that operate on the aggregate data flow between two RFCOMM entities; i.e. all DLCIs are affected.

The Modem Status command is the flow control mechanism that operates on individual DLCI.

6.3.4 Port Emulation Entity : Serial Flow Control

On Type 1 devices some port drivers (Port Emulation Entities plus RFCOMM) will need to provide flow control services as specified by the API they are emulating. An application may request a particular flow control mechanism like XON/XOFF or RTS/CTS and expect the port driver to handle the flow control.

On type 2 devices the port driver may need to perform flow control on the non-RFCOMM part of the communication path; i.e. the physical RS-232 port. This flow control is specified via the control parameters sent by the peer RFCOMM entity (usually a type 1 device). The description of flow control in this section is for port drivers on type 1 devices.

Since RFCOMM already has its own flow control mechanism, the port driver does not need to perform flow control using the methods requested by the application. In the ideal case, the application sets a flow control mechanism and assumes that the COMM system will handle the details. The port driver could then simply ignore the request and rely on RFCOMM's flow control. The application is able to send and receive data, and does not know or care that the port driver did not perform flow control using the mechanism requested. However, in the real world some problems arise :

The RFCOMM-based port driver is running on top of a packet-based protocol where data may be buffered somewhere in the communication path. Thus, the port driver cannot perform flow control with the same precision as in the wired case.

The application may decide to apply the flow control mechanism itself in addition to requesting flow control from the port driver.

These problems suggest that the port driver must do some additional work to perform flow control emulation properly. Here are the basic rules for flow control emulation.

The port driver will not solely rely on the mechanism requested by the application but use a combination of flow control mechanisms.

The port driver must be aware of the flow control mechanisms requested by the application and behave like the wired case when it sees changes on the non-data circuits (hardware flow control) or flow control characters in the incoming data (software flow control). For example, if XOFF and XON characters would have been stripped in the wired case they must be stripped by the RFCOMM based port driver.

If the application sets a flow control mechanism via the port driver interface and then proceeds to invoke the mechanism on its own, the port driver must behave in a manner similar to that of the wired case (e.g. If XOFF and XON characters would have been passed through to the wire in the wired case the port driver must also pass these characters).

6.3.5 Credit Based Flow Control

This is a mandatory feature that did not exist in RFCOMM in Bluetooth specifications 1.0B and earlier. Therefore, its use is subject to negotiation before the first DLC establishment.

Implementations conforming to this specification must support it, and must try to use it when connecting to other devices.

The credit based flow control feature provides flow control on a per - DLC basis. When used, both devices involved in a RFCOMM session will know, for each DLC, how many RFCOMM frames the other device is able to accept before it buffers fill up for that DLC. A sending entity may send as many frames on a DLC as it has credits; if the credit count reaches zero, the sender must stop and wait for further credits from the peer. It is always allowed to send frames containing no user data (length field = 0) when credit based flow control is in use. This mechanism operates independently for each DLC, and for each direction. It does not apply to DLCI 0 or to non-UIH frames.

6.4 Other Entity Interaction

6.4.1 Port Emulation and Port Proxy Entities

This section defines how the RFCOMM protocol should be used to emulate serial ports.

Type 1 devices are communication endpoints such as computers and printers.Type 2 devices are part of a communication segment; e.g. modems.

Port Emulation Entity : The port emulation entity maps a system specific communication interface (API) to the RFCOMM services.

Port Proxy Entity : The port proxy entity relays data from RFCOMM to an external RS-232 interface linked to a DCE. The communications parameters of the RS-232 interface are set according to received RPN commands,

6.4.3 Service Registration and Discovery

Registration of individual applications or services, along with the information needed to reach those (i.e. the RFCOMM Server Channel) is the responsibility of each application respectively (or possibly a Bluetooth configuration application acting on behalf of legacy applications not directly aware of Bluetooth).

6.4.3 Reliability

RFCOMM uses the services of L2CAP to establish L2CAP channels to RFCOMM entities on other devices. An L2CAP channel is used for the RFCOMM/TS 07.10 multiplexer session.

Some frame types (SABM and DISC) as well as UIH frames with multiplexer control commands sent on DLCI 0 always require a response from the remote entity, so they are acknowledged on the RFCOMM level (but not retransmitted in the absence of acknowledgement). Data frames do not require any response in the RFCOMM protocol, and are thus unacknowledged.

For the purposes of RFCOMM, a complete communication path involves two applications running on different devices (the communication endpoints) with a communication segment between them. RFCOMM is intended to cover applications that make use of the serial ports of the devices in which they reside.

Therefore, RFCOMM must require L2CAP to provide channels with maximum reliability, to ensure that all frames are delivered in order, and without duplicates. Should an L2CAP channel fail to provide this, RFCOMM will expect a link loss notification, which should be handled by RFCOMM.

6.4.4 Low Power Modes

If all L2CAP channels towards a certain device are idle for a certain amount of time, a decision may be made to put that device in a low power mode i.e hold, sniff or park mode. This will be done without any interference from RFCOMM. RFCOMM can state its latency requirements to L2CAP. This information may be used by lower layers to decide which low power mode(s) to use.

The RFCOMM protocol as such does not suffer from latency delays incurred by low power modes, and consequentially, this specification does not state any maximum latency requirement on RFCOMM's behalf. Latency sensitivity inherently depends on application requirements, which suggests that an RFCOMM service interface implementation could include a way for applications to state latency requirements, to be aggregated and conveyed to L2CAP by the RFCOMM implementation. (That is if such procedures make sense for a particular platform.)

Note , the above text contains excerpts from the Bluetooth SIG's Specification, as well as various interpretations of the Specs. For complete details of the various sections, consult the actual Bluetooth Specification.

APPLICATION OF PROFILES

HUMAN INTERFACE DEVICE

1 Introduction

HID: Human Interface Device Class

- The **Human Interface Device Class (HID)** is mainly used for devices that allow human control over a PC.
- Using these devices, the host is able to react on human input (e.g. movements of a mouse or keypresses).
- This response has to happen quickly, so that the computer user does not notice a significant delay between his action and the expected feedback.

Typical examples for HID class devices are:

- Keyboards and pointing devices (such as mouse devices, joysticks and trackballs)
- Front-panel controls (for example buttons, knobs, sliders, and switches)
- Simulation devices (such as steering wheels, pedals, other VR input devices)
- Remote controls and telephone keypads
- Other low data-rate devices that provide for example environmental data (such as thermometers, energy meters or even bar-code readers)

- A **Bluetooth HID device** is a device providing the service of human or other data input and output to and from a Bluetooth HID Host. Examples of Bluetooth HID devices are keyboards, mice, joysticks, gamepads, remote controls, and also voltmeters and temperature sensors.
- A **Bluetooth HID Host** is a device using or requesting the services of a Bluetooth HID device. Examples would be a personal computer, handheld computer, gaming console, industrial machine, or data-recording device.

1.1 User Scenarios
1.2 1.1.1 Desktop Computing Scenario

1.1.2 Living Room Scenario

1.1.3 Conference Room Scenario

1.1.4 Remote Monitoring Scenario

1.1.5 Mobile Phones Scenario

- Keyboards, keypads, and keyboards with integrating pointing devices such as trackballs or touchpads
- Trackballs, mice, and other pointing devices
- Game controllers such as gamepads, joysticks, steering wheels, throttles, rudder pedals, data gloves, etc., some including force feedback actuators, motion detectors, and/or accelerometers Front-panel controls; for example, knobs, switches, buttons, and sliders
- Controls that might be found on telephones
- Remote controls for consumer electronics devices and universal remote controls
- Simple alphanumeric and bit-mapped remote displays
- Bar code or RFID scanners

3.1 HID Roles In HID,

and by extension in HOGP, there are two defined roles.

These roles are the HID Host and the HID device.

An HID device can only be actively connected to a single HID host at time.

An HID host can be connected to many HID devices.

The HID Host implements the Bluetooth central role. The HID Host is the device that receives the input information and does further processing with the provided data. For instance, a computer would be an HID Host since this is the device that receives the HID data.

The HID device implements the Bluetooth peripheral role. The HID device transmits input information to a central computer. For instance, a mouse or keyboard would be HID devices because they send input data to the computer device.

Bluetooth HID Profile Architecture

2.2 Bluetooth HID Software Stacks

- Below is an illustration of the typical software layers that reside in the Bluetooth HID Host and the Bluetooth HID device.
- In this example, the Bluetooth HID Host is a personal computer (PC) and has the upper layers of the Bluetooth software running on its native processor. The PC is connected to a Bluetooth Controller via an HCI transport bus such as USB.

- The hardware and software layers above the HCI transport together form the Bluetooth Host, as defined in the Bluetooth Core Specification [6]. To enable a low-cost implementation, the Bluetooth HID device in this example has the Bluetooth Host and Bluetooth Controller functions running on the same CPU. Other implementations are possible.

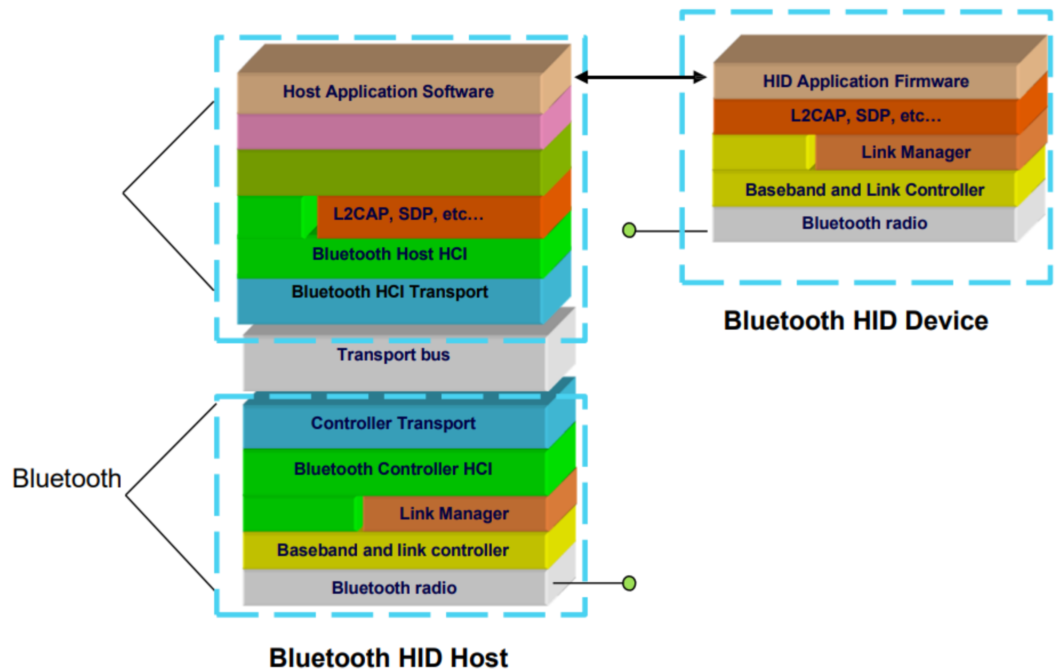


Figure 2.2: Typical Bluetooth HID Software Stacks

A human interface device (HID) is a method by which a human interacts with an electronic information system either by inputting data or providing output. A myriad of HID devices exist. The most common are the keyboards, mice, computer speakers, webcams and headsets. All devices providing an interface between the user and computer machines are considered HIDs.

- It is an integral part of a user interface, which more academically may be called a human-computer interface (HCI) or more generically as a interface device.
- The HID over GATT profile defines the procedures and features to be used by Bluetooth low energy HID Devices using GATT and Bluetooth HID Hosts using GATT.
- This profile is an adaptation of the USB HID specification [2] to operate over a Bluetooth low energy wireless link.
- This profile shall operate over an LE transport only.
- This profile requires the Generic Attribute Profile (GATT), the Battery Service, the Device Information Service, and the Scan Parameters Profile.

2.1 Roles

This profile defines three roles:

HID Device, Boot Host, and Report Host.

• The HID Device shall be a GATT server.

• The Boot Host shall be a GATT client.

• The Report Host shall be a GATT client.

Use of the term HID Host refers to both host roles: Boot Host, and Report Host.

- A Report Host is required to support a HID Parser and be able to handle arbitrary formats for data transfers (known as Reports)
- whereas a Boot Host is not required to support a HID Parser as all data transfers (Reports) for Boot Protocol Mode are of predefined length and format.

2.2 Role/Service Relationships

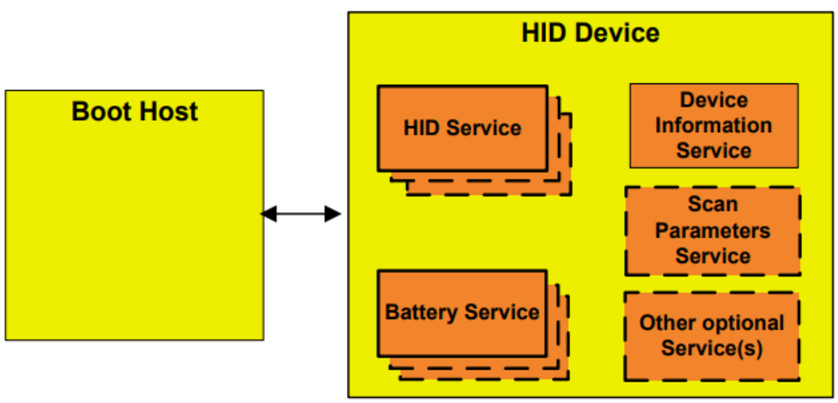


Figure 2.1: Boot Host and HID Device Roles/Service Relationship

Note: Profile roles are represented by yellow boxes and services are represented by orange boxes.

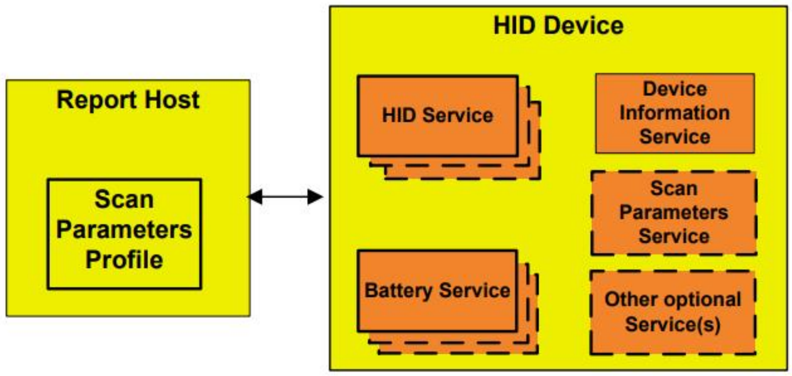


Figure 2.2: Report Host and HID Device Roles/Service Relationships

2.3 Concurrent Role Limitations and Restrictions

A Boot Host shall not concurrently be a Report Host. A Report Host shall not concurrently be a Boot Host. There are no concurrency limitations on either HID Host roles from also being a HID Device.

2.4 Topology Limitations and Restrictions

The HID Device shall use the GAP Peripheral role. The Boot Host shall use the GAP Central role. The Report Host shall use the GAP Central role.

2.5 Multiple Service Instances

Multiple service instances of the Battery Service may be supported. • Multiple service instances of the following may be supported, but are not considered as a part of this profile: • Any Service other than HID Service, Device Information Service, or Scan Parameters Service.

3 HID Device Requirements

Table 3.1 shows service requirements for the HID Device

Service	Requirement
HID Service	M
Battery Service	M
Device Information Service	M
Scan Parameters Service	O

Table 3.1: HID Device Service Requirements

4 HID Host Requirements and Behaviors

Procedure	Section Reference	Boot Host Requirement	Report Host Requirement
Service Discovery	4.3/4.5	M	M
- HID Service Discovery	4.3.1/4.5.1	M	M
- Device Information Service Discovery	4.3.2/4.5.2	O	M
- Battery Service Discovery	4.3.3/4.5.3	O	M
Characteristic Discovery	4.4/4.6	O	M
- HID Service Characteristic Discovery	4.6.1	O	M
- Device Information Service Characteristic Discovery	4.6.2	O	M
- Battery Service Characteristic Discovery	4.6.3	O	M
Report Map	4.7	X	M
Report	4.8	X	M
Boot Keyboard Input Report	4.4.1.2/4.12	C.2/C.3	X
Boot Keyboard Output Report	4.4.1.3/4.13	C.2/C.3	X
Boot Mouse Input Report	4.4.1.4/4.14	C.2	X
HID Information	4.10	X	M
HID Control Point	4.9	X	C.1
Protocol Mode	4.4.1/4.11	M	O
Non-HID Service characteristic defined within Report Map	4.8.1	X	M

C.1: Mandatory if the Host supports Suspend Mode, otherwise optional.
C.2: Mandatory to support at least one of these features.
C.3: If one of these features is supported, both features shall be supported.

Table 4.1: Boot Host and Report Host Requirements

5 Connection Establishment

This section describes the connection establishment and connection termination procedures used by a HID Host and HID Device in certain scenarios.

5.1 HID Device Requirements

5.1.1 Device Discovery

5.1.2 Connection Procedure for Non-bonded Devices

Advertising Duration	Parameter	Value
180 seconds (fast connection)	Advertising Interval	30 ms to 50 ms

Table 5.1: Recommended Advertising Parameters for Non-bonded Devices

5.1.3 Device-Initiated Connection Procedure for Bonded Devices

Advertising Duration	Parameter	Value
1.28 seconds (low latency) – Option 1	Advertising mode	Directed
30 seconds (higher latency) - Option 2	Advertising mode	Undirected
	Advertising Interval	20 ms to 30 ms

Table 5.2: Recommended advertising parameters for device-initiated connection of bonded devices

5.1.4 Host-Initiated Connection Procedure for Bonded Devices

Advertising Duration	Parameter	Value
Permanent (reduced power)	Advertising mode	Undirected
	Advertising Interval	1 s to 2.5 s

Table 5.3: Recommended Advertising Parameters for Host-Initiated Connection of Bonded Devices

5.1.5 Link Loss Reconnection Procedure

5.1.6 Idle Connection

5.2 Host Requirements

5.2.1 Device Discovery

5.2.2 Connection Procedure for Non-bonded Devices

Scan Duration	Parameter	Value
180 seconds (fast connection)	Scan Interval	22.5ms
	Scan Window	11.25ms

Table 5.4: Recommended Scan Parameters for Non-bonded Devices

5.2.3 Device-Initiated Connection Procedure for Bonded Devices

Scan Duration	Parameter	Value
Permanent (reduced power)	Scan Interval	1.28s
	Scan Window	11.25ms

Table 5.5: Recommended Scan Parameters for Device-Initiated Connection of Bonded Devices

5.2.4 Host-Initiated Connection Procedure for Bonded Devices

Scan Duration	Parameter	Value
30 seconds (fast connection)	Scan Interval	30ms to 60ms*
	Scan Window	30ms

Table 5.6: Recommended Scan Parameters for Host-Initiated Connection of Bonded Devices

5.2.5 Link Loss Reconnection Procedure

5.2.6 Fast Connection Interval

Parameter	Value
Minimum Connection Interval	7.5 ms
Maximum Connection Interval	50 ms

Table 5.7: Recommended Fast Connection Parameters

Generic Access Profile

- This profile defines the generic procedures related to discovery of Bluetooth devices (idle mode procedures) and link management aspects of connecting to Bluetooth devices (connecting mode procedures).
- It also defines procedures related to use of different security levels.
- Essentially this profile describes how the lower layers (LMP and Baseband) are used, along with some higher layers.

1.1 Profile Overview

1.1.1 Profile Stack

- The main purpose of this profile is to describe the use of the lower layers of the Bluetooth protocol stack (LC and LMP).
- To describe security related alternatives, also higher layers (L2CAP, RFCOMM and **OBEX) are included.**

2.1 PROFILE STACK

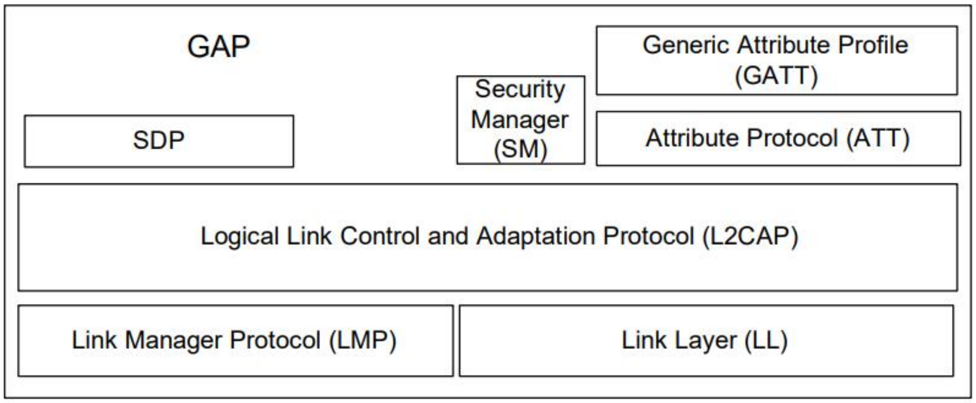


Figure 2.1: Relationship of GAP with lower layers of the Bluetooth architecture

The purpose of this profile is to describe:

- Profile roles
- Discoverability modes and procedures
- Connection modes and procedures
- Security modes and procedures

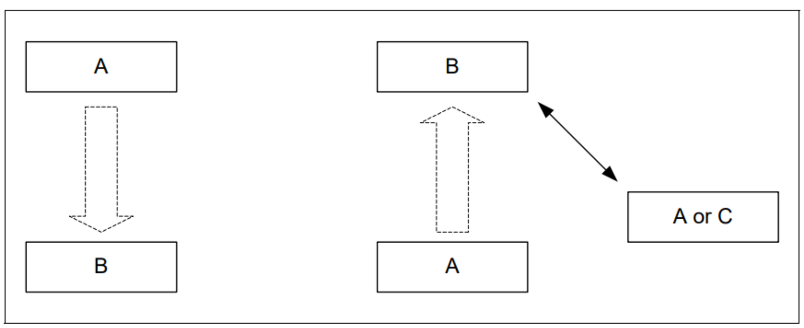


Figure 2.2: This profile covers procedures initiated by one device (A) towards another device (B) whether or not they have an existing Bluetooth link active

2.2.1 Roles when operating over BR/EDR Physical Transport

- In GAP, for describing the Bluetooth communication that occurs between two devices in the BR/EDR GAP role, the generic notation of the A-party (the paging device in case of link establishment, or initiator in case of another procedure on an established link) and the B-party (paged device or acceptor) is used.

- The A-party is the one that, for a given procedure, initiates device discovery, initiates the establishment of a physical link or initiates a transaction on an existing link.
- The initiator and the acceptor generally operate the generic procedures according to this profile or another profile referring to this profile.
- If the acceptor operates according to several profiles simultaneously, this profile describes generic mechanisms for how to handle this.

2.2.2 Roles when operating over an LE physical transport

2.3. Observers and Broadcasters vs. Centrals and Peripherals

Let's go over some advantages and disadvantages of the four different types of device: Observers, Broadcasters, Centrals, and Peripherals.

Broadcaster	Peripheral	Observer	Central
No need for a radio receiver	Needs both a receiver and transmitter	No need for a transmitter	Needs both a receiver and transmitter
No bi-directional data transfer	Supports bi-directional data transfer	No bi-directional data transfer	Supports bi-directional data transfer
Reduced hardware, reduced BLE software stack	Requires the full BLE software stack	Reduced hardware, reduced BLE software stack	Requires the full BLE software stack

Table 2: *Comparison between Observers, Broadcasters, Peripherals, and Centrals*

There are four GAP roles defined for devices operating over an LE physical transport:

• Broadcaster = A device operating in the Broadcaster role is a device that sends advertising events or periodic advertising events and may also send Broadcast Isochronous Stream (BIS) events.

as A device operating in the Broadcaster role is referred to as a Broadcaster and shall have a transmitter and may have a receiver.

• Observer = A device operating in the Observer role is a device that receives advertising events or periodic advertising events and may also receive BIS events . A device operating in the Observer role is referred to as an Observer and shall have a receiver and may have a transmitter.

• Peripheral = Any device that accepts the establishment of an LE active physical link using any of the connection establishment procedures to as being in the Peripheral role. A device operating in the Peripheral role will be in the Slave role in the Link Layer Connection state. A device operating in the Peripheral role is referred to as a Peripheral. A Peripheral shall have both a transmitter and a receiver.

Bluetooth peripheral devices are **devices you can connect to your computer using Bluetooth**. They can range from input devices like mice, keyboards to output devices like printers and even multimedia devices like digital cameras

• Central = A device that supports the Central role initiates the establishment of an LE active physical link. A device operating in the Central role will be in the Master role in the Link Layer Connection state. A device operating in the Central role is referred to as a Central. A Central shall have both a transmitter and a receiver.

4 MODES – BR/EDR PHYSICAL TRANSPORT

Procedure	Ref.	Support
Discoverability modes:	4.1	
Non-discoverable mode		C1
Discoverable mode		O
Limited discoverable mode		C3
General discoverable mode		C3
Connectability modes:	4.2	
Non-connectable mode		O
Connectable mode		M
Bondable modes:	4.3	
Non-bondable mode		O
Bondable mode		C2
Synchronizability modes:	4.4	
Non-synchronizable mode		M
Synchronizable mode		O
C1: Mandatory if limited discoverable mode is supported, otherwise optional.		
C2: Mandatory if the bonding procedure is supported, otherwise optional.		

4.1.1 Non-discoverable mode

4.1.1.1 Definition

When a Bluetooth device is in non-discoverable mode, it shall never enter the INQUIRY_SCAN state.

4.1.2 Limited Discoverable mode

4.1.2.1 Definition

The limited discoverable mode should be used by devices that need to be discoverable only for a limited period of time, during temporary conditions, or for a specific event.

4.1.3 General Discoverable mode

4.1.3.1 Definition

- The general discoverable mode shall be used by devices that need to be discoverable continuously or for no specific condition.
- Devices in the general discoverable mode will not be discovered by devices performing the limited inquiry procedure.

4.2.1 Non-connectable mode

.2.1.1 Definition

When a Bluetooth device is in non-connectable mode it shall never enter the PAGE_SCAN state.

4.2.2 Connectable mode

4.2.2.1 Definition

- When a Bluetooth device is in connectable mode it shall periodically enter the PAGE_SCAN state.
- The device performs page scan using the Bluetooth device address, BD_ADDR.
- Connection speed is a trade-off between power consumption / available bandwidth and speed.
- The Bluetooth Host is able to make these trade-offs using the Page Scan interval, Page Scan window, and Interlaced Scan parameters.

4.3 BONDABLE MODES = What does bonded mean in Bluetooth?

Bonding is **the exchange of long-term keys after pairing occurs, and storing those keys for later use** — it is the creation of permanent security between devices.

4.3.1 Non-bondable mode

4.3.1.1 Definition

- When a Bluetooth device is in non-bondable mode it shall not accept a pairing request that results in bonding.
- Devices in non-bondable mode may accept connections that do not request or require bonding.
- A device in non-bondable mode shall respond to a received LMP_IN_RAND with LMP_NOT_ACCEPTED with the reason pairing not allowed.
- When both devices support Secure Simple Pairing and the local device is in non-bondable mode, the local Host shall respond to an IO capability request where the Authentication_Requirements parameter requests dedicated bonding or general bonding with a negative response.

4.3.2 Bondable mode

4.3.2.1 Definition

- When a Bluetooth device is in bondable mode, and Secure Simple Pairing is not supported by either the local or remote device, the local device shall respond to a received LMP_IN_RAND with LMP_ACCEPTED (or with LMP_IN_RAND if it has a fixed PIN). When both devices support Secure Simple Pairing, the local Host shall respond to a user confirmation request with a positive response.

4.4 SYNCHRONIZABILITY MODES

- A Bluetooth device shall be either in non-synchronizable mode or synchronizable mode.
- When a Bluetooth device is in synchronizable mode, it transmits timing and frequency information for its active Connectionless Slave Broadcast packets.
- When a Bluetooth device is non-synchronizable, timing and frequency information is not transmitted.
- The Host is able to configure the Synchronization Train interval based on trade-offs between bandwidth, potential interference to other devices, power consumption, and the desired time for a slave to receive a synchronization train packet.

4.4.1 Non-synchronizable mode

4.4.1.1 Definition

When a Bluetooth device is in non-synchronizable mode it shall never enter the ***Synchronization Train substate.***

4.4.2 Synchronizable mode

4.4.2.1 Definition

- When a Bluetooth device is in synchronizable mode, it shall enter the Synchronization Train substate using a synchronization train interval of TGAP(Sync_Train_Interval).
- After being made synchronizable, the Bluetooth device shall be synchronizable for at least TGAP(Sync_Train_Duration).

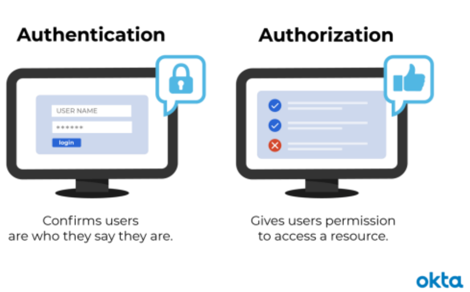
5 SECURITY ASPECTS – BR/EDR PHYSICAL TRANSPORT

	Procedure	Ref.	Support
1	Authentication	5.1	M
2	Security modes	5.2	
	Security mode 1		E
	Security mode 2		O.1
	Security mode 3		E
	Security mode 4		M
O.1: Security mode 2 may only be used for backwards compatibility when the remote device does not support Secure Simple Pairing.			

Table 5.1: Conformance requirements related to the generic authentication procedure and the security modes defined in this section

5.1 AUTHENTICATION

The generic authentication procedure describes how the LMP-authentication and LMP-pairing procedures are used when authentication is initiated by one Bluetooth device towards another, depending on if a link key exists or not and if pairing is allowed or not.



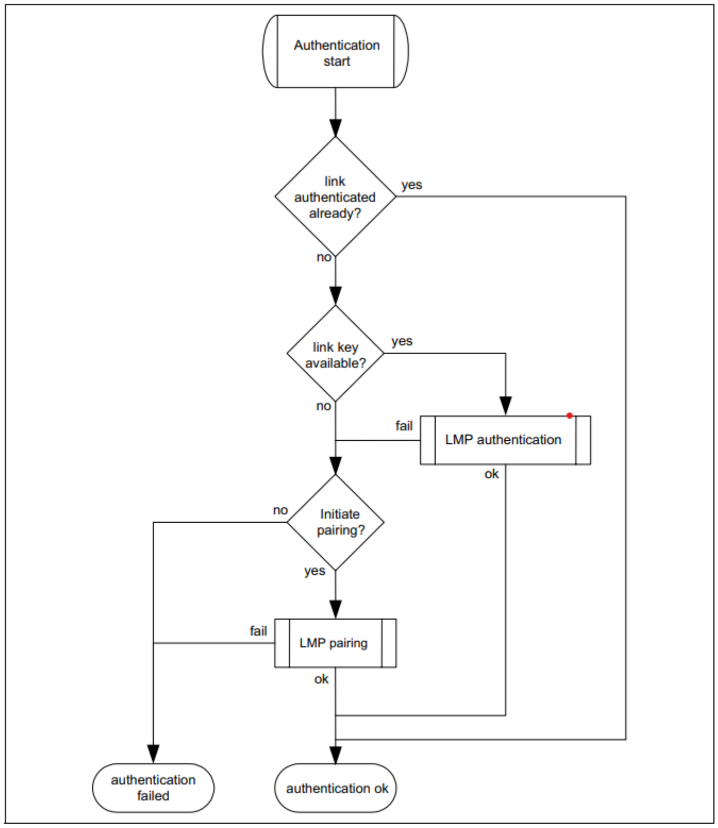


Figure 5.1: Definition of the generic authentication procedure

5.2 SECURITY MODES

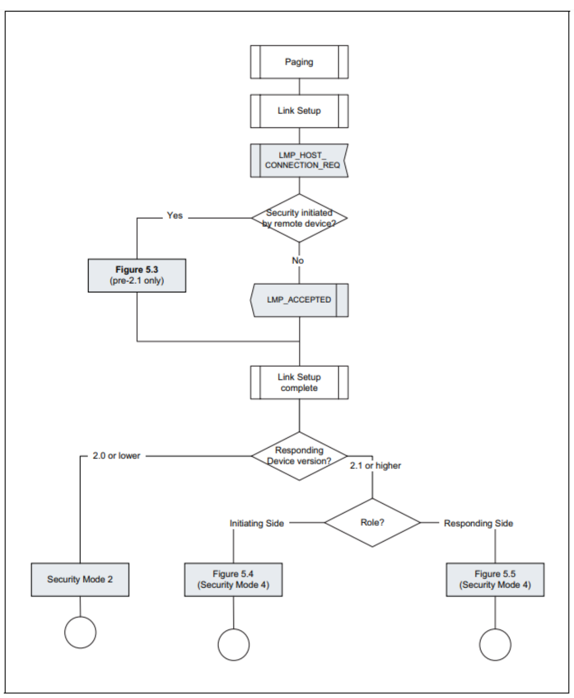


Figure 5.2: Channel establishment with security

5.2.1 Legacy security modes

Legacy security modes apply to those devices with a Controller or a Host that does not support SSP.

5.2.1.1 Security mode 1 (non-secure)

When a remote Bluetooth device is in security mode 1 it will never initiate any security procedure (i.e., it will never send LMP_AU_RAND, LMP_IN_RAND or LMP_ENCRYPTION_MODE_REQ).

5.2.1.2 Security mode 2 (service level enforced security)

- When a remote Bluetooth device is in security mode 2 it will not initiate any security procedure before a channel establishment request (L2CAP_ConnectReq) has been received or a channel establishment procedure has been initiated by itself.
- Whether a security procedure is initiated or not depends on the security requirements of the requested channel or service.

A Bluetooth device in security mode 2 should classify the security requirements of its services using at least the following attributes:

• Authorization required

• Authentication required

• Encryption required

Note: Security mode 1 can be considered (at least from a remote device point of view) as a special case of security mode 2 where no service has registered any security requirements.

5.2.1.3 Security mode 3 (link level enforced security)

When a remote Bluetooth device is in security mode 3 it will initiate security procedures before it sends LMP_SETUP_COMPLETE. A Bluetooth device in security mode 3 may reject the Host connection request (respond with LMP_NOT_ACCEPTED to the LMP_HOST_CONNECTION_REQ) based on settings in the Host (e.g., only communication with pre-paired devices allowed).

5.2.2 Security mode 4 (service level enforced security)

A Bluetooth device in security mode 4 shall classify the security requirements of its services using at least the following attributes (in order of decreasing security):

- Authenticated link key required
- Unauthenticated link key required
- Security optional; limited to specific services

6 IDLE MODE PROCEDURES – BR/EDR PHYSICAL TRANSPORT

The inquiry and discovery procedures described here are applicable only to the device that initiates them (A). The requirements on the behavior of B is according to the modes specified in Section 4 and to [2].

	Procedure	Ref.	Support
1	General inquiry	6.1	C1
2	Limited inquiry	6.2	C1
3	Name discovery	6.3	O
4	Device discovery	6.4	O
5	Bonding	6.5	O
C1: If initiation of bonding is supported, support for at least one inquiry procedure is mandatory, otherwise optional. Note: Support for LMP-pairing is mandatory [2].			

- The purpose of the general inquiry procedure is to provide the initiator with the Bluetooth device address, clock, Class of Device, page scan mode, and extended inquiry response information of devices in either general discoverable mode or limited discoverable mode.

- The purpose of the limited inquiry procedure is to provide the initiator with the Bluetooth device address, clock, Class of Device, page scan mode, and extended inquiry response information of limited discoverable devices. The latter devices are devices that are in range with regard to the initiator and set to scan for inquiry messages with the Limited Inquiry Access Code.

- The purpose of name discovery is to provide the initiator with the Bluetooth Device Name of connectable devices (i.e., devices in range that will respond to paging).

- The purpose of device discovery is to provide the initiator with the Bluetooth Device Address, clock, Class of Device, used page scan mode, Bluetooth Device Name, and extended inquiry response information of discoverable devices.

- The purpose of bonding is to create a relation between two Bluetooth devices based on a common link key (a bond). The link key is created and exchanged (pairing) during the bonding procedure and is expected to be stored by both Bluetooth devices, to be used for future authentication. In addition to pairing, the bonding procedure can involve higher-layer initialization procedures.

7 ESTABLISHMENT PROCEDURES – BR/EDR PHYSICAL TRANSPORT

	Procedure	Ref.	Support in A	Support in B
1	Link Establishment	7.1	M	M
2	Channel Establishment	7.2	O	M
3	Connection Establishment	7.3	O	O
4	Synchronization Establishment	7.5	O	O

Table 7.1: Establishment procedures

- The purpose of the Link Establishment procedure is to **establish a logical transport (of ACL type) between two Bluetooth devices** using procedures from [1] and [2].

- The purpose of the Channel Establishment procedure is to **establish a Bluetooth channel (L2CAP channel) between two Bluetooth** devices as described in

- The purpose of the Connection Establishment procedure is to **establish a connection between applications on two Bluetooth devices.**

- The purpose of the Synchronization Establishment procedure is for a device to receive **synchronization train packets using the procedures** in

9 OPERATIONAL MODES AND PROCEDURES – LE PHYSICAL TRANSPORT

Several different modes and procedures may be performed simultaneously over an LE physical transport. The following modes and procedures are defined for use over an LE physical transport:

- Broadcast mode and Observation procedure
- Discovery modes and procedures
- Connection modes and procedures
- Bonding modes and procedures
- Periodic advertising modes and procedure
- Isochronous broadcast modes and procedures

10 SECURITY ASPECTS – LE PHYSICAL TRANSPORT

10.1 REQUIREMENTS

Security Modes and Procedures	Ref.	Broadcaster	Observer	Peripheral	Central
LE Security mode 1	10.2.1	E	E	O	O
LE Security mode 2	10.2.2	E	E	O	O
LE Security mode 3	10.2.5	O	O	E	E
Authentication procedure	10.3	E	E	O	O
Authorization procedure	10.5	E	E	O	O
Connection data signing procedure	10.4.1	E	E	O	O
Authenticate signed data procedure	10.4.2	E	E	O	O

Table 10.1: Requirements related to security modes and procedures

10.2 LE SECURITY MODES

The security requirements of a device, a service or a service request are expressed in terms of a security mode and security level. Each service or service request may have its own security requirement. The device may also have a security requirement. A physical connection between two devices shall operate in only one security mode.

10.2.1 LE security mode 1 LE security mode 1 has the following security levels: 1. No security (No authentication and no encryption) 2. Unauthenticated pairing with encryption 3. Authenticated pairing with encryption 4. Authenticated LE Secure Connections pairing with encryption using a 128- bit strength encryption key.

10.2.2 LE security mode 2 LE security mode 2 has two security levels: 1. Unauthenticated pairing with data signing 2. Authenticated pairing with data signing LE security mode 2 shall only be used for connection based data signing.

10.2.5 LE security mode 3 LE security mode 3 has three security levels: 1. No security (no authentication and no encryption) 2. Use of unauthenticated Broadcast_Code 3. Use of authenticated Broadcast_Code

10.3 AUTHENTICATION PROCEDURE The authentication procedure describes how the required security is established when a device initiates a service request to a remote device and when a device receives a service request from a remote device. The authentication procedure covers LE security mode 1. The authentication procedure shall only be initiated after a connection has been established

10.5 AUTHORIZATION PROCEDURE A service may require authorization before allowing access. Authorization is a confirmation by the user to continue with the procedure. Authentication does not necessarily provide authorization. Authorization may be granted by user confirmation after successful authentication.

SERIAL PORT PROFILES

The Serial Port Profile defines the protocols and procedures that shall be used by devices using Bluetooth for RS232 (or similar) serial cable emulation.

Serial Port Profile (SPP)
It's is one of the more fundamental Bluetooth profiles (Bluetooth's original purpose was to replace RS-232 cables after all). Using SPP, each connected device can send and receive data jdcust as if there were RX and TX lines connected between them.

What is serial port in Bluetooth?

The most common type of Bluetooth socket is RFCOMM, which is the type supported by the Android APIs. RFCOMM is **a connection-oriented, streaming transport over Bluetooth**. It is also known as the Serial Port Profile (SPP).

1.2 Bluetooth Profile Structure

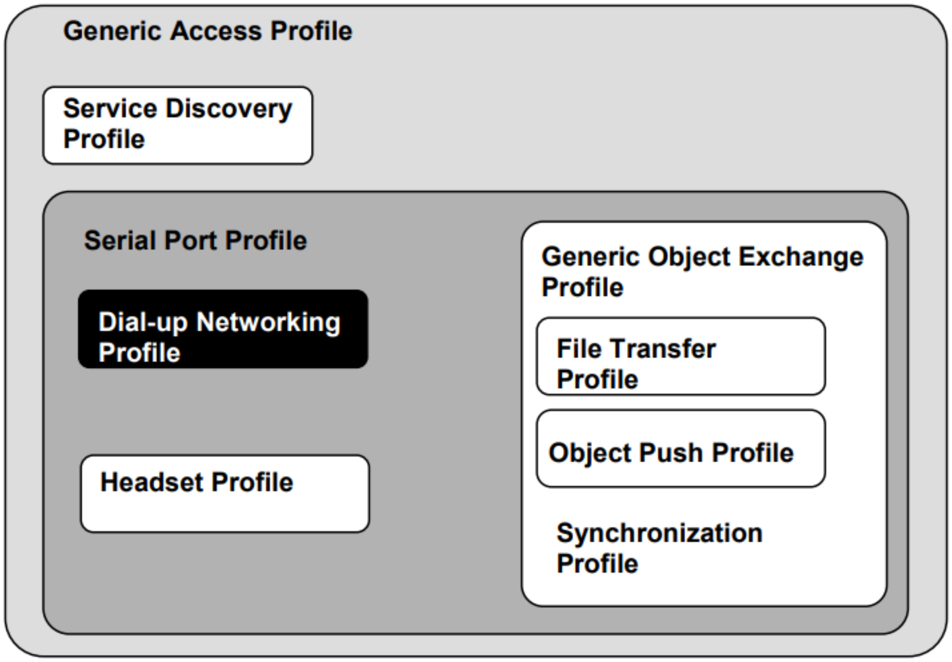


Figure 1.1: Bluetooth Profiles

2.1 Profile Stack

Figure 2.1 shows the protocols and entities used in this profile.

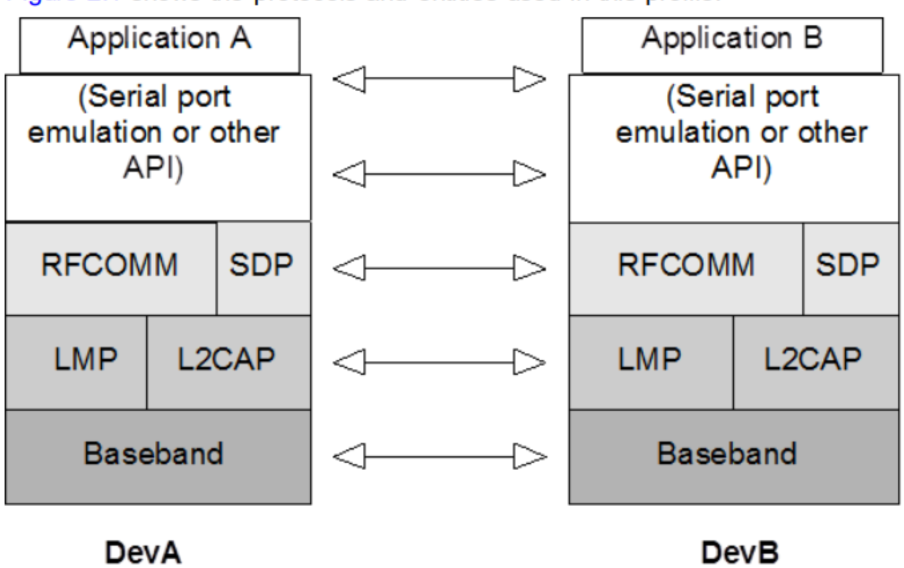


Figure 2.1: Protocol model

2.2 Configurations and roles

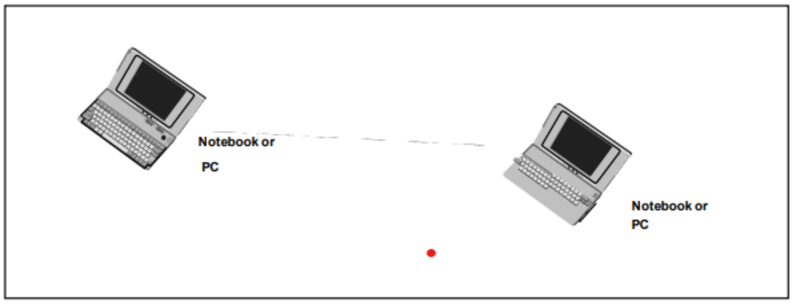


Figure 2.2: Serial Port profile, example with two notebooks.

The following roles are defined for this profile:

Device A (DevA) – This is the device that takes initiative to form a connection to another device.

Device B (DevB) – This is the device that waits for another device to take initiative to connect.

3 Application Layer

3.1 Procedure Overview

	Procedure	**Support in DevA**	**Support in DevB**
1.	Establish link and set up virtual serial connection.	M	X
2.	Accept link and establish virtual serial connection.	X	M
3.	Register Service record for application in local SDP database.	X	M

Table 3.1: Application layer procedures

3.1.1 Establish Link and Set up Virtual Serial Connection

- Submit a query using SDP to find out the RFCOMM Server channel number of the desired application in the remote device. This might include a browsing capability to let the user select among available ports (or services) in the peer device. Alternatively, if it is known exactly which service to contact, it is sufficient look up the necessary parameters using the Service Class ID associated with the desired service.
- Optionally, require authentication of the remote device to be performed. Also optionally, require encryption to be turned on.
- Request a new L2CAP channel to the remote RFCOMM entity.

- Initiate an RFCOMM session on the L2CAP channel.
- Start a new data link connection on the RFCOMM session, using the aforementioned server channel number.

3.1.2 Accept Link and Establish Virtual Serial Connection

- If requested by the remote device, take part in authentication procedure and, upon further request, turn on encryption.
- Accept a new channel establishment indication from L2CAP.
- Accept an RFCOMM session establishment on that channel.
- Accept a new data link connection on the RFCOMM session. This may trigger a local request to authenticate the remote device and turn on encryption, if the user has required that for the emulated serial port being connected to (and authentication/encryption procedures have not already been carried out).

3.1.3 Register Service Record in Local SDP Database

This procedure refers to registration of a service record for an emulated serial port (or equivalent) in the SDP database. This implies the existence of a Service Database, and the ability to respond to SDP queries.

3.2 Power Mode and Link Loss Handling

Since the power requirements may be quite different for units active in the Serial Port profile, it is not required to use any of the power-saving modes.

If sniff, park, or hold mode is used, neither RFCOMM DLCs nor the L2CAP channel is released.

4 RFCOMM Interoperability Requirements

	Procedure	Ability to initiate		Ability to respond	
		DevA	DevB	DevA	DevB
1.	Initialize RFCOMM session	M1	X1	X1	M1
2.	Shutdown RFCOMM session	M	M	M	M
3.	Establish DLC	M	X1	X1	M
4.	Disconnect DLC	M	M	M	M
5.	RS232 control signals	C1	C1	M	M
6.	Transfer information	M	M	N/A1	N/A1
7.	Test command	X	X	M	M
8.	Aggregate flow control	C1	C1	M	M
9.	Remote Line Status indication	O	O	M	M
10.	DLC parameter negotiation	O	O	M	M
11.	Remote port negotiation	O	O	M	M

Table 4.1: RFCOMM capabilities

5 L2CAP Interoperability Requirements

	Procedure	Support in DevA/DevB
1.	Channel types	
	Connection-oriented channel	M
	Connectionless channel	X1
2.	Signaling	
	Connection Establishment	M
	Configuration	M
	Connection Termination	M
	Echo	M
	Command Rejection	M
3.	Configuration Parameter Options	
	Maximum Transmission Unit	M
	Flush Timeout	M
	Quality of Service	O
	Retransmission and Flow Control	O

Table 5.1: L2CAP procedures

5.1 Channel Types

In this profile, only connection-oriented channels shall be used. This implies that broadcast data and unicast connectionless data will not be used in this profile.

5.2 Signaling

Only DevA may issue an L2CAP Connection Request within the execution of this profile. Other than that, the Serial Port Profile does not impose any additional restrictions or requirements on L2CAP signaling.

6 SDP Interoperability Requirements

6.1 SDP Service Records for Serial Port Profile

There are no SDP Service Records related to the Serial Port Profile in DevA. The following table is a description of the Serial Port related entries in the SDP database of DevB. It is allowed to add further attributes to this service record. Serial Port Profile versions earlier than v1.2 did not mandate the BluetoothProfileDescriptorList attribute. New DevA devices shall assume that a remote DevB without this attribute does not support Serial Port Profile version 1.2 or greater.

7 Link Manager (LM) Interoperability Requirements

7.1 Capability Overview

this profile also requires support for Encryption both in DevA and DevB.

7.2 Error Behavior

If a unit tries to use a mandatory feature, and the other unit replies that it is not supported, the initiating unit shall send an LMP_detach with detach reason “unsupported LMP feature.” A unit shall always be able to handle the rejection of the request for an optional feature.

7.3 Link Policy

There are no fixed master-slave roles for the execution of this profile. This profile does not state any requirements if a low-power mode is used, or when it is used. That is up to the Link Manager of each device to decide and request as seen appropriate, within the limitations of the latency requirement stated in Section 5.3.3

8 Link Control (LC) Interoperability Requirements

8.1 Inquiry

When inquiry is invoked in DevA, it shall use the General Inquiry procedure; see GAP [9], Section 6.1. Only DevA may inquire within the execution of this profile.

8.2 Inquiry Scan

For inquiry scan, (at least) the GIAC shall be used, according to one of the discoverable modes defines in GAP

That is, it is allowed to use only the Limited discoverable mode, if appropriate for the application(s) residing in DevB.

In the DevB INQUIRY RESPONSE messages, the Class of Device field will not contain any hint whether DevB is engaged in the execution of the Serial Port Profile or not (due to generalized Serial Port service for legacy applications delivered by this profile does not fit within any of the major Service Class bits in the Class Of Device field definition.)

8.3 Paging

Only DevA may page within the execution of this profile. The paging step will be skipped in DevA when execution of this profile begins when there already is a baseband connection between DevA and DevB. (In such a case, the connection may have been set up as a result of previous paging by DevB.)

8.4 Error Behavior

Since most features on the LC level have to be activated by LMP procedures, errors will mostly be caught at that layer. However, there are some LC procedures that are independent of the LMP layer; e.g., inquiry or paging. Misuse of such features is difficult or sometimes impossible to detect. There is no mechanism defined to detect or prevent such improper use.

Advanced Audio Distribution Profile (A2DP)

0.95, 1.0, 1.2, 1.3, 1.3.1, 1.3.2, 1.4

- This profile defines how multimedia audio can be streamed from one device to another over a Bluetooth connection (it is also called Bluetooth Audio Streaming).

- Advanced audio distribution profile (A2DP) defines how audio can be transmitted from one Bluetooth device to another.
- Where HFP and HSP send audio to and from both devices, A2DP is a one-way street, but the audio quality has the potential to be *much* higher.
- A2DP is well-suited to wireless audio transmissions between an MP3 player and a Bluetooth-enabled stereo.

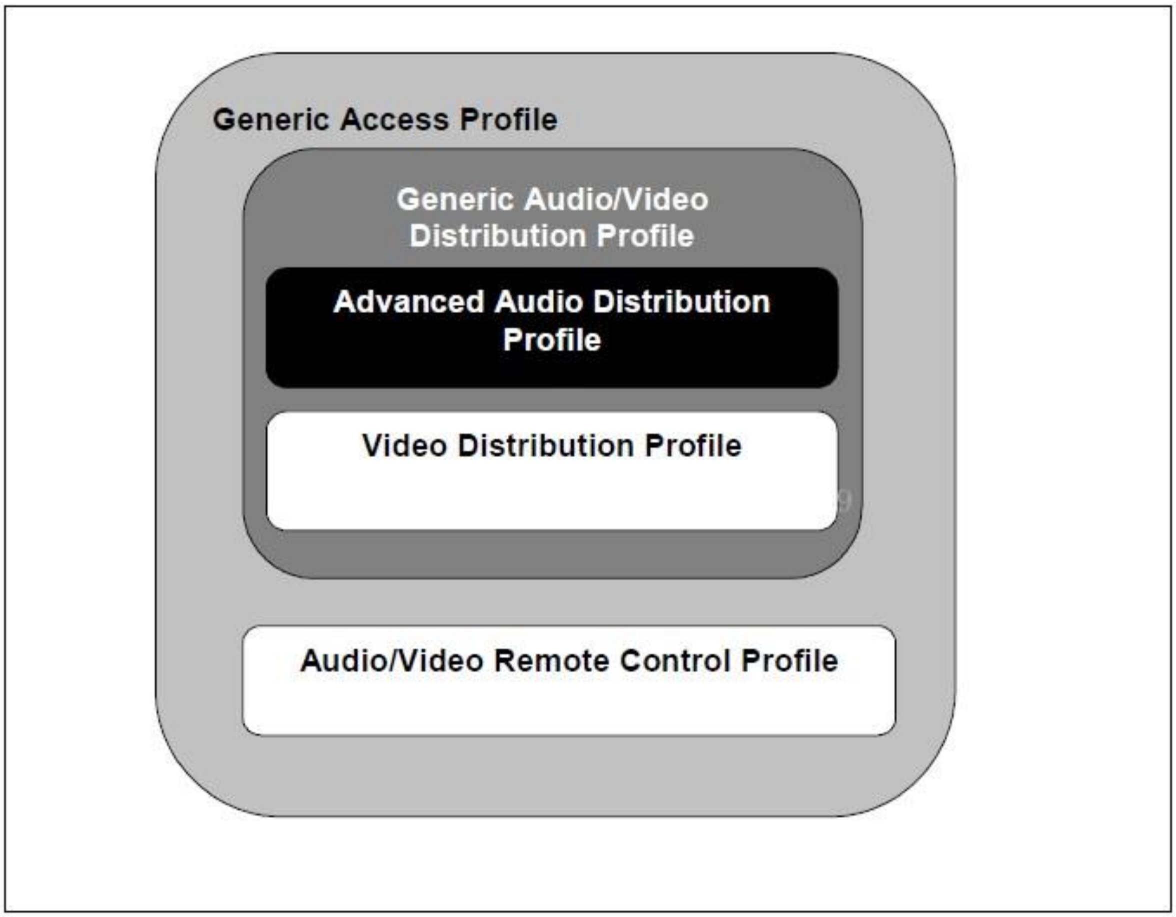
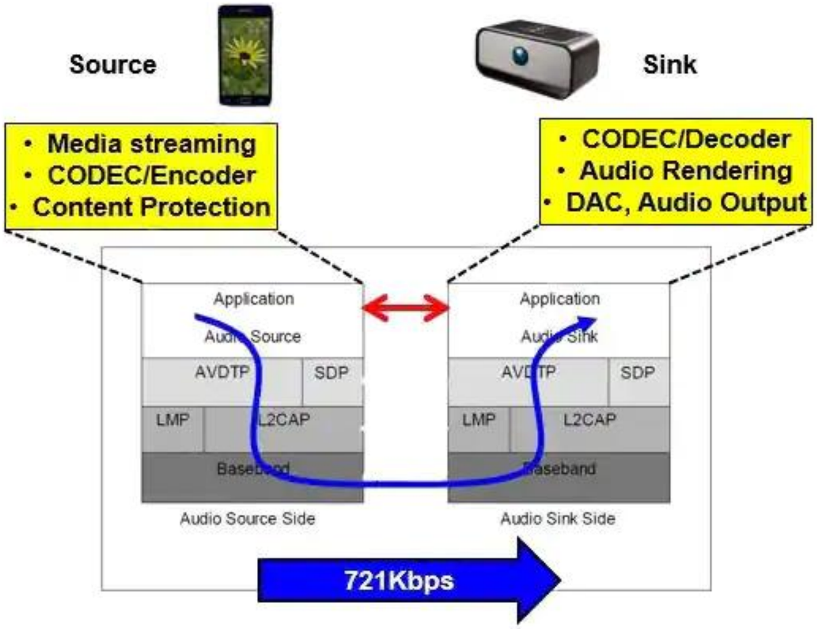


Figure 1.1: Profile Dependencies

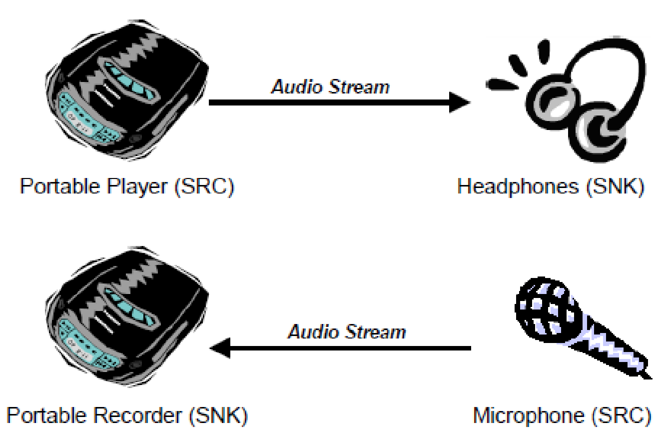


2.2 Configurations and roles

The following roles are defined for devices that implement this profile:

Source (SRC) – A device is the SRC when it acts as a source of a digital audio stream that is delivered to the SNK of the piconet.

Sink (SNK) – A device is the SNK when it acts as a sink of a digital audio stream delivered from the SRC on the same piconet.



- Source (SRC)

Source of digital audio stream that is delivered to the sink of the piconet

Media player, phone, PC

- Sink (SNK)

Acts as a sink of the digital audio stream that is delivered by the source

Stereo headset, wireless speakers, car audio system

2.3 User requirements and scenarios

The following scenario is covered by this profile:

• Setup/control/manipulate a streaming of audio data from the SRC to the SNK(s).

The following restrictions are applied to this profile:

1. The profile does not support a synchronized point-to-multipoint distribution.

2. There exists certain delay between the SRC and the SNK because of radio signal processing, data buffering, and encode/decode of the stream data.

- Countering the effects of such delays depends on implementation.

The following requirements are set in this profile:

1. The audio data rate should be sufficiently smaller than usable bit rate on the Bluetooth link. This is to allow retransmission schemes to reduce the effects of packet loss.

2. The profile does not exclude any content protection method.

2.4 Profile fundamentals

- The profile fundamentals are same as defined in the GAVDP in addition to the following requirement.
- Content Protection is provided at the application level and is not a function of the Bluetooth link level security protocol.

2.5 Conformance

- If conformance to this specification is claimed, all capabilities indicated as mandatory for this specification shall be supported in the specified manner (process-mandatory).
- This also applies for all optional and conditional capabilities for which support is indicated.

3 Application layer

3.1 Audio Streaming setup

- If the AVDTP(Audio/Video Distribution Transport Protocol) version of the remote device is unknown, the device shall perform an SDP query to get the AVDTP version on the remote device.
- This shall be performed before then GAVDP_ConnectionEstablishment procedure is performed.
- This is required because some commands in the audio streaming setup procedure depend on the AVDTP version.

3.2 Audio Streaming

- Once streaming connection is established and the Start Streaming procedure in GAVDP is executed, both the SRC and the SNK are in the STREAMING state, in which the SRC (SNK) is ready to send (receive) an audio stream.
- The SRC uses the Send Audio Stream procedure to send audio data to the SNK, which in turn employs the Receive Audio Stream procedure to receive the audio data.
- Note again that the devices shall be in the STREAMING state to send/receive audio stream.
- If the SRC/SNK is scheduled to send/receive the audio stream whereas the state is still at OPEN, the SRC/SNK shall initiate Start Streaming procedure defined in GAVDP.

3.2.1 Send Audio Stream

- In the Send Audio Stream procedure, the SRC shall, if needed, encode the data into a selected format in the signaling session.
- Then, the application layer of the SRC shall adapt the encoded data into the defined media payload format.
- When content protection is in use, a content protection header may precede encrypted audio content.

3.2.2 Receive Audio Stream

- The AVDTP entity of the SNK shall receive the stream data from the transport channel using the selected transport services and pass it to the application layer by exposed interface.
- When a content protection method is active, the application layer of the SNK shall process the retrieved AVDTP payload as described by the content protection method.
- Typically, this processing entails content protection header analysis and decryption of associated encrypted content. If applicable, the frame of audio data shall be decoded according to the selected coding format.

Figure 3.1: Block diagram of Audio Streaming procedures and the packet format

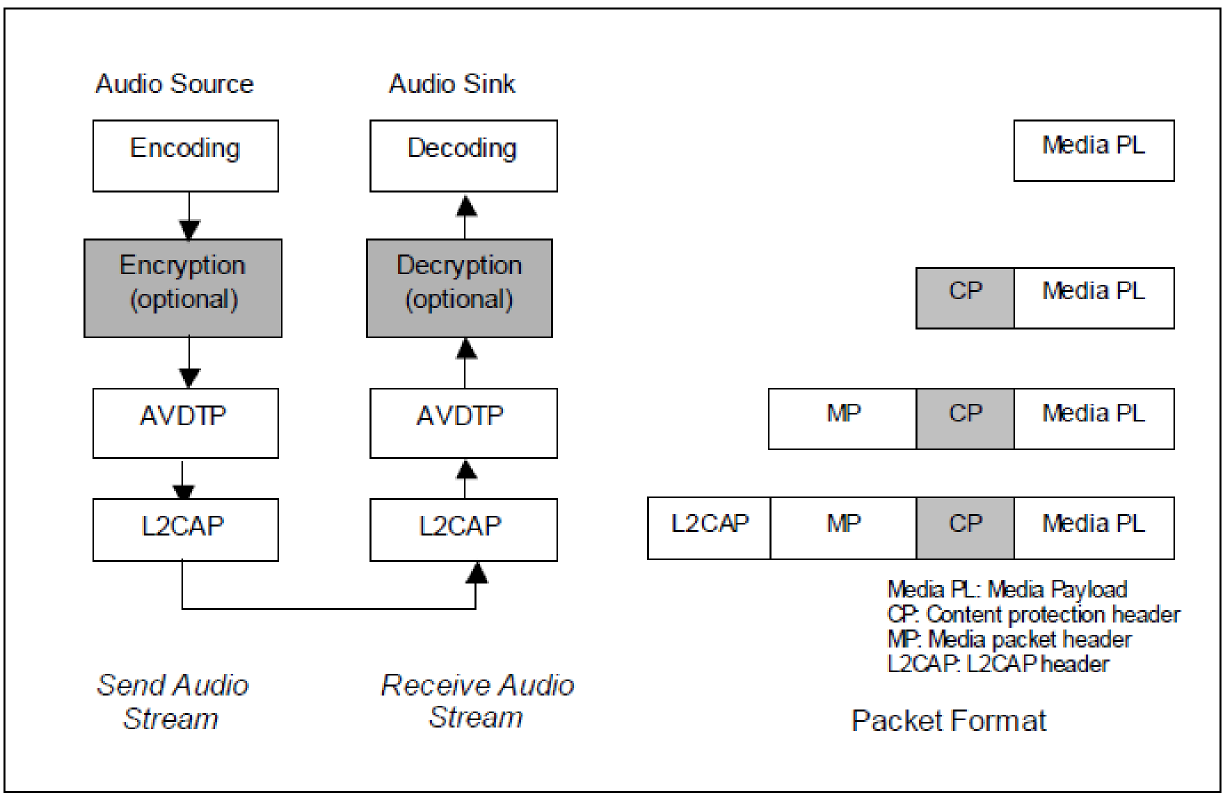


Figure 3.1: Block Diagram of Audio Streaming Procedures and the Packet Format

What is Avdtp in Bluetooth?

AVDTP **defines the signaling mechanism between Bluetooth devices for stream set-up and media streaming of audio or video using ACL links**. A/V streaming and signaling set-up messages are transported via L2CAP packets.

What is Avdtp?

AVDTP consists of **a signaling entity for negotiation of streaming parameters and a transport entity that handles the streaming**.

C.1 Audio Streaming setup

The initial states of both devices are .

- The SRC initiates the Stream Endpoint (SEP) Discovery procedure.
- This procedure serves to return the media type and Stream Endpoint Identifier (SEID) for each stream endpoint.
- The SRC finds the audio type stream endpoint.

Then, the Get All Capabilities procedure is initiated to collect service capabilities of the SNK.

- ***<u>There are two kinds of service capabilities;</u>***
- one is an application service capability, and the other is a transport service capability.
- The application service capability for A2DP consists of audio codec capability and content protection capability.
- Based on collected SEP information and service capabilities, the SRC determines the most suitable audio streaming parameters (codec, content protection and transport service) for the SNK and the SRC itself.

- Then, the SRC requests the SNK to configure the audio parameters of the SNK by using the Stream Configuration procedure.

- The SRC also configures the audio parameters of itself. Then, L2CAP channels are established as defined in the Stream Establishment procedure.

- Finally, the states of both devices are set at .

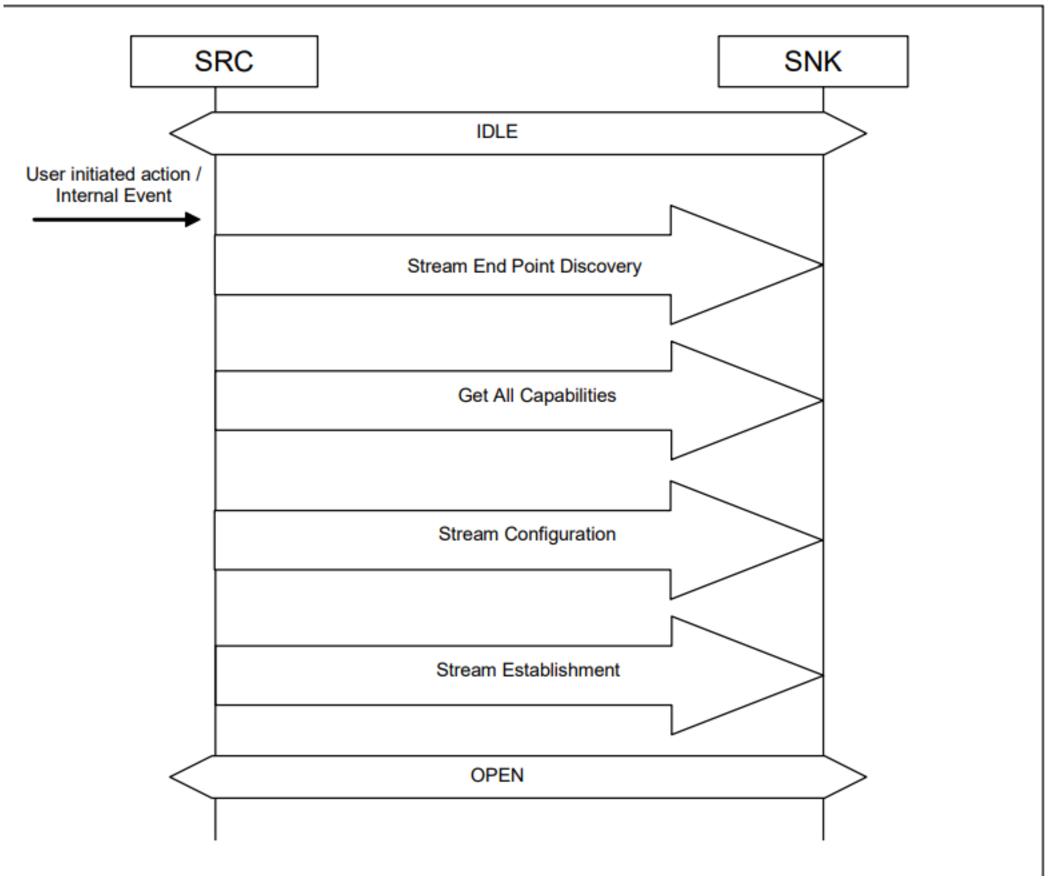


Figure 8.6: Audio Streaming setup

C.2 Audio Streaming

The SRC initiates the Start Streaming procedure by a user-initiated action or an internal event. The states of both devices are changed from <OPEN> to <STREAMING>. Audio streaming is started after this procedure is completed.

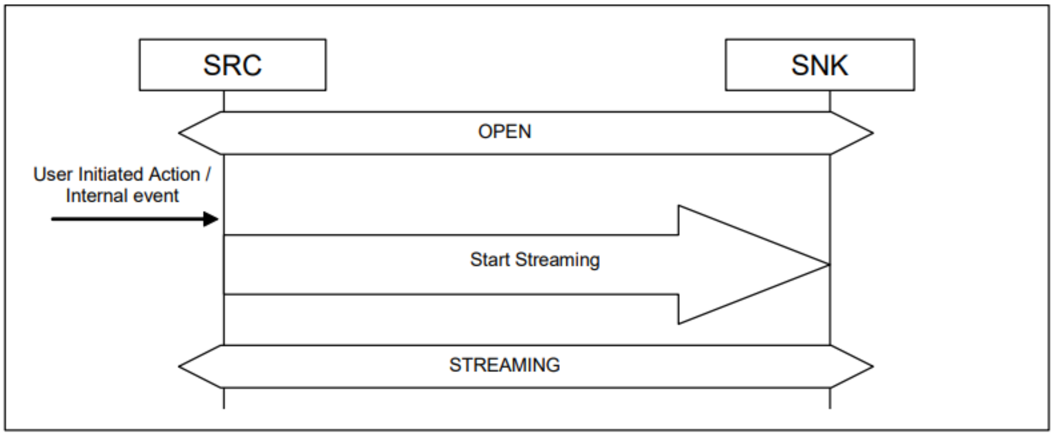
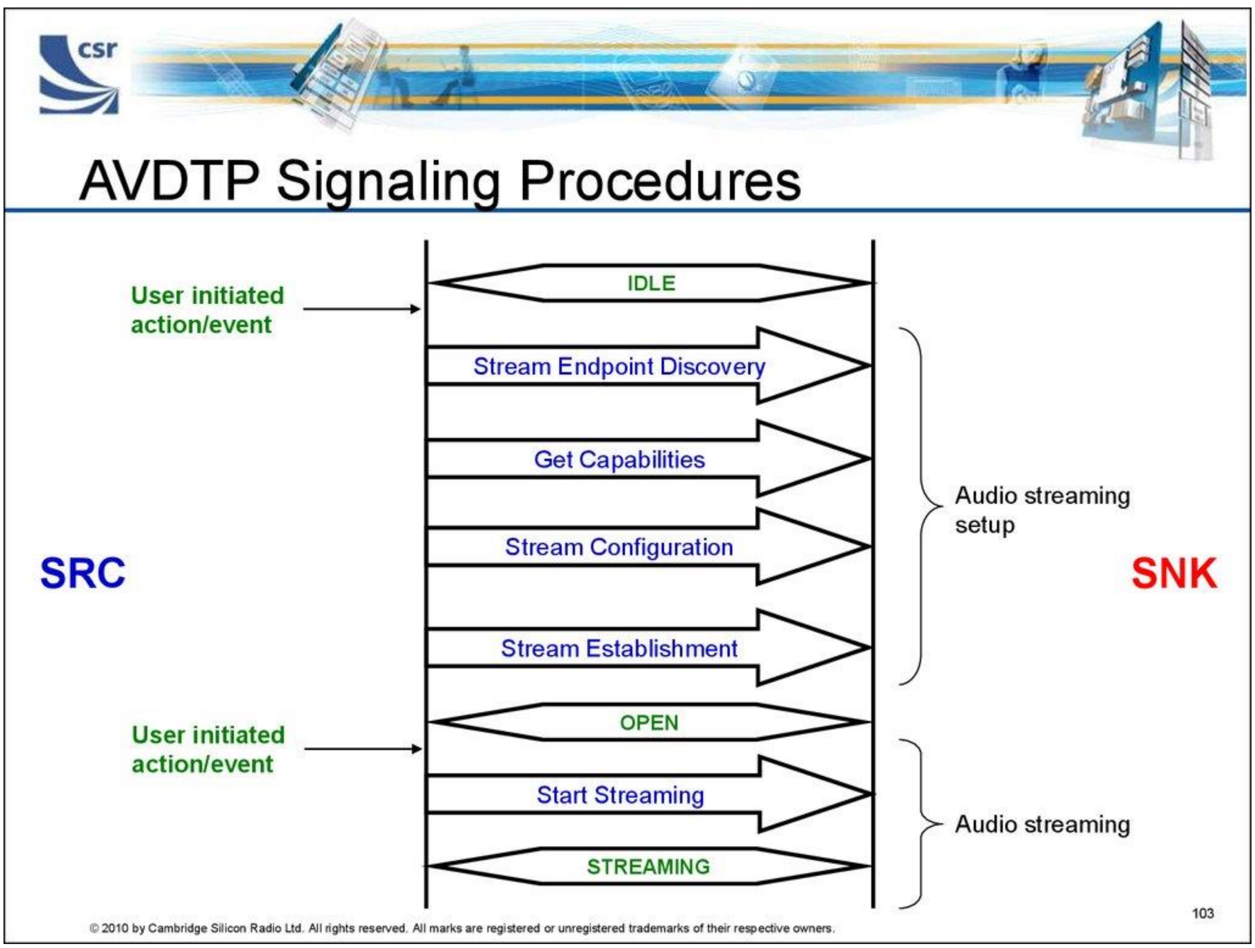


Figure 8.7: Audio Streaming



4.2 Support of codec interoperability requirements

Table 4.1 shows supported codec interoperability requirements.

Codec Interoperability Requirements	Support	Ref.
SBC	M	4.3
MPEG-1,2 Audio	O	4.4
MPEG-2,4 AAC	O	4.5
ATRAC family	O	4.6
MPEG-D USAC	O	4.8

Table 4.1: Supported codec interoperability requirements

When supporting a specific codec interoperability requirement, the device shall use the corresponding Audio Codec Type as defined in Bluetooth Assigned Numbers [5].

The following codec interoperability requirements are treated as Vendor Specific A2DP:

- Codec interoperability requirements that are not in Table 4.1 and Bluetooth Assigned Numbers [5].
- Codec interoperability requirements that do not comply with any in this specification.
- Vendor Specific A2DP codec interoperability requirements defined in Section 4.2.3 and 4.7.

4.3 SBC codec interoperability requirements

Bitstream: A bitstream is binary bits of information (1s and 0s) that can transfer from one device to another.

4.3.2 Codec Specific Information Elements

<table>
<tr><th>7</th><th>6</th><th>5</th><th>4</th><th>3</th><th>2</th><th>1</th><th>0</th><th></th></tr>
<tr><td colspan="4">Sampling Frequency</td><td colspan="4">Channel Mode</td><td>Octet0</td></tr>
<tr><td colspan="4">Block Length</td><td colspan="2">Subbands</td><td colspan="2">Allocation Method</td><td>Octet1</td></tr>
<tr><td colspan="8">Minimum Bitpool Value</td><td>Octet2</td></tr>
<tr><td colspan="8">Maximum Bitpool Value</td><td>Octet3</td></tr>
</table>

Figure 4.1: Codec Specific Information Elements for SBC

4.3.2.1 Sampling Frequency

Table 4.2 shows the value of the Sampling Frequency field for SBC. For the decoder in the SNK, the sampling frequencies 44.1 kHz and 48 kHz are mandatory to support. The encoder in the SRC shall support at least one of the sampling frequencies of 44.1 kHz and 48 kHz.

Position	Sampling Frequency (Hz)	Support in SRC	Support in SNK
Octet0; b7	16000	O	O
Octet0; b6	32000	O	O
Octet0; b5	44100	C1	M
Octet0; b4	48000	C1	M
C1: At least one of the values shall be supported			

Table 4.2: Sampling Frequency for SBC

Definition: Sampling rate or sampling frequency defines **the number of samples per second (or per other unit) taken from a continuous signal to make a discrete or digital signal**.

4.3.2.2 Channel Mode

Table 4.3 shows the value of the Channel Mode field for SBC. For the decoder in the SNK, all features shall be supported. The encoder in the SRC shall support at least MONO and one of DUAL CHANNEL, STEREO and JOINT STEREO modes.

Position	Channel Mode	Support in SRC	Support in SNK
Octet0; b3	MONO	M	M
Octet0; b2	DUAL CHANNEL	C1	M
Octet0; b1	STEREO	C1	M
Octet0; b0	JOINT STEREO	C1	M
C1: At least one of the values shall be supported			

Table 4.3: Channel Mode for SBC

4.3.2.3 Block Length

Table 4.4 shows the value of the Block Length field for SBC. Both the encoder in the SRC and the decoder in the SNK shall support all of the parameters.

Position	Block length	Support in SRC	Support in SNK
Octet1; b7	4	M	M
Octet1; b6	8	M	M
Octet1; b5	12	M	M
Octet1; b4	16	M	M

Table 4.4: Block Length for SBC

Block length = it indicates the block size with which the stream has been encoded

Note = steam: convert you video into a digital format to stream on YouTube.

Sub bands = the bit indicates the number of sub bands which the has been encoded.

ALLOCATION METHOD = this bit indicates the bit pool is allocated to different sub band. Either depends on loudness sub band signal or on the signal to notice ratio.

4.3.2.4 Subbands

Table 4.5 shows the value of Number of Subbands field for SBC. For the decoder in the SNK, all features shall be supported. The encoder in the SRC shall support at least 8 subbands case.

Position	Number of Subbands	Support in SRC	Support in SNK
Octet1; b3	4	O	M
Octet1; b2	8	M	M

Table 4.5: Number of Subbands for SBC

4.3.2.5 Allocation Method

Table 4.6 shows the value of the Allocation Method field for SBC. For the decoder in the SNK, all features shall be supported. The encoder in the SRC shall support at least the LOUDNESS method.

Position	Allocation method	Support in SRC	Support in SNK
Octet1; b1	SNR	O	M
Octet1; b0	Loudness	M	M

Table 4.6: Allocation Method for SBC

4.3.2.6 Minimum / Maximum Bit pool Value

Bit pool is **a parameter that changes encoding bitrate**: the higher it is, the higher the bitrate, and hence the quality. But exact bit pool value corresponds to exact bitrate only within exact profile.

SBC Encoder Settings*	Middle Quality				High Quality			
	Mono		Joint Stereo		Mono		Joint Stereo	
Sampling frequency (kHz)	44.1	48	44.1	48	44.1	48	44.1	48
Bitpool value	19	18	35	33	31	29	53	51
Resulting frame length (bytes)	46	44	83	79	70	66	119	115
Resulting bit rate (kb/s)	127	132	229	237	193	198	328	345
*Other settings: Block length = 16, Allocation method = Loudness, Subbands = 8								

Table 4.7: Recommended sets of SBC parameters in the SRC device

B.5.1 Frame_header

syncword -- The 8-bit string %10011100 or $9C.

sampling_frequency -- Two bits to indicate the sampling frequency with which the stream has been encoded. The sampling frequency f_s is selected conforming to Table 8.16.

sampling_frequency	fs (kHz)
00	16
01	32
10	44.1
11	48

Table 8.16: sampling_frequency

blocks -- Two bits to indicate the block size with which the stream has been encoded. The block size nrof_blocks is selected conforming to Table 8.17.

blocks	nrof_blocks
00	4
01	8
10	12
11	16

Table 8.17: blocks

channel_mode -- Two bits to indicate the channel mode that has been encoded. The variable nrof_channels is derived from this information.

channel_mode	channel mode	nrof_channels
00	MONO	1
01	DUAL_CHANNEL	2
10	STEREO	2
11	JOINT_STEREO	2

Table 8.18: channel_mode

allocation_method -- One bit to indicate the bit allocation method.

Allocation_method	bit allocation method
0	LOUDNESS
1	SNR

Table 8.19: allocation_method

subbands -- One bit to indicate the number of subbands with which the stream has been encoded. The variable nrof_subbands is derived from this information.

Subbands	nrof_subbands
0	4
1	8

Table 8.20: subbands

bitpool -- This is an 8-bit integer to indicate the size of the bit allocation pool that has been used for encoding the stream. The value of the **bitpool** field shall not exceed 16 * **nrof_subbands** for the MONO and DUAL_CHANNEL channel modes and 32 * **nrof_subbands** for the STEREO and JOINT_STEREO channel modes.

crc_check – This 8-bit parity-check word is used for error detection within the encoded stream.

4.4 MPEG-1,2 Audio codec interoperability requirements

What is the purpose of using codec?

A codec is a hardware- or software-based process that compresses and decompresses large amounts of data. Codecs are used in applications **to play and create media files for users, as well as to send media files over a network**.

What is an advantage of a codec?

Advantages: Choice of codecs mean **you can achieve a high rate compression** if you experiment. Plays in mainstream media players such as Windows media player. Can be used as a starting point to create playable DVDs.

What is a codec and how does it work?

What are Codecs? Codecs are essentially **standards of video content compression**. Codecs are made up of two components, an encoder to compress the content, and a decoder to decompress the video content and play an approximation of the original content.

SBC - **Sub-band Coding** - The mandatory and default codec for all stereo Bluetooth headphones with the Advanced Audio Distribution Profile (A2DP). It is capable of bit rates up to 328 kbps with a sampling rate of 44.1kHz.

SBC, or low-complexity subband codec, is **an audio subband codec specified by the Bluetooth Special Interest Group (SIG) for the Advanced Audio Distribution Profile (A2DP)**. *SBC is a digital audio encoder and decoder used to transfer data to Bluetooth audio output devices like headphones or loudspeakers.*

Is AAC better than SBC?

SBC offers a (slightly) better sampling rate and maximum bitrate than **AAC**, though it performs worse, especially on Apple devices. This is because AAC uses a superior compression algorithm, and Apple's phones, tablets, and headphones can run this **potentially battery-guzzling codec** so efficiently.

Is AAC a good codec?

AAC is the highest-quality codec that Apple products support, but they default to transmitting over SBC when paired headphones don't support that codec. So if you use an iPhone (or just about any Bluetooth-capable Apple product), you should make sure your headphones support AAC.

What does AAC codec do?

AAC stands for Advanced Audio Codec and is an audio coding standard used to **compress and encode digital audio files**. It was designed to be the successor to MP3 and claims to offer better audio quality.

What is AAC and SBC in Bluetooth?

AAC is the highest-quality codec that Apple products support, but they default to transmitting over SBC when paired headphones don't support that codec. So if you use an iPhone (or just about any Bluetooth-capable Apple product), you should make sure your headphones support AAC.

What is AAC and SBC codec?

This is the standard audio coding technology used by BLUETOOTH devices. All BLUETOOTH devices support SBC. **AAC**. **This is an abbreviation for Advanced Audio Coding**. AAC is mainly used by Apple products such as iPhones, and it provides higher sound quality than that of SBC.

MPEG-4 stands for "**Moving Pictures Expert Group 4**," a standard for audio and video coding compression. The method reduces the size of an audio or a video file while retaining its fidelity or quality. MPEG-4 is commonly shortened to "MP4." MP4 files can be read by computer applications such as QuickTime and VLC.

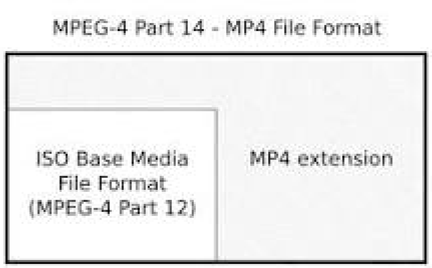
What is MPEG-4 used for?

Uses of MPEG-4 include **compression of AV data for web (streaming media) and CD distribution, voice (telephone, videophone) and broadcast television applications**.

What is MPEG-4 audio format?

What is a M4A file? MPEG-4 audio files with M4A file extension usually contain **digital audio stream encoded with AAC or ALAC (Apple Lossless Audio Codec) compression standards**. This audio container was first used by Apple to differentiate MPEG-4 audio files from the popular MP4 video file container.

Is MPEG-4 a video format?



MPEG-4 Part 14 or MP4 is **a digital multimedia container format** most commonly used to store video and audio, but it can also be used to store other data such as subtitles and still images. Like most modern container formats, it allows streaming over the Internet.

What is an ATRAC file?

Audio format used by Sony electronic devices, such as the PlayStation Portable (PSP) and Sony digital audio players; ATRAC stands for "Adaptive Transform Acoustic Coding" and refers to multiple compression algorithms developed by Sony. NOTE: ATRAC files typically have an . AA3 or.

Acronym	Description
AAC	Advanced Audio Coding
A/V	Audio/Video
A2DP	Advanced Audio Distribution Profile
ACP	Acceptor
AVDTP	Audio/Video Distribution Transport Protocol
AVRCP	Audio/Video Remote Control Profile
CP_Type	Content Protection Type
CRC	Cyclic Redundancy Check
DRC	Dynamic Range Control
ELD	Enhanced Low Delay
GAP	Generic Access Profile
GAVDP	Generic Audio/Video Distribution Profile
HE-AAC	High-Efficiency Advanced Audio Coding
ICS	Implementation Conformance Statement
IETF	Internet Engineering Task Force
INT	Initiator
LM	Link Manager
LMP	Link Manager Protocol
L2CAP	Logical Link Control and Adaptation Protocol
LSB	Least Significant Bit (Byte)
LTP	Long Term Prediction
MPEG	Moving Picture Expert Group
MSB	Most Significant Bit (Byte)
MTU	Maximum Transmission Unit
PSM	Protocol/Service Multiplexer
QoS	Quality of Service
RTP	Real-time Transport Protocol
SBC	Low Complexity Subband Coding
SDP	Service Discovery Protocol
SEID	Stream Endpoint Identifier
SEP	Stream Endpoint
SNK	sink

Acronym	Description
SUL	Sound Unit Length
SRC	source
USAC	Unified Speech and Audio Coding
UiMsbf	Unsigned integer, Most significant bit first
VBR	variable bit rate
VDP	Video Distribution Profile

Table 7.1: Acronyms and abbreviations

HFP (HANDS FREE PROFILE)

0.95, 1.0, 1.5, 1.5.1, 1.6,1.6.1, 1.7, 1.7.1, 1.7.2 ,1.8

Version History

Versions	Changes
v1.0 to v1.5	New features: • Respond and Hold • Subscriber Number Information • Enhanced Call Status • Enhanced Call Controls Incorporated errata 13, 261, 317, 549, 550, 575, 586, 635, 706, 731, 746, 819, 820, 821, 822, and 823.
v1.5 to v1.6	New features: • Wide Band Speech Codec • Codec Negotiation • Individual Indicator Activation Incorporated errata 913, 1859, 1868, 1872, 1878, 1934, 1958, 1989, 2037, 2043, 2144, 2146, 2209, 2211, 2259, 2276, 2286, 2459, 2484, 2713, 2716, 2742, 2855, 3090, 3152, 3688, 3816, and 3910.
v1.6 to v1.7	New feature: HF Indicators Incorporated errata 4718, 4893, 5213, 5336, 5806, and 8739.
v1.7.0 to v1.7.1	Incorporated erratum E6105.
v1.7.1 to v1.7.2	Incorporated errata E6544, E6628, E6835, E8620, E9009, E9034, E9089, E9119, E9122, E9123, E9127, E9158, E9168, E9169, E9170, E9174, E9203, E9204, and E10424.
v1.7.2 to v1.8	New feature: Enhanced Voice Recognition Activation Incorporated erratum E11729.

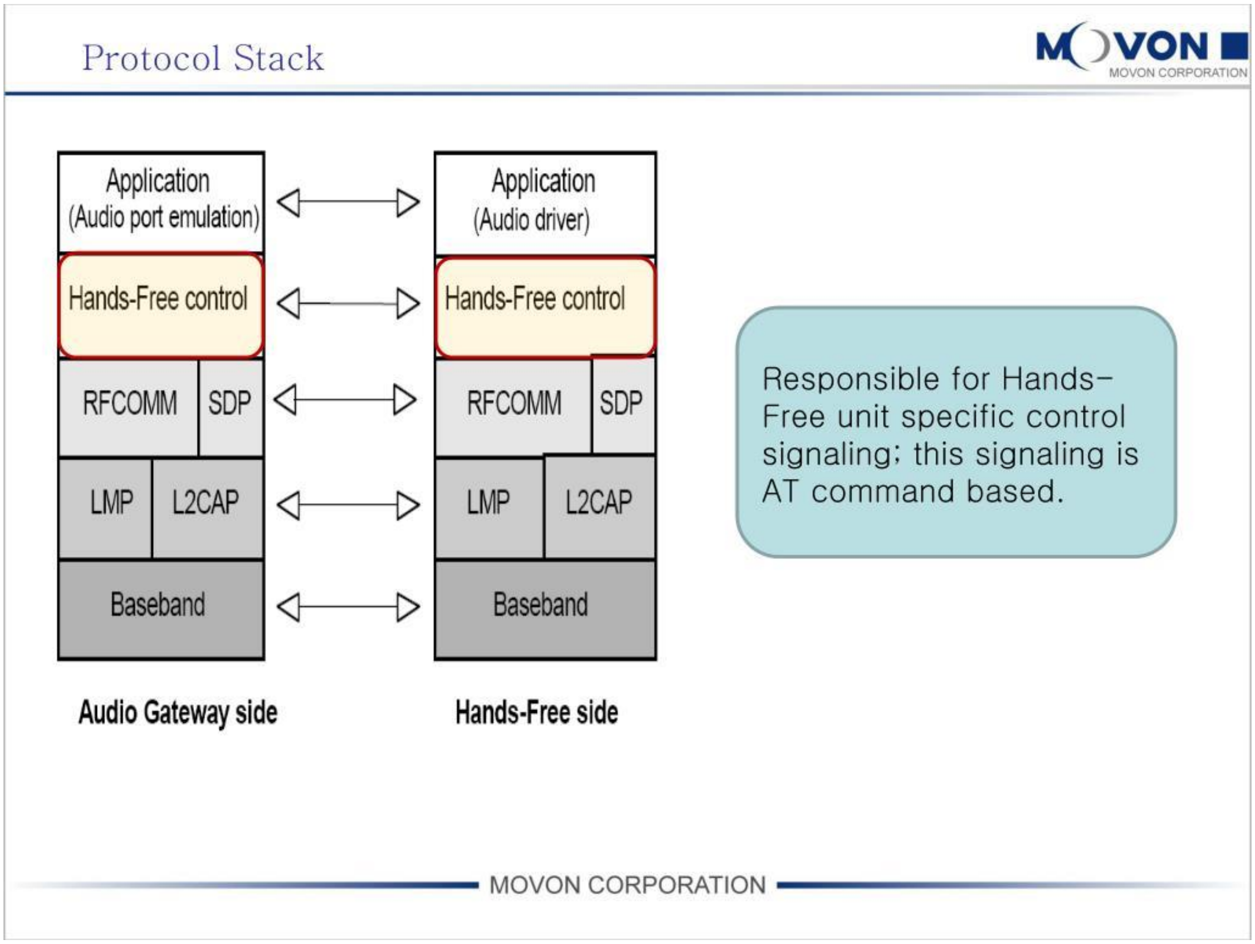
What is HFP in Bluetooth?

Hands-Free Profile. The Bluetooth Hands-Free Profile (HFP) **allows the vehicle to make and receive phone calls via a connected remote device**

- This document defines the protocols and procedures that shall be used by devices implementing the Hands-Free Profile.
- The most common examples of such devices are in-car Hands-Free units used together with cellular phones, or wearable wireless headsets.

- The profile defines how two devices supporting the Hands-Free Profile shall interact with each other on a point-to-point basis.

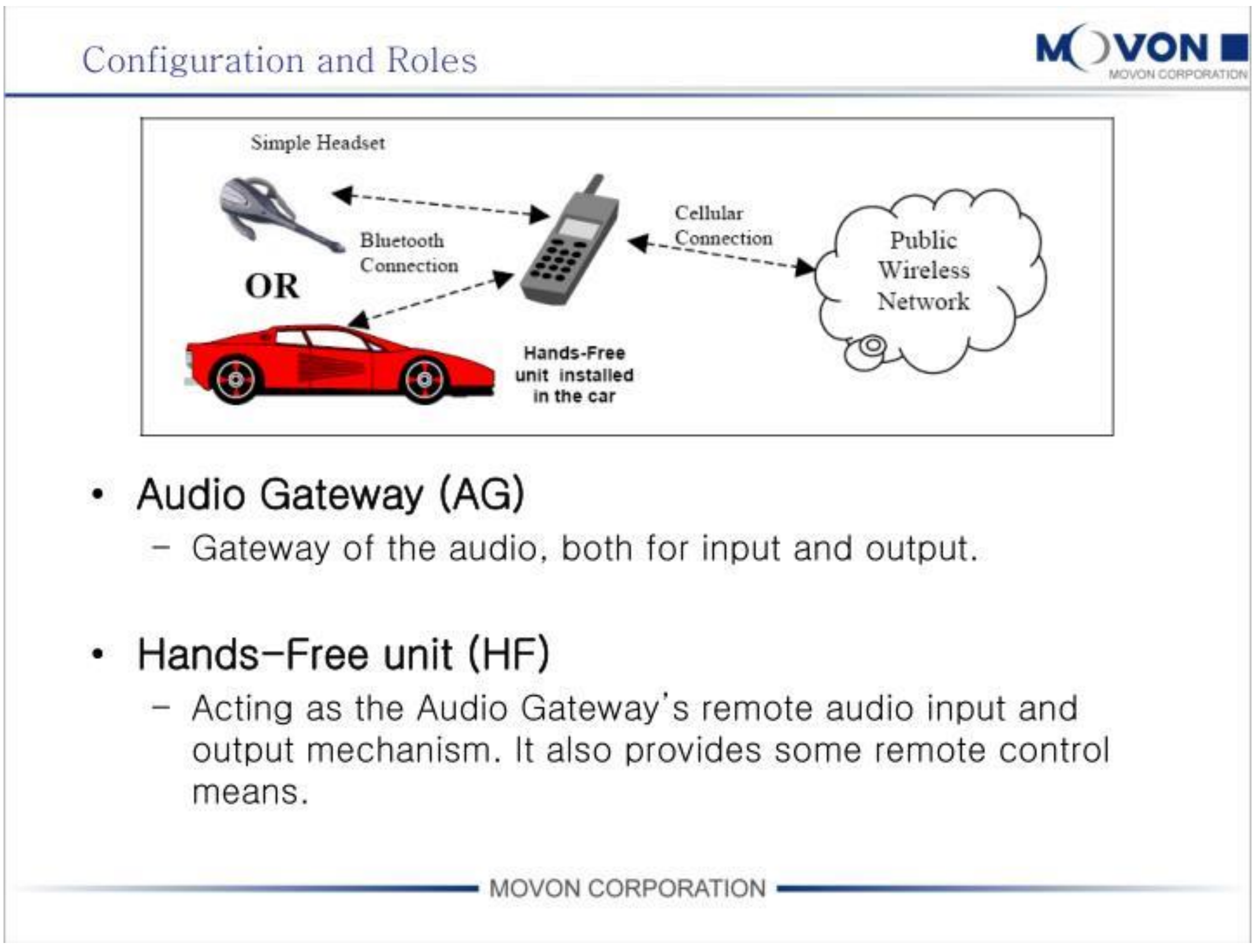
2.1 Protocol Stack



- The Baseband, LMP and L2CAP are the OSI layer 1 and 2 Bluetooth protocols.
- RFCOMM is the Bluetooth serial port emulation entity.
- SDP is the Bluetooth Service Discovery Protocol.
- Compatibility to v1.1 or later Core Specification is required.
- Hands-Free control is the entity responsible for Hands-Free unit specific control signaling.
- this signaling is AT command based. Although not shown in the model above, it is assumed by this profile that Hands-Free Control has access to some lower layer procedures (for example, Synchronous Connection establishment).

- The audio port emulation layer shown in Figure 2.1 is the entity emulating the audio port on the Audio Gateway, and the audio driver is the driver software in the Hands-Free unit.
- For the shaded protocols/entities in Figure 2.1, the Serial Port Profile [5] is used as the base standard.
- For these protocols, all mandatory requirements stated in the Serial Port Profile apply except in those cases where this specification explicitly states deviations.

2.2 Configuration and Roles



The following roles are defined for this profile:

Audio Gateway (AG) – This is the device that is the gateway of the audio, both for input and output. Typical devices acting as Audio Gateways are cellular phones.

Hands-Free unit (HF) – This is the device acting as the Audio Gateway's remote audio input and output mechanism. It also provides some remote control means. These terms are used in the rest of this document to designate these roles.

2.3 User Requirements and Scenarios

The following rules apply to this profile:

- The profile states the mandatory and optional features when the "Hands-Free Profile" is active in the Audio Gateway and the Hands-Free unit.
- The profile mandates the usage of CVSD for transmission of audio (over the Bluetooth link
- c) Between the Hands-Free unit and the Audio Gateway, only one Audio Connection per Service Level Connection at a time is supported.
- d) Both the Audio Gateway and the Hands-Free unit may initiate Audio Connection establishment and release.
- e) Whenever an "Audio Connection" exists, a related "Service Level Connection" shall also exist.
- f) The presence of a "Service Level Connection" shall not imply that an "Audio Connection" exists. Releasing a "Service Level Connection" shall also release any existing "Audio Connection" related to it.

2.4 Profile Fundamentals

- If an LMP link is not already established between the Hands-Free unit and the Audio Gateway, the LMP link shall be set up before any other procedure is performed.
- There are no fixed master or slave roles in the profile.
- The Audio Gateway and Hands-Free unit provide serial port emulation.
- For the serial port emulation, RFCOMM (see [1]) is used.
- The serial port emulation is used to transport the user data including modem control signals and AT command from the Hands-Free unit to the Audio Gateway.
- The AT commands are parsed by the Audio Gateway and responses are sent to the Hands-Free unit via the Bluetooth serial port connection.

2.5 Conformance

- If conformance to this specification is claimed, all capabilities indicated as mandatory for this specification shall be supported in the specified manner (process-mandatory).
- This also applies for all optional and conditional capabilities for which support is indicated.

1.2 Profile Dependencies

In Figure 1.1, the Bluetooth profile structure and the dependencies of the profiles are depicted. A profile is dependent upon another profile if it re-uses parts of that profile, by explicitly referencing it. Dependency is illustrated in Figure 1.1.

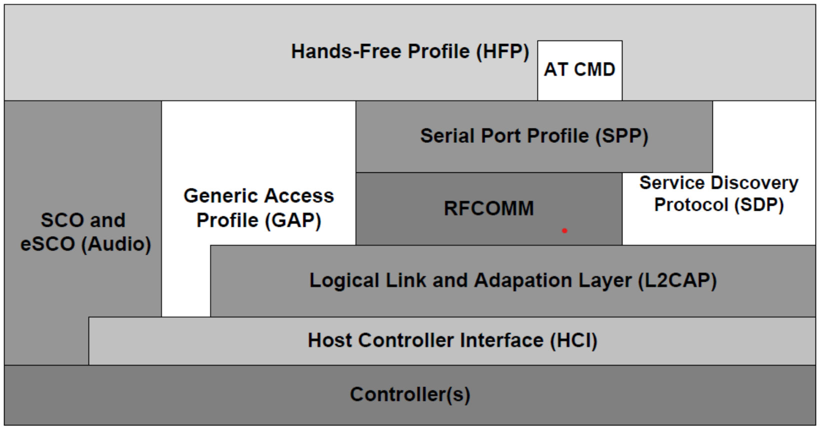


Figure 1.1: Bluetooth Profiles

3 Application Layer

shows the feature requirements for this profile.

Note 1: The HF shall support at least the two indicators "service" and "call".

Note 2: If "Three-way calling" is supported by the HF, it shall support AT+CHLD values 1 and 2. The HF may additionally support AT+CHLD values 0, 3 and 4.

Note 3: If "Three-way calling" is supported by the AG, it shall support AT+CHLD values 1 and 2. The AG may additionally support AT+CHLD values 0, 3, and 4.

Note 4: If Wide Band Speech is supported, Codec Negotiation shall also be supported.

	Feature	Procedure	Ref.
1.	Connection management	Service Level Connection establishment	4.2
		Service Level Connection release	4.3
2.	Phone status information	Transfer of Registration Status Transfer of Signal Strength Indication Transfer of Roaming Status Indication Transfer of Battery Level Indication Query of Operator Selection Extended Audio Gateway Error Codes Transfer of Call, Call Setup and Call Held Status	4.4 4.5 4.6 4.7 4.8 4.9 4.10
3.	Audio Connection handling	Audio Connection set up Audio Connection release Codec Connection set up	4.11 4.12
4.	Accept an incoming voice call	Answer an incoming call	4.13
5.	Reject an incoming voice call	Reject an incoming call	4.14
6.	Terminate a call	Terminate a call process	4.15
7.	Audio Connection transfer during an ongoing call	Audio Connection transfer towards the HF	4.16
		Audio Connection transfer towards the AG	4.17
8.	Place a call with the phone number supplied by the HF	Place a call with the phone number supplied by the HF	4.18
9.	Place a call using memory dialing	Memory dialing from the HF	4.19
10.	Place a call to the last number dialed	Last number re-dial from the HF	4.20
11.	Call waiting notification	Call waiting notification activation	4.21
12.	Three-way calling	Three-way call handling	4.22
13.	Calling Line Identification (CLI)	Calling Line Identification (CLI) notification	4.23
14.	Echo canceling (EC) and noise reduction (NR)	HF requests turning off the AG's EC and NR	4.24
15.	Voice recognition activation	Voice recognition activation	4.25
16.	Attach a phone number to a voice tag	Attach a voice tag to a phone number	4.27

	Feature	Procedure	Ref.
17.	Ability to transmit DTMF codes	Transmit DTMF code	4.28
18.	Remote audio volume control	Remote audio volume control Volume level synchronization	4.29
19.	Response and Hold	Query response and hold status Put an incoming call on hold from HF Put an incoming call on hold from AG Accept a held incoming call from HF Accept a held incoming call from AG Reject a held incoming call from HF Reject a held incoming call from AG Held incoming call terminated by caller	4.30 4.30 4.30 4.30 4.30 4.30 4.30 4.30
20.	Subscriber Number Information	Subscriber Number Information	4.31
21a.	Enhanced Call Status	Query Call List	4.32
21b.	Enhanced Call Control	Release Specified Call Private Consult Mode	4.33 4.33
22.	Individual Indicator Activation	Indicators Activation and Deactivation	4.35
23	Wide Band Speech	Wide Band Speech	9 10 11
24	Codec Negotiation	Codec Negotiation	4.11
25	HF Indicators	HF Indicators	4.36

Table 3.2: Application layer feature to procedure mapping

6 Generic Access Profile

6.1 Modes

Table 6.1 shows the support status for GAP Modes in this profile.

Procedure	Support in HF
General discoverable mode	M
Procedure	**Support in AG**
Pairable mode	M

Table 6.1: Modes

6.3 Idle Mode Procedures

Table 6.2 shows the support status for Idle mode procedures within this profile.

Procedure	Support in AG
Initiation of general inquiry	M
Initiation of general bonding	O
Initiation of dedicated bonding	O

Table 6.2: Idle mode procedures

4 HANDS-FREE CONTROL INTEROPERABILITY REQUIREMENTS

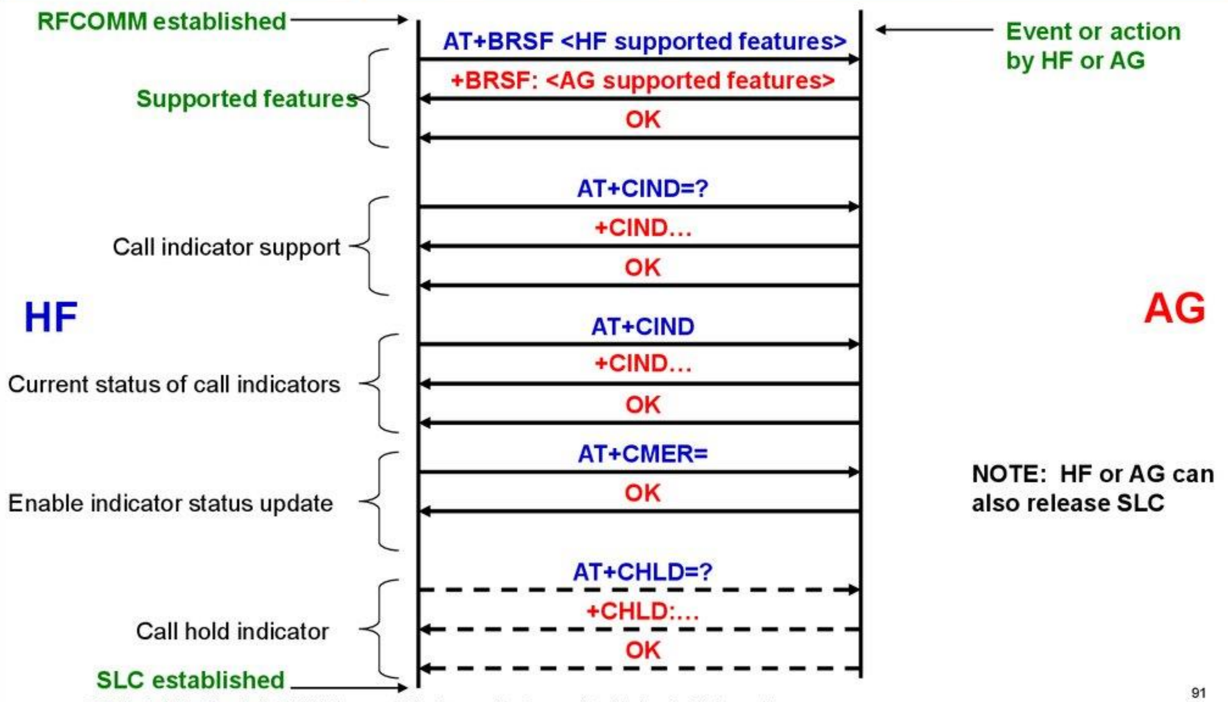
Connection management

1. Service Level Connection establishment
2. Service Level Connection release

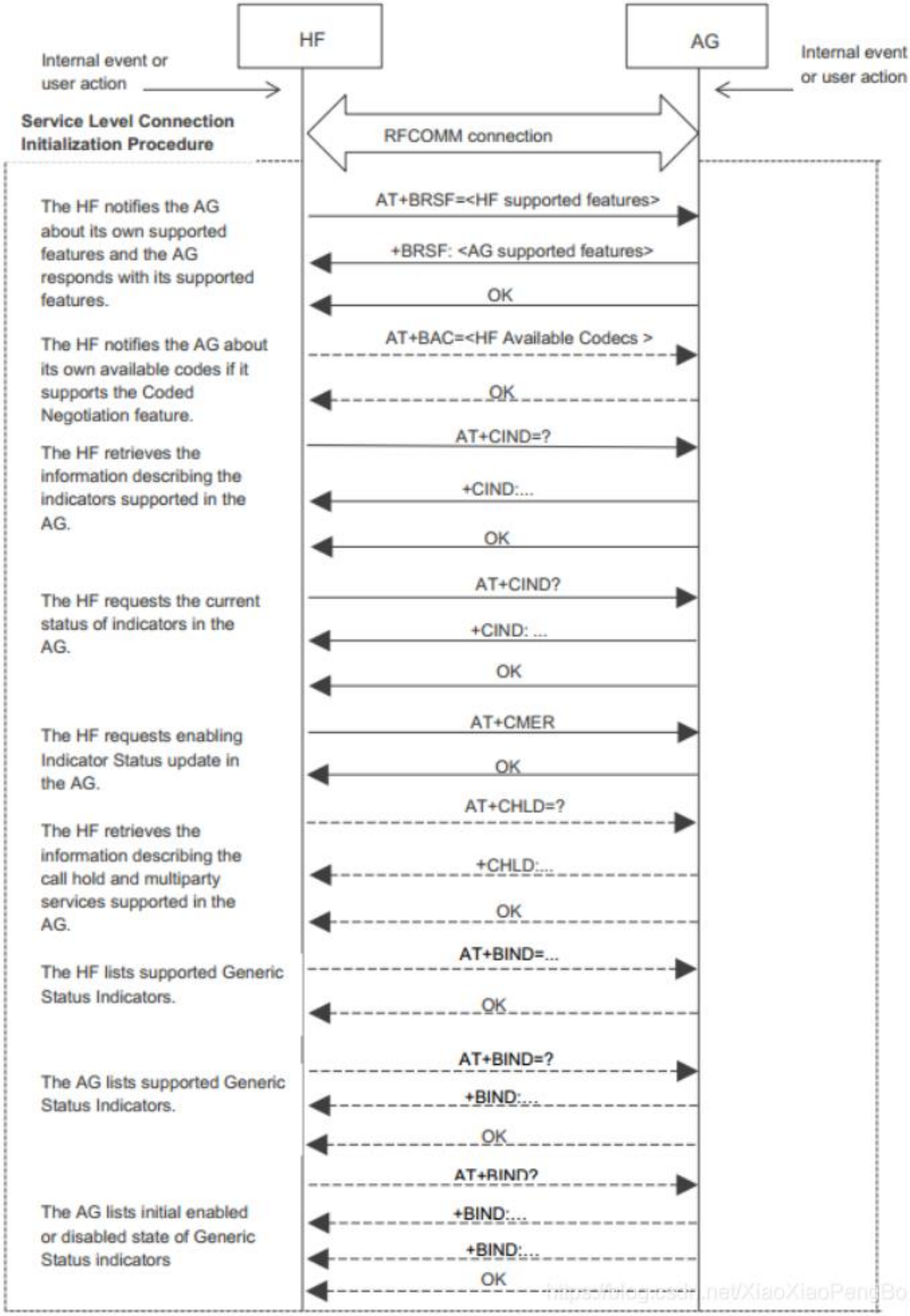
4.2 Service Level Connection Establishment

- Upon a user action or an internal event, either the HF or the AG may initiate a Service Level Connection establishment procedure.
- A Service Level Connection establishment requires the existence of a RFCOMM connection, that is, a RFCOMM data link channel between the HF and the AG.
- Both the HF and the AG may initiate the RFCOMM connection establishment.
- If there is no RFCOMM session between the AG and the HF, the initiating device shall first initialize RFCOMM.

Establishing a Service Level Connection



2


HF
AG
Internal event or user action
Internal event or user action
Service Level Connection Initialization Procedure
RFCOMM connection
The HF notifies the AG about its own supported features and the AG responds with its supported features.
AT+BRSF=<HF supported features>
+BRSF: <AG supported features>
OK
The HF notifies the AG about its own available codes if it supports the Coded Negotiation feature.
AT+BAC=<HF Available Codecs >
OK
The HF retrieves the information describing the indicators supported in the AG.
AT+CIND=?
+CIND:...
OK
The HF requests the current status of indicators in the AG.
AT+CIND?
+CIND: ...
OK
The HF requests enabling Indicator Status update in the AG.
AT+CMER
OK
The HF retrieves the information describing the call hold and multiparty services supported in the AG.
AT+CHLD=?
+CHLD:...
OK
The HF lists supported Generic Status Indicators.
AT+BIND=...
OK
The AG lists supported Generic Status Indicators.
AT+BIND=?
+BIND:...
OK
The AG lists initial enabled or disabled state of Generic Status indicators
AT+BIND?
+BIND:...
+BIND:...
OK

AT+CIND =? = current status of indicator for testing the and give the AG supported are there in the AG

- AT+CIND? = This is used for reading commands, and this for current status.
- BSRF = Bluetooth supported retrieve features
- BAC = Bluetooth available codes
- CIND = current indicators
- CMER = Current mobile termination event reporting

AT+CMER=3,0,0,1 activates “indicator events reporting”. AT+CMER=3,0,0,0 deactivates “indicator events reporting”.

- CHLD = Call hold
- BIND = Bluetooth indicator

4.3 Service Level Connection Release

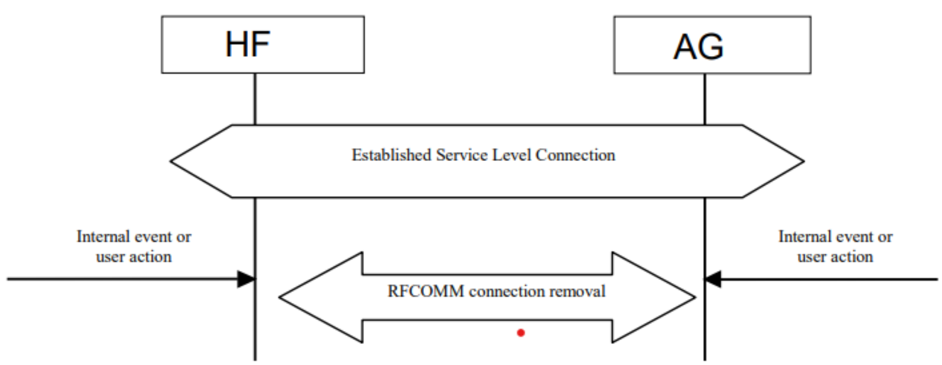


Figure 4.2: Service Level Connection removal

- The disconnection of a Service Level Connection shall result in the immediate removal of the corresponding RFCOMM data link channel between the HF and the AG.
- Also, an existing audio connection must be removed as a consequence of the removal of the Service Level Connection.
- The removal of the L2CAP and link layers is optional.
- Once’s the call will be terminated connection will be terminated.

Phone status information

1. Transfer of Registration Status
2. Transfer of Signal Strength Indication
3. Transfer of Roaming Status Indication
4. Transfer of Battery Level Indication
5. Query of Operator Selection
6. Report Extended Audio Gateway Error Codes
7. Transfer of Call, Call Setup and Call Held Status

4.4 Transfer of Registration Status

0= No calls held

1= Call is placed on hold or active/held calls swapped

(The AG has both an active AND a held call)

2= Call on hold, no active call

- signal: Signal Strength indicator, where:

  <value>= ranges from 0 to 5

- roam: Roaming status indicator, where:

  <value>=0 means roaming is not active

  <value>=1 means a roaming is active

- battchg: Battery Charge indicator of AG, where:

  <value>=ranges from 0 to 5

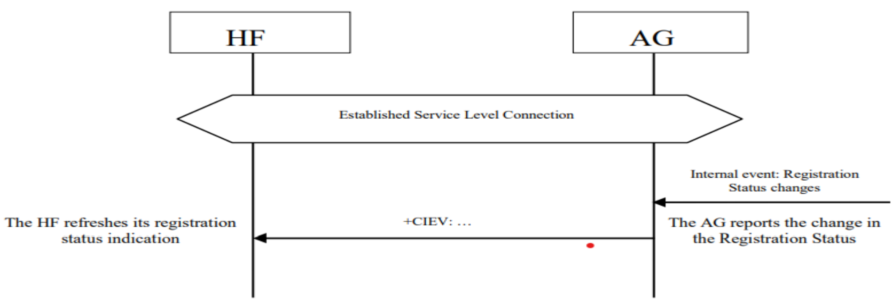


Figure 4.3: Typical Registration Status update

- +CIEV == indicator events reporting
- CMER = Current mobile termination event reporting
- BIA == Bluetooth indication activation

- The AT+CMER command, as described in Section 4.34, enables the “Registration status update” function in the AG.
- The AT+BIA command, as described in Section 4.35.1, allows the HF to deactivate/reactivate individual indicators.

- Unsolicited result codes are **messages sent from the GSM/GPRS modem or mobile phone to provide you information about the occurrence of an event**.
- When the CMER function is enabled and the registration status indicator has not been deactivated by the AT+BIA command, the AG shall send the +CIEV unsolicited result code with the corresponding service indicator and value whenever the AG's registration status changes.
- If the CMER function is disabled or the indicator has been deactivated by the AT+BIA command, the AG shall not send the unsolicited result code.

4.5 Transfer of Signal Strength Indication

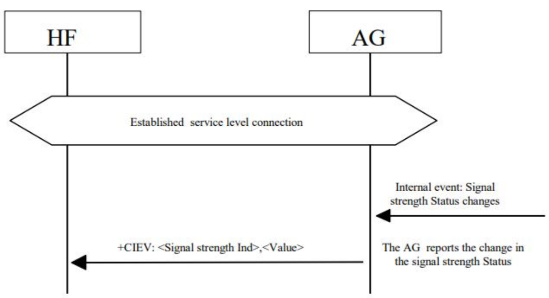


Figure 4.4: Transfer of Signal strength indication

As a result, the AG shall send the +CIEV unsolicited result code with the corresponding signal strength value whenever its signal strength changes.

- The AT+CMER command, as described in Section 4.34, enables the "Signal strength Indication" function in the AG.
- The AT+BIA command, as described in Section 4.35.1, allows the HF to deactivate/reactivate individual indicators.
- When the CMER function is enabled and the indicator has not been deactivated by the AT+BIA command, the AG shall send the +CIEV unsolicited result code with the corresponding signal strength indicator and value whenever the AG's signal strength changes.
- The HF shall be capable of interpreting the information provided by the +CIEV result code to determine the signal strength as listed in Section
- If the CMER function is disabled or the indicator has been deactivated by the AT+BIA command, the AG shall not send the unsolicited result code.
  Valve range about = 0 to 5

4.6 Transfer of Roaming Status Indication

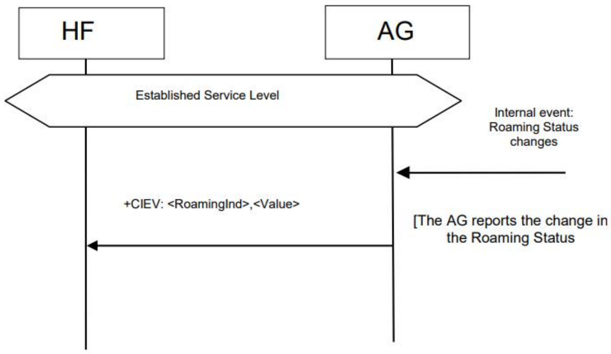


Figure 4.5: Transfer of Roaming Status Indication

- The AT+CMER command, as described in Section 4.34, enables the “Roaming Status Indication” function in the AG.
- The AT+BIA command, as described in Section 4.35.1, allows the HF to deactivate/reactivate individual indicators.
- When the CMER function is enabled and the indicator has not been deactivated by the AT+BIA command, the AG shall send the +CIEV unsolicited result code with the corresponding roaming status indicator and value whenever the AG's roaming status changes.
- The HF shall be capable of interpreting the information provided by the +CIEV result code to determine the roaming status as listed in Section 4.34.2.
- If the CMER function is disabled or the indicator has been deactivated by the AT+BIA command, the AG shall not send the unsolicited result code.

  Value 0 = not active, value 1 = active roaming status indicator

4.7 Transfer of Battery Level Indication of AG

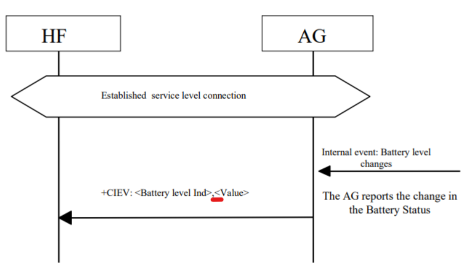


Figure 4.6: Transfer of Battery level indication

- The AT+CMER command, as described in Section 4.34 enables the “Battery Level Indication” function in the AG.

- The AT+BIA command, as described in Section 4.35.1, allows the HF to deactivate/reactivate individual indicators.
- When the CMER function is enabled and the indicator has not been deactivated by the AT+BIA command, the AG shall send the +CIEV unsolicited result code with the corresponding battery level indicator and value whenever the AG's battery level changes.
- The HF shall be capable of interpreting the information provided by the +CIEV result code to determine the battery level as listed in Section 4.34.2.
- If the CMER function is disabled or the indicator has been deactivated by the AT+BIA command, the AG shall not send the unsolicited result code. Value = 0 to 5 values.

4.8 Query Operator Selection

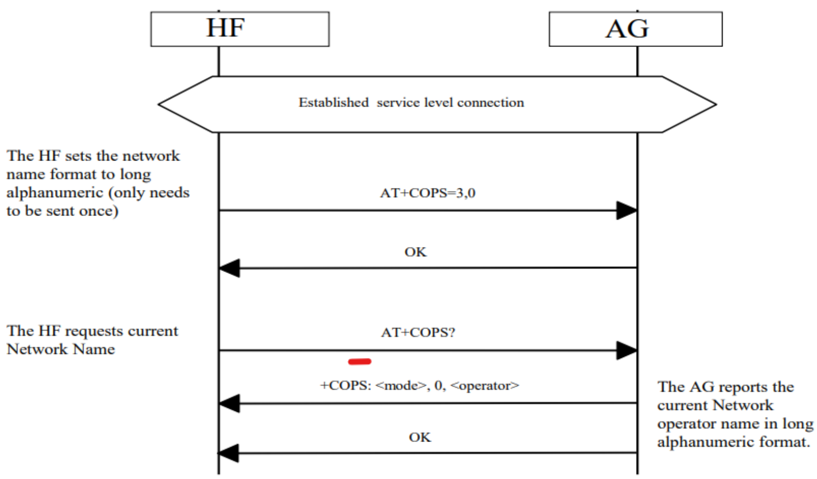


Figure 4.7: Query currently selected Network operator

- The HF shall execute this procedure to find out the name of the currently selected Network operator by AG.
- The following precondition applies for this procedure:
- The AT+COPS=3,0 shall be sent by the HF to the AG prior to sending the AT+COPS? command.
- AT+COPS=3,0 sets the format of the network operator string to the long format alphanumeric.
- The AT+COPS? command is used for reading network operator. a +COPS = Mode, format, operator.
- HF shall send AT+COPS=3,0 command to set name format to long alphanumeric.
- Long alphanumeric is defined as a maximum of 16 characters.
- The value of 3 as the first parameter indicates that this command is only affecting the format parameter (the second parameter).

- The second parameter, 0, is the value required to set the format to "long alphanumeric."
- • Upon an internal event or user-initiated action, HF shall send the AT+COPS? (Read) command to find the currently selected operator.
- • AG shall respond with +COPS response indicating the currently selected operator. If no operator is selected, and are omitted.

4.9 Report Extended Audio Gateway Error Results Code

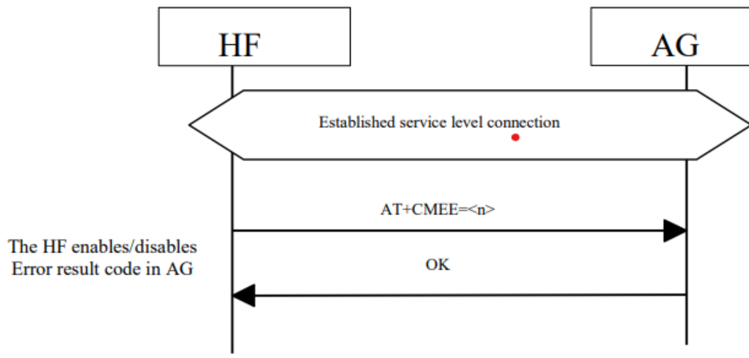


Figure 4.8: Enable/Disable AG Error result code

Standard AT command used to enable the use of result code +CME ERROR: as an indication of an error relating to the functionality of the AG. The set command AT+CMEE=1 is covered in this specification.

- The HF shall execute this procedure to enable/disable Extended Audio Gateway Error result codes in the AG.
- The following precondition applies for this procedure:
  • An ongoing connection between the AG and the HF shall exist. If this connection did not exist, the AG shall establish a connection using "Service Level Connection set up" procedure described in Section 4.2.
- Standard AT command used to enable the use of result code +CME ERROR: as an indication of an error relating to the functionality of the AG. The set command AT+CMEE=1 is covered in this specification.
- The HF shall issue the AT+CMEE command to enable/disable the "Extended Audio Gateway Error Result Code" in the AG. The parameter controls the activation/deactivation of the "Extended Error result code" notification.
- Whenever there is an error relating to the functionality of the AG as a result of AT command, the AG shall send +CME ERROR: response to the HF.

AT+CMEE = CURRENT MOBILE TERMINATION EVENT

Standard AT command used to enable the use of result code +CME ERROR: as an indication of an error relating to the functionality of the AG.

The set command AT+CMEE=1 is covered in this specification.

+CME ERROR = The ERROR response code is still allowed while using the Extended Audio Gateway Error Result Codes.

+CME ERROR: 0 – AG failure

+CME ERROR: 1 – no connection to phone

+CME ERROR: 3 – operation not allowed

+CME ERROR: 4 – operation not supported

+CME ERROR: 5 – PH-SIM PIN required

+CME ERROR: 10 – SIM not inserted

+CME ERROR: 11 – SIM PIN required

+CME ERROR: 12 – SIM PUK required

+CME ERROR: 13 – SIM failure

+CME ERROR: 14 – SIM busy

+CME ERROR: 16 – incorrect password

+CME ERROR: 17 – SIM PIN2 required

+CME ERROR: 18 – SIM PUK2 required

+CME ERROR: 20 – memory full

+CME ERROR: 21 – invalid index

+CME ERROR: 23 – memory failure

+CME ERROR: 24 – text string too long

+CME ERROR: 25 – invalid characters in text string

+CME ERROR: 26 – dial string too long

+CME ERROR: 27 – invalid characters in dial string

+CME ERROR: 30 – no network service

+CME ERROR: 31 - network Timeout.

4.10 Transfer of Call, Call Setup, and Held Call Status

Transfer of Call Status

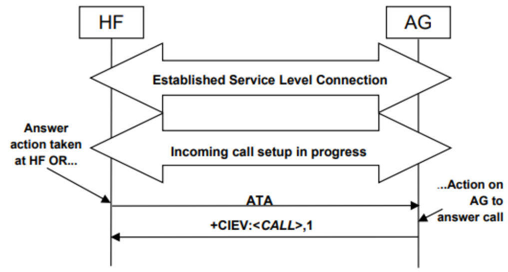


Figure 4.9: Call Status: Incoming call answered

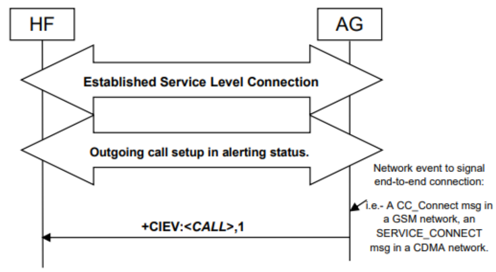


Figure 4.10: Call Status: Outgoing call

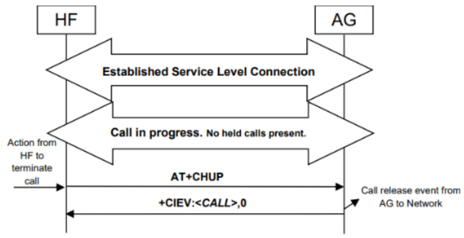
- ATA Standard call answer AT command.
  - Upon the change in status of any call in the AG, the AG shall execute this procedure to advise the HF of the current call status changes.
  - The values for the call indicator are:

    0= No calls (held or active)

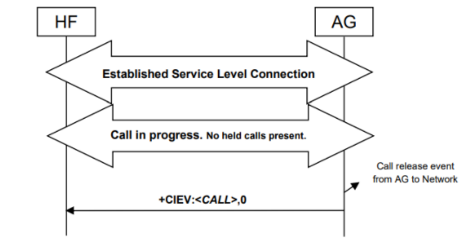
    1= Call is present (active or held)

The following precondition applies for this procedure:

- The HF shall have enabled the "Indicator’s status update" in the AG.
  - A Service Level Connection shall exist between the AG and HF devices.

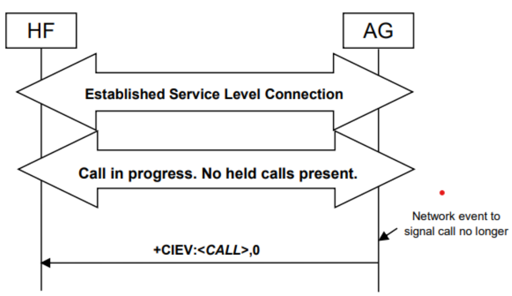


4.11: Call Status: Call Release from HF



4.12: Call Status: Call Release from AG

AT+CHUP === Standard hang-up AT command. ==AT+CHUP is also used as the command to reject any incoming call prior to answer.



13: Call Status: Call Release from Network

Transfer of Call setup Status

- Upon the change in AG's call setup status, the AG shall execute this procedure to advise the HF of the current call setup status changes.
- The values for the call setup indicator are:
  0 = No call setup in progress.
  1 = Incoming call setup in progress.

2 = Outgoing call setup in dialing state.
3 = Outgoing call setup in alerting state.

- The following pre condition applies for this procedure:
  - The HF shall have enabled the "Indicator's status update" in the AG.
  - A Service Level Connection shall exist between the AG and HF devices

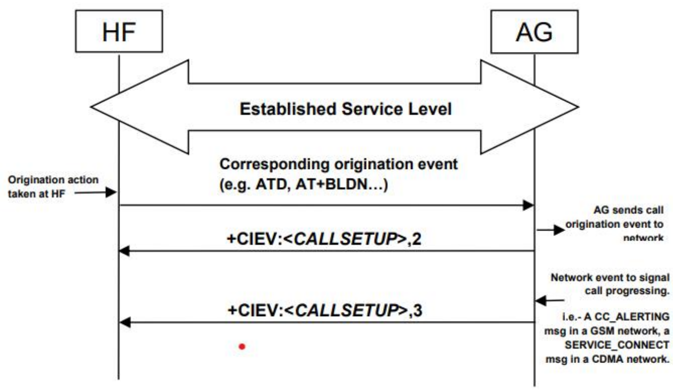
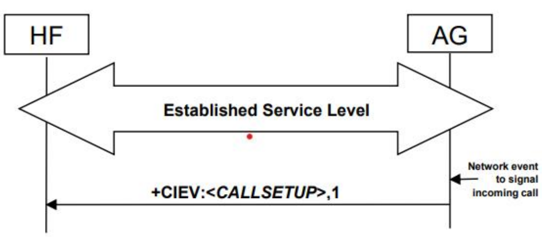


Figure 4.14: Callsetup Status: Outgoing



gure 4.15: Callsetup Status: Incoming (See Section 4.13, 4.14)

Once the AG has been advised via whatever applicable network function, that a call has reached an end to-end connected status, the call setup indicator shall be reset indicating that no call setup procedures are in progress.

Indication of Status for Held Call

- Upon the change in status of any call on hold in the AG, the AG shall execute this procedure to advise the HF of the held call status.
- The values for the call held indicator are:
  0= No calls held
  1= Call is placed on hold or active/held calls swapped (The AG has both an active AND a held call)
  2= Call on hold, no active call

The following precondition applies for this procedure:

- The HF shall have enabled the Call Status Indicators function in the AG.
- A Service Level Connection shall exist between the AG and HF devices.

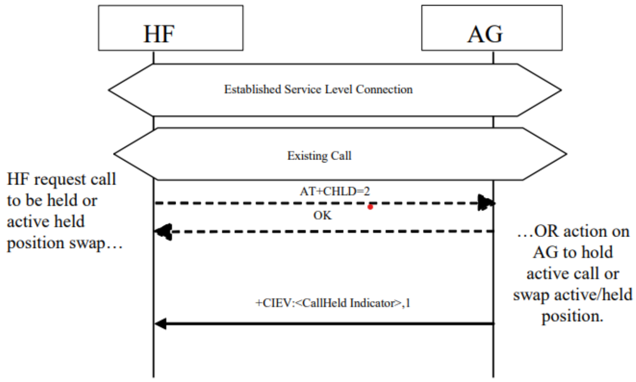


Figure 4.16: Call Held or Active/Held Position Swap

Consequently, upon the release of any call on hold by the HF, the AG or by network event, or actions by the HF or AG to retrieve a held call, the AG shall issue a +CIEV unsolicited result code with the callheld indicator value of “0”.

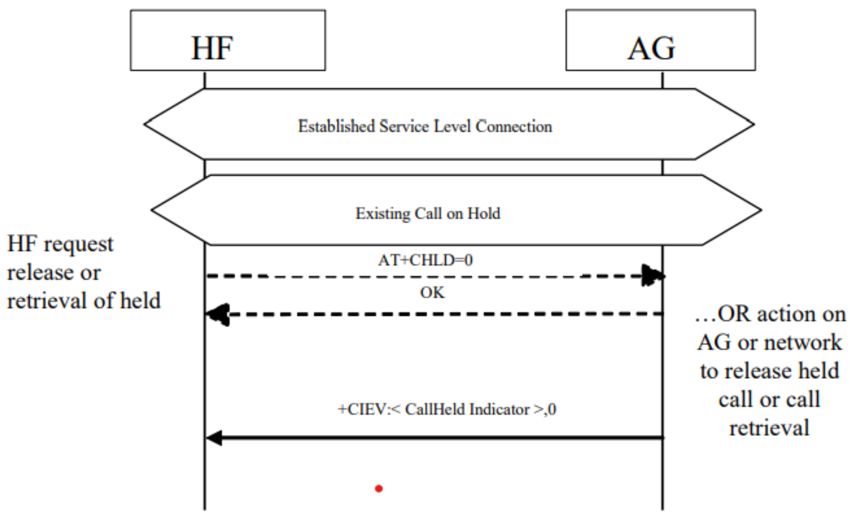


Figure 4.17: Held Call Release

If a call is still on hold when an active call is terminated or a single active call is put on hold, the AG shall issue a +CIEV unsolicited result code with the callheld indicator value of “2”.

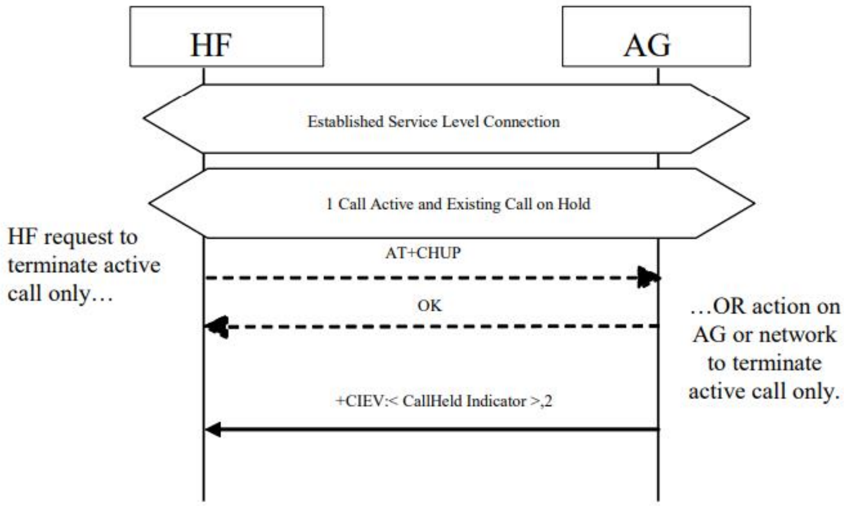


Figure 4.18: Active Call Terminated/Call Remains Held

[redacted]

Audio Connection handling

1. Audio Connection set up
2. Audio Connection release
3. Codec Connection set up

4.11 Audio Connection Setup

- Upon user action or an internal event, either the HF or the AG may initiate the establishment of an audio connection as needed.
- Further internal actions may be needed by the HF or the AG to internally route, modify sample rate, frame and/or sample alignment of the audio paths.
- More formally stated, the requirements for audio connection setup are:

• The HF shall be capable of initiating an audio connection during a call process.

• The HF may be capable of initiating an audio connection while no call is in process.

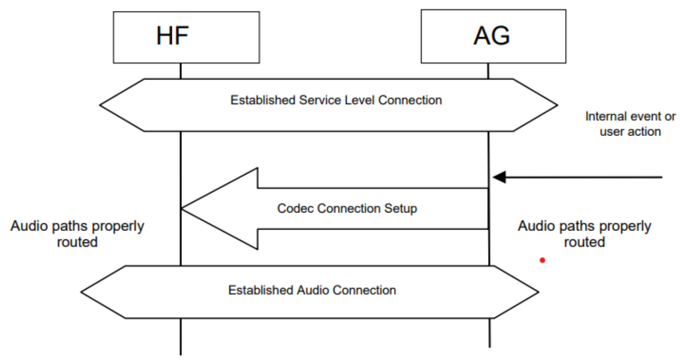
• The AG shall be capable of initiating an audio connection during a call process.

• The AG may be capable of initiating an audio connection while no call is in process.

- An Audio Connection set up procedure always means the establishment of a Synchronous Connection and it is always associated with an existing Service Level Connection.

Audio Connection Setup by AG

- When the AG is to setup an Audio Connection, it shall initiate the Codec Connection establishment procedure if the Service Level Negotiation indicated that both sides support this feature.



ıre 4.19: Procedure for Establishment of Audio Connection from AG

Audio Connection Setup by HF

- For all HF initiated audio connection establishments for which both sides support the Codec Negotiation feature, the HF shall trigger the AG to establish a Codec Connection.
- This is necessary because only the AG knows about the codec selection and settings of the network.

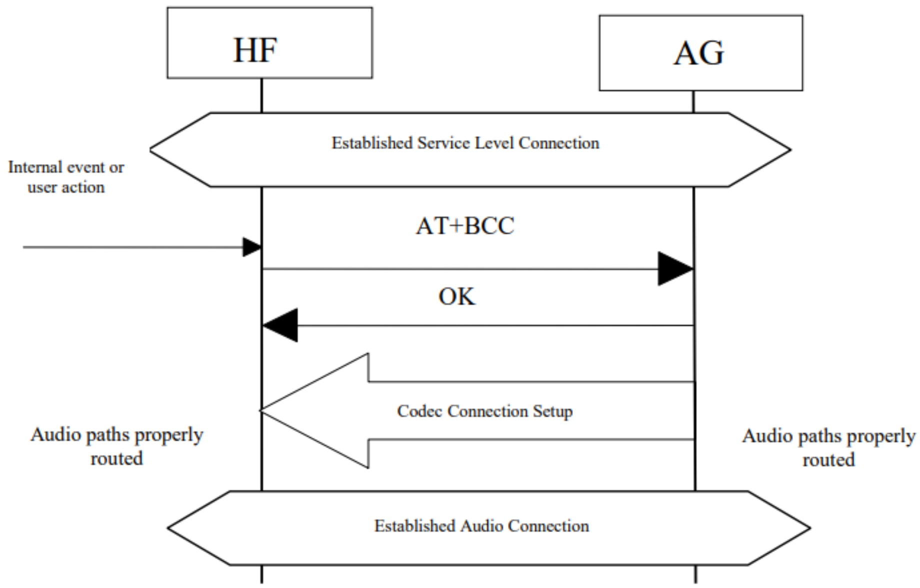
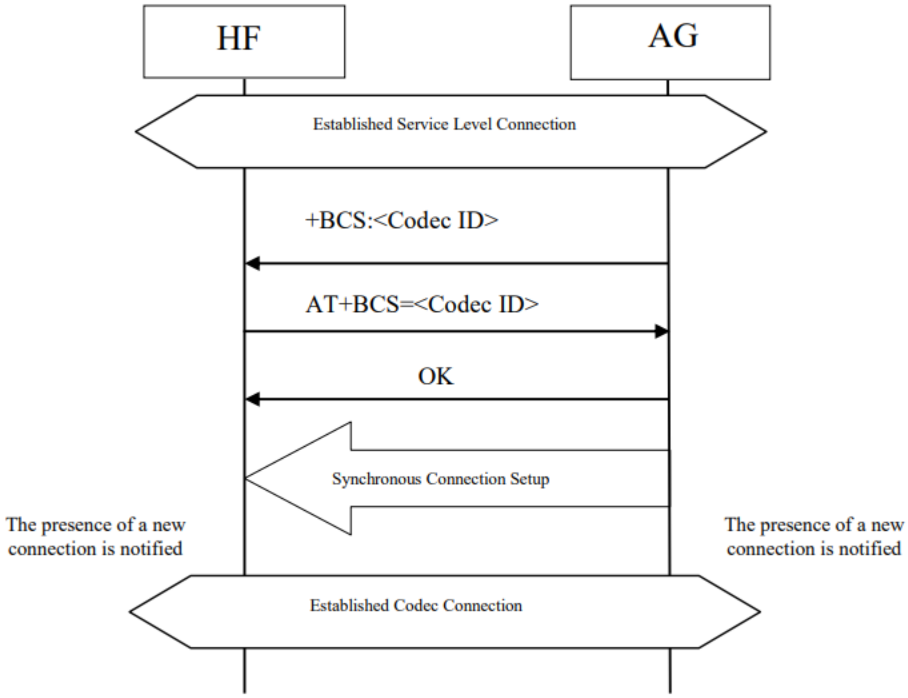


Figure 4.20: Procedure for Establishment of Audio Connection from HF

- When the HF triggers the establishment of the Codec Connection it shall send the AT command AT+BCC to the AG.
- The AG shall respond with OK if it will start the Codec Connection procedure, and with ERROR if it cannot start the Codec Connection procedure.
- After the AG has sent the OK response, the AG shall initiate the Codec Connection Setup procedure.

AT+BCC (Bluetooth Codec Connection) == Syntax: AT+BCC Description: This command is used by the HF to request the AG to start the codec connection procedure.

Codec Connection Setup



gure 4.21: Procedure for the Establishment of Codec Connection

4.12 Audio Connection Release

- Upon a user action or an internal event, either the HF or the AG may release an existing Audio Connection. More formally stated, the requirements for audio connection release are:

- The HF shall be capable of releasing an audio connection during a call process.

  - If the HF has the ability to set up an audio connection with no call-in process, the HF shall be capable of releasing the audio connection while no call is in process.
  - The AG shall be capable of releasing an audio connection during a call process.
  - If the AG has the ability to set up an audio connection with no call-in process, the AG shall be capable of releasing the audio connection while no call is in process.
- As a precondition for this procedure, an ongoing Audio Connection between the AG and the HF shall exist.
- An Audio Connection release always means the disconnection of its corresponding Synchronous Connection.
- When the audio connection is released, the audio path shall be routed to the AG.3

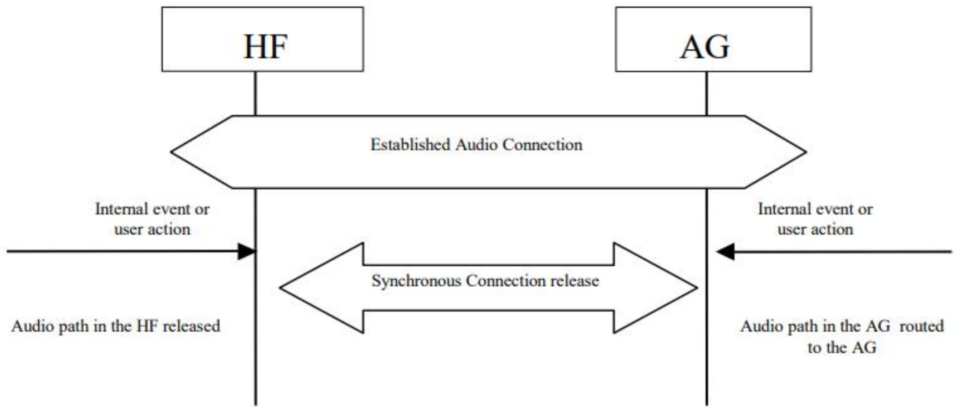


Figure 4.23: Audio Connection release

[redacted]

Accept an incoming voice call

4.13 Answer an Incoming Call

Answer Incoming Call from the HF – In-Band Ringing

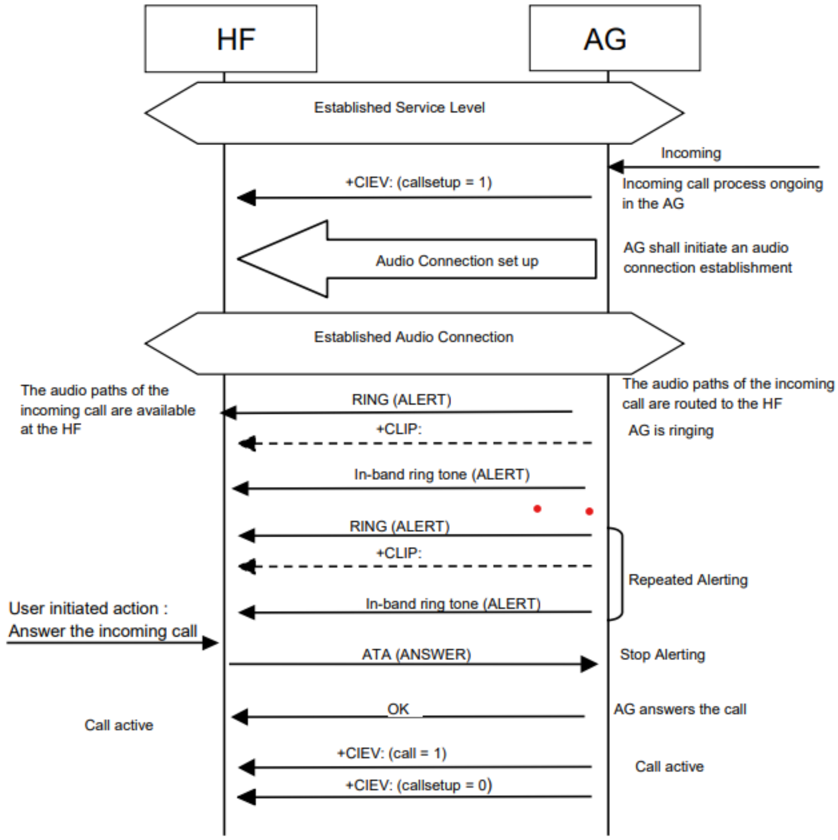


Figure 4.24: Answer an incoming call from the HF – in-band ring tone

- Optionally, the AG may provide an in-band ring tone. This case is described in Figure 4.24 below and implies as a precondition that an ongoing Service Level Connection between the AG and the HF shall exist.
- If this connection does not exist, the AG shall autonomously establish the Service Level Connection using the proper procedure as described in Section 4.2.
- As the figure below shows, if an in-band ring tone is used, the AG shall send the ring tone to the HF via the established Audio Connection.

- The user accepts the incoming voice call by using the proper means provided by the HF.

- The HF shall then send the ATA command (see Section 4.34) to the AG.

- The AG shall then begin the procedure for accepting the incoming call. If the normal incoming call procedure is interrupted for any reason, the AG shall issue the +CIEV result code, with the value indicating (callsetup=0) to notify the HF of this condition (see also Section 4.14.2).

Answer Incoming Call from the HF – No In-Band Ringing

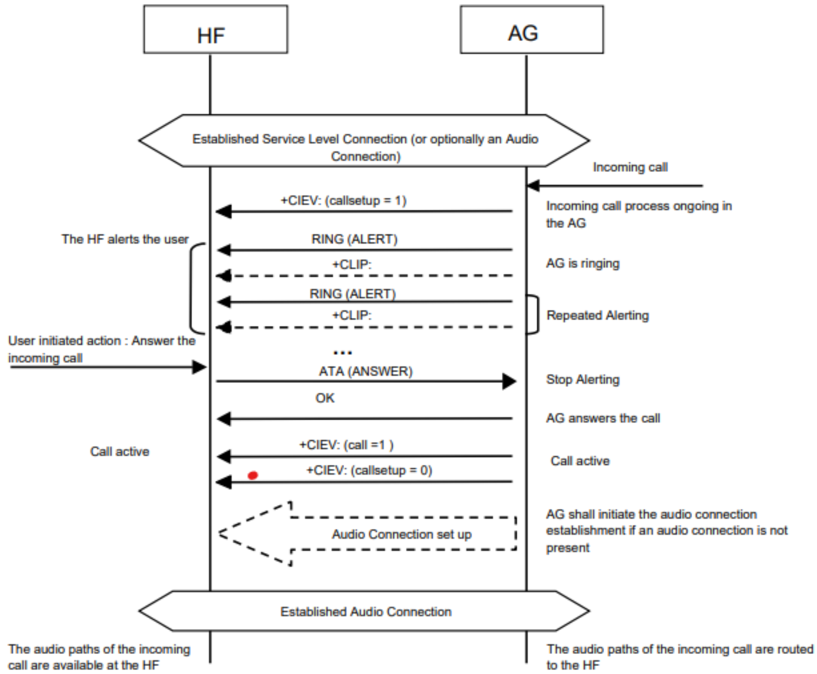


Figure 4.25: Answer an incoming call from the HF – no in-band ring tone

- As a precondition, an ongoing Service Level Connection between the AG and the HF shall exist.
- If this connection does not exist, the AG shall autonomously establish the Service Level Connection using the proper procedure as described in Section 4.2. As the figure below shows, if no in-band ring tone is used and an Audio Connection does not exist, the AG shall set up the Audio Connection and route the audio paths to the HF upon answering the call.

- The user accepts the incoming voice call by using the proper means provided by the HF.
- The HF shall then send the ATA command (see Section 4.34) to the AG, and the AG shall start the procedure for accepting the incoming call and establishing the Audio Connection if an Audio Connection does not exist (refer to Section 4.10.1).
- If the normal incoming call procedure is interrupted for any reason, the AG shall issue the +CIEV result code, with the value indicating (callsetup=0) to notify the HF of this condition (see also Section 4.14.2).

Answer Incoming Call from the AG

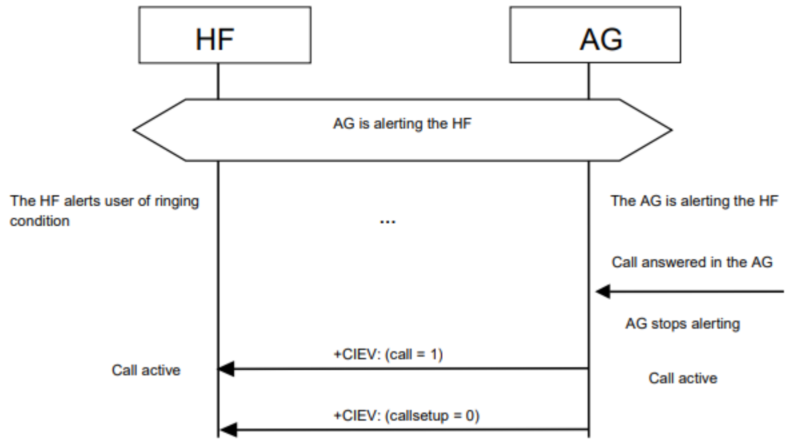


Figure 4.26: Answer an incoming call from the AG

- The following preconditions apply for this procedure:
    - As a precondition for this procedure, an ongoing Service Level Connection between the AG and the HF shall exist.
    - The AG shall alert the HF using either of the two procedures described in Sections 4.13.1 and 4.13.2.
    - The HF shall alert the user.

- The user accepts the incoming call by using the proper means provided by the AG.
- If the normal incoming call procedure is interrupted for any reason, the AG shall issue the +CIEV result code, with the value indicating (callsetup=0) to notify the HF of this condition (see also Section 4.14.2).

Change the In-Band Ring Tone Setting

- The SDP record entry “In-band ring tone” of the “SupportedFeatures” record (see Table 5.4) informs the HF if the AG is capable of sending an in-band ring tone or not.
- If the AG is capable of sending an in-band ring tone, it shall send the in-band ring tone by default.
- The AG may subsequently change this setting.
- If the AG wants to change the in-band ring tone setting during an ongoing service level connection, it shall use the unsolicited result code +BSIR (Bluetooth Set In-band Ring tone) to notify the HF about the change. See Figure 4.27 for details. See Section 4.34 for more information on the +BSIR unsolicited result code.
- The in-band ring tone setting may be changed several times during a Service Level Connection. As a precondition for this procedure, an ongoing Service Level Connection between the AG and the HF shall exist.

- If this connection does not exist, the AG shall autonomously establish the Service Level Connection using the proper procedure as described in Section 4.2.

Hands-Free Profile / Profile Specification

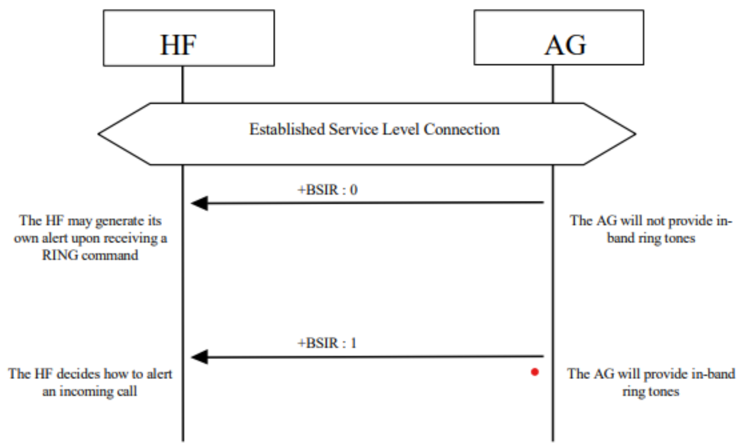


Figure 4.27: Change of the in-band ring tone setting initiated by the AG

If the HF does not want to use the AG's in-band ring tone, it may mute the Audio Connection after it has received +CIEV:(callsetup=1). The HF shall un-mute the Audio Connection upon receiving the +CIEV:(callsetup=0) indication.

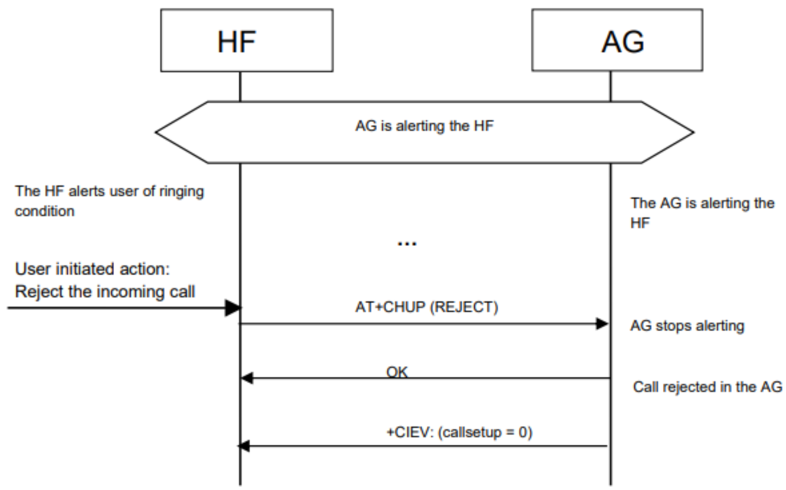
[redacted]

Reject an incoming voice call

4.14 Reject an Incoming Call

Reject an Incoming Call from the HF

- As a precondition to this procedure, the AG shall alert the HF using either of the two procedures described in Sections 4.13.1 and 4.13.2.
- The user rejects the incoming call by using the User Interface on the Hands-Free unit.
- The HF shall then send the AT+CHUP command (see Section 4.34) to the AG.
- This may happen at any time during the procedures described in Sections 4.13.1 and 4.13.2. The AG shall then cease alerting the HF of the incoming call and send the OK indication followed by the +CIEV result code, with the value indicating (callsetup=0).



ure 4.28: Reject an incoming call from the HF

Rejection/Interruption of an Incoming Call in the AG

- As a precondition to this procedure, the AG shall alert the HF using either of the two procedures described in Sections 4.13.1 and 4.13.2.
- The user rejects the incoming call by using the User Interface on the AG. Alternatively, the incoming call process may be interrupted in the AG for any other reason.
- As a consequence of this, the AG shall send the +CIEV result code, with the value indicating (callsetup=0).
- The HF shall then stop alerting the user. This may happen at any time during the procedures described in Sections 4.13.1 and 4.13.2.

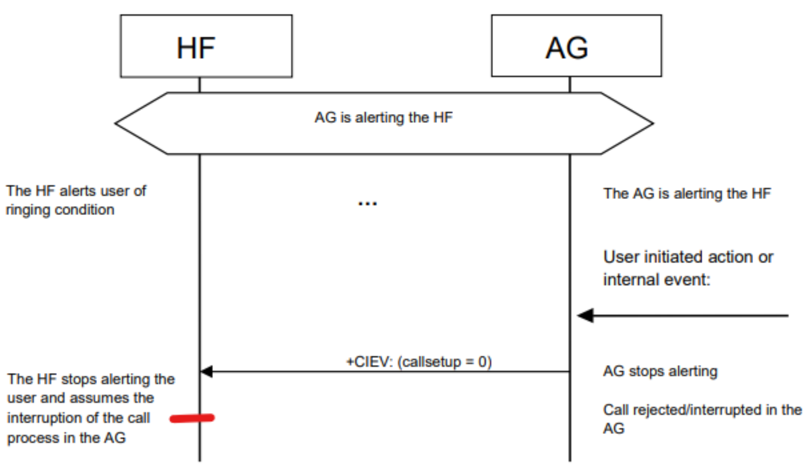


Figure 4.29: Rejection/interruption of an incoming call in the AG

[redacted]

4.15 Terminate a Call Process

Terminate a Call Process from the HF

- The following preconditions apply for this procedure:
  • An ongoing Service Level Connection between the AG and the HF shall exist. If this connection does not exist, the HF shall autonomously establish the Service Level Connection using the proper procedure as described in Section 4.2.
  • A call-related process is ongoing in the AG. Although not required for the call termination process, an Audio Connection is typically present between the HF and AG.

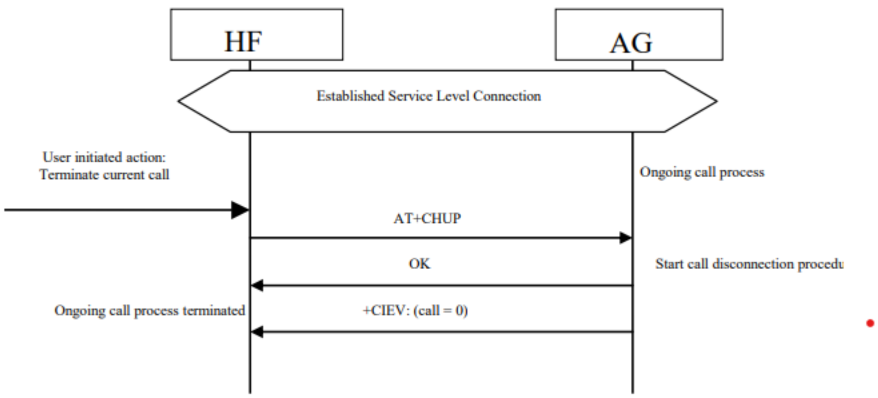


Figure 4.30: Terminate ongoing call - HF initiated

- The user may abort the ongoing call process using whatever means provided by the Hands-Free unit. The HF shall send AT+CHUP command (see Section 4.34) to the AG, and the AG shall then start the procedure to terminate or interrupt the current call procedure.
- The AG shall then send the OK indication followed by the +CIEV result code, with the value indicating (call=0).
- Performing a similar procedure, the AT+CHUP command described above may also be used for interrupting a normal outgoing call set-up

Terminate a Call Process from the AG process.

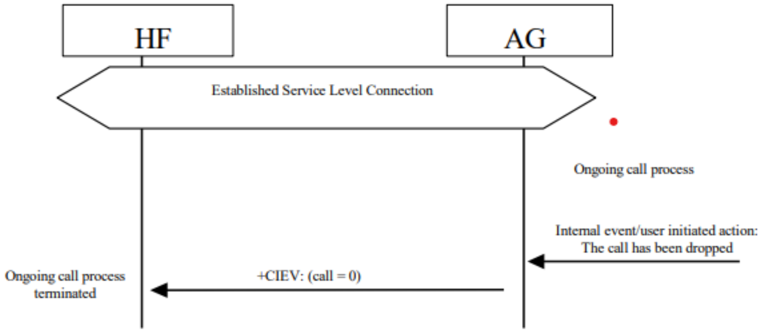


Figure 4.31: Terminate ongoing call - AG initiated

- The following preconditions apply for this procedure:
  - An ongoing Service Level Connection between the AG and the HF shall exist. If this connection does not exist, the AG shall autonomously establish the Service Level Connection using the proper procedure as described in Section 4.2.
  - A call related process is ongoing in the AG.
- This procedure is fully applicable for cases in which an ongoing call process is interrupted in the AG for any reason. In this case, the AG shall send the +CIEV result code, with the value indicating (call=0).

[redacted]

Audio Connection transfer during an ongoing call

4.16 Audio Connection Transfer towards the HF

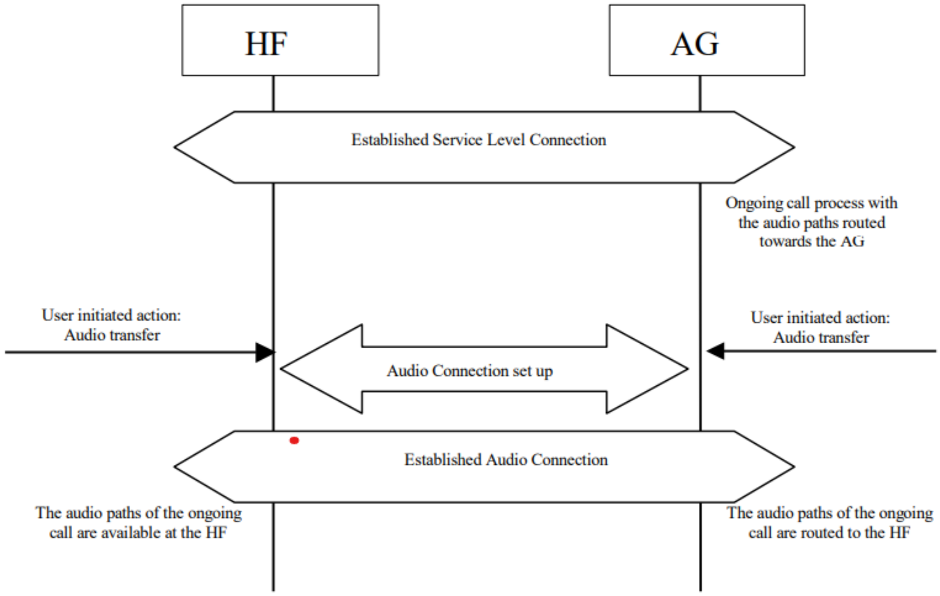


Figure 4.32: Audio Connection transfer to the HF

- The audio paths of an ongoing call may be transferred from the AG to the HF.
- This procedure represents a particular case of an "Audio Connection set up" procedure, as described in Section 4.10.1.
- The call connection transfer from the AG to the HF is initiated by user action either on the HF or on the AG.
- This shall result in either the HF or the AG, respectively, initiating an "Audio Connection set up" procedure with the audio paths of the current call being routed to the HF.
- This procedure is only applicable if there is no current Audio Connection established between the HF and the AG.
- In fact, if the Audio Connection already exists, this procedure is not necessary because the audio path of the AG is assumed to be already routed towards the HF.
- The following preconditions apply for this procedure:
  - An ongoing Service Level Connection between the AG and the HF shall exist. If this connection does not exist, the initiator of the "Audio Connection transfer towards the HF" procedure shall autonomously establish the Service Level Connection using the proper procedure as described in Section 4.2.
  - An ongoing call exists in the AG, with the audio paths routed to the AG means.

4.17 Audio Connection Transfer towards the AG

The audio paths of an ongoing call may be transferred from the HF to the AG. This procedure represents a particular case of an "Audio Connection release" procedure, as described in Section 4.12. The call connection transfer from the HF to the AG is initiated by a user action in the HF or due to an internal event or user action on the AG side. This results in an "Audio

Connection release" procedure being initiated either by the HF or the AG respectively, with the current call kept and its audio paths routed to the AG. If, as a consequence of an HF initiated "Audio Connection transfer towards the AG" procedure, the existing Service Level Connection is autonomously removed by the AG, the AG shall attempt to re-establish the Service Level Connection once the current call ends. As a precondition for this procedure, an ongoing call process shall exist in the AG. The audio paths of the ongoing call shall be available in the HF via an Audio Connection established between the AG and the HF.

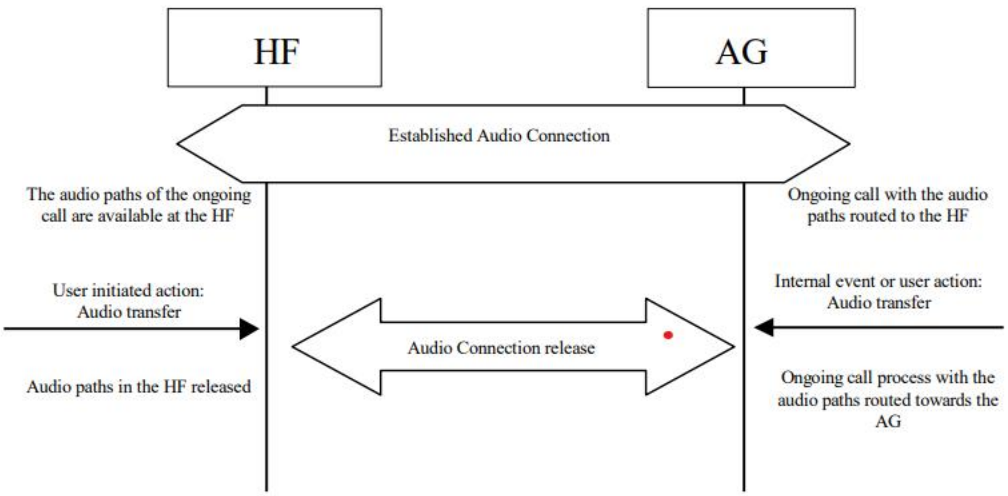


Figure 4.33: Audio Connection transfer to the AG

[redacted]

4.18 Place a Call with the Phone Number Supplied by the HF

The HF may initiate outgoing voice calls by providing the destination phone number to the AG. To start the call set-up, the HF shall initiate the Service Level Connection establishment (if necessary) and send a proper ATDdd…dd; command to the AG. The AG shall then start the call establishment procedure using the phone number received from the HF and issues the +CIEV result code, with the value (callsetup=2) to notify the HF that the call set-up has been successfully initiated. See Section 4.34 for more information on the ATDdd…dd; command. As a precondition for this procedure, an ongoing Service Level Connection between the AG and the HF shall exist. If this connection does not exist, the HF shall autonomously establish the Service Level Connection using the proper procedure as described in Section 4.2. If an Audio Connection is not established, the AG shall establish the proper Audio Connection and route the audio paths of the outgoing call to the HF immediately following the commencement of the ongoing call set up procedure. Once the AG is informed that the alerting of the remote party has begun, the AG shall issue the +CIEV result code, with the value indicating (callsetup=3). If the wireless network does not provide the AG of an indication of alerting the remote party, the AG may not send this indication. Upon call connection the AG shall send the +CIEV result code, with the value indicating (call=1). If the normal outgoing call establishment procedure is interrupted for any reason, the AG shall issue the +CIEV result code, with the value indicating (callsetup=0), to notify the HF of this condition (see Section 4.15.2). If the AG supports the "Three-way calling" feature and if a call is already ongoing in the AG, performing this procedure shall result in a new call being placed to a third party with the current ongoing call put on hold. For details on how to handle multiparty calls refer to Section 4.22.2

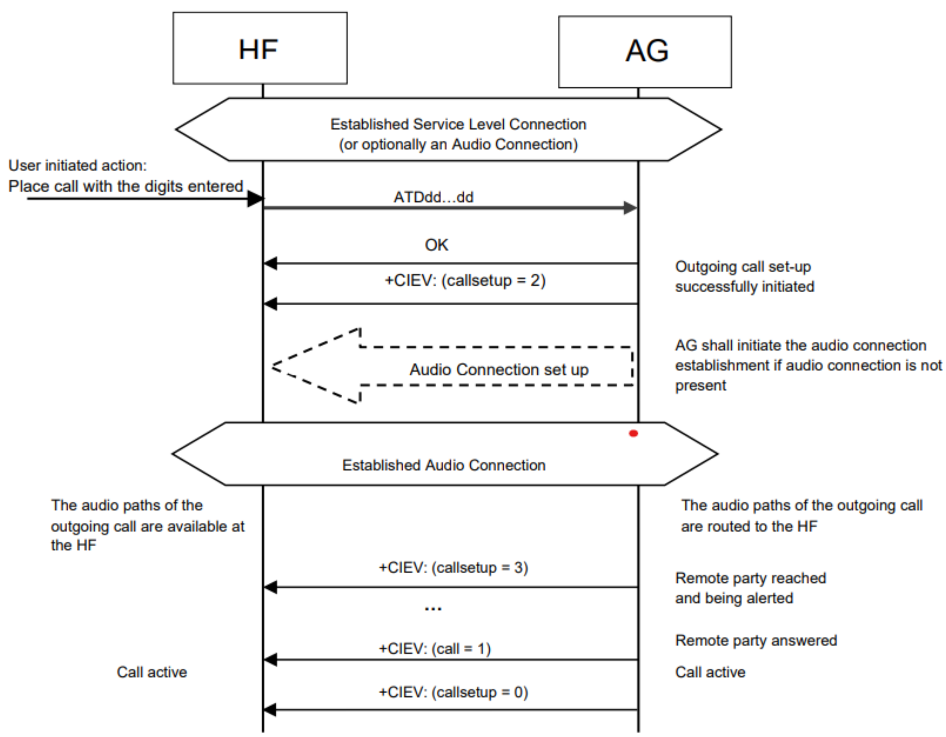


Figure 4.34: Place an outgoing voice call with the digits entered in the HF

[redacted]

4.19 Memory Dialing from the HF

- The HF may initiate outgoing voice calls using the memory dialing feature of the AG.
- To start the call setup, the HF shall initiate the Service Level Connection establishment (if necessary) and send an ATD>Nan…; command to the AG.
- The AG shall then start the call establishment procedure using the phone number stored in the AG memory location given by Nan…; and issue the +CIEV result code, with the value (callsetup=2) to notify the HF that the call set-up has been successfully initiated.
- See Section 4.34 for more information on the ATD>Nan… command. As a precondition for this procedure, an ongoing Service Level Connection between the AG and the HF shall exist.
- If this connection does not exist, the HF shall autonomously establish the Service Level Connection using the proper procedure as described in Section 4.2. If an Audio Connection is not established, the AG shall establish the proper Audio Connection and route the audio paths of the outgoing call to the HF immediately following the commencement of the ongoing call set up procedure.
- Once alerting of the remote party begins, the AG shall issue the +CIEV result code, with the value indicating (callsetup=3).
- Upon call connection the AG shall send the +CIEV result code, with the value indicating (call=1).
- If the normal outgoing call establishment procedure is interrupted for any reason, the AG shall issue the +CIEV result code, with the value indicating (callsetup=0), to notify the HF of this condition (see Section 4.15.2).

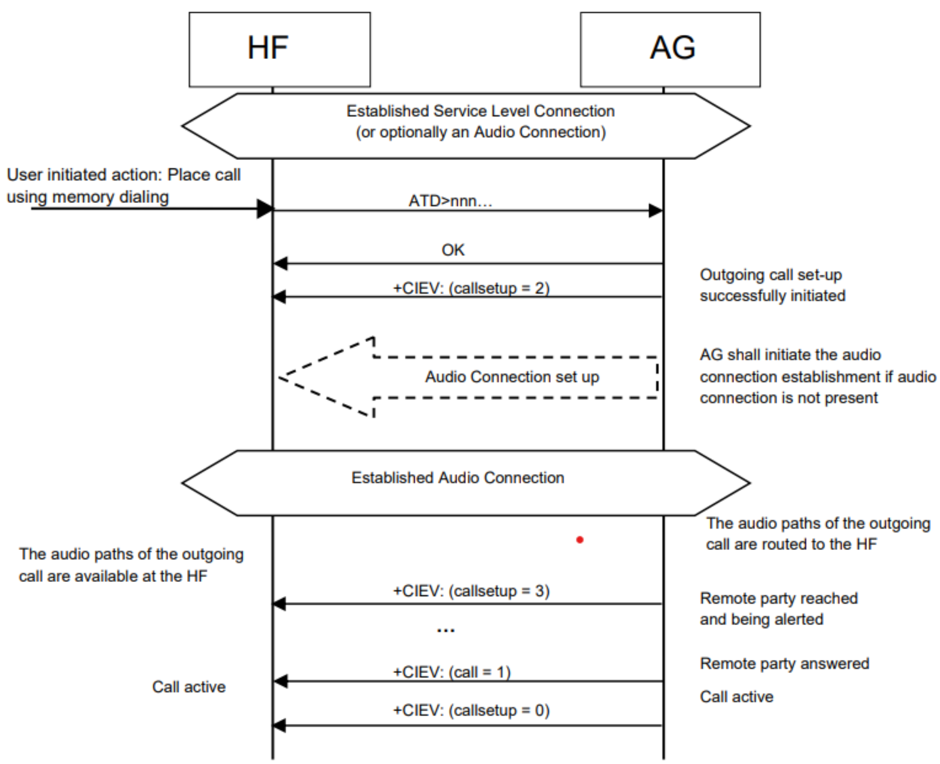


Figure 4.35: Place an outgoing voice call using memory dialing

- If the AG supports the “Three-way calling” feature and if a call is already ongoing in the AG, performing this procedure shall result in a new call being placed to a third party with the current ongoing call put on hold.
- For details on how to handle multiparty calls refer to Section 4.22.2. If there is no number stored for the memory location given by the HF, the AG shall respond with ERROR.

[redacted]

4.20 Last Number Re-Dial from the HF

- The HF may initiate outgoing voice calls by recalling the last number dialed by the AG.
- To start the call set-up, the HF shall initiate the Service Level Connection establishment (if necessary) and send an AT+BLDN command to the AG.
- The AG shall then start the call establishment procedure using the last phone number dialed by the AG, and issues the +CIEV result code, with the value (callsetup=2), to notify the HF that the call set-up has been successfully initiated. See Section 4.34 for more information on the AT+BLDN command.
- As a precondition for this procedure, an ongoing Service Level Connection between the AG and the HF shall exist.
- If this connection does not exist, the HF shall autonomously establish the Service Level Connection using the proper procedure as described in Section 4.2.
- If an Audio Connection is not established, the AG shall establish the proper Audio Connection and route the audio paths of the outgoing call to the HF immediately following the commencement of the ongoing call set-up procedure.
- Once alerting of the remote party begins, the AG shall issue the +CIEV result code, with the value indicating (callsetup=3).
- Upon call connection the AG shall send the +CIEV result code, with the value indicating (call=1). If the normal outgoing call establishment procedure is interrupted for any reason, the AG shall issue the +CIEV result code, with the value indicating (callsetup=0), to notify the HF of this condition (see Section 4.15.2).
- If the AG supports the “Three-way calling” feature and if a call is already ongoing in the AG, performing this procedure shall result in a new call being placed to a third party with the current ongoing call put on hold.

- For details on how to handle multiparty calls refer to Section 4.22.2. If there is no number stored for the memory location given by the HF, the AG shall respond with ERROR

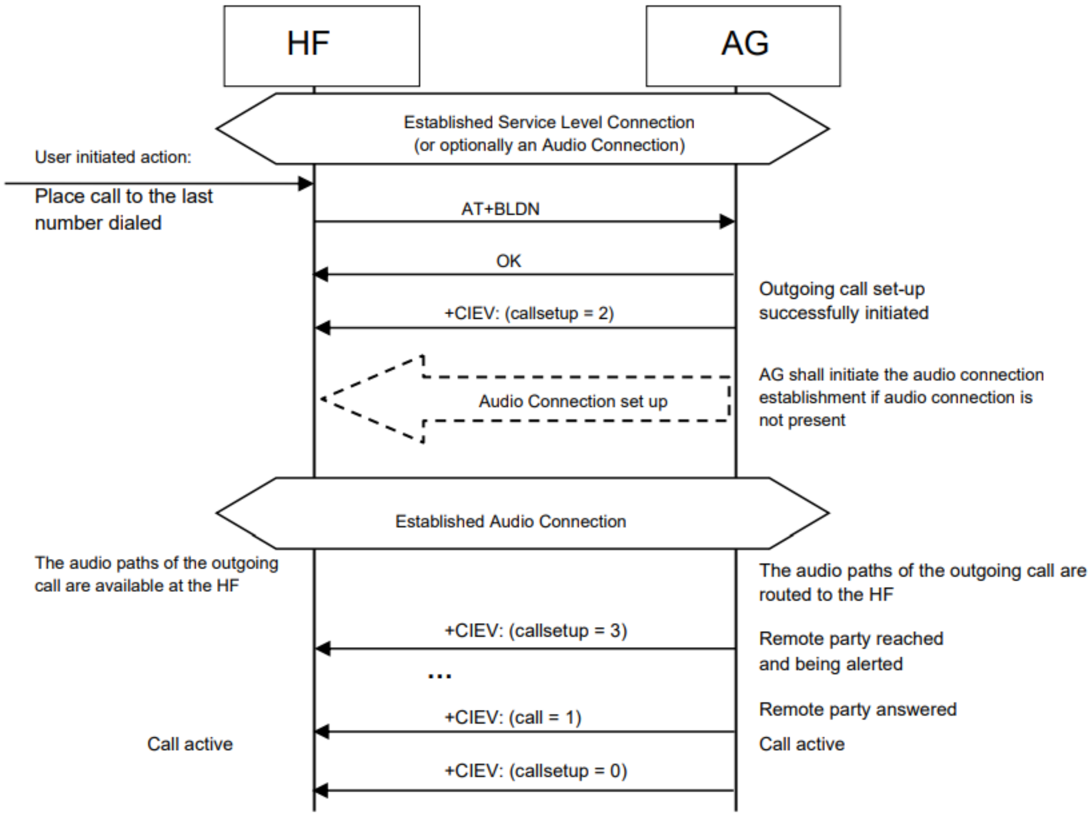


Figure 4.36: Place an outgoing voice call with the last number dialed

[redacted]

4.21 Call Waiting Notification Activation

- The HF may issue the AT+CCWA command to enable the “Call Waiting notification” function in the AG.
- Once the “Call Waiting notification” is enabled, the AG shall send the corresponding +CCWA unsolicited result code to the HF whenever an incoming call is waiting during an ongoing call.
- It is always assumed that the “call waiting” service is already active in the network.

- Once the HF issues the AT+CCWA command, the AG shall respond with OK.
- It shall then keep the “Call Waiting notification” enabled until either the AT+CCWA command is issued to disable “Call Waiting notification,” or the current Service Level Connection between the AG and the HF is dropped for any reason.
- See Section 4.34 for more information on the AT+CCWA command.
- As a precondition for this procedure, an ongoing Service Level Connection between the AG and the HF shall exist.
- If this connection does not exist, the HF shall autonomously establish the Service Level Connection using the proper procedure as described in Section 4.2.

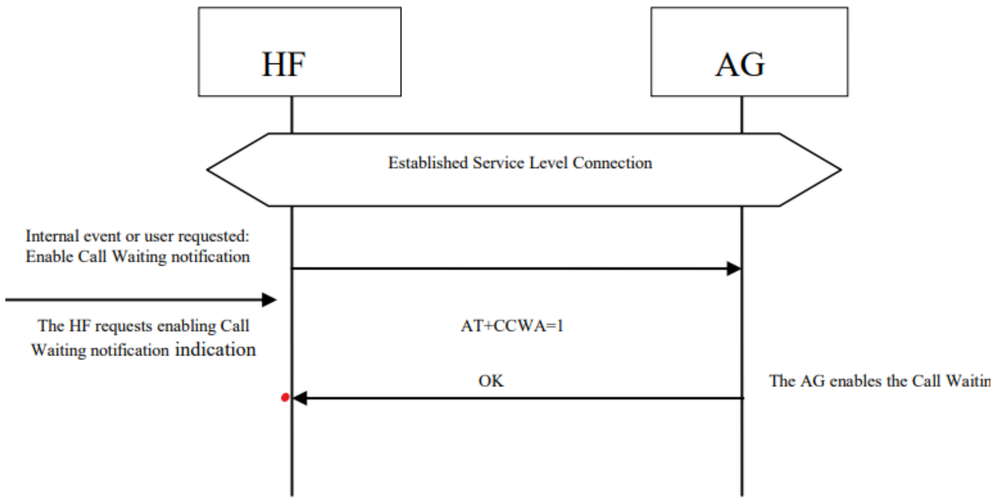


Figure 4.37: Activation of Call waiting notification

[redacted]

4.22 Three-Way Call Handling

Proper management of several concurrent calls shall be accomplished by performing the procedures described in [2] but with some limitations stated in this specification. For more details, refer to Section 4.34. The HF device cannot always assume that the "call hold and/or multiparty" services are available in the network. If the AG determines that a requested action by the HF device cannot be performed due to the inability of the network to support that feature or lack of subscriber subscription, the AG shall return a +CME error. There are two +CME ERROR codes that are used to indicate network related failure reasons to the HF: • 30 - No Network Service. Indicates that an AT+CHLD command cannot be implemented due to network limitations. • 31 - Network Timeout. Indicates that an AT+CHLD command cannot be implemented due to network problems. In general, when the user deals with multiple concurrent calls, the HF shall issue the corresponding AT+CHLD command as a result of user actions. This command allows the control of multiple concurrent calls and provides means for holding calls, releasing calls, switching between two calls, and adding a call to a multiparty conference. When this feature is supported, the HF and AG are only mandated to implement the "basic Three-Way calling" commands AT+CHLD = 1 and 2. This section covers two cases. In one case, the third party call is received in the AG, and notification is sent to the HF via a Call Waiting notification. In the second case, the third party call is placed from the HF. See Section 4.34 for more information on the AT+CHLD command.

The following preconditions apply for these procedures: • As a precondition for this procedure, an ongoing Service Level Connection between the AG and the HF shall exist. If this connection does not exist, the initiator of the procedure shall autonomously establish the Service Level Connection using the proper procedure as described in Section 4.2

Three-Way Calling—Call Waiting Notification

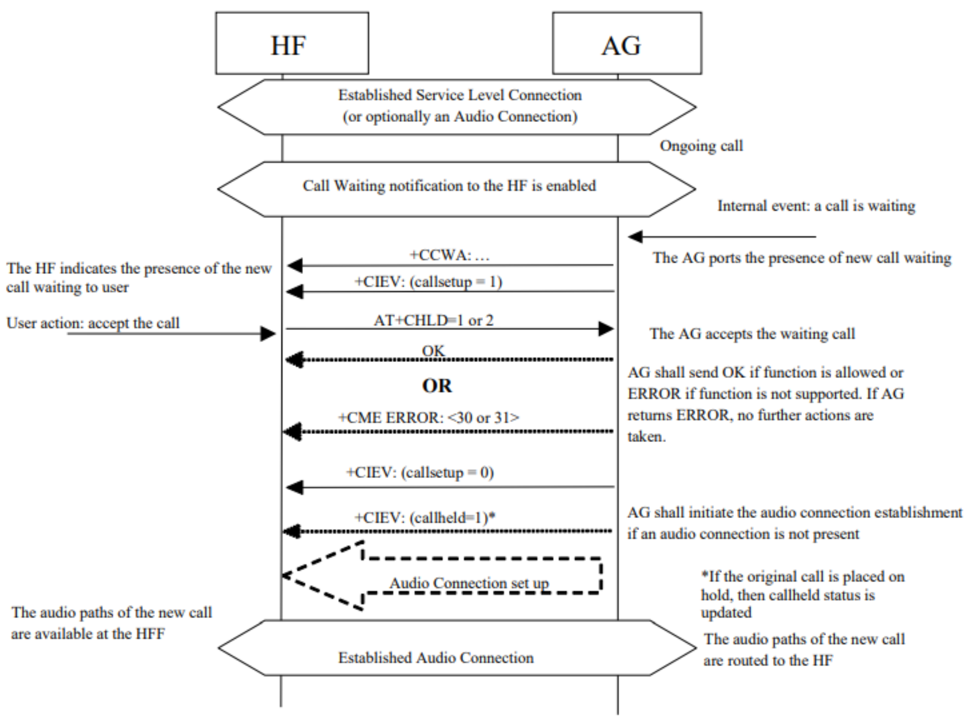


Figure 4.38: Typical Call Waiting indication followed by HF possible response

In addition to the two previously stated preconditions, the Call Waiting notification to the HF shall already be enabled in the AG (that is, the procedure stated in Section 4.21 has been performed).

If the AG receives a third party call, it shall send the call waiting notification +CCWA and +CIEV result code, with the value indicating (callsetup=1), to the HF. If the user rejects the call at the HF, the HF shall send the AT+CHLD command with parameter 0 to the AG. The AG shall then reject the call and respond with OK, and issue the +CIEV result code with the value indicating (callsetup=0).

If the user accepts the call at the HF, the HF shall send the AT+CHLD with parameter 1 or 2 to the AG. The HF cannot cause the waiting call to be added as a conference call via a single AT+CHLD command; but if this is desired the HF can achieve this by first issuing an AT+CHLD=2 command, and then issuing an AT+CHLD=3 command. The AG shall then accept the waiting call and respond with OK, and issue the +CIEV result code with the value indicating (callsetup=0). If the HF elects to send AT+CHLD=2 (placing the original call on hold), then the AG shall send the +CIEV result code with the value indicating a held call (callheld=1). Optionally, the HF may then use the AT+CHLD command, in order to change the status of the held and active calls. If the normal incoming call procedure is interrupted for any reason, the AG shall issue the +CIEV result code, with the value indicating (callsetup=0), to notify the HF of this condition (see Section 4.14.2).

Three-Way Calls – Third Party Call Placed from the HF

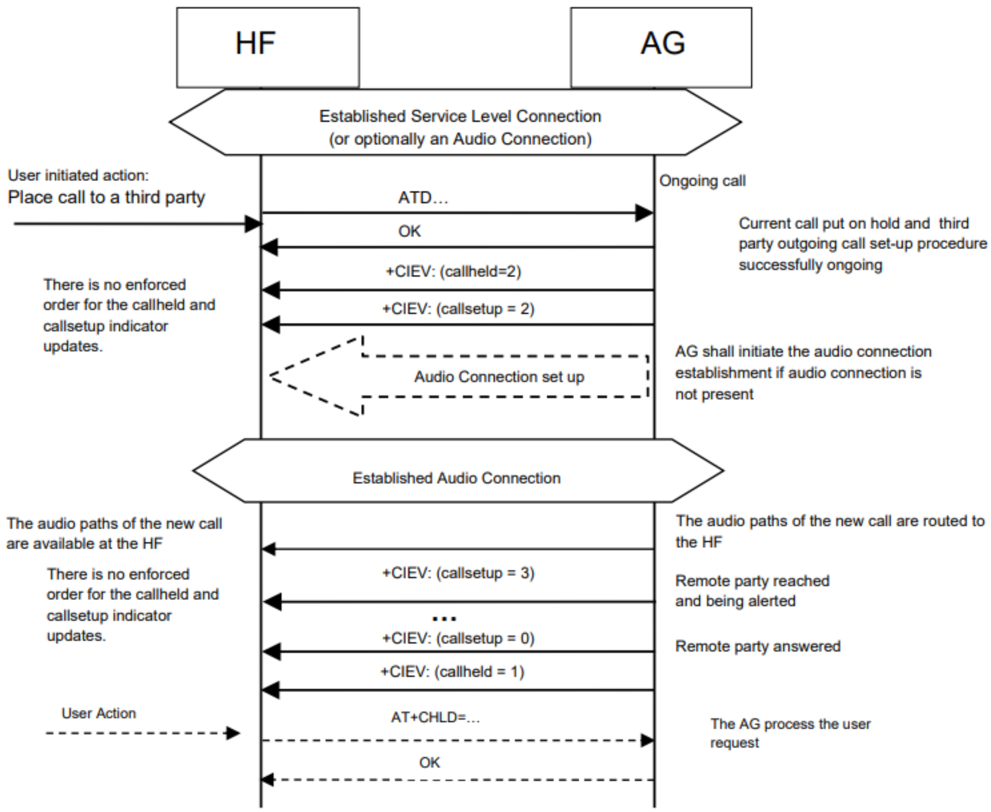


Figure 4.39: Three-way call handling when the third party call is placed from the HF

If a third party call is placed from the HF using the ATD command, the AG shall send the OK indication and two +CIEV result codes, one with the value indicating (callsetup=2), and one with the value indicating (callheld=2) to the HF. It is permissible for the AG to send these two +CIEV result codes in either order as the timing of events in the AG may differ between implementations and network types. If the remote party is reached and alerted, the AG shall issue the +CIEV result code with the value indicating (callsetup=3). Ifthe wireless network does not provide the AG of an indication of alerting the remote party, the AG may not send this indication. If the remote party answers the call, the AG shall issue the +CIEV result codes with the values indicating (callsetup=0) and (callheld=1). As before, there is no enforced order to these two +CIEV result codes. Optionally, the HF may then use the AT+CHLD command in order to change the status of the held and active calls. If the AT+CHLD command results in the change in a held call status the AG shall provide the status indication using the +CIEV result code with the value indicating the call held status (callheld=). If the normal outgoing call procedure is interrupted for any reason, the AG shall issue the +CIEV result code, with the value indicating (callsetup=0), to notify the HF of this condition (see Section 4.15.2). The AG shall then update the callheld status to indicate the change of status of the original (held) call based upon one of the two following scenarios: • The AG may choose to leave the original call on hold. In this case, the AG shall issue the +CIEV result code with the value indicating (call held=2). • Alternatively the AG may autonomously retrieve the held call, thus changing the status and shall send the +CIEV indicator (call held=0). In either case, the +CIEV response code (call=1) shall remain unchanged.

[redacted]

4.23 Calling Line Identification (CLI) Notification

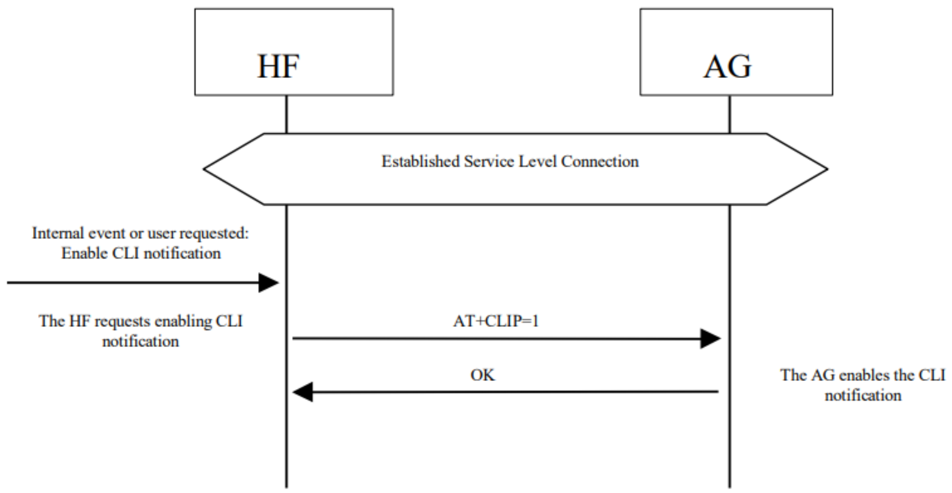


Figure 4.40: Activation of CLI notification

- The HF may issue the AT+CLIP command to enable the “Calling Line Identification notification” function in the AG.
- If the calling subscriber number information is available from the network, the AG shall issue the +CLIP unsolicited result code just after every RING indication when the HF is alerted in an incoming call. See Section 4.13 for more details.
- Once the HF issues the AT+CLIP command, the AG shall respond with OK. The AG shall then keep the “Calling Line Identification notification” enabled until either the AT+CLIP command is issued by the HF to disable it, or the current Service Level Connection between the AG and the HF is dropped for any reason.
- See Section 4.34 for more information on the AT+CLIP command. As a precondition for this procedure, an ongoing Service Level Connection between the AG and the HF shall exist.
- If this connection does not exist, the HF shall autonomously establish the Service Level Connection using the proper procedure as described in Section 4.2.

[redacted]

4.24 The HF Requests Turning off the AG's EC and NR

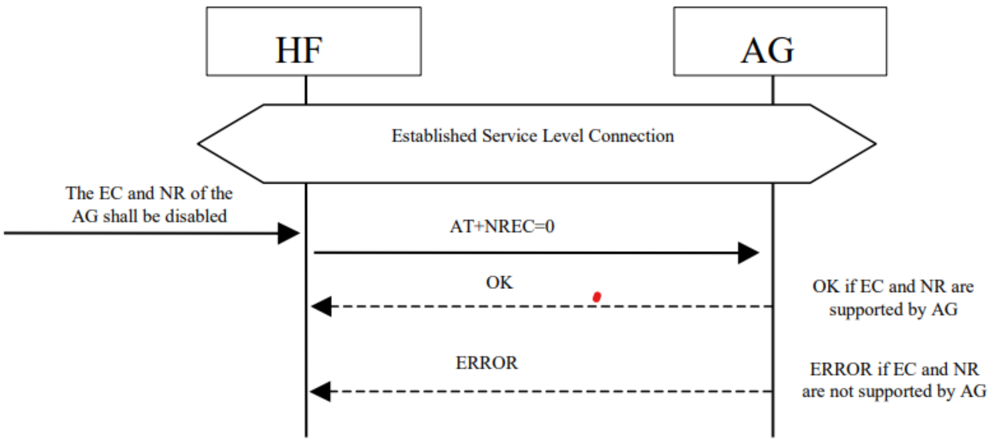


Figure 4.41: NR and EC functions available in the AG

The HF sends the AT+NREC command and AG confirms with either OK or ERROR indication.

- The HF may disable the echo canceling and noise reduction functions resident in the AG via the AT+NREC command.
- If the HF supports embedded EC and/or NR functions, it shall support the AT+NREC command as described in the procedures in this section.
- Moreover, if the HF has these functions enabled, it shall perform this procedure before any Audio Connection between the HF and the AG is established.
- By default, if the AG supports its own embedded echo canceling and/or noise reduction functions, it shall have them activated until the AT+NREC command is received.
- From then on, and until the current Service Level Connection between the AG and HF is dropped for any reason, the AG shall disable these functions every time an Audio Connection between the HF and the AG is used for audio routing.
- If the AG does not support any echo canceling and noise reduction functions, it shall respond with the ERROR indicator on reception of the AT+NREC command.
- See Section 4.34 for more information on the AT+NREC command.
- As a precondition for this procedure, an ongoing Service Level Connection between the AG and the HF shall exist.
- If this connection does not exist, the HF shall autonomously establish the Service Level Connection using the proper procedure as described in Section 4.2

[redacted]

4.25 Voice Recognition Activation / Enhanced Voice Recognition Activation

4.26 Enhanced Voice Recognition Activation with textual representation

representation

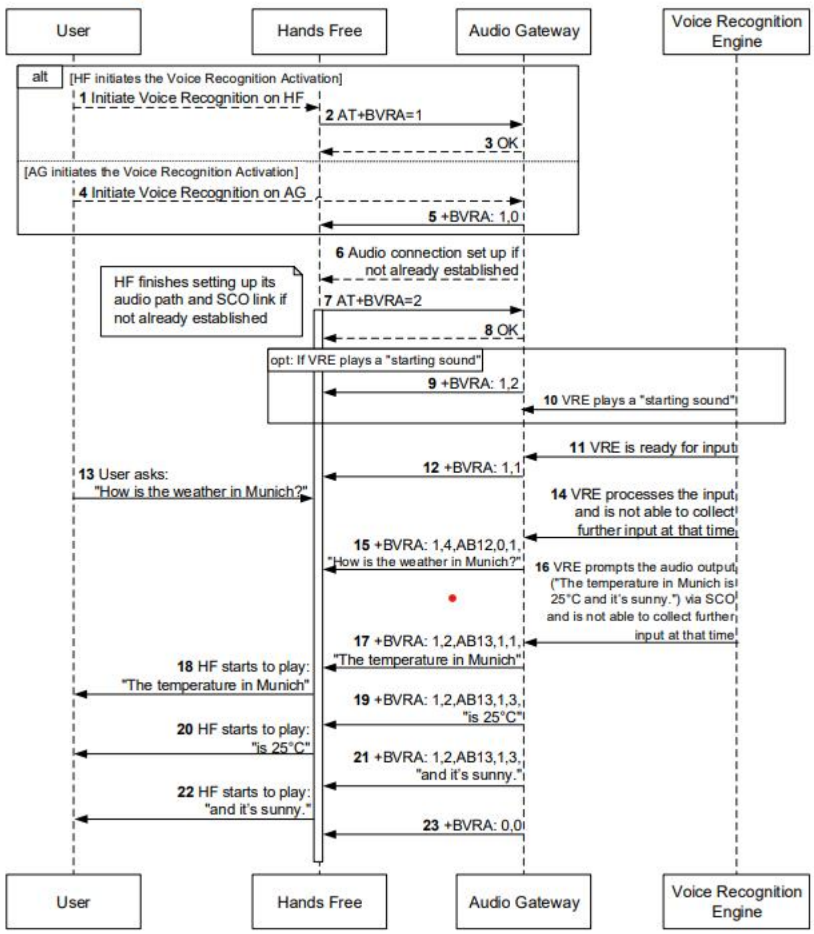
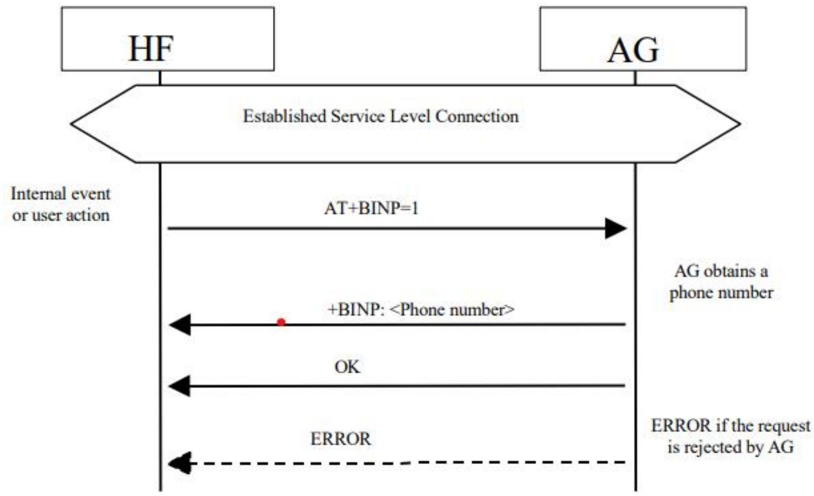


Figure 4.47: Sequence diagram with textual representation

[redacted]

4.27 Attach a Phone Number to a Voice Tag

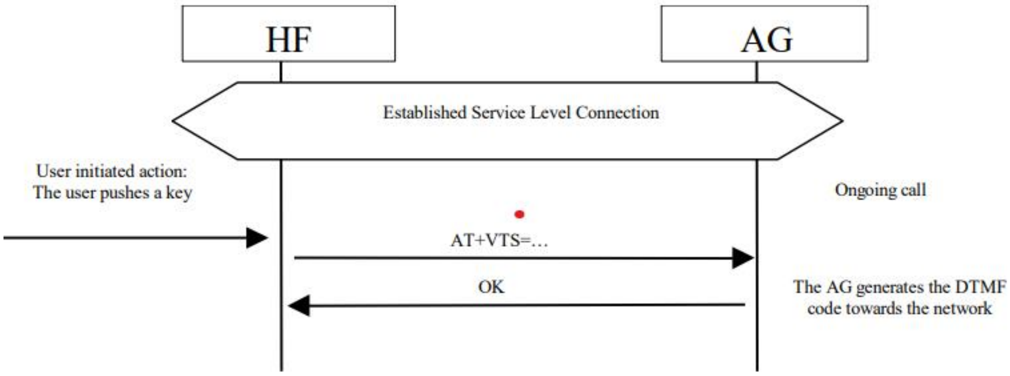


ure 4.48: Request phone number to the AG

- This procedure is applicable to HFs supporting internal voice recognition functionality.
- It provides a means to read numbers from the AG for the purpose of creating a unique voice tag and storing the number and its linked voice tag in the HF's memory.
- The HF may then use its internal Voice Recognition to dial the linked phone numbers when a voice tag is recognized by using the procedure "Place a call with the phone number supplied by the HF" described in Section 4.18.
- Upon an internal event or user action, the HF may request a phone number from the AG by issuing the AT+BINP=1 command.
- Depending on the current status of the AG, it may either accept or reject this request.
- If the AG accepts the request, it shall obtain a phone number and send the phone number back to the HF by issuing the +BINP response.
- If the AG rejects the request from the HF, it shall issue the ERROR result code to indicate this circumstance to the HF.

[redacted]

4.28 Transmit DTMF Codes



ɪure 4.49: Transmit DTMF code

- During an ongoing call, the HF transmits the AT+VTS command to instruct the AG to transmit a specific DTMF code to its network connection.
- See Section 4.34 for more information on the AT+VTS command.
- The following preconditions apply for this procedure:
- • An ongoing Service Level Connection between the AG and the HF shall exist. If this connection does not exist, the HF shall autonomously establish the Service Level Connection using the proper procedure as described in Section 4.2.
- • An ongoing call in the AG exists.

[redacted]

4.29 Remote Audio Volume Control

Audio Volume Control

- This procedure enables the user to modify the speaker volume and microphone gain of the HF from the AG.
- The AG may control the gain of the microphone and speaker of the HF by sending the unsolicited result codes +VGM and +VGS respectively.
- There is no limit in the amount and order of result codes.
- If the remote audio volume control feature is supported in the HF device, it shall support at least remote control of the speaker volume.
- As a precondition for this procedure, an ongoing Service Level Connection between the AG and the HF shall exist.

- If this connection does not exist, the AG shall autonomously establish the Service Level Connection using the proper procedure as described in Section 4.2. An audio connection is not a necessary precondition for this feature.

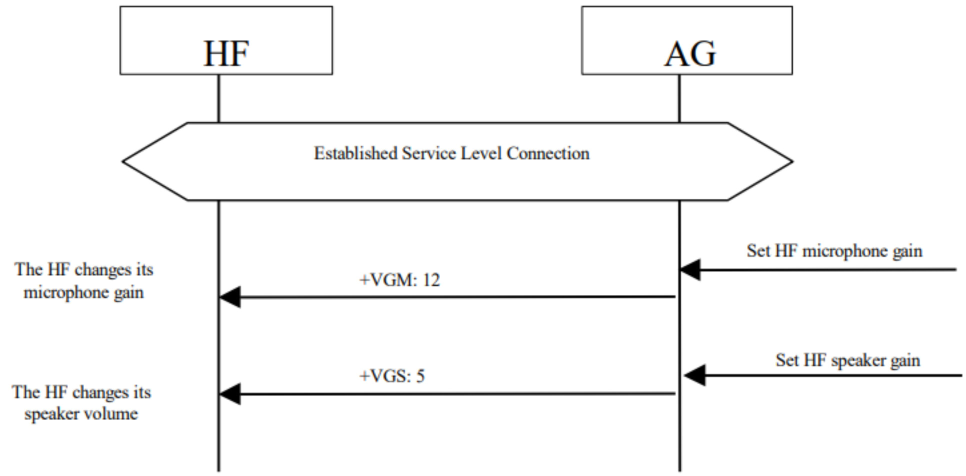


Figure 4.50: Typical example of audio volume control

- Both the speaker and microphone gains are represented as parameter to the +VGS and +VGM, on a scale from 0 to 15.
- The values are absolute values, and relate to a particular (implementation dependent) volume level controlled by the HF.
- See Section 4.34 for more information on these commands and unsolicited result codes.

- This procedure allows the HF to inform the AG of the current gain settings corresponding to the HF's speaker volume and microphone gain.
- On Service Level Connection establishment, the HF shall always inform the AG of its current gain settings by using the AT commands AT+VGS and AT+VGM.
- If local means are implemented on the HF to control the gain settings, the HF shall also use the AT commands AT+VGS and AT+VGM to permanently update the AG of changes in these gain settings.
- In all cases, the gain settings shall be kept stored, at both sides, for the duration of the current Service Level Connection. Moreover, if the Service Level Connection is released as a consequence of an HF initiated "Audio Connection transfer towards the AG" as

stated in Section 4.17, the HF shall also keep the gain settings and re-store them when the Service Level Connection is re-established.

- The HF shall support speaker gain synchronization when it supports remote speaker gain control.
- The HF shall support microphone gain synchronization when it supports remote microphone gain control.
- As a precondition for this procedure, an ongoing Service Level Connection between the AG and the HF shall exist.
- If this connection does not exist, the HF shall autonomously establish the Service Level Connection using the proper procedure as described in Section 4.2.

Volume Level Synchronization

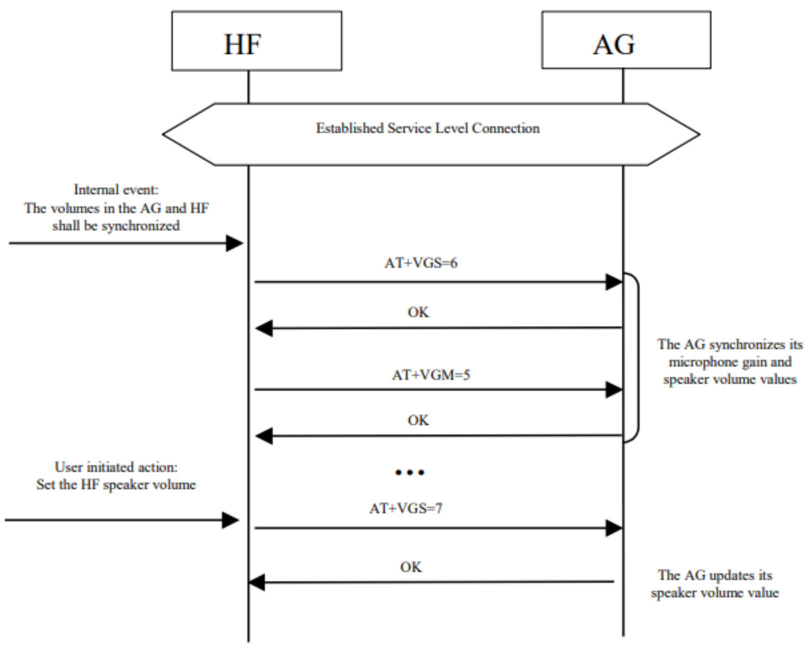


Figure 4.51: Typical example of volume level synchronization

See Section 4.34 for more information on these commands and unsolicited result codes.

- This procedure allows the HF to inform the AG of the current gain settings corresponding to the HF's speaker volume and microphone gain.
- On Service Level Connection establishment, the HF shall always inform the AG of its current gain settings by using the AT commands AT+VGS and AT+VGM.

- If local means are implemented on the HF to control the gain settings, the HF shall also use the AT commands AT+VGS and AT+VGM to permanently update the AG of changes in these gain settings.
- In all cases, the gain settings shall be kept stored, at both sides, for the duration of the current Service Level Connection.
- Moreover, if the Service Level Connection is released as a consequence of an HF initiated "Audio Connection transfer towards the AG" as stated in Section 4.17, the HF shall also keep the gain settings and re-store them when the Service Level Connection is re-established.
- The HF shall support speaker gain synchronization when it supports remote speaker gain control.
- The HF shall support microphone gain synchronization when it supports remote microphone gain control.
- As a precondition for this procedure, an ongoing Service Level Connection between the AG and the HF shall exist.
- If this connection does not exist, the HF shall autonomously establish the Service Level Connection using the proper procedure as described in Section 4.2.

[redacted]

4.30 Response and Hold

1. Query Response and Hold Status

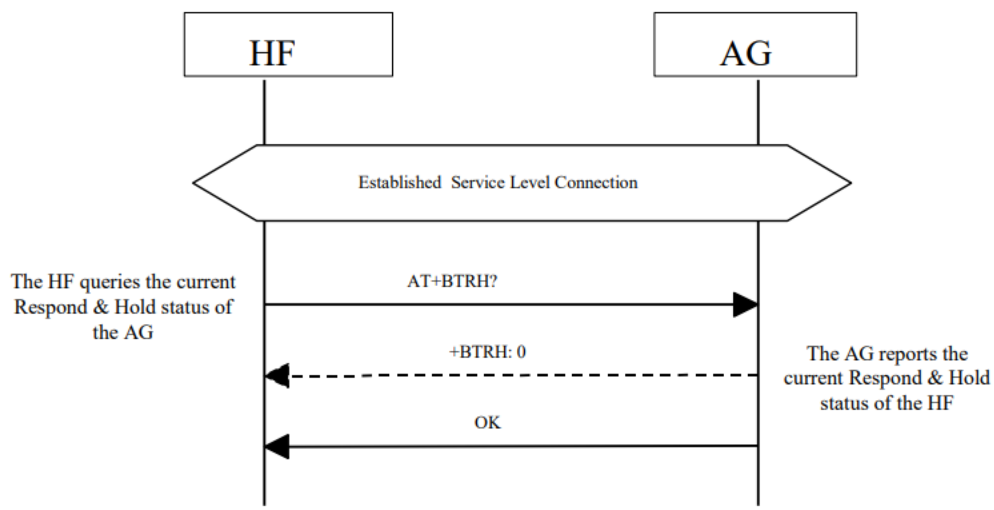


Figure 4.52: Query Response and Hold State of AG

2. Put an Incoming Call on Hold from HF

4.30.2 Put an Incoming Call on Hold from HF

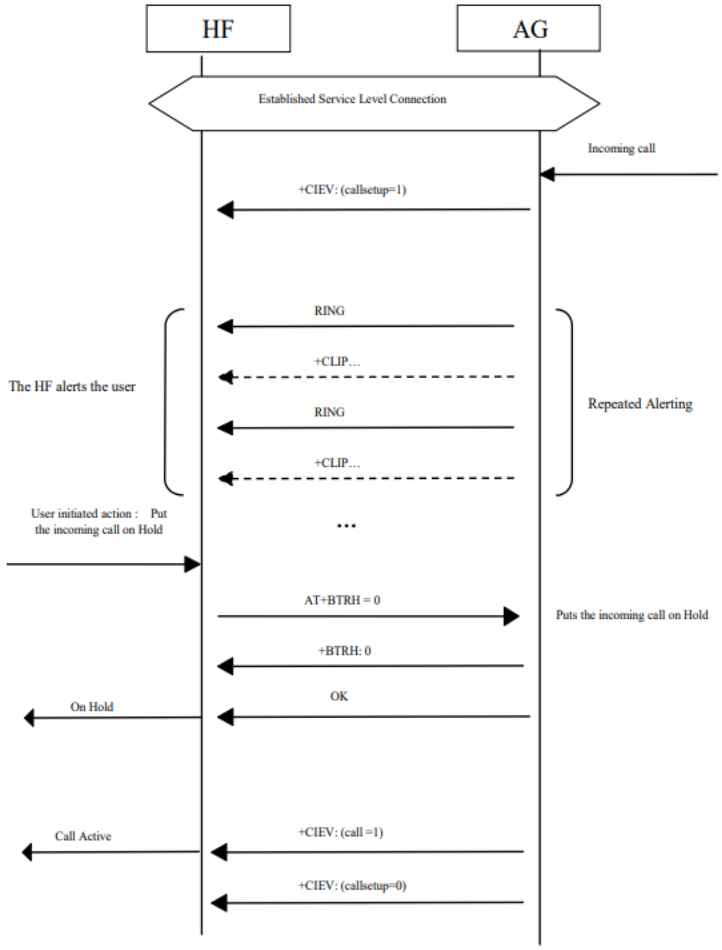


Figure 4.53: Put an incoming call on Hold from HF

3. Put an Incoming Call on Hold from AG

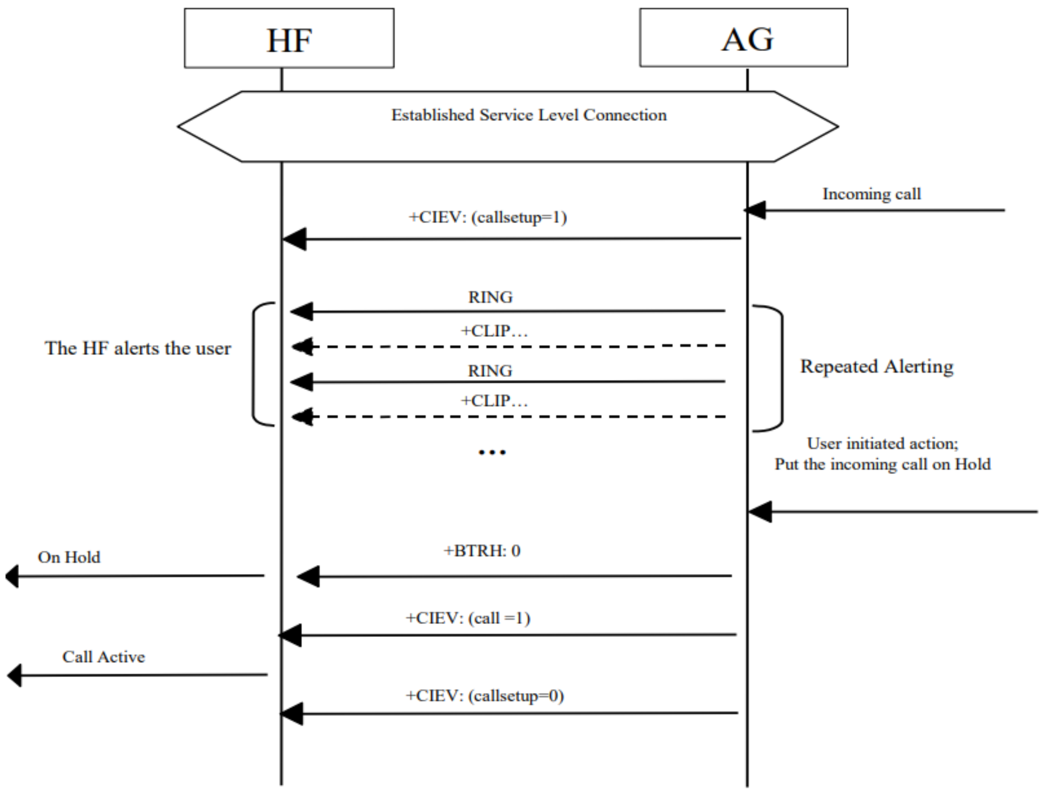


Figure 4.54: Put an incoming call on Hold from AG

4. Accept a Held Incoming Call from HF

- An incoming call was put on hold.

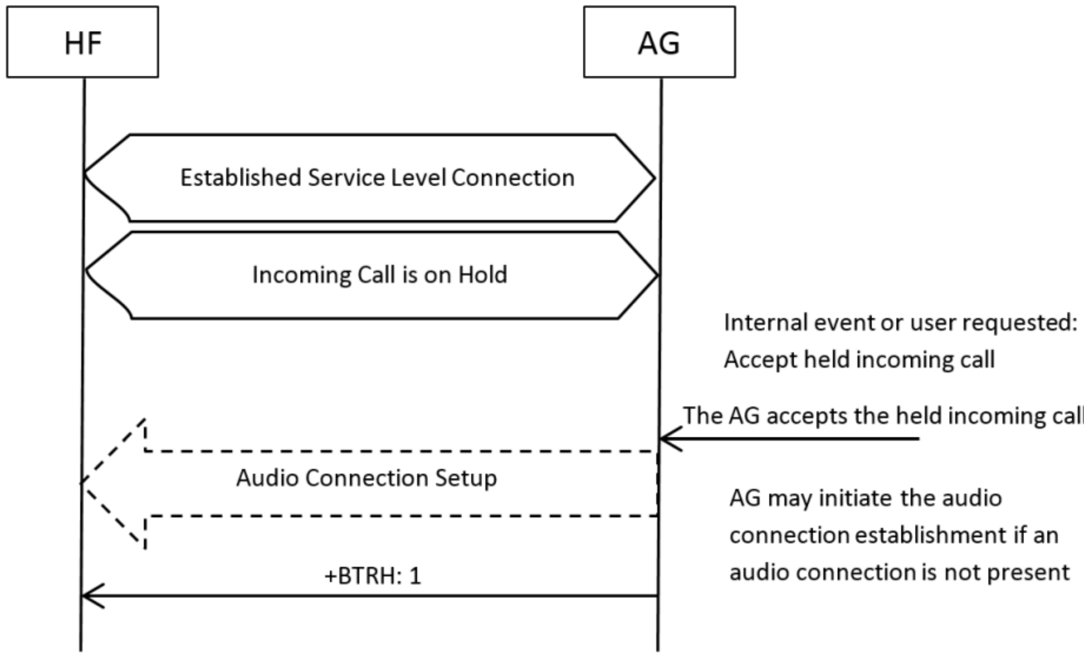


Figure 4.56: Accept a held incoming call from AG

5. Accept a Held Incoming Call from AG

- An incoming call was put on hold.

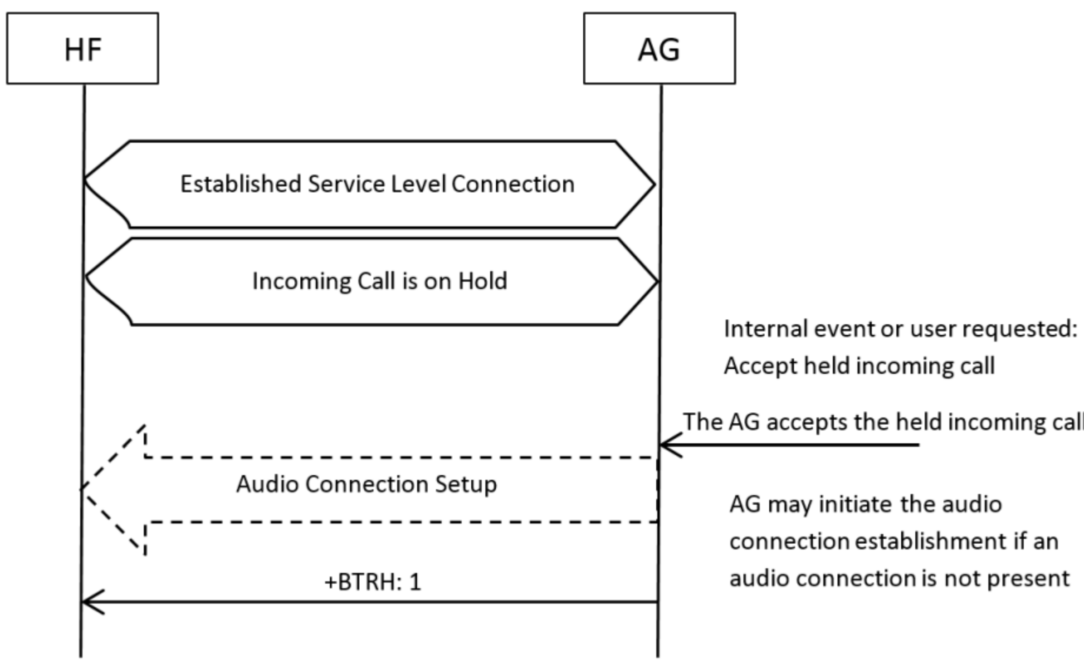


Figure 4.56: Accept a held incoming call from AG

6. Reject a Held Incoming Call from HF

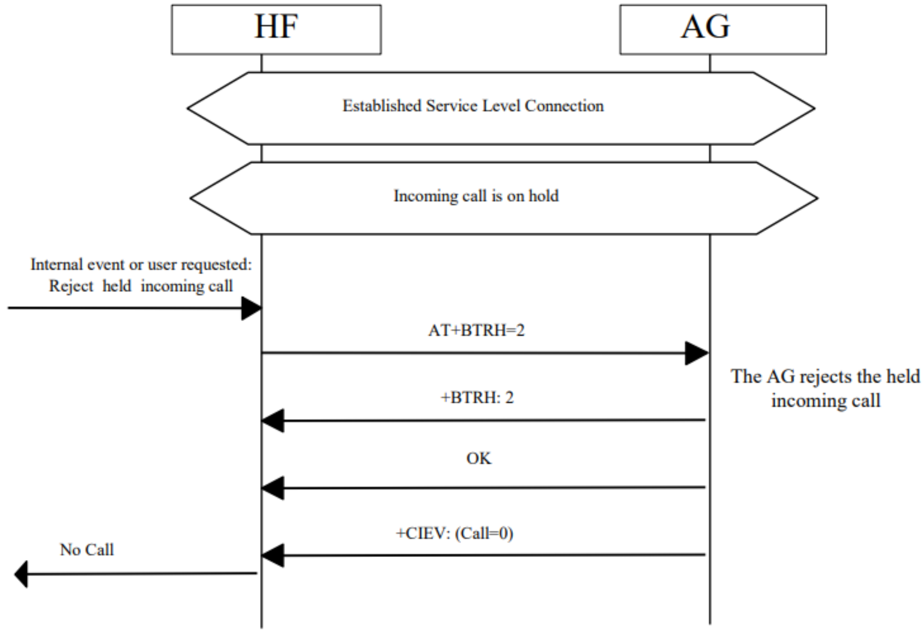


Figure 4.57: Reject a held incoming call from HF

7. Reject a Held Incoming Call from AG

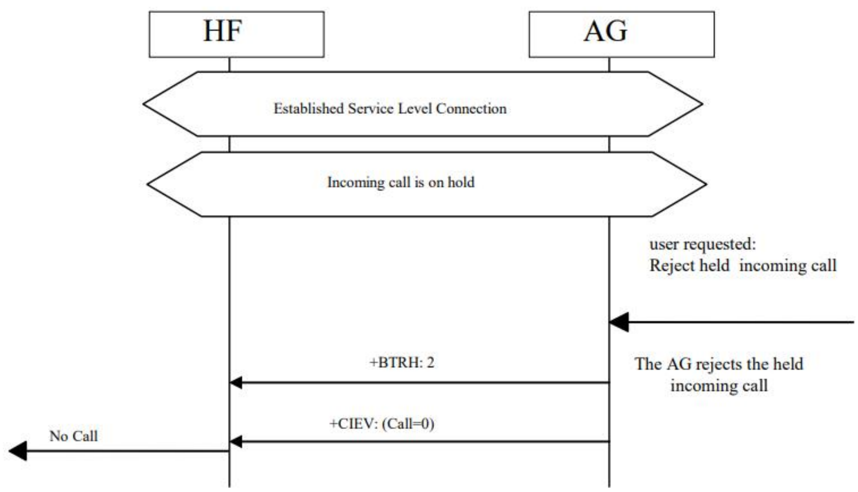


Figure 4.58: Reject a held incoming call from AG

8. Held Incoming Call Terminated by Caller

- An incoming call was put on hold.

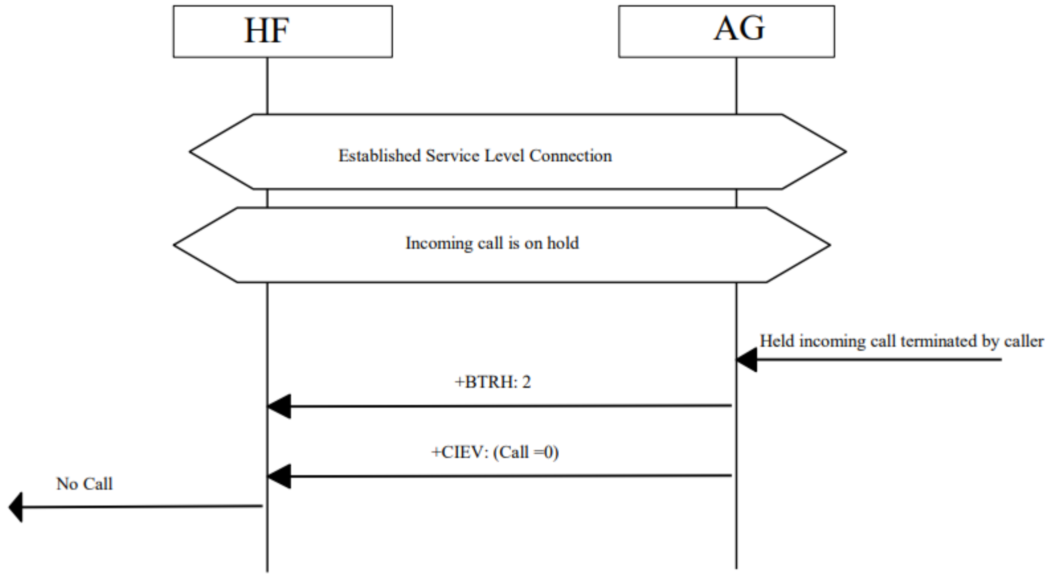


Figure 4.59: Held incoming call terminated by caller

[redacted]

4.31 Subscriber Number Information

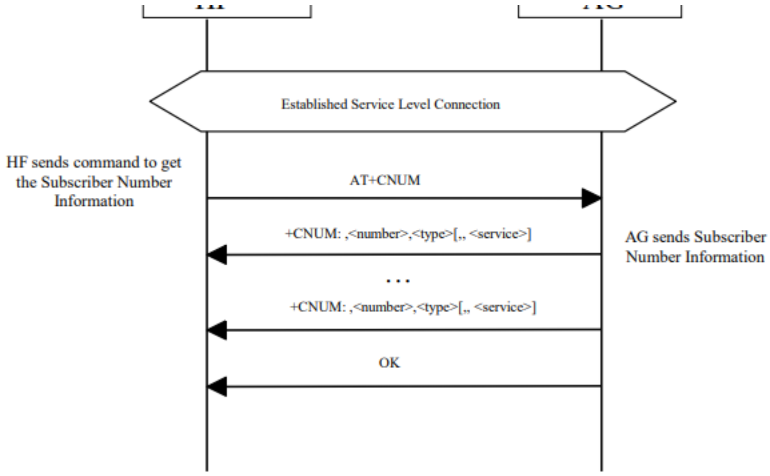


Figure 4.60: Query Subscriber Number Information of AG

This procedure illustrates AG response to the query of an empty subscriber number.

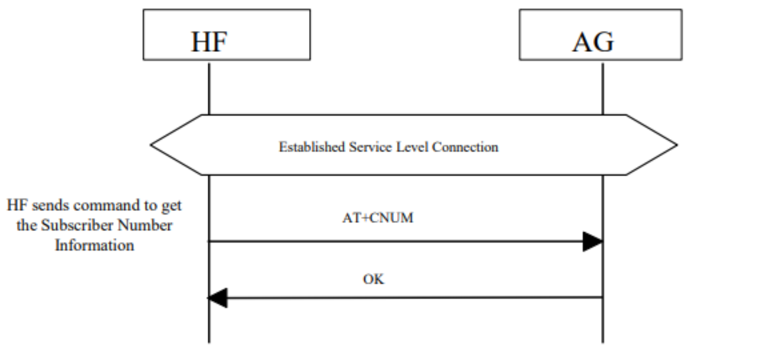


Figure 4.61: Empty Subscriber Number Information from AG

The following precondition applies for this procedure: • An ongoing Service Level Connection between the HF and AG shall exist. If this connection does not exist, the HF shall establish a connection using the "Service Level Connection set up" procedure described in Section 4.2. • The HF shall send the AT+CNUM command to query the AG subscriber number information. • If the subscriber number information is available, the AG shall respond with the +CNUM response. If multiple numbers are available, the AG shall send a separate +CNUM response for each available number. • The AG shall signal the completion of the AT+CNUM action command with an OK response. The OK will follow zero or more occurrences of the +CNUM response. (See Figure 4.57 and Figure 4.58).

[redacted]

4.32 Enhanced Call Status Mechanisms

4.32.1 Query List of Current Calls in AG

The HF shall execute this procedure to query the list of current calls in AG.

The following precondition applies for this procedure:

- A Service Level Connection exists between the AG and HF devices. If no current Service Level Connection exists, the HF shall first initiate one.

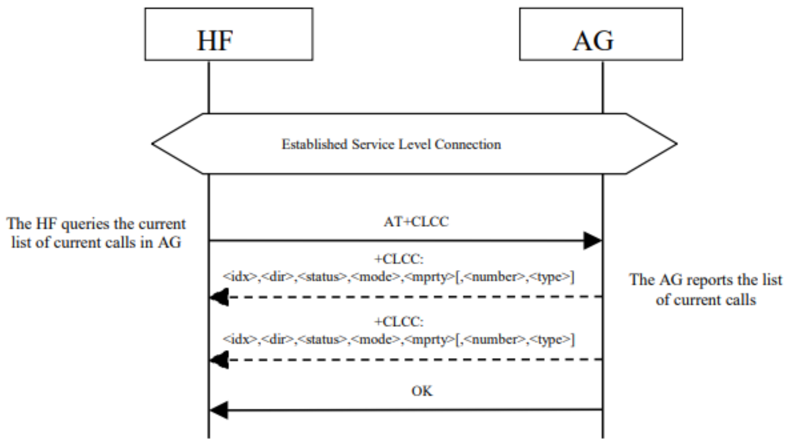


Figure 4.62: Query List of Current Calls

- HF shall find out the list of current calls in AG by sending the AT+CLCC command.
- If the command succeeds and if there is an outgoing (Mobile Originated) or an incoming (Mobile Terminated) call in AG, AG shall send a +CLCC response with appropriate parameters filled in to HF.
- If there are no calls available, no +CLCC response is sent to HF.
- The AG shall always send OK response to HF.

[redacted]

4.33 Enhanced Call Control Mechanisms

4.33.1 Release Specified Call Index

The HF shall execute this procedure to release a specific call in the AG.

The following precondition applies for this procedure:

- A Service Level Connection exists between the AG and HF devices. If no current Service Level Connection exists, the HF shall first initiate one.

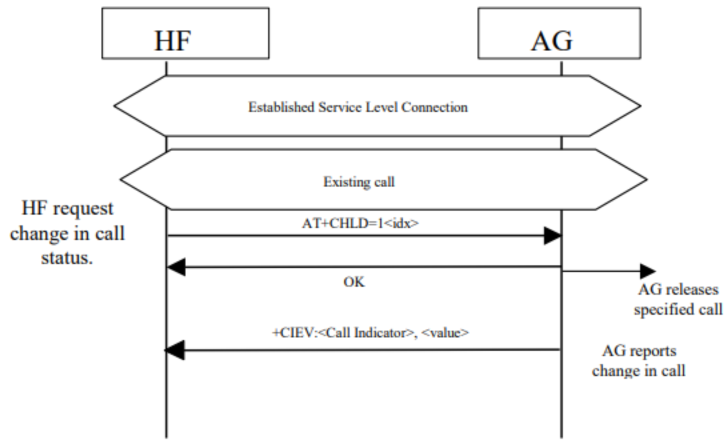


Figure 4.63: Release Specified Active Call

- The HF shall send the AT+CHLD=1<idx> command to release a specific active call.
- The AG shall release the specified call.

If there is a change in the call status, the AG shall report the change in call status. If there is a change in the held call status, the AG shall report the change in call held status.

If the index (<idx>) is not valid, the AG shall report the proper error code.

4.33.2 Private Consultation Mode

The HF shall execute this procedure to place all parties of a multiparty call on hold with the exception of the specified call.

The following precondition applies for this procedure:

- A Service Level Connection exists between the AG and HF devices. If no current Service Level Connection exists, the HF shall first initiate one.
- Existing multiparty call is active in AG.

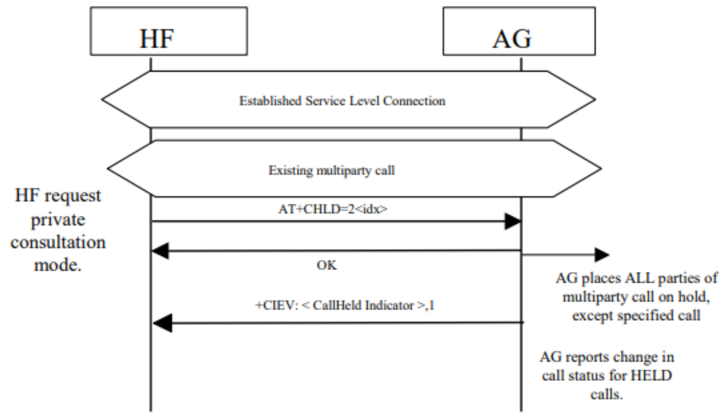


Figure 4.64: Request Private Consultation Mode

- HF shall send the AT+CHLD=2<idx> command to request private consultation mode.
- AG shall place all other parties of call on hold.
- AG shall report the change in status of the held parties.
- If the index (<idx>) is not valid, the AG shall respond with the proper error code.

[redacted]

4.34 AT Command and Results Codes

[redacted]

4.35 Indicators Activation and Deactivation

[redacted]

4.36 HF Indicators

[redacted]

EXPLAIN ABOUT CODEC IN HFP

Table 3.3 shows the codec requirements for this profile.

	Codec	Support in HF	Support in AG
1.	CVSD	M	M
2.	mSBC	C.1	C.1
C.1: Mandatory if Wide Band Speech is supported, or excluded otherwise			

Table 3.3: Requirements on supported codecs

Table 3.4 shows a summary of the mapping of codec requirements on link features for this profile.

	Feature	Support in HF	Support in AG
1.	D0 – CVSD on SCO link (HV1)	M	M
2.	D1 – CVSD on SCO link (HV3)	M	M
3.	S1 – CVSD eSCO link (EV3)	M	M
4.	S2 – CVSD on EDR eSCO link (2-EV3)	M	M
5.	S3 – CVSD on EDR eSCO link (2-EV3)	M	M
6.	S4 – CVSD on EDR eSCO link (2-EV3)	M	M
7.	T1 – mSBC on eSCO link (EV3)	C1	C1
8.	T2 – mSBC on EDR eSCO link (2-EV3)	C1	C1
C1: Mandatory if Wide Band Speech is supported, excluded otherwise			

Table 3.4: Codec to link feature mapping

What is CVSD codec?

Continuously Variable Slope Delta modulation, or CVSD, is **a codec which was designed for use in military wireless communication systems**. This codec supports various bitrates, depending on the sampling rate.

CVSD is used in both commercial and military communications

Features List

- Full Duplex codec
- Functions are C-callable
- Designed for Multi-channel operation

What is the best codec for audio?

AAC

Best Audio Codec for Online Video and Live Streaming

Here's why we think **AAC** is the ideal codec for video-on-demand and streaming content: AAC is supported across a wide range of devices, including Android, MAC, iOS, Windows, and more. This codec provides better sound quality than its counterparts like MP3.

What is CVSD in Bluetooth?

This paper presents the **continuously variable slope delta** (CVSD) speech coding method which is widely used in Bluetooth applications. It is a nonlinear sampled data feedback system which accepts a band limited analog signal and encodes it into binary form for transmission through a digital channel.

SBC audio codec

Following are the features of SBC bluetooth audio codec.

• It is short form of Sub Band Coding.

• It provides reasonably good audio quality at medium bit rates while keeping low computational complexity.

• It is the only mandatory codec used in A2DP profile specified by Bluetooth SIG.

• It is digital audio encoder and decoder.

• It is used to transfer data to bluetooth audio output devices such as headphones, speakers etc.

• SBC supports both mono and stereo streams.

• It uses sampling frequencies up to 48 kHz

• It supports bit rates upto 198 Kbps (mono stream) and upto 345 Kbps (stereo stream).

• It uses 4 sub bands or 8 sub bands. It uses adaptive bit allocation algorithm along with adaptive block PCM quantizers.

• It is used by A2DP profile version 1.3.

mSBC audio codec

Following are the features of mSBC (modified SBC) bluetooth audio codec.

• It is 16KHz monaural configuration of SBC codec.

• It is used by Hands free profile v1.6.

• It is also known as WBS (Wide Band Speech).

• It has been developed to support mobile networks with HD voice.

What is wide band speech?

Wideband audio, also known as wideband voice or HD voice, is **high definition voice quality for telephony audio**, contrasted with standard digital telephony "toll quality". It extends the frequency range of audio signals transmitted over telephone lines, resulting in higher quality speech

High Definition

1 AT+BRSF= Bluetooth supported features(HF sends AT+BRSF command to AG and AG notify the supported features in HF)

2 AT+BAC= Bluetooth available codecs(in HFP there are 2 available codecs they are 1.mSBC 2.CVSD)

3 AT+CIND=? call indicators (test command,

AT+CIND? Read command

4 AT+CMER call merge

5 AT+BIA Bluetooth indicaton activation

6 AT+BINP bluetooth input

7 AT+BLDN Bluetooth last dialed number

8 AT+BVRA Bluetooth Voice Recognition Activation

9 AT+NREC Noise Reduction and Echo Canceling

10 AT+VGM Gain of Microphone

11 AT+VGS Gain of Speaker

12 +BSIR Bluetooth Setting of In-band Ring tone

13 AT+BTRH Bluetooth Response and Hold Feature

14 AT+BCC Bluetooth Codec Connection

15 AT+BCS Bluetooth Codec Selection

16 AT+BIND Bluetooth indicator

17 AT+BIEV Bluetooth HF Indicators Feature

18 AT+CLCC List Current Calls command

19 AT+COPS used for network reading operator

20 AT+CLIP Calling Line Identification notification

21 +CIEV indicator events reporting

8 List of Acronyms and Abbreviations

Abbreviation or Acronym	Meaning
AG	Audio Gateway
AT	Attention
BTR	Bluetooth Reference Point
BVRA	Bluetooth Voice Recognition Activation
CLI	Calling Line Identification
CODEC	COder DECoder
CVSD	Continuously Variable Slope Delta modulation
DTMF	Dual Tone Multi-Frequency
EC	Echo Cancellation
EDR	Enhanced Data Rate
eSCO	Extended Synchronous Connection Oriented
GAP	Generic Access Profile
GSM	Global System for Mobile communication
HF	Hands-Free unit
L2CAP	Logical Link Control and Adaptation Protocol
LMP	Link Manager Protocol
mSBC	Modified Sub Band Codec
NR	Noise Reduction
OSI	Open System Interconnection
PLC	Packet Loss Concealment
PIN	Personal Identification Number
POI	Point Of Interconnection
RFCOMM	Serial port transport protocol over L2CAP
SBC	Sub Band Codec
SCO	Synchronous Connection Oriented
SDP	Service Discovery Protocol
UI	User Interface
UUID	Universally Unique Identifier
VR	Voice Recognition
WBS	Wide Band Speech

AT Command HF→ AG	Function	Result Codes HF ← AG
ATA	Answer call	
ATD	Dial call	
AT+CCWA	Call Waiting	+CCWA
AT+CHLD	Call hold & multiparty	
AT+CHUP	Hang up (and reject)	
AT+CIND	Call indicators	+CIND
AT+CLIP	Calling line identification	+CLIP
AT+CMER	Event reporting activation	+CIEV
AT+BLDN	Last dialed number	
AT+BVRA	Voice recognition	+BVRA
AT+BRSF	Retrieve supported features	+BRSF
AT+NREC	Enable/disable NR/EC	
AT+VGM	Set gain of microphone	+VGM
AT+VGS	Set gain of speaker	+VGS
AT+BTRH	Response & hold	+BTRH

- (Bluetooth Retrieve Supported Features = BRSF
- +BCS (Bluetooth Codec Selection)
- +BTRH (Bluetooth Response and Hold Feature)
- +BSIR (Bluetooth Setting of In-band Ring tone)
- BAC = Bluetooth available codes
- CIND = Current indicator control
- AT = Attention
- +BIND (Bluetooth HF Indicators Feature)
- CMER Mobile Termination Event Reporting